IBM Z NetView
Version 6 Release 3

Installation: Configuring Additional Components

GC27-2851-05

Note

Before using this information and the product it supports, read the information in "Notices" on page 243.

This edition applies to version 6, release 3 of IBM Z NetView (product number 5697-NV6) and to all subsequent versions, releases, and modifications until otherwise indicated in new editions.

This edition replaces GC27-2851-04.

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About this publication

The IBM Z® NetView® product provides advanced capabilities that you can use to maintain the highest degree of availability of your complex, multi-platform, multi-vendor networks and systems from a single point of control. This publication, *IBM Z NetView Installation: Configuring Additional Components*, provides information about configuring additional functions beyond the base functions of the NetView program for their enterprise.

Intended audience

This publication is for system programmers who install additional components of the NetView program, beyond the base program.

Publications

This section lists publications in the IBM Z NetView library and related documents. It also describes how to access NetView publications online and how to order NetView publications.

IBM Z NetView library

The following documents are available in the IBM Z NetView library:

- *Administration Reference*, SC27-2869, describes the NetView program definition statements required for system administration.
- *Application Programmer's Guide*, SC27-2870, describes the NetView program-to-program interface (PPI) and how to use the NetView application programming interfaces (APIs).
- *Automation Guide*, SC27-2846, describes how to use automated operations to improve system and network efficiency and operator productivity.
- *Command Reference Volume 1 (A-N)*, SC27-2847, and *Command Reference Volume 2 (O-Z)*, SC27-2848, describe the NetView commands, which can be used for network and system operation and in command lists and command procedures.
- *Installation: Configuring Additional Components*, GC27-2851, describes how to configure NetView functions beyond the base functions.
- *Installation: Configuring the NetView Enterprise Management Agent*, GC27-2853, describes how to install and configure the IBM Z NetView Enterprise Management Agent.
- *Installation: Getting Started*, GI11-9443, describes how to install and configure the base NetView program.
- *Installation: Migration Guide*, GC27-2854, describes the new functions that are provided by the current release of the NetView product and the migration of the base functions from a previous release.
- *IP Management*, SC27-2855, describes how to use the NetView product to manage IP networks.
- *Messages and Codes Volume 1 (AAU-DSI)*, GC27-2856, and *Messages and Codes Volume 2 (DUI-IHS)*, GC27-2857, describe the messages for the NetView product, the NetView abend codes, the sense codes that are included in NetView messages, and generic alert code points.
- *Programming: Pipes*, SC27-2859, describes how to use the NetView pipelines to customize a NetView installation.
- *Programming: REXX and the NetView Command List Language*, SC27-2861, describes how to write command lists for the NetView product using the Restructured Extended Executor language (REXX) or the NetView command list language.
- *Security Reference*, SC27-2863, describes how to implement authorization checking for the NetView environment.

- *Troubleshooting Guide*, GC27-2865, provides information about documenting, diagnosing, and solving problems that occur in the NetView product.
- *Tuning Guide*, SC27-2874, provides tuning information to help achieve certain performance goals for the NetView product and the network environment.
- *User's Guide: Automated Operations Network*, SC27-2866, describes how to use the NetView Automated Operations Network (AON) component, which provides event-driven network automation, to improve system and network efficiency. It also describes how to tailor and extend the automated operations capabilities of the AON component.
- *User's Guide: NetView*, SC27-2867, describes how to use the NetView product to manage complex, multivendor networks and systems from a single point.
- *User's Guide: NetView Enterprise Management Agent*, SC27-2876, describes how to use the NetView Enterprise Management Agent.
- *Using Tivoli System Automation for GDPS/PPRC HyperSwap Manager with NetView*, GI11-4704, provides information about the Tivoli® System Automation for GDPS®/PPRC HyperSwap® Manager with NetView feature, which supports the GDPS and Peer-to-Peer Remote Copy (PPRC) HyperSwap Manager services offering.
- *Licensed Program Specifications*, GC31-8848, provides the license information for the NetView product.
- *Program Directory for IBM Z NetView US English*, GI11-9444, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Japanese*, GI11-9445, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Enterprise Management Agent*, GI11-9446, contains information about the material and procedures that are associated with installing the IBM Z NetView Enterprise Management Agent.

The following books are archived:

- *Customization Guide*, SC27-2849, describes how to customize the NetView product and points to sources of related information.
- *Data Model Reference*, SC27-2850, provides information about the Graphic Monitor Facility host subsystem (GMFHS), SNA topology manager, and MultiSystem Manager data models.
- *Installation: Configuring Graphical Components*, GC27-2852, describes how to install and configure the NetView graphics components.
- *Programming: Assembler*, SC27-2858, describes how to write exit routines, command processors, and subtasks for the NetView product using assembler language.
- *Programming: PL/I and C*, SC27-2860, describes how to write command processors and installation exit routines for the NetView product using PL/I or C.
- *Resource Object Data Manager and GMFHS Programmer's Guide*, SC27-2862, describes the NetView Resource Object Data Manager (RODM), including how to define your non-SNA network to RODM and use RODM for network automation and for application programming.
- *SNA Topology Manager Implementation Guide*, SC27-2864, describes planning for and implementing the NetView SNA topology manager, which can be used to manage subarea, Advanced Peer-to-Peer Networking, and TN3270 resources.
- *User's Guide: NetView Management Console*, SC27-2868, provides information about the NetView management console interface of the NetView product.

Related publications

You can find additional product information on the IBM Z NetView web site at https://www.ibm.com/us-en/marketplace/ibm-tivoli-netview-for-zos.

For information about the NetView Bridge function, see *Tivoli NetView for OS/390® Bridge Implementation*, SC31-8238-03 (available only in the V1R4 library).

Terminology in this Library

The following terms are used in this library:

CNMCMD
For the CNMCMD member and the members that are included in it using the %INCLUDE statement

CNMSTYLE
For the CNMSTYLE member and the members that are included in it using the %INCLUDE statement

DSIOPF
For the DSIOPF member and the members that are included in it using the %INCLUDE statement

IBM® Tivoli Netcool®/OMNIbus
For either of these products:

- IBM Tivoli Netcool/OMNIbus
- IBM Tivoli OMNIbus and Network Manager

MVS™
For z/OS® operating systems

MVS element
For the base control program (BCP) element of the z/OS operating system

NetView
For the following products:

- IBM Z NetView version 6 release 3
- IBM Tivoli NetView for z/OS version 6 release 2 modification 1
- NetView releases that are no longer supported

PARMLIB
For SYS1.PARMLIB and other data sets in the concatenation sequence

VTAM®
For Communications Server - SNA Services

Unless otherwise indicated, topics to programs indicate the latest version and release of the programs. If only a version is indicated, the topic is to all releases within that version.

When a topic is made about using a personal computer or workstation, any programmable workstation can be used.

Using IBM Z NetView online help

The following types of IBM Z NetView mainframe online help are available, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions

Accessing publications online

IBM posts publications for this and all other products, as they become available and whenever they are updated, to the IBM Knowledge Center at https://www.ibm.com/support/knowledgecenter. You can find IBM Z NetView documentation on IBM Z NetView Knowledge Center.

Note: If you print PDF documents on other than letter-sized paper, set the option in the **Print** window that enables Adobe Reader to print letter-sized pages on your local paper.

Ordering publications

You can order many Tivoli publications online at http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss

You can also order by telephone by calling one of these numbers:

- In the United States: 800-426-4968
- In Canada: 800-879-2755

In other countries, contact your software account representative to order Tivoli publications. To locate the telephone number of your local representative, perform the following steps:

1. Go to http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss.
2. Select your country from the list and click the grey arrow button beside the list.
3. Click **About this site** to see an information page that includes the telephone number of your local representative.

Accessibility

Accessibility features help users with a physical disability, such as restricted mobility or limited vision, to use software products successfully. Standard shortcut and accelerator keys are used by the product and are documented by the operating system. Refer to the documentation provided by your operating system for more information.

For additional information, see the Accessibility appendix in the *User's Guide: NetView*.

Tivoli user groups

Tivoli user groups are independent, user-run membership organizations that provide Tivoli users with information to assist them in the implementation of Tivoli Software solutions. Through these groups, members can share information and learn from the knowledge and experience of other Tivoli users.

Support information

If you have a problem with your IBM software, you want to resolve it quickly. IBM provides the following ways for you to obtain the support you need:

Online
Please follow the instructions located in the support guide entry: https://www.ibm.com/support/home/pages/support-guide/?product=4429363.

Troubleshooting information
For more information about resolving problems with the IBM Z NetView product, see the *IBM Z NetView Troubleshooting Guide*. You can also discuss technical issues about the IBM Z NetView product through the NetView user group located at https://groups.io/g/NetView. This user group is for IBM Z NetView customers only, and registration is required. This forum is also monitored by interested parties within IBM who answer questions and provide guidance about the NetView product. When a problem with the code is found, you are asked to open an official case to obtain resolution.

Conventions used in this publication

This section describes the conventions that are used in this publication.

Typeface conventions

This publication uses the following typeface conventions:

Bold

- Lowercase commands and mixed case commands that are otherwise difficult to distinguish from surrounding text
- Interface controls (check boxes, push buttons, radio buttons, spin buttons, fields, folders, icons, list boxes, items inside list boxes, multicolumn lists, containers, menu choices, menu names, tabs, property sheets), labels (such as **Tip:**, and **Operating system considerations**:)
- Keywords and parameters in text

Italic

- Citations (examples: titles of publications, diskettes, and CDs
- Words defined in text (example: a nonswitched line is called a *point-to-point line*)
- Emphasis of words and letters (words as words example: "Use the word *that* to introduce a restrictive clause."; letters as letters example: "The LUN address must start with the letter *L*.")
- New terms in text (except in a definition list): a *view* is a frame in a workspace that contains data.
- Variables and values you must provide: ... where *myname* represents...

`Monospace`

- Examples and code examples
- File names, programming keywords, and other elements that are difficult to distinguish from surrounding text
- Message text and prompts addressed to the user
- Text that the user must type
- Values for arguments or command options

Operating system-dependent variables and paths

For workstation components, this publication uses the UNIX convention for specifying environment variables and for directory notation.

When using the Windows command line, replace **$***variable* with **%***variable***%** for environment variables and replace each forward slash (**/**) with a backslash (****) in directory paths. The names of environment variables are not always the same in the Windows and UNIX environments. For example, %TEMP% in Windows environments is equivalent to $TMPDIR in UNIX environments.

Note: If you are using the bash shell on a Windows system, you can use the UNIX conventions.

Syntax diagrams

The following syntax elements are shown in syntax diagrams. Read syntax diagrams from left-to-right, top-to-bottom, following the horizontal line (the main path).

- “Symbols” on page xv
- “Parameters” on page xvi
- “Punctuation and parentheses” on page xvi
- “Abbreviations” on page xvii

For examples of syntax, see “Syntax examples” on page xvii.

Symbols

The following symbols are used in syntax diagrams:

▶▶─

Marks the beginning of the command syntax.

→◀
: Marks the end of the command syntax.

→
: Indicates that the command syntax is continued on the next line.

▶—
: Indicates that a statement is continued from the previous line.

|
: Marks the beginning and end of a fragment or part of the command syntax.

Parameters

The following types of parameters are used in syntax diagrams:

Required
: Required parameters are shown on the main path.

Optional
: Optional parameters are shown below the main path.

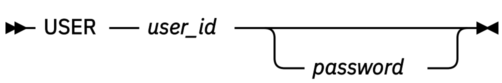
Default
: Default parameters are shown above the main path. In parameter descriptions, default parameters are underlined.

Syntax diagrams do not rely on highlighting, brackets, or braces. In syntax diagrams, the position of the elements relative to the main syntax line indicates whether an element is required, optional, or the default value.

When you issue a command, spaces are required between the parameters unless a different separator, such as a comma, is specified in the syntax.

Parameters are classified as keywords or variables. Keywords are shown in uppercase letters. Variables, which represent names or values that you supply, are shown in lowercase letters and are either italicized or, in NetView help, displayed in a differentiating color.

In the following example, the USER command is a keyword, the *user_id* parameter is a required variable, and the *password* parameter is an optional variable.



Punctuation and parentheses

You must include all punctuation that is shown in the syntax diagram, such as colons, semicolons, commas, minus signs, and both single and double quotation marks.

When an operand can have more than one value, the values are typically enclosed in parentheses and separated by commas. For a single value, the parentheses typically can be omitted. For more information, see "Multiple operands or values" on page xviii.

If a command requires positional commas to separate keywords and variables, the commas are shown before the keywords or variables.

When examples of commands are shown, commas are also used to indicate the absence of a positional operand. For example, the second comma indicates that an optional operand is not being used:

```
COMMAND_NAME opt_variable_1,,opt_variable_3
```

You do not need to specify the trailing positional commas. Trailing positional and non-positional commas either are ignored or cause a command to be rejected. Restrictions for each command state whether trailing commas cause the command to be rejected.

Abbreviations

Command and keyword abbreviations are listed in synonym tables after each command description.

Syntax examples

The following examples show the different uses of syntax elements:

- "Required syntax elements" on page xvii
- "Optional syntax elements" on page xvii
- "Default keywords and values" on page xvii
- "Multiple operands or values" on page xviii
- "Syntax that is longer than one line" on page xviii
- "Syntax fragments" on page xviii

Required syntax elements

Required keywords and variables are shown on the main syntax line. You must code required keywords and variables.

```
►► REQUIRED_KEYWORD  — required_variable  ►◄
```

A required choice (two or more items) is shown in a vertical stack on the main path. The items are shown in alphanumeric order.

```
►►┬─ REQUIRED_OPERAND_OR_VALUE_1 ─┬─►◄
  └─ REQUIRED_OPERAND_OR_VALUE_2 ─┘
```

Optional syntax elements

Optional keywords and variables are shown below the main syntax line. You can choose not to code optional keywords and variables.

```
►►┬──────────────────┬─►◄
  └─ OPTIONAL_OPERAND ┘
```

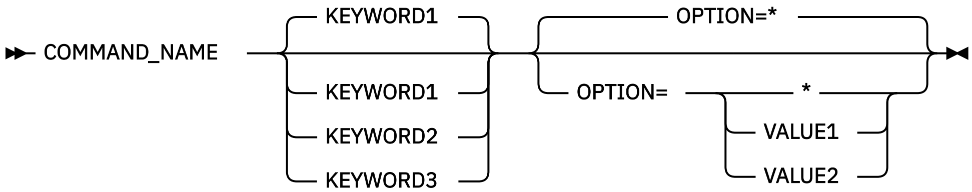
A required choice (two or more items) is shown in a vertical stack below the main path. The items are shown in alphanumeric order.

```
►►┬───────────────────────────────┬─►◄
  ├─ OPTIONAL_OPERAND_OR_VALUE_1 ─┤
  └─ OPTIONAL_OPERAND_OR_VALUE_2 ─┘
```

Default keywords and values

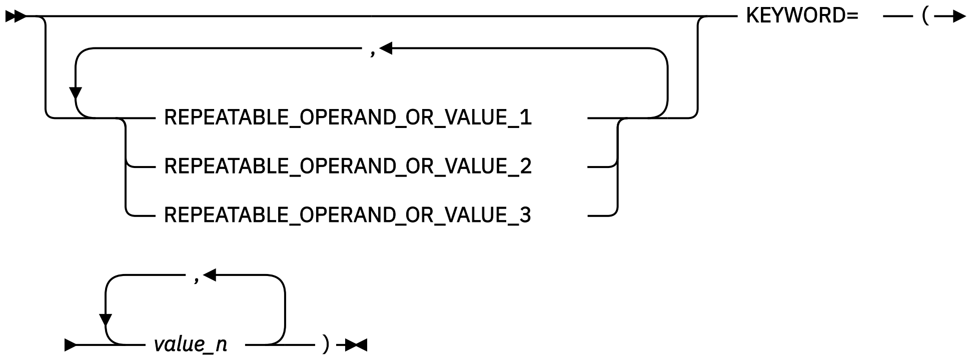
Default keywords and values are shown above the main syntax line in one of the following ways:

- A default keyword is shown only above the main syntax line. You can specify this keyword or allow it to default. The following syntax example shows the default keyword KEYWORD1 above the main syntax line and the rest of the optional keywords below the main syntax line.
- If an operand has a default value, the operand is shown both above and below the main syntax line. A value below the main syntax line indicates that if you specify the operand, you must also specify either the default value or another value shown. If you do not specify the operand, the default value above the main syntax line is used. The following syntax example shows the default values for operand OPTION=* above and below the main syntax line.



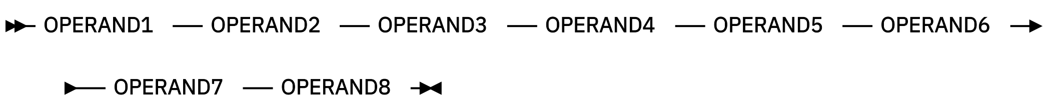
Multiple operands or values

An arrow returning to the left above a group of operands or values indicates that more than one can be selected or that a single one can be repeated.



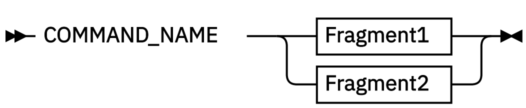
Syntax that is longer than one line

If a diagram is longer than one line, each line that is to be continued ends with a single arrowhead and the following line begins with a single arrowhead.



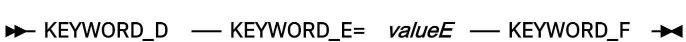
Syntax fragments

Some syntax diagrams contain syntax fragments, which are used for lengthy, complex, or repeated sections of syntax. Syntax fragments follow the main diagram. Each syntax fragment name is mixed case and is shown in the main diagram and in the heading of the fragment. The following syntax example shows a syntax diagram with two fragments that are identified as Fragment1 and Fragment2.



Fragment1

Fragment2



Chapter 1. Introduction

Using the NetView program, you can manage complex networks and systems from multiple independent software vendors from a single point. This chapter is an overview of the major components of the NetView program as they relate to the installation and configuration steps described in this book. See Figure 1 on page 1 for the NetView host components.

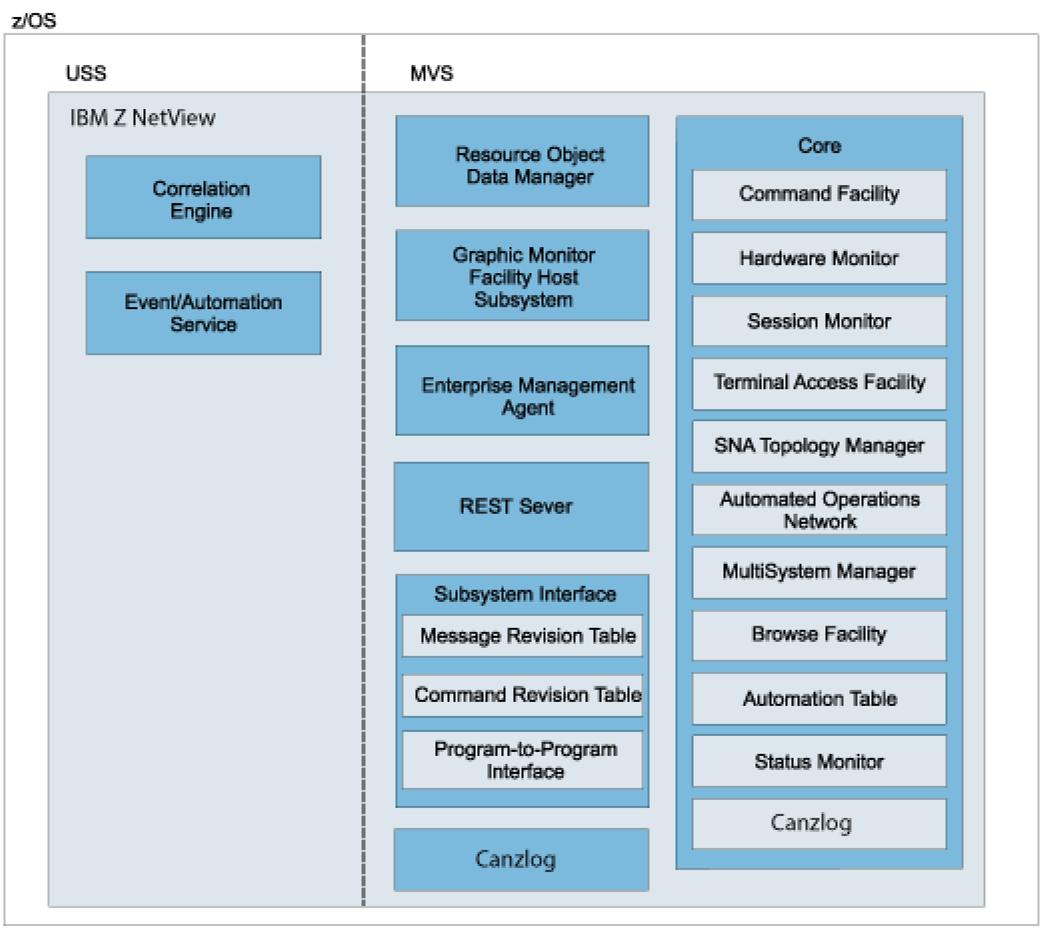


Figure 1. NetView Program Host Components

Command Facility

The command facility is used to send commands and receive messages. The command facility also provides base functions and services for other components such as intercomponent communication, presentation services, database services, and automation facilities.

If you want information about...	Refer to...
Installation considerations for the command facility	“Defining the Command Facility” on page 7

Hardware Monitor

The hardware monitor component collects and displays events and statistical data for both hardware and software to identify failing resources in a network. It provides probable cause and recommended actions so that operators can perform problem determination more efficiently.

If you want information about...	Refer to...
Installation considerations for the hardware monitor	"Defining the Hardware Monitor" on page 28

Session Monitor

The session monitor component provides information about SNA sessions (subarea and Advanced Peer-to-Peer Networking) including session partner identification, session status, connectivity of active sessions, and response time data. The session monitor also provides session trace data, route data, and Virtual Telecommunications Access Method (VTAM) sense code information for problem determination.

If you want information about...	Refer to...
Installation considerations for the session monitor	"Defining the Session Monitor" on page 30

Terminal Access Facility

The terminal access facility (TAF) provides operator control of any combination of CICS®, IMS, TSO, and other subsystems from one terminal. The operator does not have to log off or use a separate terminal for each subsystem. The subsystem can be in the same domain or in another domain.

If you want information about...	Refer to...
Defining TAF	"Defining the Terminal Access Facility" on page 183

SNA Topology Manager

The SNA topology manager dynamically collects topology and status of Advanced Peer-to-Peer Networking and subarea resources. This data is stored in the Resource Object Data Manager (RODM) for display by the NetView management console.

The topology agent supplies information consisting of the SNA nodes in a network, the Advanced Peer-to-Peer Networking transmission groups (TGs) between them, and the underlying logical links and ports supporting the TGs, in response to requests from the manager application.

If you want information about...	Refer to...
Installation considerations for the SNA topology manager and its agent	*IBM Z NetView Installation: Configuring Graphical Components*

Automated Operations Network

Automated Operations Network (AON) uses NetView automation facilities to automate the monitoring and recovery of both TCP/IP and SNA network resources. AON can monitor messages and alerts, and then automatically perform recovery actions. AON also provides an automated help desk to assist with resolving network problem, and generates reports so that you can monitor how well your automation is working.

AON provides default policy definitions that enable automation, without lengthy configuration, as soon as AON is enabled.

If you want information about...	Refer to...
Installation considerations for AON	"Defining AON" on page 37
Using AON	*IBM Z NetView User's Guide: Automated Operations Network*

MultiSystem Manager

MultiSystem Manager provides for the management of distributed resources from the NetView program. The NetView operator can use MultiSystem Manager to view and manage resources that are identified and managed locally by products such as IBM Tivoli Network Manager. The topology and status of these resources are dynamically managed through RODM and the graphical workstation components of the NetView program.

If you want information about...	Refer to...
Installation considerations for the MultiSystem Manager and its agents	*IBM Z NetView Installation: Configuring Graphical Components*
Using the MultiSystem Manager	*IBM Z NetView User's Guide: NetView Management Console* and *IBM Z NetView IP Management*

Browse Facility

The browse facility is used to view local or remote NetView data set members including the NetView log, NetView parameter files, and NetView panels. You can also browse the consolidated audit, NetView, and z/OS log (Canzlog).

If you want information about...	Refer to...
BROWSE command	*IBM Z NetView User's Guide: NetView*

Automation Table

With the NetView automation table, you can specify processing options for incoming messages and MSUs and issue automatic responses. The table contains a sequence of statements that define the actions that the NetView program can take in various circumstances. The automation table is one of several components that provide automation capabilities.

Users can also create an automation table in IBM Z NetView. For more details about IBM Z NetView REST Server, see *Automation Table* in Chapter *NetView Program Automation Facilities* in the *IBM Z NetView Automation Guide* or the RESTful Server book.

If you want information about...	Refer to...
Automation	*IBM Z NetView Automation Guide*

Status Monitor

The status monitor component provides status information about SNA subarea network resources.

If you want information about...	Refer to...
Installation considerations for the status monitor	"Defining the Status Monitor" on page 17

Resource Object Data Manager

Resource Object Data Manager (RODM) is an object-oriented data cache. Objects in RODM can represent resources in your network. The data cache is located entirely in the memory of the host processor for fast access to data and high transaction rates. RODM can contain approximately 2 million objects, providing support for large and growing networks.

The MultiSystem Manager and SNA topology manager components of the NetView program populate RODM with information such as the topology and status of resources they monitor, and maintain that information as changes occur. Using data in RODM, the Graphic Monitor Facility host subsystem component dynamically builds graphical views for display by the NetView management console. When the topology or status changes in RODM, methods automatically update the views that include the affected resources.

Additionally, authorized operators can use the RODMVIEW command to display, create, update, and delete classes, objects, fields, and relationships in RODM.

If you want information about...	Refer to...
Installation considerations for RODM	*IBM Z NetView Installation: Configuring Graphical Components*

Graphic Monitor Facility Host Subsystem

The NetView Graphic Monitor Facility host subsystem component maintains the status of resources in RODM and supplies the NetView Management Console with information about RODM resources. It works with RODM and the NetView Management Console to display graphic views of networks and to issue commands to resources that you select from a NetView Management Console view.

The Graphic Monitor Facility host subsystem works with the SNA topology manager and the NetView Management Console to manage SNA resources. It works with MultiSystem Manager and the NetView Management Console to manage non-SNA resources.

If you want information about...	Refer to...
Installation considerations for GMFHS	*IBM Z NetView Installation: Configuring Graphical Components*

IBM Z NetView Enterprise Management Agent

The IBM Z NetView Enterprise Management Agent enables management of your network from the Tivoli Enterprise Portal using sampled and real-time data. The sampled data can provide information about network resources and outages, using situations and expert advice. It can also indicate trends in your network when historical data is used. Additionally, NetView, VTAM, and z/OS commands can be issued directly from the Tivoli Enterprise Portal to provide instant display and troubleshooting capabilities. The Z NetView Enterprise Management Agent enables management of both availability and performance data from the Tivoli Enterprise Portal using cross-product links to selected z/OS OMEGAMON® XE agents.

If you want information about...	Refer to...
Z NetView Enterprise Management Agent	*IBM Z NetView User's Guide: NetView Enterprise Management Agent*

Subsystem Interface

The subsystem interface is used to receive system messages and to enter system commands. With extended multiple console support (EMCS) consoles, the subsystem interface is used to receive commands, but not messages. In a single system, multiple NetView programs can use the subsystem interface. Each NetView program that uses the subsystem interface requires a NetView subsystem address space in addition to the NetView application address space.

You can use the message revision table to intercept z/OS messages before they are displayed, logged, automated, or routed through your sysplex. With this table, you can make decisions about a message based on its message ID, job name, and other properties and can revise or suppress a message or take certain actions. The message revision table is one of several components that provide automation capabilities.

You can use the command revision table to intercept z/OS commands and make simple modifications inline, without needing to transfer the command to the NetView application address space. Commands can be deleted; parameters and keywords can be added, removed, or modified; nicknames can be expanded (such as creating new command or parameter synonyms); and explanatory WTO messages can be issued. The command revision table is one of several components that provide automation capabilities.

The program-to-program interface (PPI) is an address space provided by the NetView program to enable application programs to communicate with the NetView program and other applications running in the same host. When an application calls the PPI using its application program interface (API), the request is synchronous.

If you want information about...	Refer to...
PPI	*IBM Z NetView Application Programmer's Guide*
Message revision table	*IBM Z NetView Automation Guide*

Correlation Engine

The correlation engine correlates multiple events over time, based on duplicates, thresholds, presence or absence of specific events, and other user-specified criteria. The correlation engine is one of several components that provide automation capabilities.

If you want information about...	Refer to...
Automation	*IBM Z NetView Automation Guide*
Installation considerations for the NetView UNIX System Services	Chapter 11, “Setting Up UNIX System Services for the NetView Program,” on page 195

Event/Automation Service

The Event/Automation Service (E/AS) serves as a gateway for event data between the Z NetView management environment, managers and agents that deal with Event Integration Facility (EIF) events, and SNMP managers and agents. With this gateway function, you can manage all network events from the management platform of your choice.

If you want information about...	Refer to...
Installation considerations for the Event/Automation Service	“Enabling the Event/Automation Service” on page 203
Installation considerations for the NetView UNIX System Services	Chapter 11, “Setting Up UNIX System Services for the NetView Program,” on page 195

UNIX System Services

The IBM Z NetView program uses UNIX System Services for the following functions:

- UNIX command server
- AON/TCP functions
- Event/Automation Service
- Event correlation

NetView operators or programs can interact with z/OS UNIX System Services using the PIPE UNIX stage, the IPCMD command, and the UNIX command server.

If you want information about...	Refer to...
Installation considerations for the NetView UNIX System Services	Chapter 11, “Setting Up UNIX System Services for the NetView Program,” on page 195

Help Facility

The Z NetView mainframe online help is available for the following areas, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions
- Helpdesk

If you want information about...	Refer to...
Customizing help panels	*IBM Z NetView Customization Guide*

Chapter 2. Defining NetView Components

This chapter describes setting up NetView components, including the following topics:

- "Defining the Command Facility" on page 7
- "Defining the Status Monitor" on page 17
- "Defining the Hardware Monitor" on page 28
- "Defining the Session Monitor" on page 30
- "Defining AON" on page 37
- "Enabling TCP/IP Connection and Packet Trace Data Collection" on page 129

Defining the Command Facility

You can customize the command facility for your environment:

- "Including Additional Task Statements" on page 7
- "Defining Command Facility Panel Format" on page 7
- "Assembling and Link-Editing the NetView Constants Module" on page 8
- "Defining Generic Automation Receiver Support" on page 13
- "Reviewing System Definitions" on page 13
- "Defining Buffer Pools" on page 14
- "Defining VSAM Performance Options" on page 16
- "Defining the MEMSTORE Function" on page 17

Including Additional Task Statements

If you have written any tasks besides those supplied on the distribution tape, include the following task statements in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.task_name.MOD=task_module
TASK.task_name.MEM=task_init_member
TASK.task_name.PRI=task_priority
TASK.task_name.INIT=initialize_task(Y/N)
```

where *task_name* is the name of your task. Replace the other variables in the statements with information to define your task to the NetView program.

If a task you have written uses VSAM, allocate the VSAM data sets. If the parameter that defines the task contains a DSTINIT statement specifying FUNCT=CNMI or FUNCT=BOTH, add a VTAM APPL statement to A01APPLS (CNMS0013).

If you want information about...	Refer to...
Designing functions	*IBM Z NetView Customization Guide*
Writing your own user subtasks	*IBM Z NetView Programming: Assembler*
TASK statement and reserved task names	*IBM Z NetView Administration Reference*

Defining Command Facility Panel Format

CNMSCNFT lets you define the screen colors, prefix data, and prefix display order for message formatting. The SCRNFMT keyword on the **DEFAULTS** command specifies the beginning of the screen format definitions. The **OVERRIDE** command also has a SCRNFMT keyword for overriding the current screen format definitions. Each set of SCRNFMT definitions results in a complete replacement of all values for all attributes. If you do not code any operands, the NetView default values are used.

If you want information about...	Refer to...
The definition statements	*IBM Z NetView Administration Reference*
Customizing the NetView command facility panel	*IBM Z NetView Customization Guide*

Assembling and Link-Editing the NetView Constants Module

The NetView constants module, DSICTMOD, contains timeout values for various NetView functions. The constants module also contains values for storage sizes, sense codes, and storage management performance options.

The CNMS0055 job assembles and link-edits the module. Run this sample to change the NetView default values for the constants described in this section.

You can modify the module using a system service aid, such as AMASPZAP, or replace it by reassembling DSICTMOD using CNMS0055. Your new copy of DSICTMOD must reside in a user-defined library that is concatenated before NETVIEW.V6R3M0.CNMLINK in your NetView start procedure CNMPROC (CNMSJ009). Whenever you modify values in DSICTMOD or replace the module, restart the NetView program to activate the new values.

Boundary Function Trace Initialization Timeout

If the session monitor is started with the TRACE function active, it sends a TRACE data request to each PU type 4 node after becoming aware of it through session awareness (SAW) data. After the request is sent, the session monitor waits for the response. If it does not receive a response within the specified time, message AAU114I is sent to the authorized receiver.

The default value is 180 seconds.

Connectivity Test Timeout

The connectivity test is selected from the Session List panel of the session monitor. Each test can consist of one or more route test requests. For each route test request, the session monitor waits for the response. If it does not receive a response within the specified time, the entire connectivity test fails. Message AAU114I is sent to the authorized receiver and message AAU947I is sent to the operator who requested the test.

The default value is 180 seconds.

Gateway TRACE Initialization Timeout

If the session monitor is started with the TRACE function active, it sends a gateway (GW) TRACE data request to each GW after becoming aware of it through SAW data. After the request is sent, the session monitor waits for the response. If it does not receive a response within the specified time, message AAU114I is sent to the authorized receiver.

The default value is 180 seconds.

Gateway Boundary Function Trace Request Timeout

When GW TRACE data is requested for display, the session monitor sends a request to GW NCP for TRACE data. If it does not receive a response within the specified time, the session monitor sends message AAU114I to the authorized receiver and message AAU947I to the operator who made the request.

The default value is 180 seconds.

LINEMAP Command Timeout

The session monitor LINEMAP command issues a line map request to the destination PU. The session monitor waits for a response. If it does not receive a response within the specified time, message AAU114I is sent to the authorized receiver and message AAU947I is sent to the operator who issued the request.

The default value is 180 seconds.

NCP Boundary Function Trace Data Request Timeout

A boundary function trace request is sent every time an operator requests a boundary function trace display. The session monitor waits for a response. If it does not receive a response within the specified time limit, the session monitor sends message AAU114I to the authorized receiver and message AAU947I to the operator requesting the display.

The default value is 180 seconds.

Nonpersistent Sessions Timeout Value

The timeout interval specifies, in seconds, the time between messages during which a nonpersistent session stays active. If the time between conversations is greater than this amount, the session ends. If you do not change the default value of 0, and the LUC session is nonpersistent, message DSI624I is issued and the session is persistent.

The default value is 000 seconds.

Query PSID Request Timeout

If the session monitor is started with the TRACE function active, the session monitor sends a QUERY PSID request to each subarea node for its release level. After the request is sent, the session monitor waits for a response. If it does not receive a response within the specified time, the session monitor sends message AAU114I to the authorized receiver.

The default value is 180 seconds.

Route Test Initialization Timeout

The session monitor issues a route test request for each new route it knows of through SAW data. The route test is issued with a time limit. If the response to the test request is not received within the specified time, the session monitor sends message AAU114I to the authorized receiver.

The default value is 180 seconds.

RTM Collection Request Timeout

When an operator issues the COLLECT RTM command, the session monitor sends a message to the operator to indicate successful start of the command. For each destination PU located, the command processor then drives another process in the session monitor to send an RTM data request. The PU sends RTM data to the session monitor in response to that request. If it does not receive the data within the specified time, the session monitor sends message AAU114I to the authorized receiver.

The default value is 180 seconds.

RTM Initialization Request Timeout

To determine the RTM capabilities of a device, an NMVT RU is sent to the PU. RTM INIT specifies the amount of time allowed for the PU to respond.

The default value is 180 seconds.

Service Point Control Interface Commands Timeout

This is the timeout value for a command to complete to a service point. If the command does not respond in this interval, it is canceled. Provide appropriate timeout values to prevent commands to service points from restricting the use of critical resources (such as DSRBs) when the command fails.

Use this constant to set the default for the COSTIME keyword on the NetView **DEFAULTS** command. The minimum value is 0, which specifies that the timeout value is determined by the value on the DEFAULTS RCVREPLY keyword. X'FFFFFFFF' specifies that the timeout value is determined by the DEFAULTS MAXREPLY keyword. The maximum value is the value assigned to the DEFAULTS MAXREPLY keyword.

If you want information about...	Refer to...
The DEFAULTS command and its keywords	*IBM Z NetView Command Reference Volume 1 (A-N)*

TRACE NCP Command Timeout

The session monitor sends an NCP TRACE START/END request in response to the TRACE START/STOP command. The NCP processes the request and sends a response back to the session monitor. If it does not receive a response within the specified time, the session monitor sends message AAU114I to the authorized receiver and message AAU947I to the operator who issued the TRACE command.

The default value is 180 seconds.

VR Status Request Timeout

In response to a route status request from an operator, the session monitor sends a request for route status data. If it does not receive a response within the specified time, the session monitor sends message AAU114I to the authorized receiver. A timeout condition does not cause the entire route status request to fail. Thus, the requesting operator can receive a partial route status display or a data service failure message.

The default value is 180 seconds.

Hardware Monitor Remote Data Retrieval Timeout

An operator at a focal point is logged on to the NetView program and wants to obtain detailed data about an event that generated an alert. The operator issues a request to the distributed host. If the response is not received within the specified time, the operator receives a timeout notification. This timeout is also used for a FOCALPT CHANGE command to change an alert focal point using LU conversation (LUC).

The default value is 120 seconds.

Hardware Monitor Solicited Commands Timeout

This is the timeout value for hardware monitor solicited commands. The following commands are timed by the hardware monitor timeout value:

- NPDA TEST
- REQMS

On timeout, the BNJ093I and BNJ992I messages are sent to the NPDA TEST command. The BNJ093I message is a component message line, and the BNJ992I message is sent to the authorized receiver. The REQMS command receives the BNJ992I message on timeout, and the message is sent to the authorized receiver.

The default value is 180 seconds.

HLL Default Stack Size

The STACK runtime option is used for PL/I dynamic storage allocation.

You can use the HLLENV statement (PSTACK keyword) in the CNMSTUSR or C*xx*STGEN member to preinitialize the HLL environment. The default value for PSTACK is 131072 bytes. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

HLL Default HEAP Area

HEAP specifies storage that is used to allocate controlled and based variables. It also specifies how that storage is to be managed.

You can use the HLLENV statement (PHEAP keyword) in the CNMSTUSR or C*xx*STGEN member to preinitialize the HLL environment. The default value for PHEAP is 131072 bytes. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Task Public Message Queue Thresholds

The following group has three pairs of threshold values. Each pair consists of a task threshold value and a reissue threshold value. Table 1 on page 11 shows the threshold values for the groups. When the number of buffers in the public message queue exceeds the task threshold value for a particular type of task, message DSI374A is issued. This condition can indicate that the buffers on the public message queue are not being processed. Message DSI374A is issued thereafter every time the reissue threshold is exceeded. For example, if the task is an OST, DSI374A is issued if the number of buffers in the public message queues exceeds 1000. Message DSI374A is reissued when the count reaches 1100, 1200, 1300, and so on.

Table 1. Thresholds for Task Types

Task Type	Default Threshold	Default Reissue Threshold
PPT	1000	100
OST/AUTO	1000	100
DST/OPT/HCT	3000	500

Maximum Number of APPCCMD Retries

The APPCCMD retries constant specifies the maximum number of times that the NetView program attempts to issue an LU 6.2 command to the NetView management console server. Note that the command might fail because of a temporary error. Only those errors that VTAM defines as temporary are eligible to be retried.

The default value is three times.

Entry for LU 6.2 Transport Support

This constant specifies the maximum number of LUs with which the MS transport function or high performance transport function can be expected to have sessions. Changing this constant changes the size of the internal tables used by the transport functions, and can affect storage used by the transport functions.

The default value is 2000 LUs.

Timeout Value for CSCF

This constant specifies the number of seconds the NetView program waits for a reply when a central site control facility (CSCF) request is sent to the target physical unit (PU). If a reply is not received from the PU within the specified number of seconds, a timeout occurs and the CSCF session is stopped. Certain commands issued on the requested PU can take several seconds to complete and can directly relate to the characteristics of that PU. Be sure to adjust this timeout value appropriately to accommodate both communication errors, which can result in no reply for a given request, and PU commands, which can take several seconds to run and return a reply.

The default is 30 seconds.

CSCF Application Idle Timeout

This constant is the number of minutes that a CSCF session is allowed to remain active in the NetView system without an operator having any interaction with the PU with which the operator is in session. Each time CSCF interacts with the PU in any way (for example, an attention key or command entered on the command line is passed to the PU), the timer is reset to this number of minutes. If an operator has no interaction with the PU for this number of minutes, a timeout occurs and the NetView program ends the session. This timeout is allowed because there can be only one CSCF session for each PU at any time. If an operator establishes a CSCF session with a PU, no other operators can have a CSCF session with that PU until the active session ends.

The default value is 20 minutes.

Storage Management Performance

This field determines whether NetView storage management keeps the first allocation of storage below the 16 Mb line for an individual subpool and size when it is no longer in use. If the storage is not freed, NetView performance is enhanced, while below-the-line storage use is increased. If the storage is freed, performance during later requests for below-the-line storage is slower, but storage use is smaller.

The default, X'00', keeps below the line storage.

Note: If you have less than 300 users logged on at any one time, or if you have a large amount of user-written code that runs below-the-line, use the default.

Automation Table Loading

This field determines whether an automation table successfully loads if commands or command lists that are called out of the automation table are missing. If a command or command list is missing, an error message is issued, regardless of how this field is set. If you set the byte to X'01' and errors other than the missing commands or command lists do not occur, the table you specified on the AUTOTBL command is activated and replaces the current active automation table in storage.

The default is X'00' (missing commands and command lists prevent the automation table from loading).

RTM Initialization Retry and Interval

Use this field to specify the maximum number of retries and the number of seconds between each try for the response time monitor.

The default is 5 retries with 60 seconds between each try.

Expected Number of Task Global Variables

By specifying the number of task global variables you expect to use, you can improve the access time for retrieving task global variables.

You can use the QRYGLOBL command to determine the total number of task global variables currently defined by each task.

The default is 100 variables.

If you want information about...	Refer to...
Storage requirements	*IBM Z NetView Tuning Guide*

Expected Number of Common Global Variables

By specifying the number of common global variables you expect to use, you can improve the access time for retrieving common global variables.

The QRYGLOBL command can be used to determine the total number of common global variables currently defined. Depending on which AON components and functions you are using, you might want to increase this number.

Note: You can also update the COMMON statement to set common global variables in the CNMSTUSR or C*xx*STGEN member. The variables are set before any autotasks are started and before automation is enabled. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
QRYGLOBL command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Sense Code Filtering

You can modify session monitor sense code filtering. For more information, see "Adding a Sense Code for Filtering" on page 33 and "Stopping Sense Code Filtering" on page 34.

Defining Generic Automation Receiver Support

The generic automation receiver allows an MDS-MU to be sent to a NetView system with the generic automation receiver. The generic automation receiver then submits the MDS-MU to NetView automation.

CNMCMSYS contains the following command definition statements. These statements are required to start the generic automation receiver support in the NetView program:

```
CMDDEF.DSIREGGR.MOD=DSIREGGR
CMDDEF.DSIREGGR.SEC=BY
CMDDEF.DSILOGGR.MOD=DSILOGGR
CMDDEF.DSILOGGR.SEC=BY
CMDDEF.DSINVGRP.MOD=DSINVGRP
CMDDEF.DSINVGRP.PARSE=N
CMDDEF.DSINVGRP.RES=N
CMDDEF.DSINVGRP.SEC=BY
```

If you expect use of the generic automation receiver to be heavy, add the following CMDDEF to CNMCMDU:

```
CMDDEF.DSINVGRP.RES=Y
```

You can define the generic automation receiver as its own task by issuing the following command:

```
'AUTOTASK OPID=DSINVGR'
```

Note: Consider adding this command to the CNMSTUSR or C*xx*STGEN member so that it is issued at NetView initialization. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

This statement points to the following operator statement in DSIOPF in the DSIPARM data set:

```
DSINVGR     OPERATOR    PASSWORD=GENREC
            PROFILEN    DSIPRFGR
```

The generic automation receiver also uses the following profile statement in DSIPRFGR in the DSIPRF data set:

```
DSIPRFGR    PROFILE  IC=DSIREGGR
            AUTH     MSGRECVR=NO,CTL=GLOBAL
```

Reviewing System Definitions

Review and adjust the system parameters by preparing a message processing facility (MPF) list that blocks unnecessary messages sent to the NetView program for automation. This can have a significant effect on performance if you are using NetView automation.

Note: Do not send messages that are not automated to the NetView program. Each message that the NetView program receives causes a search of the automation table.

Defining Buffer Pools

Use job CNMSJM01 in NETVIEW.V6R3M0.CNMSAMP to define your buffer pools.

The VSAM local shared resources (LSR) performance option is the sharing of common control blocks, such as input/output (I/O) control blocks, buffers, and channel programs. When running the NetView program, LSR is the default. LSR also causes VSAM to search buffers for direct record retrievals. Without LSR, VSAM carries out I/O for direct retrievals regardless of whether the control interval containing the record you want is in storage.

Deferred write (DFR) option causes VSAM to defer the write I/O action when records are directly inserted or replaced in direct mode. Without DFR, VSAM does not defer the I/O for direct inserts or replacements of records. With DFR, the buffers are written in these instances:

- When no more buffers are available to do a retrieve
- When the application issues the WRTBFR macro indicating that VSAM is to write out the modified buffers
- When the database is closed

If the NetView program stops without closing the databases, the records in the DFR buffers are not written to the databases.

With the LSR or DFR options, VSAM uses a resource pool for buffering. The NetView program creates this resource pool during initialization when the NetView program issues the VSAM BLDVRP macro. The resource pool is divided into buffer pools based on the VSAM control interval (CI) sizes passed to the BLDVRP macro in the BLDVRP parameter list. For the NetView program, the DSIZVLSR module is the BLDVRP parameter list that is passed to the BLDVRP macro. By using the resource pool, you can show VSAM how many and what size buffers to allocate. This resource pool is in extended storage.

Note: Run CNMSJM01 to link-edit the DSIZVLSR module.

The BLDVRP macro has been specified with values that separate the index and data control intervals into separate pools. Separating the INDEX and DATA intervals allows the critical index records to remain resident in memory without having to allocate an excessive number of buffers. The VSAM index and data control interval sizes have been selected so that similar function share-pool sizes reduce contention.

Buffer Pool Sizes

When you open a database and specify LSR or DFR, VSAM looks for a buffer pool for the INDEX and DATA components, depending on their control interval sizes. A buffer pool that is the same size as the control interval size is chosen. If a buffer pool with the same size has not been defined, the next higher buffer pool size is chosen. If compatible buffer pools are not defined, the open fails with a VSAM error code of X'DC'. If no resource pool is defined, the open fails with a VSAM error code of X'E4'. Databases with the same control interval sizes share the same buffer pool. Allocate enough buffers of a particular size to satisfy all users sharing the buffer pool.

VSAM performance is affected by the buffer allocations. Use the DSIZVLSR module to specify the size and number of buffers to allocate.

Changes to the following parameters for defining clusters can affect the values specified for the LSR pool built by CNMSJM01. If these values are modified, refer to the *IBM Z NetView Tuning Guide* to verify that the parameters specified for the LSR pool are still valid.

- CONTROLINTERVALSIZE (CISZ)
- CYLINDERS
- KEYS
- KILOBYTES
- MEGABYTES

- RECORDS
- TRACKS

The operands specified in the examples have been selected based on using an IBM 3390 DASD (using ICF catalogs). If other types of devices are used to allocate these clusters, these operands might need to be adjusted for optimal use of the device. For 3380 DASD, refer to the *IBM Z NetView Tuning Guide* for CISIZE recommendations.

If you use the recommended VSAM cluster definitions, the buffer sizes in CNMSJM01 are required.

Note: Values in CNMSJM01 reflect the number of bytes recommended for each VSAM buffer size.

Minimum Buffer Allocations

Use the following formulas to determine the minimum number of buffers you need to define for the INDEX and DATA pools for each NetView VSAM data services task (DST).

- INDEX buffers: allocate (2 X DSRBO + 2)
- DATA buffers: allocate (DSRBO + 3)

Note: DSRBO is a DSTINIT parameter for NetView VSAM DST initialization members. The DSRBO parameter shows how many consecutive VSAM requests, operator requests, or both, can be scheduled. For an example, see AAUPRMLP (session monitor) and BNJMBDST (hardware monitor) initialization members.

Define one INDEX buffer for each DSRBO and one additional INDEX buffer for control interval splits and for the highest level INDEX.

Define one DATA buffer for each DSRBO and three additional DATA buffers for control interval splits and control area splits.

Additional Buffer Allocations

Consider the following information to determine how many additional buffers you can define for INDEX and DATA for each VSAM DST:

- Allocate enough INDEX buffers to get the entire INDEX in storage, plus two additional buffers.

  Note: The IDCAMS LISTCAT command or the NetView LISTCAT command displays the number of index records. Start with 20 INDEX buffers and then monitor.

- Allocate enough DATA buffers so VSAM can read an entire hard disk drive track of control intervals for each DSRBO specified. In sequential mode, VSAM reads ahead an entire track of data if enough buffers are available.

  For example, the session monitor has a 24.5K data control interval (CI), and a 3390 DASD has a 56K track size. Therefore, two CIs can fit on one track. Multiplying the number of CIs for each track by the session monitor DSRBO (default of 10) gives you 20 DATA buffers.

- The MACRF=DFR statement uses the LSR and DFR VSAM options to reduce the number of I/O accesses to the VSAM database by the hardware monitor. All VSAM buffers used by the hardware monitor are 18.4K. The hardware monitor is the only DST to use buffers from this size pool. Therefore, the global buffer definitions in the DSIZVLSR CSECT need to allocate enough 18.4K buffers for the hardware monitor. Calculate the number of buffers needed using the formula:

  ```
  ((2 X DSRBO value) + 3)
  ```

  For example, if your DSRBO value is 5, use:

  ```
  ((2 x 5) + 3)
  ```

  to give you a required total of 13 buffers.

- Experiment with additional buffers on larger systems. However, allocating excessive buffers can degrade performance. Eventually it takes VSAM longer to find a record in a buffer than it does to read it.

Monitor the processor utilization, paging, real storage, hard disk drive utilization, and the NetView program response time when experimenting with buffer sizes.

Note: The NetView VSAMPOOL command displays VSAM buffer pool allocation and usage.

Changes to CNMSJM01

Use the VSAM LSR performance option to increase the efficiency of record processing.

To use the VSAM LSR performance option:

1. Edit CNMSJM01.

   The DSIZVLSR CSECT contains two LSR Pools. The first pool defines resources for the DATA component. The second pool defines resources for the INDEX component.

   See "Minimum Buffer Allocations" on page 15 for information about selecting the proper values for the BUFFERS keyword.

   Refer to the *IBM Z NetView Tuning Guide* for a detailed example on determining tuned values for the LSR pools.

   Allocate at least two buffers for each DSTINIT statement that uses `MACRF=LSR` or `MACRF=DFR`. Of these two buffers, one is used for index records and the other is used for the data records. Use the CISIZE information found in the VSAM catalog for the index and data components to select appropriate buffer sizes. Also, allocate enough buffers to store most (or all) index records.

   Refer to *IBM Z NetView Tuning Guide* for additional information concerning how to calculate your buffer values.
2. Run CNMSJM01 to allocate and link-edit the DSIZVLSR CSECT.
3. Ensure that the return code is 0 before proceeding to the next step.
4. Your new copy of DSIZVLSR must reside in a user-defined library that is concatenated before NETVIEW.V6R3M0.CNMLINK in your NetView start procedure CNMPROC (CNMSJ009). Whenever you modify values in DSIZVLSR or replace the module, restart the NetView program to activate the new values.

Defining VSAM Performance Options

Two VSAM performance options, LSR and DFR, can be defined for NetView VSAM DSTs to improve VSAM processing and reduce I/O and storage.

LSR is the sharing of common control blocks such as I/O control blocks, buffers, and channel programs. LSR also causes VSAM to search buffers for direct record retrievals. Without LSR, VSAM carries out I/O for direct retrievals regardless of whether the control interval containing the preferred record is in storage.

DFR causes VSAM to defer the write I/O when records are directly inserted or replaced in direct mode. Without DFR, VSAM does not defer the I/O for direct inserts or replacement of records. With DFR the buffers are written in these instances:

- When no more buffers are available to do a retrieve
- When the application issues the WRTBFR macro indicating that VSAM is to write out the modified buffers
- When the database is closed

If the NetView system stops without closing the databases, the records in the DFR buffers are not written to the databases. The exposure is minimized by the extended specify task abnormal exits (ESTAEs) that trap abends and close the databases. However, if the system operator stops the NetView program with the MVS FORCE command, the ESTAEs are not driven. Do not cancel the NetView program, except as a last resort. If you issue a FORCE command, try to close the databases by issuing the NetView SWITCH command with the T option. This does not perform a switch; it just closes the active database. If this procedure does not work, issue the NetView STOP FORCE command for each active VSAM task. If you use the MVS FORCE command to bring down the NetView system and you have specified DFR, you might have to delete and redefine the affected databases.

If you specify DFR, you get both the LSR and DFR options.

LSR and DFR values are defined with the DSTINIT statement:

```
DSTINIT MACRF=xxx
```

Where:

LSR
: Specifies that the LSR option is used for VSAM performance.

DFR
: Specifies that DFR option is used for VSAM performance.

Table 2 on page 17 lists the DFR and LSR values for the NetView components and facilities.

Table 2. LSR and DFR Values for NetView Facilities and Components

Component	Member	Value
Central site control facility	DSIKINIT	Specify LSR
Hardware monitor	CNMSTYLE	Specify LSR
Network log	DSILOGBK	Do not specify DFR or LSR
Save/restore database	DSISVRTD	Specify LSR
Session monitor	CNMSTYLE	Specify LSR
TCP/IP Connection Management	CNMSTYLE	Specify LSR

Defining the MEMSTORE Function

To improve NetView performance and reduce disk I/O, you can let NetView monitor its access of PDS members and keep high-access members in storage. The NetView program includes a MEMSTORE CLIST (CNME1054). The NetView program is shipped with MEMSTORE enabled in the CNMSTYLE member.

Note that operators can see the members loaded in storage by using the BROWSE and LIST commands. For more information, refer to the *IBM Z NetView Programming: Pipes*.

Note: The memStore statements in the CNMSTYLE member specify the thresholds for automatic retention of members in storage. The inStore statements in the CNMSTYLE member specify the members that are to remain in storage regardless of their usage. You can use the RESTYLE MEMSTORE command to enable changes without recycling the NetView program.

Defining the Status Monitor

With the status monitor, you can perform activities such as the following activities:

- Process command lists
- Provide status information for automated recovery of failing devices
- Specify the initial status for resources not known to VTAM

This section describes how to define the status monitor to suit your requirements.

Note: The status monitor monitors only the SNA resources that were defined in the VTAMLST data sets when you started the NetView program. You can use the NetView management console to dynamically discover and monitor resources (both SNA and IP). The NetView management console also provides a graphical interface. For more information, refer to *IBM Z NetView Installation: Configuring Graphical Components*.

Processing Command Lists from the Status Monitor

You can process command lists from the Status Detail panels of the status monitor. These are ordinary command lists that can be processed without any operands or with the node name as the only operand. The following command lists are supplied in DSICNM:

```
C AUTOTR
C NODE
C EVENTS
C INACTF
C MONOFF
C MONON
C RECYCLE
C REDIAL
C SESS
C STATIONS
C STATS
```

The C is in position 1 and the command list name starts in position 3. You can add command lists to the existing set of lists or you can replace them with those that you define. A maximum of 16 command lists is allowed.

If you want information about...	Refer to...
The C statement	*IBM Z NetView Administration Reference*

Specifying the Designated Interface with VTAM

The following statement in DSICNM specifies that the status monitor of this NetView system runs as a secondary network resource status monitor and does not receive unsolicited messages from VTAM:

```
* O SECSTAT
```

Use this statement if you have more than one active NetView status monitor. O SECSTAT is commented out in DSICNM. Uncomment this statement for the status monitor that is not monitoring the network's status.

The O (letter O, not zero) is in position 1 and the SECSTAT starts in position 3. If you want to run the status monitor with the primary interface to receive unsolicited messages, either leave this statement commented out, or delete it.

If you do not specify O SECSTAT, the first status monitor initialized is given the network status updates from VTAM.

If you code O SECSTAT in multiple NetView programs in one LPAR, neither one receives the updates from VTAM. For more information, see Appendix A, "Running Multiple NetView Programs in the Same LPAR," on page 233.

Specifying the Automatic Reactivation of Failing Nodes

The following statement in DSICNM specifies that failing nodes can be reactivated if they are defined for reactivation by a STATOPT statement:

```
O MONIT
```

The O (alphabetical O, not zero) is in position 1 and the `MONIT` starts in position 3. If you do not want this feature, delete this statement.

Note: Disable the `O MONIT` statement if you are using AON to automate your SNA resources.

The following statement in DSICNM can be used to specify the maximum number of times that the status monitor MONIT function is to activate a particular resource:

```
M MAXREACT      00
```

The default value is 00, which means that an unlimited number of activation attempts are made for the resource. The value specified applies to all resources monitored by the status monitor. Maximum reactivation counters for resources are set to zero at status monitor initialization.

Note: The value specified is used to limit the number of times the status monitor accepts a resource for reactivation by the MONIT function and does not limit the number of reactivation attempts made for the resource after it is accepted by the MONIT function for reactivation.

The following statement in DSICNM can be used to specify a time interval for the MONIT function reactivation attempts:

```
M REACTINT       00
```

The time interval is specified in minutes. The default value is 00, which means that reactivation attempts for resources is made at 1-minute intervals. The value specified applies to all resources monitored by the status monitor.

Modifying the Message Indicator Settings

VTAM, MVS, JES, and NetView messages and responses are recorded in the network log. You can assign panel color codes, highlighting, and alarms to show to an operator when certain messages occur.

To emphasize a message, use message alert settings (A statements) in DSICNM. For example, if you specify A3 PYBYN, the following alert settings are used for message indicator 3:

P
: The alert is colored pink.

Y
: The alert indicator is set on authorized receiver.

B
: The alert blinks.

Y
: The alarm sounds when the alert is received.

N
: A copy of the unsolicited message is not sent to the system console.

For more information, refer to *IBM Z NetView Administration Reference*.

Providing Status Information for Automated Recovery of Failing Devices

You can automate message CNM094I using NetView automation to provide automatic reactivation of failing resources. The SENDMSG statement specifies the resource types for which the NetView program issues message CNM094I when those resources change status. Message CNM094I provides status information for all resources defined to the status monitor.

Note: If a resource has several status changes in rapid succession, CNM094I might not be issued for the intermediate statuses.

Use the SENDMSG statement to specify each type of resource for which you need additional status information.

The following statements are valid SENDMSG statements in DSICNM:

```
*SENDMSG HOST
*SENDMSG NCP/CA MAJOR NODES
*SENDMSG LINES
*SENDMSG PUS/CLUSTERS
*SENDMSG LUS/TERMINALS
*SENDMSG SWITCHED MAJOR NODES
*SENDMSG SWITCHED PUS
*SENDMSG SWITCHED LUS
*SENDMSG XCA MAJOR NODES
*SENDMSG XCA LINES
*SENDMSG XCA PUS
*SENDMSG LOCAL SNA MAJOR NODES
```

```
*SENDMSG LOCAL PUS
*SENDMSG LOCAL LUS/TERMS
*SENDMSG APPL MAJOR NODES
*SENDMSG APPLICATIONS
*SENDMSG CDRM MAJOR NODES
*SENDMSG CDRMS
*SENDMSG CDRSC MAJOR NODES
*SENDMSG CDRSCS
```

The O SENDMSG statement specifies that the NetView program is to issue message CNM094I at status monitor initialization for the resource types specified on the SENDMSG statement.

The SENDMSG statement must start in column 1 and the resource type must start in column 9.

Code a SENDMSG statement for each resource type for which you need additional status information.

Note: You cannot specify individual resources on the SENDMSG statement. You can specify only a resource type.

To avoid degradation of system performance, carefully select the type of resources for which you want status information.

If you request additional information about a resource type and if your network has many instances of that resource type, the status monitor issues many corresponding CNM094I messages, which can slow down the system

You can use CNM094I with NetView automation and the NetView management console to enhance the recovery of resources in your network. The automation table entry for this message in DSITBL01 suppresses the display and logging of this message.

Specifying the Initial Status for Resources Not Known to VTAM

The following statement in DSICNM specifies whether the status monitor sets an initial status of RESET for any resources that are known to the status monitor but are unknown to the VTAM associated with the status monitor:

```
* O RESET
```

Uncomment this statement for any status monitor that sends status to the SNA topology manager to enable the SNA topology manager to resolve the status of multiply owned resources. The O must be coded in position 1 and the RESET must be coded in position 3.

If you do not specify O RESET, the status monitor sets an initial status of NEVER ACTIVE for all resources not known to the VTAM associated with this status monitor. If this status monitor sends status to the SNA topology manager, the SNA topology manager might not be able to resolve the status of these resources.

Defining SNA Resources to the Status Monitor

In the status monitor, you can assign a descriptive name to each resource or node. This helps the operations staff better understand the network they control, which reduces the education needed and the time required to quickly identify and correct a problem in the network. The status monitor can also increase node availability by automatically reactivating failed nodes when possible.

The status monitor helps control network nodes and display their status. It groups the resources of your network into major and minor node categories the same way they are defined in the VTAM definitions. The approach the status monitor takes in structuring its view of the network is similar in concept and nomenclature to that used by VTAM. The following terms used by the status monitor have the same meanings as they do in VTAM:

Resource
: A generic term for any named entity defined to VTAM.

Node
: A generic term for resource, but it implies a hierarchical relationship.

Major node
An aggregate of minor node definitions represented by a VTAM definition data set member.

Minor node
A resource in a VTAM definition within a major node.

Defining the Nodes

Before the status monitor can monitor a network in the domain where it runs, define the nodes that constitute the network and the relationships between these nodes. Each minor node must belong to a major node. Generally duplicate node names are not used.

The following are some exceptions for which it might be feasible to use duplicate names:

- When defining CDRSC/APPL LUs, the preprocessor checks the resource if the CDRSC was a duplicate of an APPL. For example, you might want to define a CDRSC for one host system by the same name as an APPL LU for another host system.
- Switched LU names can be duplicated under two different major node names.
- Switched LU names and nonswitched LU names can be duplicates of each other.

If duplicate resource names are found, the preprocessor puts a warning message in the print member and sends a return code of 4.

The network definition is held in DSINDEF. This data set member is created by program CNMNDEF (CNMSJ007), which is the status monitor preprocessor. The input to this program comes from the major nodes (VTAM definition members) that together define the total network nodes of the domain where the status monitor is running.

Note:

1. The status monitor preprocessor detects resources that are in the VTAMLST data sets but are not known to VTAM. These resources are still placed in DSINDEF but are automatically omitted from monitoring during status monitor initialization.
2. The status monitor recognizes a maximum of 999999 resources, including the host. If you have more than this number of resources defined to VTAM, code STATOPT=OMIT for some of your resources or define a subset of your VTAMLST resources as the input member to the status monitor preprocessor. If you do not limit your resources to 999999, a message is issued during NetView initialization, and only the first 999999 resources are known to the status monitor. The host is considered a resource; therefore STATMON panels show a maximum of 999998 resources. The host name is displayed in the upper left corner of the panel.
3. You can modify the VTAMLST to monitor an independent LU. Remove the independent LU from under its associated PU and add it underneath a cross-domain resource (CDRSC) major node. For example, suppose your independent LU was previously defined in the following way:

   ```
   A01L01    LINE
   A01P2A0   PU PUTYPE=2
   A01A2L01  LU LOCADDR=0
   ```

   It is now defined under a CDRSC major node with its associated PU name, as shown in the following example:

   ```
                                                    VBUILD    TYPE=CDRSC
   A01A2L01  CDRSC ALSLIST=(A01P2A0)
   ```

   If you do not modify the VTAMLST, the independent LU does not show its correct status.

Defining the Network and Creating CNMCONxx

The CNMCON*xx* data set member of your VTAMLST contains a list of the major nodes known to VTAM that are not included in the ATCCON*xx* (CNMS0003) data set member of your VTAMLST. If all the major nodes that make up your network are specified in the ATCCON*xx*, go to "Defining Resources by Names You Choose" on page 22.

If all of your major nodes are not in ATCCON*xx*, define these major node names to the status monitor. The resources in ATCCON*xx* are automatically started when VTAM starts. If you want resources defined to the status monitor, but not started at initialization, perform the following steps:

1. Create a member named for your major node that contains the VTAM definitions for the node.
2. Define the name of the member by one of the following methods:
   - Specify the name of the member in ATCCON*xx* on a status monitor STATOPT statement. An asterisk (*) must occur in position 1 of this statement, and STATOPT must start in position 16.
   - Specify the name in a member called CNMCON*xx* in VTAMLST on VTAM or STATOPT statements and ensure that CNMCON=*xx* is a part of the parameter list in CNMNDEF that the preprocessor passes to CNMPP.

     Note: CNMS0084 is included in the sample network as an example of a CNMCON*xx* member.

After inserting STATOPT statements, run the status monitor preprocessor CNMNDEF (CNMSJ007). Specify all the major node names that together define the domain's resources with an ATCCON*xx* list or a CNMCON*xx* list.

The CNMCON*xx* list must include the major and minor nodes that are not usually part of the resources for a domain, but can be acquired. Any node that can be a part of the domain, but is not yet acquired, is displayed as NEVACT on the status monitor panels, if it is defined to the status monitor and its higher-level node is not in RESET or RELSD status. All resources that are downstream of a resource in RESET or RELSD status are shown as OTHER on the status monitor panels.

If you want information about...	Refer to...
The STATOPT statement	*IBM Z NetView Administration Reference*

Defining Resources by Names You Choose

You can define a resource (such as a line or a physical unit) with a name of your choice. To do this, insert a STATOPT statement directly after the VTAM or NCP resource definition. An asterisk (*) must occur in position 1 of this statement, and STATOPT must start in position 16. After inserting STATOPT statements, run the status monitor preprocessor CNMNDEF (CNMSJ007).

The following examples show you some uses of STATOPT statements for your production environment.

A01APPLS (CNMS0013)

You can find the following STATOPT statement in A01APPLS (CNMS0013):

```
&CNMDOMN.001 APPL  AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.,EAS=4,          X
               MODTAB=AMODETAB,DLOGMOD=DSILGMOD
*              STATOPT='NETVIEW 001'
```

Where:

STATOPT
Specifies that the description NETVIEW 001 is assigned to APPL CNM01001. This description is displayed on the DESCRIPT form of the Status Detail panels.

You can change this STATOPT statement to the following statement:

```
&CNMDOMN.001 APPL  AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.,EAS=4,          X
               MODTAB=AMODETAB,DLOGMOD=DSILGMOD
*              STATOPT=('NETVIEW 001',NOACTY)
```

Where:

NOACTY
Excludes the node from activity recording.

You can also change this STATOPT statement to the following statement:

```
&CNMDOMN.001 APPL  AUTH=(NVPACE,SPO,ACQ,PASS),PRTCT=&CNMDOMN.,EAS=4,           X
               MODTAB=AMODETAB,DLOGMOD=DSILGMOD
*              STATOPT=OMIT
```

Where:

OMIT
Excludes this node and all the dependent lower nodes that follow from the status monitor network definition.

A04A54C (CNMS0065)

You can find the following example of the STATOPT statement in A04A54C (CNMS0065):

```
A04F0020 LINE  ADDRESS=(020,FULL), ** LINK ADDRESS                 ** X
               SPEED=56000         ** LINK SPEED                   **
*              STATOPT=('LINE020',NOMONIT)
```

Where:

'LINE020'
Specifies that the description LINE020 is assigned to resource A04F0020.

NOMONIT
Excludes the node from automatic reactivation.

You can find the following example of the STATOPT statement in A04A54C (CNMS0065):

```
A04F1028 LINE  ADDRESS=(1028,FULL), ** LINK ADDRESS                ** X
               SPEED=1843200        ** LINK SPEED                  **
*              STATOPT='LINK ADDR=1028'
```

Where:

LINK ADDR=1028
Specifies that the description LINK ADDR=1028 is assigned to line A04F1028. You can change this description to something more significant, such as the name of the destination (for example, ATLANTA).

A01CDRM (CNMS0014)

You can find the following example of the STATOPT statement in A01CDRM (CNMS0014):

```
A01M     CDRM  CDRDYN=YES,        ** AUTHORIZE DYNAMIC CD              X
               CDRSC=OPT,         ** AUTHORIZE DYNAMIC CD              X
               ELEMENT=1,         ** DEFAULT                           X
               ISTATUS=ACTIVE,    ** DEFAULT                           X
               RECOVERY=YES,      ** DEFAULT                           X
               SUBAREA=1,         ** NETWORK UNIQUE SUBAREA ADDRESS    X
               VPACING=63         ** DEFAULT
*              STATOPT='NETA CDRM'
```

Where:

NETA CDRM
Specifies that the description NETA CDRM is assigned to cross-domain resource manager A01M. This node is included for automatic reactivation and activity recording.

A01SNA (CNMS0073)

You can find the following example of the STATOPT statement in A01SNA (CNMS0073):

```
A01P7A0  PU    CUADDR=7A0,         ** PHYSICAL UNIT ADDRESS          **X
               DLOGMOD=M23278I,    ** DEFAULT LOGON MODE ENTRY NAME  **X
               MODETAB=AMODETAB,   ** LOGON MODE TABLE NAME          **X
               USSTAB=AUSSTAB,     ** USS DEFINITION TABLE NAME      **X
               MAXBFRU=15,         ** VTAM BUFFERS TO RECEIVE DATA   **X
               PUTYPE=2,           ** TYPE 2 PHYSICAL UNIT           **X
               VPACING=0           ** NO PACING FOR LU SESSIONS      **
*              STATOPT='SNALOCALTERM'
```

```
**
A01A7A02 LU    LOCADDR=2           ** LOGICAL UNIT                     **
A01A7A03 LU    LOCADDR=3           ** LOGICAL UNIT                     **
A01A7A04 LU    LOCADDR=4           ** LOGICAL UNIT                     **
A01A7A05 LU    LOCADDR=5,          ** LOGICAL UNIT                     **X
               DLOGMOD=M3287SCS,   ** DEFAULT LOG MODE ENTRY NAME      **X
               SSCPFM=USSSCS       ** VTAM - SNA SCS PRINTER           **
*              STATOPT=('PRINTER',NOMONIT)
```

Where:

SNALOCALTERM
: Specifies that this STATOPT statement applies to PU A01P7A0 only.

PRINTER
: Specifies that this STATOPT statement applies to LU A01A7A05 only.

NOMONIT
: Specifies that NOMONIT excludes only the printer from automatic reactivation.

Add STATOPT statements to define your resources.

If you want information about...	Refer to...
The STATOPT statement	*IBM Z NetView Administration Reference*

Defining a Channel to the Status Monitor

You can assign names of a channel and its link station dynamically on the **VARY NET,ACT,ID=*n*** command. However, the dynamic names you create are not known to the status monitor unless you define them in VTAMLST. Refer to CTCA0102 (CNMS0038) and CTNA0104 (CNMS0081) for examples. In certain configurations, you can define a channel-attachment major node that specifies the name to the channel and the link station.

If you want information about...	Refer to...
Defining major nodes	The z/OS Communications Server library

Defining a Host Physical Unit Name for Status Forwarding

Specify the HOSTPU parameter as a unique name within its network in ATCSTR*xx* (CNMS0007). This enables the NetView system to assign a name to the physical unit for the host.

If you want information about...	Refer to...
Assigning names to the physical unit for the host	The z/OS Communications Server library

Running the Status Monitor Preprocessor

Run the CNMNDEF (CNMSJ007) status monitor preprocessor. After inserting the STATOPT statements or after changing any VTAM or NCP definitions, run the preprocessor.

Note:

1. The status monitor preprocessor automatically substitutes any system symbolics that it finds in the VTAM startup file (ATCSTR*xx*) and other VTAMLST members that it processes. To substitute values for symbolics that are not defined or that are defined differently than the definitions currently in effect, run the CNMSJM12 sample job against the members that contain those symbolics to create a set of members with the symbolics resolved for use by the status monitor preprocessor.
2. The status monitor preprocessor expects the VTAM and NCP definitions to be working definitions. When defining an APPL major node, a VBUILD statement is required by the NetView program even though this statement is not required by VTAM. The status monitor preprocessor detects certain errors in the definitions when these errors affect information required by the status monitor. Various configurations require that you use the NCP/EP definition facility (NDF) utility to modify and create new

statements and keywords in the NCP definitions. In these situations, provide the output from the NDF utility to the status monitor preprocessor to ensure accuracy.

3. If you are using the NetView sample network, run the NCP generation definitions provided by the NetView program through the NCP network definition facility (NDF) utility before using these definitions in the sample network. The NDF utility generates the correct NCP major node that is referenced by VTAM. Run the NDF utility before running the status monitor preprocessor CNMNDEF (CNMSJ007) or unpredictable results can occur.

The status monitor preprocessor processes the VTAM NETID keyword in various types of major node definition files. This creates a list of network identifiers. This list is added to the end of DSINDEF with a new record type. If you receive message CNM048E BACKLEVEL DSINDEF - STATUS MONITOR MAIN TASK IS TERMINATING, update your DSINDEF member by running the status monitor preprocessor. The NetView program requires that a network identifier list be included at the end of DSINDEF. This list is created automatically when you run the status monitor preprocessor.

CNMNDEF (CNMSJ007) was copied to your system PROCLIB. The preprocessor is a job in the sample network. However, you can change it to a system-started procedure by following the instructions in the member.

Before you run the preprocessor, review the parameters that are passed to program CNMPP. The syntax for the parameter statement is:

```
//   PARM='&START,LIST=&BOTH&LIST,CONFIG
=&BOTH&CONFIG,CNMCON=&CNMCON'
```

Where:

CNMCON
Use the same value specified or implied for CNMCON when you started VTAM. Include this parameter if you created a CNMCON*xx* member for major nodes that are not included in ATCCON*xx*. This is an optional parameter and no default value exists. This value can be any two alphanumeric or national (@, #, $) characters.

CONFIG
Use the value specified or implied for CONFIG when you started VTAM. If you do not specify a CONFIG value in the PARM statement, the preprocessor uses the CONFIG value specified in ATCSTR*xx*. If you do not specify CONFIG in ATCSTR*xx*, the default is 00, which points to configuration list ATCCON00 (CNMS0006). This value can be any two alphanumeric or national (@, #, $) characters.

Note:

1. The last two characters in ATCCON*xx* are set to the value of CONFIG.
2. If you use the default of 00, make sure ATCCON00 (CNMS0006) is not empty.

HOSTSA
Use the same value specified or implied for HOSTSA when you started VTAM. The default is 1.

HOSTSA can be any 1- to 5-character numeric value from 1 to the value specified for RACSASUP in the VTAM constants module, ISTRACON.

HOSTPU
Use the same value implied for the host PU name when you started VTAM. If you do not specify HOSTPU in the parameter statement, the preprocessor uses the HOSTPU value specified in ATCSTR*xx*. If HOSTPU is not specified in ATCSTR*xx*, the NetView program uses ISTPUS as the default. This is an optional parameter.

LIST
Use the value specified or implied for LIST when you started VTAM (CNMS0010). This value can be any two alphanumeric or national (@, #, $) characters.

Note: The last two characters in ATCSTR*xx* are set to the value of LIST.

START
Values can be COLD or WARM. Use COLD to run the preprocessor. WARM bypasses the preprocessor.

If you want information about...	Refer to...
RACSASUP	The z/OS Communications Server library

Determining the Program Region Size for the Status Monitor Preprocessor

The preprocessor requires a region size greater than or equal to:

```
(N x 80 bytes) / 1000 = S
```

Where:

N
Is the approximate number of nodes in the network.

S
Is the region size, rounded up to the next 1K bytes, with a minimum value of 1K bytes.

Put this value in the JCL region parameter. If you code the region value as 0 (default), results are unpredictable.

Increase the space required in the DSIPARM library by 160 bytes per node. This includes room for compressing the partitioned data set.

Starting the Status Monitor

You can start the status monitor using the **STARTCNM STATMON** command. This command starts the following optional tasks:

- *domain_name*VMT (for example, CNM01VMT)
- *domain_name*BRW (for example, CNM01BRW)

Task *domain_name*VMT uses DSICNM. DSICNM is a task initialization member in the DSIPARM data set.

You can also start these tasks automatically during NetView initialization. To do this, use the following task statement in the CNMSTUSR or C*xx*STGEN member, and update the INIT parameter. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.&DOMAIN.VMT.INIT=Y
```

The Browse task is already set to INIT=Y in the CNMSTYLE member. Recycle the NetView program for these changes to take effect.

Testing the Status Monitor

To go to the status monitor, enter this command at the command line:

```
STATMON
```

You see a panel similar to Figure 2 on page 27.

```
STATMON.DSS                     DOMAIN STATUS SUMMARY   (REFRESH=ON)    09:54 A
HOST: HOST01           *1*   *2*   *3*   *4*
                       ACTIVE  PENDING   INACT    MONIT    NEVACT   OTHER
 .....3 NCP/CA/LAN/PK  ......   ......   ......   ......   .....3   ......
 ....28  LINES         ......   ......   ......   ......   ....28   ......
 ....31  PUS/CLUSTERS  ......   ......   ......   ......   ....31   ......
 ....61  LUS/TERMS     ......   ......   ......   ......   ....61   ......
 .....1 SWITCHED MAJ   ......   ......   ......   ......   .....1   ......
 .....6  SWITCHED PUS  ......   ......   ......   ......   .....6   ......
 ....24  SWITCHED LUS  ......   ......   ......   ......   ....24   ......
 .....2 LOCAL MAJ NDS  .....1   ......   ......   ......   .....1   ......
 .....1  PUS           ......   ......   ......   ......   .....1   ......
 ....14  LUS/TERMS     .....6   ......   ......   ......   .....8   ......
 .....4 APPL MAJ NDS   .....3   ......   ......   ......   .....1   ......
 ....86  APPLICATIONS  ....15   ......   ......   ......   ....48   ....23
 .....2 CDRM MAJ NDS   .....1   ......   ......   ......   .....1   ......
 .....4  CDRMS         .....1   ......   ......   ......   .....3   ......
 .....2 CDRSC MAJ NDS  .....1   ......   ......   ......   .....1   ......
 ....22  CDRSCS        ......   ......   ......   ......   ....22   ......
 ------ -------------  ------   ------   ------   ------   ------   ------
 ...291 TOTAL NODES    ....28   ......   ......   ......   ...240   ....23

CMD==>
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
```

Figure 2. Status Monitor Domain Status Summary Panel

You can browse the network log for any of the message alert settings, *1* through *4*.

Position your cursor before message alert setting *1* and enter s.

You see a panel similar to Figure 3 on page 27.

```
 STATMON.BROWSE     ACTS  NETWORK LOG FOR 2/1/19 (95136) COLS 017 094  09:56
 HOST: HOST01         s *1*   *2*   *3*   *4*               SCROLL ==> CSR
 ---2----+----3----+----4----+----5----+----6----+----7----+----8----+----9---
  CNM01    10:44:52   IST080I  DSIKREM  ACTIV      CNM01VPD CONCT      DSIROVS
  CNM01    10:44:52   IST080I  CNM01000 ACTIV      CNM01001 ACTIV      CNM01002
  CNM01    10:44:52   IST080I  CNM01003 ACTIV      CNM01004 ACT/S      CNM01005
  CNM01    10:44:52   IST080I  CNM01006 CONCT      CNM01007 CONCT      CNM01008
  CNM01    10:44:52   IST080I  CNM01009 CONCT      CNM01010 CONCT      CNM01011
  CNM01    10:44:52   IST080I  CNM01012 CONCT      CNM01013 CONCT      CNM01014
  CNM01    10:44:52   IST080I  CNM01015 CONCT      CNM01016 CONCT      CNM01017
  CNM01    10:44:52   IST080I  CNM01018 CONCT      CNM01019 CONCT      TAF01OPT
  CNM01    10:44:52   IST080I  TAF01000 CONCT      TAF01001 CONCT      TAF01002
  CNM01    10:44:52   IST080I  TAF01003 CONCT      TAF01004 CONCT      TAF01F00
  CNM01    10:44:52   IST080I  TAF01F01 CONCT      TAF01F02 CONCT      TAF01F03
  CNM01    10:44:52   IST080I  TAF01F04 CONCT      TAF01F05 CONCT      TAF01F06
  CNM01    10:44:52   IST080I  TAF01F07 CONCT      TAF01F08 CONCT      TAF01F09
  CNM01    10:44:52   IST080I  TAF01F10 CONCT      TAF01F11 CONCT      TAF01F12
  CNM01    10:44:52   IST080I  TAF01F13 CONCT      TAF01F14 CONCT      TAF01F15
  CNM01    10:44:52   IST080I  TAF01F16 CONCT      TAF01F17 CONCT      TAF01F18
  CNM01    10:44:52   IST080I  TAF01F19 CONCT
  CNM01    10:44:52   IST089I  A01USER  TYPE = APPL SEGMENT     , ACTIV

CMD==>
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
```

Figure 3. Status Monitor Network Log

On the top line of this panel, the abbreviation ACTS tells you that you are browsing the active secondary network log.

Note: Low system activity can cause the data presented in the log display to lag a few moments behind real events in the network.

Stopping the Status Monitor

You can stop the status monitor by using the **STOPCNM STATMON** command.

Defining the Hardware Monitor

The CNMSTYLE member defines the hardware monitor initialization values in those statements that begin with the characters NPDA.

Review the default settings in the CNMSTYLE member and make any changes necessary for your environment in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. You can use the RESTYLE NPDA command to enable changes without recycling the NetView program. The BNJDSERV task is recycled. For more information, refer to the *IBM Z NetView Administration Reference*.

Table 3. NPDA Statements

Function	CNMSTYLE Statement
Databases	NPDA.PDDNM NPDA.SDDNM NPDA.ALERTLOG
Data services tasks	NPDA.DSRBU NPDA.DSRBO NPDA.MACRF
Logging options	NPDA.REPORTS NPDA.ALRTINFP.RECORD
PPI receiver name for alerts routed to the event console	NPDA.TECROUTE
Storage for alerts (ALCACHE)	NPDA.ALCACHE
Hardware alerts panel data	NPDA.ALT_ALERT NPDA.MDSIND
Wrap count	NPDA.W
Error-to-traffic (E/T) ration	NPDA.R
Thresholds for line quality and impulse hits for leased lines connected to IBM LPDA-2 modems	NPDA.LQTHRESH NPDA.IHTHRESH
Rate at which events can be logged	NPDA.RATE
MSUs blocked by the RATE filter can pass to the automation table	NPDA.AUTORATE
Basic Encoding Rules (BER) data	NPDA.PRELOAD_BER
Thresholding factor for messages that are generated by incorrect alerts	NPDA.ERR_RATE
Alert forwarding protocol	NPDA.ALERTFWD
Recording filters	NPDA.PDFILTER

Defining Passwords

The hardware monitor databases are defined using job CNMSJ004 using input member CNMSI101.

To define security passwords for the hardware monitor databases:

1. Stop the hardware monitor.
2. Modify the definition statements in CNMSI101 that define the hardware monitor databases, changing them to include the specification of VSAM cluster passwords. Rerun job CNMSJ004 using these modified statements to delete and redefine the hardware monitor databases.
3. Update the CNMSTPWD member in DSIPARM to include the passwords that you specified when redefining the hardware monitor databases. The following example shows the PWD statements that define the passwords for the hardware monitor databases:

   ```
   PWD.BNJDSERV.P = p_password
   PWD.BNJDSERV.S = s_password
   ```

   Where:

   p_password
   Is the 1- to 8-character password for the primary database.

   s_password
   Is the 1- to 8-character password for the secondary database.
4. Restart the hardware monitor.

Defining Additional Generic Alert Code Points

The hardware monitor allows for the definition of additional generic alert code points and for additional resource types carried in the X'05' subvector.

The code point tables are installed in BNJPNL1. These tables are read during NetView initialization and must have the following names:

- BNJ92TBL
- BNJ93TBL
- BNJ94TBL
- BNJ95TBL
- BNJ96TBL
- BNJ81TBL
- BNJ82TBL
- BNJ85TBL
- BNJ86TBL

While reading the tables during initialization, the NetView program allows syntax errors in the code point entries and builds the tables if possible. Any major errors (for example, a member not found or a control line that is not valid) result in an empty table being built. This can cause undefined code point text seen by callers and end users.

The user can change the code point tables before or after NetView initialization. If the tables are changed after initialization, the user can issue the CPTBL command to dynamically start the changes.

If you want information about...	Refer to...
Migrating and customizing generic alert code points	*IBM Z NetView Customization Guide*
The CPTBL command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Changing the Colors of the Sample Network

The hardware monitor panels and color maps are defined by DD statements BNJPNL1 and BNJPNL2 in CNMPROC (CNMSJ009). BNJPNL1 is searched for the requested panel and BNJPNL2 is used for the related color map.

If you want information about...	Refer to...
How to change the color map to suit your requirements	*IBM Z NetView Customization Guide*

Starting the Hardware Monitor

You can start the hardware monitor using the STARTCNM NPDA command. This command starts the following optional tasks:

- BNJDSERV
- BNJMNPDA
- DSICRTR
- DSI6DST
- *domain_name*LUC

You can also start these tasks automatically during NetView initialization. To do this, use the following task statements in the CNMSTUSR or C*xx*STGEN member, and update the INIT parameter. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.BNJDSERV.INIT=Y
TASK.BNJMNPDA.INIT=Y
TASK.DSICRTR.INIT=Y
TASK.&DOMAIN.LUC.INIT=Y
```

The DSI6DST optional task is already set to INIT=Y in the CNMSTYLE member. For these changes to the CNMSTYLE member to take effect, recycle the NetView program.

Stopping the Hardware Monitor

You can stop the hardware monitor by using the **STOPCNM NPDA** command.

Defining the Session Monitor

The CNMSTYLE member defines the session monitor initialization values in the statements that begin with the characters NLDM. For more information, see the *IBM Z NetView Administration Reference*.

Table 4. NLDM Statements

Function	CNMSTYLE Statement
Databases	NLDM.PDDNM NLDM.SDDNM
Data services task parameters	NLDM.DSRBO NLDM.MACRF
Other domains	NLDM.AUTODOM NLDM.AUTHORIZ.*suffix* NLDM.CDRMDEF
Session awareness data collection	NLDM.SAW NLDM.SAWNUM NLDM.SAWSIZE

Table 4. NLDM Statements (continued)	
Function	**CNMSTYLE Statement**
PIU trace parameters	NLDM.KEEPDISC NLDM.KEEPPIU NLDM.MAXEND NLDM.PIUTNUM NLDM.PIUTSIZE
RTM parameters	NLDM.RTM NLDM.RTMDISP NLDM.KEEPRTM NLDM.PERFMEM
Trace started at initialization	NLDM.TRACEGW NLDM.TRACELU NLDM.TRACESC
Network parameters	NLDM.NETID NLDM.LUCOUNT
Timers	NLDM.AMLUTDLY NLDM.CDTIME NLDM.DRDELAY NLDM.FCTIME NLDM.RETRY
External logging	NLDM.LOG
Session availability	NLDM.SESSTATS
Session wrapping	NLDM.KEEPSESS
ER data	NLDM.RTDASD NLDM.ERCOUNT
Keep class definitions	NLDM.KEEPMEM
Display settings	NLDM.SESSMAX
Exception list	PEXLSTxx

Defining Passwords

The session monitor databases are defined using job CNMSJ004 using input member CNMSI101.

To define security passwords for the session monitor databases:

1. Stop the session monitor.
2. Modify the definition statements in CNMSI101 that define the session monitor databases, changing them to include the specification of VSAM cluster passwords. Rerun job CNMSJ004 using these modified statements to delete and redefine the session monitor databases.

3. Update the CNMSTPWD member in DSIPARM to include the passwords that you specified when redefining the session monitor databases. The following example shows the PWD statements that define the passwords for the session monitor databases:

```
PWD.AAUTSKLP.P = p_password
PWD.AAUTSKLP.S = s_password
```

Where:

p_password
Is the 1- to 8-character password for the primary database.

s_password
Is the 1- to 8-character password for the secondary database.

4. Restart the session monitor.

Defining Sense Code Filtering

The most efficient method to filter sessions based on sense codes is to use the VTAM VARY command.

You can also filter sessions using the session monitor. You can analyze a session monitor VSAM data set and print the results using job CNMSJM10. The results show how many times each unique sense code occurred. You can use this information to decide which sense codes to filter.

The following sections explain how to:

- Decide which sense codes to filter.
- Add a sense code for filtering.
- Stop sense code filtering.

Deciding Which Sense Codes to Filter

To analyze the session monitor VSAM data set and filter sense codes:

1. Run job CNMSJM10.

   The job generates a report that is sent to the printer. This report contains the sense code (which includes the reason code) and frequency count for each unique sense code. Figure 4 on page 32 shows an example of such a report. In this report, the sense codes and the reason codes are combined in the column labeled SENSE CODE. The frequency counts are given in the column labeled TOTAL.

   The report can contain a total of 200 sense code entries. If more than 200 sense codes exist, the 200th sense code entry contains the frequency counts for the remaining sense codes not displayed in the report.

```
SENSE CODE COUNTS:

ITEM#        SENSE CODE     TOTAL          PERCENT
----------   ----------     ----------     -----------
         1     00000000              3           25.0%
         2     087D0001              8           66.6%
         3     80200007              1            8.3%
----------   ----------     ----------     -----------
TOTAL                               12           99.9%
```

Figure 4. Sense Code Report

2. Analyze the report.

   Look at the report to determine if any of the sense codes can be filtered. Referring to Figure 4 on page 32, suppose you decide to filter sense code 087D0001; go to the next step.

3. Consult sample CNMS0055 which is shown in Figure 5 on page 33.

```
*
* ENTRIES FOR DATA RECORDING SENSE CODE FILTERING
*
NSENSE   DC    F'0'      NUMBER OF SENSE CODE ENTRIES IN TABLE
*                        (BE SURE NUMBER IS NOT GREATER THAN 25)
*
SENSE01  DC    XL4'00000000'  SENSE CODE # 1 TO BE FILTERED
SLEN01   DC    AL1(0)         NUMBER OF SIGNIFICANT LEADING BYTES
*                             FOR SENSE CODE # 1 COMPARISON
*
SENSE02  DC    XL4'00000000'  SENSE CODE # 2 TO BE FILTERED
SLEN02   DC    AL1(0)         NUMBER OF BYTES TO SENSE CODE # 2
```

Figure 5. Sample (as Supplied with the NetView Program)

If the sense code is in the sample, the sense code is being filtered (that is, it is not recorded on the session monitor VSAM data set) and you do not have to do anything. Otherwise, the sense code is NOT being filtered (it is recorded on the session monitor VSAM data set).

Adding a Sense Code for Filtering

Continuing with the preceding example: The sense code you want to filter is not in the sample. You need to add the sense code to the sample that you want to filter.

To add the sense code to the sample:

1. Modify DSICTMOD and reassemble using CNMS0055 to change filter status.

   If the sample was the one shown in Figure 5 on page 33:

   a. Change NSENSE to the number of sense codes in the sample. Here, it is 1 because you want to filter only one sense code. This number cannot be greater than 25 because the table holds only 25 sense code entries. See the results of this change in Figure 6 on page 33.
   b. Change SENSE01 to the 2-byte sense code followed by the 2-byte reason code. See the sense code report (in this example, Figure 4 on page 32) to get the sense code and reason code of 087D0001. See the results of this change in Figure 6 on page 33.

      Note: To filter all sense codes with the same first 2 hexadecimal digits, fill in the first 2 digits (1 byte) and fill the remaining 6 hexadecimal digits (3 bytes) with zeros.

      To filter all sense codes with the same first 4 hexadecimal digits, fill in the first 4 digits (2 bytes) and fill the remaining 4 hexadecimal digits (2 bytes) with zeros.

      Change the number of significant bytes (SLEN01) to correspond with your decision.
   c. Change SLEN01 to the length of significant bytes. In this example, you want to filter 087D0001, so you enter 4. See the results of this change in Figure 6 on page 33.

   ```
   NSENSE DC F'1'               (1 sense code in sample)
   SENSE01 DC XL4'087D0001'     (Sense code/reason code)
   SLEN01  DC AL1(4)            (Use the entire 4 bytes)

   SENSE02 DC XL4'00000000'
   SLEN02  DC AL1(0)
   ```

   Figure 6. Sample (with Sense Code 087D0001 Added)
2. Run CNMS0055 to reassemble DSICTMOD which initiates the filtering process.

   Note: If the NetView system is currently active and values in DSICTMOD are modified, restart the NetView program to use the new values.

   In the example in Figure 6 on page 33, sense code 087D0001 is now filtered.
3. Repeat the steps to filter additional sense codes. Remember to change NSENSE by the number of sense codes in the sample.

Note: If you run sample CNMSJM10 again without clearing the VSAM data set of the sense codes you just filtered, the sense codes remain in the VSAM data set. However, the sense codes that were updated in DSICTMOD are being filtered.

Stopping Sense Code Filtering

If you decide that you no longer want to filter a specific sense code, you can change the sample by either:

- Changing the length of the sense code to 0; that is, AL1(0)
- Deleting the sense code entry from the table. (Deleting the sense code entries that you no longer want to filter can help performance.)

If you change the length (SLEN01 in Figure 6 on page 33) to 0, that sense code is skipped. Run CNMS0055 to reassemble DSICTMOD to change filtering status.

Note: If the NetView system is currently active and the values in DSICTMOD are modified, restart your NetView program to use the new values.

Whether you change the length of the sense code to 0 or delete it from the table (DSICTMOD), maintain 25 two-line entries (place holders) in the table.

If you delete an entry (two lines), replace that entry in the table to maintain the required 25 two-line entries in the table. To replace the entry:

1. Add the two-line entry beneath the last sense code in the table being filtered. For example, assume you deleted the following from the table:

   ```
   SENSE01 DC XL4 '08D70001'
   SLEN01  DC AL1 (4)
   ```

   Add the following as a place holder after the last sense code in the table being filtered:

   ```
   SENSE01 DC XL4 '00000000'
   SLEN01  DC AL1 (0)
   ```

   This procedure ensures that the filtered sense codes stay together at the top of the table, and also maintains 25 entries in the table.
2. Change NSENSE to the number of filtered sense codes.
3. Run CNMS0055 to reassemble DSICTMOD to change filtering status.

Defining Session Awareness (SAW) Data

Decide how much session awareness (SAW) data and trace data you want to collect and keep. You can keep data for all sessions, or just for specific sessions. For each session for which SAW data is collected, decide the following items:

- The number of PIUs to keep
- Whether to keep session history data

For performance reasons, do SAW filtering at VTAM rather than at the NetView program, although filtering at the NetView program is supported.

Filtering decides whether particular SAW data is collected at all. You can choose what to do with the SAW data collected by the NetView program, as described below:

- Review the defaults that are coded in the CNMSTYLE member.
- Review the KCLASS and MAPSESS statements supplied in AAUKEEP1.
- Make any necessary changes to the defaults.

If you want information about...	Refer to...
Setting up SAW data collection	*IBM Z NetView Tuning Guide*

If you want information about...	Refer to...
Coding SAW in VTAM	The z/OS Communications Server library

Coding KCLASS and MAPSESS Statements in the NetView Program

A keep member is defined in the samples. You can alter it to define your keep classes, or you can use the defined sample member by uncommenting the NLDM.KEEPMEM statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. To uncomment the KEEPMEM statement, remove the asterisk from the following statement:

```
*NLDM.KEEPMEM=AAUKEEP1
```

This keep member contains two types of statements, KCLASS statements and MAPSESS statements. The KCLASS statements define the restrictions to the amount of data that is kept. You can have multiple KCLASS statements to define different restrictions.

The KCLASS statements must occur at the beginning of the data set member. KCLASS statements must precede all MAPSESS statements.

To change the keep member, create KCLASS statements to define your restrictions.

A sample KCLASS statement in AAUKEEP1 is:

```
* SAMPK2     KCLASS   SAW=YES,+
*                     KEEPPIU=10,                                    +
*                     DASD=YES,                                        +
*                     KEEPSESS=10,                                 +
*                     DGROUP=TSO
```

Where:

SAMPK2
Is the name of the keep class being defined.

SAW=YES
Specifies that session awareness data is kept. SAW=YES is the default.

KEEPPIU=10
Specifies the PIUs kept for each session in this keep class. This value can be from 0 – 999, and the default is 7. In this example, 10 PIUs are kept.

DASD=YES
Specifies that sessions are always recorded to the session monitor VSAM database.

KEEPSESS=10
Indicates the DASD session wrap count (0–999) for all sessions mapping into this KCLASS. If the value is 0, session wrapping does not occur until the count of sessions for this KCLASS exceeds 32767. Use the keyword DASD=NO to prevent recording of sessions for this KCLASS. If KEEPSESS is not coded, the global KEEPSESS value is used for sessions mapping into this KCLASS. If the global wrap count is 0 in the CNMSTYLE member, wrapping does not occur, regardless of the value of KEEPSESS. Also, sessions are not recorded by DGROUPs.

DGROUP=TSO
Specifies the grouping characteristics of all the MAPSESS sessions mapping to this KCLASS statement.

Note: To remove session data from the session monitor VSAM database, use the PURGEDB command. Restrict the use of the PURGEDB command to nonpeak times.

After you create your KCLASS statements, create MAPSESS statements. The MAPSESS statements define the sessions to which a KCLASS statement apply.

In the samples, the first MAPSESS statement in AAUKEEP1 is:

```
MAP1   MAPSESS   KCLASS=SSCPSSCP,
PRI=A??M,SEC=A??M
```

Where:

MAP1
Identifies the MAPSESS statement in any related error messages.

KCLASS
Specifies that any session that matches all the other MAPSESS operands is to be assigned to keep class SSCPSSCP.

PRI=A??M
Specifies all primary session partner names that have a first character of *A* and a fourth character of *M*.

SEC=A??M
Specifies all secondary session partner names that have a first character of *A* and a fourth character of *M*.

In this example, any session where the primary session partner and the secondary session partner have names with a first character of *A* and a fourth character of *M* have SAW data stored as specified by KCLASS SSCPSSCP.

If you want information about...	Refer to...
The PURGEDB command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Discarding SAW Data

Code the information in the following sections in VTAM. VTAM includes a default table, ISTMGC10 in VTAMLIB, where session awareness data (SAW) can be filtered.

If you want information about...	Refer to...
Filtering SAW data from VTAM	The z/OS Communications Server library

Specifying That SAW Data Is to Be Discarded

You can save storage by discarding SAW data for selected SSCP-LU and LU-LU sessions. To do this, add the following KCLASS statement to AAUKEEP1:

```
NOSAW    KCLASS  SAW=NO
```

You can then create MAPSESS statements to define those sessions for which you want to discard the SAW data. An example of such a statement follows:

```
MAPD  MAPSESS  KCLASS=NOSAW,PRI=TSO*,SEC=B??B????
```

In this example, the SAW data is discarded for any session with:

- A KCLASS named NOSAW
- A primary end point name beginning with TSO
- A secondary end point name with B as both the first and the fourth characters

Starting the Session Monitor

You can start the session monitor using the **STARTCNM NLDM** command. This command starts the following optional tasks:

- AAUTCNMI
- AAUTSKLP

- DSIAMLUT
- *domain_name*LUC
- DSICRTR

You can also start these tasks automatically during NetView initialization. To do this, use the following task statements in the CNMSTUSR or C*xx*STGEN member, and update the INIT parameter. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.AAUTCNMI.INIT=Y
TASK.AAUTSKLP.INIT=Y
TASK.DSIAMLUT.INIT=Y
TASK.&DOMAIN.LUC.INIT=Y
TASK.DSICRTR.INIT=Y
```

For these changes to the CNMSTYLE member to take effect, recycle the NetView program.

Stopping the Session Monitor

You can stop the session monitor by using the **STOPCNM NLDM** command.

Defining AON

The Automated Operations Network (AON) provides a way to provide automation across multiple network protocols. AON intercepts alerts and messages that indicate problems with network resources and attempts to recover failed resources. Installation activities include the following activities:

- "Setting Up Base AON " on page 37
- "Setting Up AON/TCP Support" on page 45
- "Setting Up AON/SNA Support" on page 59

If you are running AON and IBM Z System Automation in the same NetView address space, refer to "Enabling Workload Management to Manage the NetView Program" on page 190. For more information about AON customization, refer to *IBM Z NetView User's Guide: Automated Operations Network*.

Setting Up Base AON

You can use Table 5 on page 37 as you work through the AON installation tasks described in this section.

Table 5. Base AON Installation Summary

Task	**Job or Member Name**	**Reference**
Update the CNMSTUSR or C*xx*STGEN member to enable the AON tower and subtowers, as necessary. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.	DSIPARM (CNMSTYLE)	• "Updating CNMSTYLE" on page 38 • *IBM Z NetView Installation: Getting Started*
Allocate VSAM clusters	CNMSAMP (CNMSJ004)	"Allocating VSAM Clusters" on page 39
If you are using an SAF product, add gateway and automation operator definitions and passwords	DSIPARM (DSIOPF)	"Adding Gateway and Automation Operator Definitions and Passwords (SAF only)" on page 40
Change the domain ID	CNMCLST (EZLEISP1, EZLEISP2)	"Changing the Domain ID" on page 40
Update the NetView startup procedure	CNMPROC (CNMSJ009)	"Updating the NetView Startup Procedure" on page 41

Table 5. Base AON Installation Summary (continued)

Task	Job or Member Name	Reference
Update policy information.	DSIPARM (CNMSTYLE)	"Updating CNMSTYLE" on page 38
	• DSIPARM (EZLCFG01) • DSIPARM (FKXCFG01) • DSIPARM (FKVCFG01)	• "Updating the Control File Policy Definitions" on page 41 • *IBM Z NetView Automation Guide*
Restrict access to AON commands and menu selections.	• CNMSAMP (CNMSCAT2)	• "Restricting Access to AON Commands and Menu Selections" on page 45 • *IBM Z NetView Security Reference*
Tune the REXX environment.	DSIPARM (CNMSTYLE)	"Adding REXX Environment Blocks" on page 45

Updating CNMSTYLE

To enable AON, update the AON TOWER statement in the CNMSTUSR or C*xx*STGEN member. Also update subtower statements in the CNMSTUSR or C*xx*STGEN member for additional functions that you are implementing. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Subtower
Description

SNA
SNA automation (AON/SNA)

To also enable AON/SNA X.25 support, remove the asterisk (*) from the following statement:

```
*TOWER.AON.SNA = X25
```

TCP
TCP/IP automation (AON/TCP)

To also enable Intrusion Detection Services (IDS) support, remove the asterisk (*) from the following statement:

```
*TOWER.AON.TCP = IDS
```

AON uses policy definitions to provide automation of your network resources. The NetView program loads the DSIPARM member EZLCFG01 control file during initialization. This file contains values such as the notification operator IDs, the automation operator IDs, threshold values for resources, monitoring values, and recovery values. By default, the policy definitions are shipped enabled.

If you are migrating from a previous release of the NetView product, you might want to change the policy definition member name. If so, locate the POLICY statement in the CNMSTYLE member:

```
POLICY.AON = EZLCFG01
```

Use the POLICY.AON statement in the CNMSTUSR or C*xx*STGEN member, and change the name of the policy definition member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
AON tower and subtower statements	*IBM Z NetView Installation: Getting Started*
AON policy file	• "Updating the Control File Policy Definitions" on page 41 • *IBM Z NetView Automation Guide*

Allocating VSAM Clusters

Table 6 on page 39 lists the VSAM databases that were allocated by sample job CNMSJ004 during base NetView installation.

Table 6. AON VSAM databases

VSAM database	Description
NETVIEW.CNM01.STATS	AON status database
NETVIEW.CNM01.LOGP NETVIEW.CNM01.LOGS	AON log database
NETVIEW.CNM01.PASSWORD	AON password database for gateway operator IDs and passwords

If changes are needed to these databases (for example, defining passwords or allocating additional space), rerun CNMSJ004 to delete and define these databases.

Note:

1. Place the VSAM database INDEX and DATA components on different devices for better performance.
2. Because each AON component uses these VSAM clusters, you might need to reallocate these clusters again if the initial space allocation is not large enough. The AON VSAM clusters for the status database are allocated by default with 4 cylinders of hard disk drive. In very large networks, you might need to define additional space.
3. The AON status database is allocated as REUSE so the DBMAINT facility can properly perform database maintenance. The way the status databases are allocated must match the value of the DBMAINT keyword in the ENVIRON SETUP control file entry in EZLCFG01. By default, the ENVIRON SETUP control file uses DBMAINT=REUSE.

   If the values do not match, errors occur. For more information about the ENVIRON SETUP control file entry, refer to the *IBM Z NetView Administration Reference*.

Defining Passwords

The AON databases are defined using job CNMSJ004 using input member EZLSI101.

To define security passwords for the AON databases:

1. Stop AON.
2. Modify the definition statements in EZLSI101 that define the AON databases, changing them to include the specification of VSAM cluster passwords. Rerun job CNMSJ004 using these modified statements to delete and redefine the AON databases.

3. Update the CNMSTPWD member in the DSIPARM data set to include the passwords that you specified when redefining the AON databases. The following example shows the PWD statements that define the passwords for the AON databases:

```
PWD.EZLSTAT.P = ps_password
PWD.EZLLOG.P = pl_password
PWD.EZLLOG.S = sl_password
```

Where:

ps_password
Is the 1- to 8-character password for the primary AON status database.

pl_password
Is the 1- to 8-character password for the primary AON log database.

sl_password
Is the 1- to 8-character password for the secondary AON log database.

4. Restart AON.

Note: The use of a secondary AON status VSAM database is supported, but not recommended.

Adding Gateway and Automation Operator Definitions and Passwords (SAF only)

If you are using an SAF product such as RACF® for security, define all gateway and automation operator IDs to that security product. The operator IDs are located in DSIOPF include members EZLOPF, FKVOPF, and FKXOPF. Ensure that you allocated a VSAM password data set (NETVIEW.CNM01.PASSWORD) that is used to manage the RACF-required user IDs and passwords of the gateway operators logging on to other NetView domains.

Changing the Domain ID

AON members are copied to the following NetView user data sets by sample job CNMSJ003 during base NetView installation:

- NETVIEW.V6R3USER.CNM01.DSIPARM
- NETVIEW.V6R3USER.CNM01.CNMPNL1

To change the domain ID in AON members without having to edit the individual members:

1. Copy the EZLEISP1 and EZLEISP2 members from the CNMCLST data set to a data set in the SYSPROC concatenation of your TSO procedure. EZLEISP1 is the program that changes the domain ID in the AON members. EZLEISP2 is a macro called by EZLEISP1.
2. From TSO, enter the following command:

```
EZLEISP1 dataset olddomain newdomain
```

where:

dataset
The data sets that contain the members to be changed, which are typically, the following data sets:

- NETVIEW.V6R3USER.CNM01.DSIPARM
- NETVIEW.V6R3USER.CNM01.CNMPNL1

For fully qualified data set names, include single quotation marks (') around the data set name.

Note: Do not run EZLEISP1 against the SMP/E target or distribution libraries.

olddomain
The domain ID that you want to change (the default domain ID is CNM01).

newdomain
The new domain ID.

For example, to change all occurrences of domain ID CNM01 to domain ID CNM44 for the AON members in data set NETVIEW.V6R3USER.CNM01.DSIPARM , enter:

```
EZLEISP1 'NETVIEW.V6R3USER.CNM01.DSIPARM' CNM01 CNM44
```

EZLEISP1 issues the following output messages:

```
  time Processed dsn member, Modified.
  time Processed dsn member, unchanged.
  time Processed dsn member, ERROR RC = rc
```

Updating the NetView Startup Procedure

Make sure the following AON data sets are uncommented in CNMPROC (CNMSJ009):

- Automation Status File data sets:

```
//* AON AUTOMATION STATUS FILE
//*
//*EZLSTAT  DD   DSN=NETVIEW.CNM01.STATS,
//*         DISP=SHR,AMP='AMORG,BUFNI=10,BUFND=5'
```

- Automation Password data sets:

```
//* AON PASSWORD DATASET - FOR GATEWAY SESSION PASSWORD MANAGEMENT
//*
//*EZLPSWD  DD   DSN=NETVIEW.CNM01.PASSWORD,
//*         DISP=SHR,AMP='AMORG,BUFNI=10,BUFND=5'
```

- Automation Log data sets:

```
//* AON AUTOMATION LOG DATASETS
//*
//*EZLLOGP  DD   DSN=NETVIEW.CNM01.LOGP,
//*         DISP=SHR,AMP='AMORG,BUFNI=10,BUFND=5'
//*EZLLOGS  DD   DSN=NETVIEW.CNM01.LOGS,
//*         DISP=SHR,AMP='AMORG,BUFNI=10,BUFND=5'
```

Note:

1. The data set names on the DD statement in the NetView startup procedure also are shown in the VSAM cluster definitions for the logs and status file. If you changed the data set names, also make sure that the cluster definitions use the new names.
2. If you changed the DD name, change all occurrences of the DD name in the verify step of the NetView procedure. Also verify that the DD name in the EZLLOGM and EZLSTSM members of the DSIPARM data set are the same as the name you are using.

Updating the Control File Policy Definitions

The AON policy definitions are loaded when the NetView program is initialized. AON provides minimum automation functions. Update the following policy members in DSIPARM with additional information, such as TCP/IP for MVS stack information:

- EZLCFG01 (AON base)
- FKXCFG01 (AON/TCP)
- FKVCFG01 (AON/SNA)

If you want information about...	Refer to...
Loading AON policy	"Updating CNMSTYLE" on page 38
AON policy definitions	*IBM Z NetView Automation Guide*
AON policy definition statements	*IBM Z NetView Administration Reference*

Overview of AON Policy Definitions

The following table provides an overview of the AON policy definitions, whether they are new or have changed for this release, whether they are required, and which automation component they use.

Table 7. Control File Entries. An asterisk (*****) next to the entry in the Component column indicates that the function no longer requires AON; for more information, see Chapter 6, "Configuring IP Management," on page 119.

Entry Description	Entry Name	New (N), Change (C), or No Change (NC)?	Required?	Component
Active monitoring	ACTMON	NC	No	Base*
Adjacent NetViews	ADJNETV	NC	No	Base
Automation operators	AUTOOPS	NC	Yes	Base
Automation of cross-domain logons	CDLOG	NC	No	Base
Dynamic Display Facility (DDF)	DDF	NC	Yes	Base
Generic DDF	DDFGENERIC	NC	Yes	Base
Grouping DDF resources	DDFGROUP	NC	No	Base
Environment AIP status	ENVIRON AIP	NC	No	Base
DDF environment	ENVIRON DDF	NC	Yes	Base
Environment exit	ENVIRON EXIT	NC	No	Base
Environment RACF	ENVIRON RACF	NC	No	Base
Environment setup	ENVIRON SETUP	C	Yes	Base
Environment timeout	ENVIRON TIMEOUT	C	Yes	Base
Automation log	EZLTLOG	NC	Yes	Base
Notification forwarding for focal point services	FORWARD FOCALPT	NC	No	Base
Application definition for focal point services	FULLSESS	NC	No	Base
Notification forwarding	GATEWAY	NC	No	Base
Defining installed components	INSTALLOPT	NC	No	Base
Large-scaling thresholds	LSTHRESH	NC	No	Base
Monitor intervals	MONIT	NC	Yes	Base
Monitor mode	MONITOR	NC	No	Base
Notification operators	NTFYOP	NC	Yes	Base
Recovery automation flag	RECOVERY	C	Yes	Base
Defining sessions to monitor	SESSION	NC	No	Base
Error thresholds	THRESHOLDS	NC	Yes	Base

Table 7. Control File Entries. An asterisk (*****) next to the entry in the Component column indicates that the function no longer requires AON; for more information, see Chapter 6, "Configuring IP Management," on page 119. *(continued)*

Entry Description	Entry Name	New (N), Change (C), or No Change (NC)?	Required?	Component
Timer automation	TIMER	NC	No	Base
Include members	%INCLUDE	NC	No	Base
Notification policy	NOTIFY	NC	Yes	Base
Identify control points	CPCPSESS	NC	No	SNA
SNBU environments	ENVIRON SNBU	NC	No	SNA
NCP recovery	NCPRECOV	NC	No	SNA
Monitor sessions	SESSION	NC	No	SNA
Switched Network Backup automation	SNBU	NC	No	SNA
SNBU default automation parameters	SNBU DEFAULTS	NC	No	SNA
SNBU default PU parameters	SNBU PU	NC	No	SNA
SNBU modem pool definition	SNBUPOOL	NC	No	SNA
Subsystem for NetView access	SUBSYSTEM	NC	No	SNA
Switch to backup line	TGSWITCH	NC	No	SNA
X.25 switched virtual circuit (SVC) definitions	X25MONIT	NC	No	SNA
AON/TCP TSO Servers	TSOSERV	NC	No	TCP
Load CLIST into storage	RESIDENT	NC	No	Base
Critical AON/TCP Resource Def	TCPIP	NC	No	TCP
TCP/IP for 390 Host Def	IPHOST	NC	No	TCP*
TCP/IP for 390 Interface Def	IPINFC	NC	No	TCP*
TCP/IP for 390 Router Def	IPROUTER	NC	No	TCP*
TCP/IP for 390 Socket Def	IPPORT	NC	No	TCP*
TCP/IP for 390 NameServer Def	IPNAMESERV	NC	No	TCP*
TCP/IP for 390 Telnet Server Def	IPTELNET	N	No	TCP*
TCP/IP for 390 TN3270 Server Def	IPTN3270	NC	No	TCP*

Before going to the next step, compare the contents of the AON control files with your existing control files to determine what is required to merge these files. Merge your customization into the new level of EZLCFG01, FKVCFG01, or FKXCFG01 in the DSIPARM data set.

If you want information about...	Refer to...
Control file entries	*IBM Z NetView Administration Reference*

Setting the Automation Log Switch

The AON automation log has automatic switching capabilities. When the automation log is full, the EZLTLOG entry in the control file specifies whether the automation log must be automatically switched. You can also specify a job to run when the logs switch by modifying the EZLTLOG entry in EZLCFG01 and uncommenting the JOB= parameter.

Figure 7 on page 44 shows the EZLTLOG statements that are shipped with AON.

```
EZLTLOG       PRIMARY,AUTOFLIP=YES,
                        LIT='PRIMARY AUTOMATION LOG'        * CODE , *
*                       JOB=USER.PROCLIB(EZLSJ007)
EZLTLOG       SECONDARY,AUTOFLIP=YES,
                        LIT='SECONDARY AUTOMATION LOG'      * CODE , *
*                       JOB=USER.PROCLIB(EZLSJ009)
```

Figure 7. EZLTLOG statements

Where:

PRIMARY
Specifies the primary automation log.

SECONDARY
Specifies the secondary automation log.

AUTOFLIP
Specifies whether the log switches to the other log when the current log fills up. The default as shipped is YES.

LIT
Specifies the text for the message that is used to notify operators of a log switch.

JOB
Specifies the job to run when the logs switch.

To automatically switch to the secondary automation log, set the AUTOFLIP keyword to YES on the *primary* EZLTLOG statement. To automatically switch back to the primary automation log, set the AUTOFLIP keyword to YES on the *secondary* EZLTLOG statement. To deactivate automation log functions, replace the two EZLTLOG entries in Figure 7 on page 44 with the following single EZLTLOG entry:

```
EZLTLOG    NONE
```

Using a Sequential Data Set to Back Up Log Files

To use the a sequential data set to automatically back up the automation logs, do these steps:

1. Run job EZLSJ005 in the NETVIEW.V6R3USER.INSTALL data set to allocate the NETVIEW.LOGHIST data set used by the automation log backup. Update the EZLSJ005 job to reflect the correct DASD type, data set name, and any other information that is unique to your environment. The NETVIEW.LOGHIST data set is a sequential data set to which AON appends log data when the primary or secondary automation log file is full.
2. Copy the EZLSJ007 and EZLSJ009 jobs from the CNMSAMP data set to the PROCLIB data set. These JCL jobs reproduce the automation log files into the NETVIEW.LOGHIST data set before it is cleared. Review these jobs to make sure that the cluster names and VSAM data set names are correct. Table 8 on page 45 contains the names of the automation log, the name of the JCL job to run when the log switches, and the name of the DSIPARM member that contains the IDCAMS commands for the log.

Table 8. Information for sequential data set backup of log files		
Automation Log	**JCL Job**	**DSIPARM Member**
NETVIEW.CNM01.LOGP	EZLSJ007	EZLSUP01
NETVIEW.CNM01.LOGS	EZLSJ009	EZLSUS01

3. Uncomment the entry with the JOB= keyword on the primary and secondary EZLTLOG statements, making sure that the correct data set name is given for the EZLSJ007 and EZLSJ009 sample jobs.

   Note:

   a. Each line of an entry must end with a comma unless it is the last line of the entry.
   b. For automatic job submission, the subsystem interface (SSI) must be active.

Restricting Access to AON Commands and Menu Selections

You can use the NetView command authorization table or a system authorization facility (SAF) product such as RACF to restrict access to commands and menu selections.

AON displays the following message for unauthorized menu selections:

```
EZL215I OPTION opt NOT PROCESSED - ACCESS NOT AUTHORIZED
```

and the following message for unauthorized commands:

```
DSI213I ACCESS TO 'object' IS NOT AUTHORIZED
```

The initial NetView security settings are defined by the SECOPTS statements in the CNMSTYLE member. Initially, you can use the default security settings and restrictions for AON commands. As needed, identify commands or menu selections to which you want to further restrict user access. For a list of AON commands, keywords, and values that can be protected, refer to *IBM Z NetView Security Reference*. Most menus (panels) have corresponding commands that drive them. For example, a menu that begins with the characters FKXK have a corresponding command named FKXE.

Adding REXX Environment Blocks

Consider tuning the REXX environment. You might need to allocate additional blocks (300-400) depending on the number of subsystem and automation operators that you define. You can use the DEFAULTS command or the DEFAULTS.REXXSLMT statement in the CNMSTUSR or C*xx*STGEN member to increase the storage associated with a REXX environment. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If more blocks are required than are available, the NetView program issues the CNM416I REXX environment initialization error messages. If you receive this message, review the number of blocks that you are allocating.

If you want information about...	Refer to...
REXX Environment	“Using Language Processor (REXX) Environments in the NetView Environment” on page 93

Setting Up AON/TCP Support

The NetView program provides the following TCP/IP functions. (TCP390 definitions are required only for remote TCP/IPs and to override discovered data.)

- Stack management

  You can use SNMPView to manage the stacks. Both monitoring and management functions support local and remote stacks.

- IP connection management

Using the 3270, Web browser, or NetView management console, you can manage IP connections into your stacks. All connections are supported, including TN3270, FTP, CICS, and SMTP. You can use operator filters to assist in reducing the amount of data. You can view connection byte counts and take action (for example, breaking the connection).

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- IP commands

  Using the 3270, Web browser, or NetView management console, you can issue various IP commands. AON/TCP on the 3270 provides IP command panels, including the Ping, tracerte, and generic commands.

- IP server management

  With the IP server management function, you can monitor and manage TSO and UNIX servers from a 3270 panel. The panel displays the current status of the servers and provides options to start and stop them.

- SNMP functions

  Using 3270 panels, you can issue SNMP commands such as GET, SET, WALK, and Remote Ping. You can also use the MIB groups function to define several MIBs to be part of a group and use the NetView program to retrieve that group. Sample MIB groups are provided.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- IP resource manager

  Using the IP resource manager, you can display any IP resource, including stacks that are discovered by the discovery manager, and the current status of the resource from 3270 panels. You can manage those resources by using options that are provided on the panels.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- Integration with CISCOWorks Blue Inter-network Status Monitor

  Using 3270 panels, you can navigate to the CISCOWorks Blue Inter-network Status Monitor, if it is installed.

- Proactive monitoring of IP resources

  You can proactively monitor critical IP resources, including the following resources:

  - IP stacks
  - Hosts
  - Interfaces
  - Routers
  - TN3270 servers

  For example, you can define a performance MIB to query within a router and define a threshold. When you exceed that threshold, AON/TCP notifies you of the problem.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- IP connection monitoring and thresholding

  You can monitor IP connections such as printer connections and apply policy definitions to determine if they are unavailable. You can choose to notify an operator or use automation to break the connection.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- Intrusion Detection Services

  You can use Intrusion Detection Services (IDS) support from the z/OS Communications Server to delete the following IDS events and take actions:

  - Scan detection
  - Attack detection

  - Traffic regulation for TCP connections
  - UDP receive queues

  Using the notification and inform policies, you can issue a message or generate an alert, an Event Integration Facility (EIF) event, email, or report based on a particular event type.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

- IP trace

  Using 3270 panels or the Web browser, you can start and stop both component and packet traces.

  Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

You can use Table 9 on page 47 as you work through the AON/TCP installation tasks described in this section.

Table 9. AON/TCP Installation Summary

Task	Job or Member Name	Reference
Update the CNMSTUSR or C*xx*STGEN member to enable the AON/TCP subtower. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.	DSIPARM (CNMSTYLE)	• "Updating CNMSTYLE" on page 38 • *IBM Z NetView Installation: Getting Started*
Enable AON/TCP IP390 automation	DSIPARM (FKXTABLE)	"Enabling IP390 Automation" on page 49
Configure a UNIX command server	• DSIPARM (CNMSTYLE) • CNMSAMP (CNMSJUNX or CNMSSUNX) • DSIPARM (CNMPOLCY) • DSIPARM (FKXCFG01)	"Configuring UNIX Command Servers" on page 49
Configure a TSO command server	• DSIPARM (CNMSTYLE) • CNMSAMP (CNMSJTSO or CNMSSTSO) • DSIPARM (CNMPOLCY) • DSIPARM (FKXCFG01)	"Configuring TSO Command Servers" on page 50
Configure NetView SNMP support	DSIPARM (CNMSTYLE)	"Configuring NetView SNMP Support" on page 51
Configure SNMP support	• DSIPARM (CNMPOLCY)	"Configuring AON/TCP SNMP Support" on page 51
Define at least one local MVS stack to AON/TCP	DSIPARM (CNMPOLCY)	"Defining Local TCP/IP Stacks" on page 52

Table 9. AON/TCP Installation Summary (continued)		
Task	**Job or Member Name**	**Reference**
As needed, define stacks for remote NetView domains	• DSIPARM (CNMSTYLE) • DSIPARM (CNMPOLCY)	“Setting Up Cross-Domain Communication” on page 53
Set up AON/TCP to monitor IP resources	DSIPARM (FKXCFG01)	“Setting Up for Proactive Monitoring of IP Resources” on page 55
If needed, update community names	DSIPARM (CNMSCM and CNMPOLCY)	“Resolving Community Names (Optional)” on page 56
Enable TCP/IP connection management	DSIPARM (FKXCFG01)	“Managing TCP/IP Connections” on page 56
Define the TCP/IP connection monitoring policy	DSIPARM (FKXCFG01)	“Defining TCP/IP Connection Monitoring Policy” on page 57
Enable Intrusion Detection Services	DSIPARM (CNMSTIDS)	“Enabling Intrusion Detection Services” on page 57
Enable TCP/IP component trace	DSIPARM (CNMSTYLE)	“Customizing TCP/IP Tracing” on page 58

Defining AON/TCP Functions

Table 10 on page 48 summarizes the required and optional tasks for implementing a particular function.

Table 10. Customization by Function											
Task	**TCP/IP Stack**	**UNIX Cmd Srvr**	**TSO Cmd Srvr**	**Rmt Stacks**	**Rmt Gtwys**	**Rmt Srvr Setup**	**Auto tasks**	**TN 3270 Srvr**	**SNMP Setup**	**SNMP Comm Name“1” on page 49**	**Moni-toring Method**
Stack management	RD	O	O	O	O	O	R/O		R	O	
Ping command	RD			O	O	O					
Tracerte command	RD		O	O	O	O					
Generic IP commands	O	O	O	O	O						
Manage IP connections	RD			O	O			O	R	O	
SNMP GET, SET, and similar tasks	RD			O					R	O	
SNMP MIB groups	RD			O					R	O	
SNMP remote Ping	RD			O					R	O	

Table 10. Customization by Function (continued)											
Task	**TCP/ IP Stack**	**UNIX Cmd Srvr**	**TSO Cmd Srvr**	**Rmt Stacks**	**Rmt Gtwys**	**Rmt Srvr Setup**	**Auto tasks**	**TN 3270 Srvr**	**SNMP Setup**	**SNMP Comm Name"1" on page 49**	**Moni-toring Method**
IP resource manager	RD	O	O	O					R	O	
IP trace support	RD		O	O							
SNMP View	RD			O	O	O			R	O	
Proactive monitoring	RD			O			R/O	O	R	O	O
IP connection monitoring and thresholding	RD						R/O		R	O	
Intrusion detection automation	RD	R					R/O				

Note:

1. If you are using a community name, you must define it to the NetView program.
2. The abbreviations in the table have the following meanings:
   - R: Required task
   - RD: Required definition, coding by user is optional
   - O: Optional task
   - R/O: Required task, customization is optional

Enabling IP390 Automation

Review DSIPARM member FKXTABLE and verify that `EZLOPT IP390,ENABLE=Y` is coded.

Note: Be sure, when you change FKXTABLE, that it does not contain any sequence numbers. Sequence numbers in FKXTABLE can cause unpredictable results.

Defining Command Servers to AON/TCP

When you define stacks using the TCP390 policy definition, determine the command servers that you need:

- UNIX command server (see "Configuring UNIX Command Servers" on page 49)
- TSO command server (see "Configuring TSO Command Servers" on page 50)

You can specify both UNIX and TSO command servers.

Configuring UNIX Command Servers

For information about the requirements for the UNIX command server setup, see "Enabling the UNIX Command Server" on page 201.

The following AON functions require a UNIX command server:

- Issuing UNIX commands (AON/TCP Option 2.4)
- Intrusion detection, if the syslog automation option or UNIX command types, or both, are being used

To set up a UNIX command server:

1. Allocate an MVS initiator for the UNIX command server. If the command server is to be run as a started task, an MVS initiator is not required. Refer to the online command help for DEFAULTS and START for more information about running the UNIX command server as a started task.

   Note: The DEFAULTS.STRTSERV statement in the CNMSTYLE member specifies how the command server must be run.
2. Customize CNMSSUNX in CNMSAMP to enable the UNIX command server to run as a started task. This is the default.

   Note: If necessary, you can customize CNMSJUNX in CNMSAMP to enable the UNIX command server to run as a submitted job.
3. Create additional CNMSJxxx or CNMSSxxx jobs for multiple TCP/IP stacks.
4. Authorize all AON/TCP autotasks and any other operators to use the UNIX command servers. The default autotask names are AUTTCP1 through AUTTCP10 and are defined using the AUTOOPS policy definition in the FKXCFG01 member. See "Defining TCP/IP Autotasks" on page 54 for task names that need to be changed for your installation.

For the UNIX command server to start automatically for each stack during initialization, specify UNIXSERV=YES on the TCP390 policy definition in DSIPARM member CNMPOLCY. For more information, see "Defining Local TCP/IP Stacks" on page 52.

Configuring TSO Command Servers

AON supports multiple TSO command servers for improved performance. To set up multiple TSO command servers:

1. A TSO ID is required for each command server. TSO IDs for the command servers must use the following naming convention:
   - TSO IDs for TSO command servers must have the same name differentiated by a trailing number.
   - The trailing numbers are sequential and must start at 1.
   - The base name must match the *servname* in the SERVER parameter of the TCP390 statement.
   - The count in the SERVER parameter is the highest numbered TSO command server.
2. Allocate an MVS initiator for each TSO command server. If the command servers are going to start as started tasks, MVS initiators are not required. Refer to the online command help for DEFAULTS and START for more information about starting the TSO command servers as started tasks.

   Note: The DEFAULTS.STRTSERV statement in the CNMSTYLE member specifies how the command server must be run.
3. Customize CNMSSTSO in CNMSAMP to enable the TSO command server to run as a started task. This is the default.

   Note:

   a. You can use the job name of the started task to qualify the RACF started-class profile name. This provides additional granularity with RACF protection than if a NetView user starts a TSO command server. You can use the following command to display which operators started a TSO command server:

      ```
      MVS D A,L
      ```
   b. If the additional RACF qualification is not needed, you can customize CNMSJTSO in CNMSAMP to enable the TSO command server to run as a submitted job.
4. Create additional CNMSJxxx or CNMSSxxx jobs for multiple TCP/IP stacks.
5. Define a TSOSERV policy definition for *servname* and reference CNMSJTSO (or equivalent name).
6. Authorize all AON/TCP autotasks and any other operators to use the TSO command servers. The default autotask names are AUTTCP1 through AUTTCP10 and are defined using the AUTOOPS policy

definition in the FKXCFG01 member. See “Defining TCP/IP Autotasks” on page 54 for task names that need to be changed for your installation.

For the TSO command server to automatically start for each stack during initialization, update the following POLICY definitions:

- TCP390 (DSIPARM member CNMPOLCY)
- TSOSERV (DSIPARM member FKXCFG01)

For example, consider the following policy definitions:

```
TCP390       NMPIPL10,
             IPADDR=9.67.50.52,
             COMMUNITYNAME=private,
             DOMAIN=LOCAL,
             UNIXSERV=YES,
             SERVER=(NV2TS,3),
             TCPNAME=TCPIP,
             HOSTNAME=NMPIPL10.raleigh.ibm.com,
                ...
TSOSERV      NV2TS,PROC=CNMSJTSO
```

These policy definitions set up three TSO command servers: NV2TS1, NV2TS2, and NV2TS3. When started, AON/TCP uses CNMSJTSO as the procedure. These policy definitions can also automatically start the UNIX command server (UNIXSERV=YES). For more information, see “Defining Local TCP/IP Stacks” on page 52.

Configuring NetView SNMP Support

Most AON/TCP functions are SNMP-based. See the comments in the CNMSTYLE member in DSIPARM to configure the NetView SNMP command.

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

To make updates, use the following statements in the CNMSTUSR or C*xx*STGEN member, and make any necessary modifications. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
COMMON.CNMSNMP.MIBS = all
COMMON.CNMSNMP.MIBPATH = /etc/netview/mibs:/usr/lpp/netview/v6r3/mibs
COMMON.CNMSNMP.timeout = 1
COMMON.CNMSNMP.retries = 5
```

For security considerations, see the *IBM Z NetView Security Reference*.

Configuring AON/TCP SNMP Support

AON/TCP uses SNMP requests for many functions. Several policy definitions can affect those requests.

Table 11 on page 52 contains policy definitions for SNMP requests that are defined in EZLCFG01 and CNMPOLCY. If AON/TCP SNMP requests experience frequent timeout errors, you can adjust the values.

Table 11. Control File Entries		
Control File Entry	**Parameter**	**Value to Adjust**
TCP390 DEFAULTS	SNMPRETRY=2	Specifies the number of times the SNMP request is to be retried, for example: `SNMP GET … -r 2 …`
	SNMPTO=3	Specifies the number of seconds that SNMP waits for a response, for example: `SNMP GET … -t 3 …`
	NONREP=0	Specifies the number of non-repeating variables for a BULK request, for example: `SNMP GETBULK … -B 0 xx …` This parameter is sets up default values for the NVSNMP GETBULK operator request.
	MAXREP=10	Specifies the maximum number of repetitions of repeating variables for a BULK request, for example: `SNMP GETBULK … -B xx 10 …` This parameter is sets up default values for the NVSNMP BULKWALK operator request.
ENVIRON TIMEOUT	SNMP=29	Specifies the time in seconds for AON/TCP to wait for a response to an SNMP request within a PIPE (for example, a SOCKET command).

Defining Local TCP/IP Stacks

At a minimum, define at least one local MVS stack to AON/TCP. If you do not define this, it is defined automatically by the NetView discovery manager.

Use the TCP390 policy definition in DSIPARM member CNMPOLCY to define a local TCP/IP stack. The entire stack definition, or portions of it, are optional. NetView dynamically determines information about your stack, such as its IP address, host name, and TCP process name. If you want to run with default values for all functions (such as Intrusion Detection Automation Services), you do not need to define a TCP390 statement for your stack. You need to define your stack if you want to override the default settings. The following example defines a local stack named NMPIPL10:

```
TCP390      NMPIPL10,
            IPADDR=9.67.50.52,
            COMMUNITYNAME=private,
            DOMAIN=LOCAL,
            UNIXSERV=YES,
            SERVER=(NV2TS,3),
            TCPNAME=TCPIP,
            HOSTNAME=NMPIPL10.raleigh.ibm.com,
                ...
```

Note:

1. The stack NMPIPL10.raleigh.ibm.com is known as NMPIPL10.
2. SNMP requests (for example, stack monitoring processes) uses a community name of `private`. The default community name is `public`.
3. The NetView domain managing this stack is the LOCAL domain. This value can be set to the local domain ID. The local domain is the same domain where this policy definition resides.

4. This policy defines both the UNIX command server and three TSO command servers. For more information, see “Defining Command Servers to AON/TCP” on page 49.
5. You can specify either the IPADDR (recommended) or HOSTNAME parameter. If you do not specify the IPADDR, AON/TCP dynamically determines the IP address of the stack based on the HOSTNAME parameter. This takes additional processor cycles.
6. For security, if you specify COMMUNITYNAME, restrict access to the member containing your policy definitions.
7. TCPNAME defines the TCP/IP start procedure name. AON/TCP supports system symbolics. You can use TCPNAME=&CNMTCPN.

For more information about the TCP390 policy definition, refer to *IBM Z NetView Administration Reference*.

Setting Up Cross-Domain Communication

Some AON/TCP functions (for example connection management, SNMP functions, monitoring functions, IP tracing functions, and commands) support communication with remote NetView domains.

The discovery manager discovers all local stacks on a given system. Code TCP390 definitions for stacks on systems that are outside of your sysplex.

For example, to set up for cross-domain communication from NMPIPL10 (domain NTV70) to NMPIPL27 (domain NTV9D):

- In NTV70, define TCP/IP stacks on the remote domain. For example, to define a stack named NMPIPL27 in NetView domain NTV9D, use the following definition:

```
TCP390    NMPIPL27,
          IPADDR=9.67.50.41,
          DOMAIN=NTV9D,
          UNIXSERV=YES,
          TCPNAME=TCPIP,
          HOSTNAME=NMPIPL27.raleigh.ibm.com,
            ...
```

- In NTV9D, define the policy for the local stack:

```
TCP390    NMPIPL27,
          IPADDR=9.67.50.41,
          DOMAIN=LOCAL,
          UNIXSERV=YES,
          TCPNAME=TCPIP,
          HOSTNAME=NMPIPL27.raleigh.ibm.com,
           ...
```

For more information about TCP390 policy definitions, refer to *IBM Z NetView Administration Reference*.

For cross-domain communication, set up a remote gateway session. For example, to establish a remote gateway from NMPIPL10 (domain NTV70) to NMPIPL27 (domain NTV9D):

- On NTV70, define a gateway operator autotask using the AUTOOPS policy definition. For example, on NTV70:

```
AUTOOPS    GATOPER,ID=GATNTV70
```

  defines GATNTV70 as the gateway autotask in domain NTV70.
- To automatically start remote gateway sessions when the GATNTV70 autotask is logged on, define CDLOG entries for your GATOPER autotask. Make sure that each operator in the target domain is defined in DSIOPF.

  For example, define a CDLOG policy definition in NTV70 for each target domain. To connect to domain NTV9D, use the following CDLOG entry:

```
CDLOG  GATNTV70.NTV9D,
       SESSTYPE=RMT,
       INIT=YES,
```

```
        TARGOP=RMTNTV70,
        DESC='RMT GATEWAY TO NTV9D'
```

In this case, when operator GATNTV70 logs on, a RMTCMD session is automatically started to NTV9D as RMTNTV70.

For more information about CDLOG, refer to *IBM Z NetView Administration Reference*.

For each remote NetView domain and each TCP/IP stack that you plan to use, add the following statement to the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
auxInitCmd.IP = EXCMD AUTO1,FKXERINI  spname  servername  count  proc
```

where:

spname
The name of your *local* TCP/IP stack.

servername
The name of the TSO or UNIX command server on the MVS host. *Servername* is the root TSO command server ID when defining multiple TSO command servers. When defining a UNIX command server, set *servername* to YES. If AON IP390 is installed and this is a TSO command server, then *servername* must match the root TSO command server ID defined for the TCP/IP stack :

```
TCP390 .... SERVER=(servername,count)
```

count
If defining TSO command servers, the *count* parameter is the number of TSO command servers that are defined for this TCP/IP stack. The minimum is 1 and the maximum is 5. If defining a UNIX command server, set *count* to UNIX.

If AON IP390 is installed and this is a TSO command server, the *count* parameter must match the *count* defined for the TCP/IP stack :

```
TCP390 .... SERVER=(servername,count)
```

proc
The name of the job to start the command servers.

The default job for TSO command servers is CNMSJTSO for submitted jobs and CNMSSTSO for started tasks. If AON IP390 is installed, *proc* must match the job found on the TSOSERV definition for the corresponding *servername*. For example, `TSOSERV servername,PROC=proc`

The default job for the UNIX command server is CNMSJUNX for submitted jobs and CNMSSUNX for a started task.

FKXERINI initializes:

- The TSO or UNIX command server used by AON IP390 functions in the remote domain
- Global variables that are used by AON IP390 functions

FKXERINI is run during NetView startup in the local and remote domains. FKXERINI must run in *all* domains against each local stack.

As needed, define TSO or UNIX command servers.

Defining TCP/IP Autotasks

AON/TCP provides policy definitions for 10 NetView autotasks that can be used for TCP/IP automation and management. Review the following statement in DSIPARM member FKXCFG01:

```
AUTOOPS    TCPOPER,ID=(AUTTCP,10)
```

This defines AUTTCP1 through AUTTCP10 as the autotasks used by AON/TCP. Modify this statement as necessary for your installation (for example, change AUTTCP to AONTCP). If you change the autotask IDs, also make changes to DSIOPF to define your task IDs.

For more information about the AUTOOPS statement, refer to *IBM Z NetView Administration Reference*.

Setting Up for Proactive Monitoring of IP Resources

You can use AON/TCP to proactively monitor your critical IP resources based on policy definitions. You can use the following monitoring methods:

- "Pinging a Resource" on page 55
- "MIB Polling" on page 55
- "MIB Thresholding" on page 56

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

When AON/TCP proactive monitoring detects a failure:

- AON/TCP uses the NOTIFY policy to determine the type of notification to be sent.
- AON/TCP schedules a recovery monitoring timer. This timer is scheduled as specified on the MONIT policy definitions. The timer checks the resource status and optionally sends a notification that the resource is still down.

Table 12 on page 55 shows the IP resource types you can monitor and the required policy definitions.

Table 12. Control File Entries

Resource	**Policy Definition**
Host	IPHOST
Stack	TCP390
Router	IPROUTER
Interface	IPINFC
Socket	IPPORT
Server	IPNAMESERV
Telnet server	IPTELNET
TN3270 server	IPTN3270
IP connection	IPCONN As shipped, IPCONN definitions are commented out.

For more information about policy definitions, refer to *IBM Z NetView Administration Reference*.

Pinging a Resource

To perform a ping of a resource, code `FORMAT=PING` on its policy definition. If the ping response works, the resource status is NORMAL. If not, the resource status is DOWN.

MIB Polling

MIB polling uses SNMP to poll the interface table (ifTable) for the defined resource. The administration status is compared to the operational status. If one or more interfaces are expected to be up and are not, the resource is marked as degraded. Degraded does not mean that the resource is down.

To enable MIB polling, code FORMAT=SNMP, as shown in the following example:

```
IPHOST HOST01,SP=NMPIPL10,
OPTION=IP390
ACTMON=ALLHOSTS,
FORMAT=SNMP,
STATUS=(NORMAL,DEGR*),
HOSTNAME=host01.raleigh.ibm.com
```

The ACTMON=ALLHOSTS statement refers to the following statement that contains a common definition for monitoring all the IP hosts:

```
ACTMON ALLHOSTS,OPTION=IP390,INTVL=00:30,STATUS=NORMAL,
       FORMAT=PING
```

Using this definition, HOST01 is monitored every 30 minutes using a ping. The expected status is NORMAL (resource is active).

Additionally, you can add the following status parameter so that a status of degraded is not treated as a resource failure: STATUS=(NORMAL,DEGR*).

MIB Thresholding

MIB thresholding uses SNMP to query MIBs defined in the policy definition for the resource. You can define the MIB, its threshold value, and the threshold operator (greater than, less than, equal). When the proactive monitoring timer pops, AON/TCP retrieves the MIB values and compares them to the policy definition for the resource. For example, to add a MIB threshold to the HOST01 resource and use the ipRoutingDiscards.0 MIB, code the following statements:

```
IPHOST HOST01,SP=NMPIPL10,
OPTION=IP390
ACTMON=ALLHOSTS,
FORMAT=SNMP,
STATUS=(NORMAL,DEGR*,THRESH*),
MIBVAR=(ipRoutingDiscards.0,GE,3),
HOSTNAME=host01.raleigh.ibm.com
```

In this example, if the MIB value is greater than or equal to 3, the resource status is set to THRESH. Notice also that THRESH is an acceptable status and therefore is not treated as a resource failure.

Resolving Community Names (Optional)

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

When monitoring resources using SNMP, the NetView program might need access to the community name for a resource. NetView SNMP reads MIB data from community-name protected resources using DSIPARM member CNMSCM.

To use CNMSCM for community name resolution, add an entry line for each host name to be resolved to a community name and then save the file. For more information, see the *IBM Z NetView Administration Reference*. To prevent unauthorized viewing or modification of CNMSCM, see the *IBM Z NetView Security Reference*.

For information about defining community names to TCP/IP, see the *z/OS Communications Server IP Configuration Reference*.

Note: This definition is for resources that are not TCP/IP stacks.

Managing TCP/IP Connections

TCP/IP connections can be established for any socket and can be established using a TN3270 server. You can manage these connections from the 3270 interface, the NetView management console, or the Web browser.

Note: This function is now provided in the base NetView program and no longer requires AON/TCP. However, AON/TCP is still required for the IBM 2210, the IBM 2216, and the CISCO CIP.

To enable connection management, see Table 13 on page 57.

Table 13. Connection Management	
Connection	**See**
To a local TCP/IP stack	"Defining Local TCP/IP Stacks" on page 52 Set up your local stack definition with SNMP capability.
From a TN3270 server	"Setting Up for Proactive Monitoring of IP Resources" on page 55 (TN3270 servers)

Because most stacks have numerous connections, consider limiting the amount of data that your operator must view. To do this, you can use one of the following methods:

- Code SESSTAT=NO on the IPPORT definition. All connection data for that port is ignored. Code this for ports that you do not want to manage.

  For example, if you do not want to manage connections with the NetView Web server interface task:

  ```
  IPPORT DSIWBTSK,SP=NMPNET10,
                  PORT=80,
                  PROTOCOL=TCP,
                  TCPNAME=NVPROCN,
                  STATUS=NORMAL,
                  FORMAT=PORT,
                  SESSTAT=NO,
                  DESC="NetView Web Browser Socket"
  ```

- Use a session filter. You can define filters for each operator to restrict the data based on IP address, local MVS port, LU, and application. For more information about session filters, refer to the *IBM Z NetView User's Guide: Automated Operations Network*.
- For the Web browser, adjust the MAXCONN parameter. In the Web browser, you can limit the number of connections shown based on the value of the MAXCONN parameter on the TP390 DEFAULTS definition. Sample FKXCFG01 sets the limit to 1000. This limit applies to all connection types.

Defining TCP/IP Connection Monitoring Policy

You can monitor IP connections such as printer connections and apply policy definitions to determine if they are stopped. You can choose to notify an operator or use automation to break the connection.

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

To do this, customize the sample policy definitions in FKXCFG01:

```
*AUTOOPS  CONNOPER,ID=AUTCMON
*ACTMON IPCONN,INTVL=00:01:00
*NOTIFY IPCONN,ALERT=NO,MSG=YES,DDF=NO,INFORM=NO
*IPCONN TCPIP*,
```

Enabling Intrusion Detection Services

To enable Intrusion Detection Services (IDS), review the IDS statements in the CNMSTIDS member and update AON policy definitions in the CNMSTUSR or C*xx*STGEN member (see Table 14 on page 58). For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

Table 14. IDS Support	
Policy Definition	**Update**
TCP390	On the TCP390 stack policy definition, define your local TCP/IP stack and specify IDSAUTO=Y and optionally specify the IDSINTVL parameter.

NOTIFY	By default, the NOTIFY IDSAUTO policy is set up to issue alerts and messages for IDS events. You can optionally update this policy to enable IDS events to be forwarded to a designated event server. For example:
	`NOTIFY   IDSAUTO,`
	`      **ALERT=TEC,**`
	`         MSG=YES,`
	`         DDF=NO`

NTFYOP	Use the NTFYOP policy definition to define which NetView operators receive IDS messages (class=64). For example:
	`NTFYOP   OPER1,`
	`   **OPER='IDS-AUTO-SVCS',**`
	`      CLASS=(64),HELDMSG=(I)`

INFORM	Update the Inform policy for email notifications (reports and responses to commands). For example:
	`GROUP    IDSOPERS,LIST=OPER1,OPER2,OPER3;`
	`**INFORM     OPER1**;`
	`  CONTACT CONNECTION=EMAIL,`
	`        INTERFACE=EZLESMTP,`
	`        ROUTE=IBPERSC@US.IBM.COM,`
	`        NAME=C. PERSON;`
	`INFORM    OPER2,...`
	`INFORM    OPER3,...`
	For more information, refer to sample EZLINSMP.

For more information about the policy definitions, refer to *IBM Z NetView Administration Reference*.

If Intrusion Detection Services is configured to use the Unix syslogd daemon, the syslog configuration file will need to include the following output definition statement:

```
*.xxxxxxxx.daemon.* /dev/console
```

In this statement, xxxxxxxx is replaced by the job name that is assigned to the syslogd daemon.

For details about syslogd configuration statements, see the *z/OS Communications Server: IP Configuration Reference*.

Customizing TCP/IP Tracing

Output writers can be used to store trace data for later analysis. You can customize the default output writer names for component or packet trace in the CNMSTYLE member. Review the following statements for your environment and make changes as appropriate in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

- Source JCL definitions for component and packet trace:

  ```
  COMMON.EZLTCPcTRACEwriter = CTTCP       // Component trace writer name
  COMMON.EZLTCPpTRACEwriter = PKTCP       // Packet trace writer name
  ```

  The Open Systems Adapter (OSA) trace does not have a default writer. You can use the packet trace writer for OSA packet trace data.

- Time interval to wait for a response:

```
COMMON.EZLIPTraceJCLWait   = 2     // ... for source JCL error response
```

For more information about IP tracing, see *IBM Z NetView IP Management*. For information about the IPTRACE command, which enables trace control through a panel interface, see the NetView online help.

Note: This function is now provided in the base NetView program and no longer requires AON/TCP.

Setting Up AON/SNA Support

You can use Table 15 on page 59 as you work through the AON/SNA installation tasks described in this section.

Table 15. AON/SNA Installation Summary

Task	**Job or Member Name**	**Reference**
Update the CNMSTUSR or C*xx*STGEN member to enable the AON/SNA subtower. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.	DSIPARM (CNMSTYLE)	• "Updating CNMSTYLE" on page 38 • *IBM Z NetView Installation: Getting Started*
Remove recovery support from the status monitor for resources that AON/SNA monitors	DSIPARM (DSICNM)	"Updating the Status Monitor" on page 59
Define the subsystem interface address space	PARMLIB (MPFLST)	"Defining the Subsystem Interface Address Space" on page 60
If you do not need subarea resource automation support, disable subarea processing	DSIPARM (FKVTABLE)	"Setting Up AON/SNA Subarea Support" on page 60
Enable Advanced Peer-to-Peer Networking support	• DSIPARM (FKVTABLE) • DSIPARM (FKVCFG01)	"Enabling Advanced Peer-to-Peer Networking Monitoring Support" on page 61
Enable X.25 support	• DSIPARM (FKVTABLE) • DSIPARM (DSICRTTD) • DSIPARM (FKVCFG01)	"Implementing X.25 Monitoring" on page 61

Updating the Status Monitor

For AON/SNA support, update the STATMON statements in DSICNM in the following way:

1. Comment out the following two statements from the command list name table:

```
C   MONON
C   MONOFF
```

2. Comment out the O MONIT statement. Unless this statement is removed or commented out, AON/SNA cannot function correctly.
3. MONIT must be turned off for every resource to ensure that AON/SNA performs recovery for these resources.

4. You can still use the STATMON entry to reflect the current status of network resources in an automated environment.

Defining the Subsystem Interface Address Space

To define subsystem interface address space:

1. If you are not running with extended consoles, define a subsystem interface (SSI) address space for the NetView program. This enables AON to submit jobs for log file maintenance and to support the NetView Access Services component of the AON helpdesk facility.
2. Check the message processing facility list (MPFLST) in PARMLIB and make sure that all EZL messages can be sent to and from the NetView program. If your NOENTRY or DEFAULT entries in the MPF list are SUP(NO) AUTO(NO), specify the following entry for AON:

   ```
   EZL*,SUP(NO),AUTO(YES)
   ```

3. If you have IBM NetView Access Services (NVAS) and use the AON SNA Help Desk to help manage those sessions, make sure that all EMS messages can pass to and from the NetView program. If you use NVAS and your NOENTRY or DEFAULT entries in the MPF list are SUP(NO) AUTO(NO), specify the following entry for AON:

   ```
   EMS*,SUP(NO),AUTO(YES)
   ```

Note:

1. If you are not using a console automation package, specify all other messages with AUTO(NO) to prevent them from going to the NetView program and to improve performance.
2. If you are using a console automation package, you must code an automation table entry at the top of the table to discard extraneous messages coming from the SSI to AON. For example, if the MPFLST entry for console automation is:

   ```
   IEF*,SUP(NO),AUTO(YES)
   ```

   The corresponding automation table entry for AON is:

   ```
   IF MSGID='IEF'.
      THEN DISPLAY(N) NETLOG(N);
   ```

Setting Up AON/SNA Subarea Support

AON/SNA subarea automation is automatically enabled.

If you do not need subarea resource automation support, disable subarea processing. This also prevents AON/SNA SNBU from operating on PU thresholding exceptions. However, AON/SNA SNBU automation can still occur from alerts.

To disable subarea support, follow these procedures:

1. Edit the FKVTABLE member and locate the following statement:

   ```
   EZLOPT SA,ENABLE=Y
   ```

   Note: Be sure when you change FKVTABLE that it does not contain any sequence numbers. Sequence numbers in FKVTABLE can cause unpredictable results.

2. Change ENABLE=Y to ENABLE=N.

   These procedures cause the following actions:

   - Prevents subarea initialization
   - Disables the subarea menu by make it unavailable from the operator interface
   - Prevents message processing of subarea related automation in automation table

If you want SNA subarea support to recover your NCPs, add an NCPRECOV statement for each channel-attached NCP for this host. For more information, refer to the *IBM Z NetView Administration Reference*.

Enabling Advanced Peer-to-Peer Networking Monitoring Support

To set up AON/SNA Advanced Peer-to-Peer Networking, perform the following steps:

1. Enable AON/SNA Advanced Peer-to-Peer Networking by changing the following statement in the FKVTABLE member:

   ```
   EZLOPT APPN,ENABLE=N
   ```

   to:

   ```
   EZLOPT APPN,ENABLE=Y
   ```

   Note: Be sure when you change FKVTABLE that it does not contain any sequence numbers. Sequence numbers in FKVTABLE can cause unpredictable results.

2. Decide which control points you want to monitor. If you are not sure which control points you want to monitor, you might want to enable AON/SNA Advanced Peer-to-Peer Networking and not define any resources. After you enable AON/SNA Advanced Peer-to-Peer Networking, you can use the operator panel portion of AON/SNA Advanced Peer-to-Peer Networking to look at AON/SNA Advanced Peer-to-Peer Networking resources and issue Advanced Peer-to-Peer Networking-related VTAM commands. When you decide which resources you want to actively monitor, add an entry for each control point to FKVCFG01 as shown in the following example:

   ```
   ACTMON USIBMTA.TA1CP208,RESTYPE=CP,
          OPTION=APPN,INTVL=01:00
   ```

   This example shows that you can use network-qualified names.

3. Decide which CP-CP sessions you want to actively monitor. These sessions are defined using two statements:

   ```
   ACTMON GARTH,RESTYPE=CPCPSESS,OPTION=APPN
   CPCPSESS GARTH,CP1=USIBMNR.NR51W001.GARTH,CP2=USIBMTA.TA01
   ```

   The ACTMON control file entry defines the resource and resource types you want to monitor. An alias is used for the CPCPSESS control file entry. The interval for active monitoring can be specified on each ACTMON statement. If it is not specified, the value specified on the ACTMON APPN entry is used.

   In the preceding example, GARTH is an alias name used only by AON/SNA to refer to the session. These alias names need to be unique within AON/SNA. The CPCPSESS statement defines the actual session between the two control points specified by the CP1 and CP2 entries. You can use network-qualified names.

Implementing X.25 Monitoring

This section explains how to install and implement AON/SNA X.25 support. These instructions assume AON/SNA is already installed.

To set up X.25 support:

1. Enable X.25 support by changing the following statement in the FKVTABLE member. Change:

   ```
   EZLOPT X25,ENABLE=N
   ```

   to:

   ```
   EZLOPT X25,ENABLE=Y
   ```

   Note: Be sure when you change FKVTABLE that it does not contain any sequence numbers. Sequence numbers in FKVTABLE can cause unpredictable results.

2. Edit the DSICRTTD member of your DSIPARM data set and uncomment the following statement:

```
DSTINIT XITCI=FKVXITAN
```

Note: AON ships with the FKVXITAN XITCI exit already in CNMLINK. To modify the exit, use the FKVPITAN sample that is found in CNMSAMP.

3. Define X25MONIT entries in your control file for switched virtual circuit monitoring. The default control file member for AON/SNA is FKVCFG01.

Completing AON Tailoring

At this point, you can initialize AON and complete the installation verification procedure. You might need to make additional modifications to the control file entries to enable additional AON functions, and to maximize the performance of functions such as RECOVERY, THRESHOLDS, and MONIT.

Testing AON Automation

The following tests verify that AON automation is working properly.

Note: You must be logged on as a notification operator (your user ID must be defined as a NTFYOP) to perform this test.

Testing the Enhanced Automation

1. Log on to the NetView program.
2. Enter EZLEATST

   Sample result:

```
NetView V6-NM             NetView     CNM01 OPER1  01/10/19 11:16:22
* CNM01    EZLEATST
W CNM01    DSI039I MSG FROM AUTO1    : AONCMD TEST SUCCESSFUL
M CNM01    DSI039I MSG FROM OPER1    : AON MSG TEST SUCCESSFUL
M CNM01    DSI039I MSG FROM OPER1    : TESTING WAIT TIMEOUT FUNCTION (WAITING
           29 SECONDS)
C AON01    EZL001I REQUEST WAIT WAS SUCCESSFUL FOR TIMEOUT
```

   The EZLEATST routine calls a command list that tests NetView functions requested by the AON &WAIT, &WAIT TIMEOUT, MSG, and EXCMD functions. Verify that these functions completed successfully. If any errors are detected, the test issues a message and stops.

Verifying AON Tasks

To verify that the AON tasks are active:

1. Enter LIST STATUS=TASKS
2. Verify that the following AON tasks are active:
   - EZLTCFG
   - EZLTSTS
   - EZLTLOG
   - EZLTDDF
   - AONBASE
   - AONMSG1
   - AONMSG2
   - AUTALRT
   - AUTTRAP

Note:

a. AUTTRAP is active only if the AON/TCP tower is enabled.
b. There might be additional tasks depending on how much customization has been done and which automation components are active.

3. Enter `REGISTER QUERY=MS`.
4. Verify that the following applications are registered:

AONALERT
Required for sending MSUs to the hardware monitor

EZLMSAPL
If you are using the AON workstation interface

Verifying AON Panels

Complete the following test to verify that the AON panels display correctly.

1. Enter `AON`.

   Sample result:

```
EZLK0000            AON: Operator Commands Main Menu            CNM01


Select an option

         _ 0.   Tutorial
           1.   AON Base Functions
           2.   SNA Automation
           3.   TCP/IP Automation
```

2. Enter 1.

   Sample result:

```
EZLK0100            AON: Base Functions            CNM01


Select an option

         _ 0. Tutorial
           1. Help Desk
           2. AutoView
           3. DDF
           4. Automation Settings
           5. Cross Domain Functions
           6. Timer
           7. Task and Log Maintenance
           8. Support Functions
           9. Display the Inform Log
```

3. Enter 4.

   Sample result:

```
EZLK4000                AON: Automation Settings            CNM01


Select an option

         _ 1. Automation
           2. Notification Operators
           3. Thresholds
           4. Monitor Intervals
           5. Active Monitoring
```

4. Press F2.

   Sample result:

```
EZLK0000            AON: Operator Commands Main Menu            CNM01


Select an option

         _ 0.   Tutorial
           1.   AON Base Functions
           2.   SNA Automation
           3.   TCP/IP Automation
```

Testing AON Commands

You must be a notify operator (NTFYOP) to use many of these commands.

To test the AON commands, follow these steps:

1. Enter **SETNTFY *operid*** to verify that the EZL919I message is received, indicating the operation was successful.
2. Log on to the new notify operator ID.
3. Enter **DISNTFY** to verify that you receive the automation status of the notify operators.
4. Enter **DISAUTO** to verify that the default automation settings are loaded from the control file.
5. Enter **AONTRACE ENTRY ON DOMAIN** to verify that the EZL241W message is received, indicating that your request was unsuccessful.
6. Enter **NLOG** to verify that no startup messages are displayed on the panel.
7. Enter **POLICY REQ=STATUS** to verify that the control file is loaded.
8. Enter **POLICY REQ=GET ENTRY=NTFYOP** to list the notify operators specified in the AON control file.
9. Enter **DSPCFG NTFYOP** to verify that similar information is displayed.

Testing AON/SNA

This section provides installation verification procedures for the following functions:

- AON/SNA VTAM subarea automation
- AON/SNA Advanced Peer-to-Peer Networking monitoring
- AON/SNA X.25 monitoring

Testing AON/SNA VTAM Subarea Automation

This section shows you how to set up a test for SNA recovery.

Testing SNA Resource Recovery

To perform resource recovery for SNA:

- AON and AON/SNA installed and customized
- An available test PU
- Your ID set up as a notification operator (NTFYOP) with a message class of 20
- AON/SNA SNBU disabled during the test
- DDF customized for your environment
- Enter the `DSPCFG MONIT` command to display monitor intervals
- Monit intervals for PUs must be those shipped in the sample control file

You can cause a failure on the PU by:

- Turning off the controller
- Unplugging from the patch panel

The following message is displayed from the command facility while running the test. During the test, TA1P523A is the name of your PU. The message is:

```
EZL506I PU TA1P523A ON CNM01 INACTIVE - RECOVERY MONITORING
        HAS BEEN INITIATED
```

If you do not receive this message, check the netlog. If you find the message in the netlog, you might not be set up as a notification operator.

Checking DDF

To check DDF, perform the following steps:

1. Enter: DDF
2. Move the cursor to SNA.
3. Press F8 to page down.

   Your PU is displayed in pink.
4. Move the cursor to PU and press F8.
5. Move the cursor to your PU name and press F2 to show the details associated with the PU.

AON/SNA displays the Detail Status Display panel.

Sample result:

```
   ---- DETAIL STATUS DISPLAY ----
                                             1 OF    2


      COMPONENT: TA1P523A        SYSTEM   : CNM01

      COLOR    : PINK            PRIORITY :     270

      DATE     : 08/19/19        TIME     : 09:53:06

      REPORTER : AONMSG2         NODE     : CNM01

      DUPLICATE COUNT:

 1 '*EZL506I PU TA1P523A ON CNM01 INACTIVE - RECOVERY MONITORING HAS
 BEEN INITIATED'
```

Checking Timers

To check your timers:

1. Enter TIMER on the command line.
2. Press Enter. The NetView program displays the Timer Management panel. When you do this test, notice the timer for your PU.

   Sample result:

```
EZLK6000               TIMER MANAGEMENT    CNM01 NETOP1   08/19/19 8:28:41
                                                            1 TO    5 OF    5
Target:            Target Network ID:            Total Selected Timers:    5
                                                   Total Purged Timers:    0
Filter criteria:
Type one action code. then press enter.
  1|A=Add  2|C=Change  3|P=Purge  4=Add CHRON timer
    Timer ID  Scheduled           Type   Interval   Task      Save   Catchup
   _  EZL00002  08/19/19 10:29:27 AFTER             AONNET2
             EZLECATV TA1P523A PU 2 08/19/19 09:51
```

3. Press F3 to return to the command facility.

   After a few minutes, the following message is displayed:

```
EZL507I REMINDER: PU TA1P523A ON CNM01 HAS BEEN UNRECOVERABLE
        FOR 4 MINS.
```

4. Resolve the hardware error, which causes the following message to display:

   Sample result:

```
EZL504I PU TA1P523A IS AVAILABLE (REPORTED BY CNM01)
```

   If you check DDF, AON/SNA has deleted your PU name from the CNM01 Network Status - Physical Units panel.

   Sample result:

```
FKVPNLP                                               PAGE 1 OF 1
            CNM01 NETWORK STATUS - PHYSICAL UNITS

   PU
```

Checking the NLOG

To display the NetView automation log:

1. Enter NLOG on the command line.

Sample result:

```
LOG BROWSE - CNM01    ACTS 08/19/19 (96040)---- MSG --------- COLUMNS 062 139
COMMAND ===>                                                  SCROLL ===> PAGE

*EZL509I PU TA1P523A IS UNAVAILABLE (REPORTED BY CNM01)
*EZL506I PU TA1P523A ON CNM01 INACTIVE - RECOVERY MONITORING HAS BEEN INITIATE
*EZL507I REMINDER: PU TA1P523A ON CNM01 HAS BEEN UNRECOVERABLE FOR 4 MINS.
*EZL504I PU TA1P523A IS AVAILABLE (REPORTED BY CNM01)
```

Testing Thresholding

To test the thresholding, make the PU fail enough times to trip the critical threshold.

Note: These testing examples use the shipped defaults. If you use values other than the shipped defaults, the information shown on your panels can vary from those shown here.

To trip the critical threshold:

1. Set your critical threshold to 2 errors in 10 minutes for PUs using the **SETTHRES** command.
2. Cause the PU to fail.

When you trip the critical threshold, you see the following messages:

```
EZL509I PU TA1P523A IS UNAVAILABLE (REPORTED BY CNM01)

EZL501I RECOVERY FOR PU TA1P523A ON CNM01 HALTED - 2 ERRORS
        SINCE 09:51 ON 08/19/19 - CRITICAL ERROR THRESHOLD EXCEEDED
```

Go back to DDF before resolving the hardware error that occurred. To go to DDF:

1. Enter DDF on the command line.
2. Follow the steps in "Checking DDF" on page 65 to display the Detail Status Display panel for your PU name.

   Sample result:

```
                  ---- DETAIL STATUS DISPLAY ----
                                                              1 OF    2

        COMPONENT: TA1P523A              SYSTEM   : CNM01

        COLOR    : RED                   PRIORITY :      175

        DATE     : 08/19/19              TIME     : 11:54:34

        REPORTER : AONMSG                NODE     : CNM01

        DUPLICATE COUNT:

         1 'EZL501I RECOVERY FOR PU TA1P523A ON CNM01 HALTED - 2 ERRORS SINCE
         11:49 ON 08/19/19 - CRITICAL ERROR THRESHOLD EXCEEDED'
```

3. Resolve the hardware error that occurred during the test.

Testing NCP Recovery

You can bypass this section if you are not using AON/SNA to perform NCP recovery. To perform this test, you must have an NCP available that can be forced to fail. In addition, the NCPRECOV control file entry for your NCP must be coded in the following way:

```
NCPRECOV ncpname,HOST=domainid,DUMP=(N,N),RELOAD=(Y,N),
LINKSTA=link_sta_name,DUMPSTA=link_sta_name
```

This specifies

- No for dump
- Yes to reload for noncritical responses

Note: Specify no (N) for the AUTODMP and AUTOIPL parameters in the PCCU macro for the NCP you are using to test.

Causing a Failure on the NCP

You can cause a failure on the NCP by taking one of the following actions:

- Enter initial program load (IPL) from the Moss console.
- The initial machine load (IML) from the front panel.

If you cause the NCP to fail, you receive messages similar to:

```
EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
EZL506I NCP TA1N500 ON CNM01 INACTIVE - RECOVERY MONITORING
        HAS BEEN INITIATED
EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
EZL509I NCP TA1N500 IS UNAVAILABLE (REPORTED BY CNM01)
EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
FKV532I REPLY OF -NO- WAS ISSUED BY AUTOMATION FOR TA1N500
        FROM CNM01: NON-CRITICAL DUMP REPLY FROM RECOVERY HOST
FKV535I REPLY OF -YES- WAS ISSUED BY AUTOMATION FOR TA1N500 FROM
        CNM01: NON-CRITICAL RELOAD REPLY FROM RECOVERY HOST
FKV556I LOAD OF TA1N500 BY OPERATOR STARTED
FKV544I RELOAD WAS SUCCESSFUL FOR TA1N500 AND IS AVAILABLE
EZL504I LINKSTA 0F31-S IS AVAILABLE (REPORTED BY CNM01)
EZL504I LINKSTA 0F31-S IS AVAILABLE (REPORTED BY CNM01)
EZL504I NCP TA1N500 IS AVAILABLE (REPORTED BY CNMM01)
```

Note: The messages are displayed after the dumps and loads are completed. Therefore, a significant amount of time might pass before the messages are displayed.

Checking the NLOG

To display the NetView automation log, enter NLOG on the command line.

Sample result:

```
LOG BROWSE - CNM01    ACTS 08/19/19 (96040)---- MSG --------- COLUMNS 062 139
COMMAND ===>                                                  SCROLL ===> PAGE

EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
*EZL506I NCP TA1N500 ON CNM01 INACTIVE - RECOVERY MONITORING HAS BEEN INITIATE
EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
EZL509I NCP TA1N500 IS UNAVAILABLE (REPORTED BY CNM01)
EZL502I RECOVERY FOR NCP TA1N500 ON CNM01 CONTINUING - 1 ERRORS SINCE 12:57 ON
FKV532I REPLY OF -NO- WAS ISSUED BY AUTOMATION FOR TA1N500 FROM CNM01 : NON-CR
FKV532I REPLY OF -NO- WAS ISSUED BY AUTOMATION FOR TA1N500 FROM CNM01 : NON-CR
FKV535I REPLY OF -YES- WAS ISSUED BY AUTOMATION FOR TA1N500 FROM CNM01 : NON-C
*EZL509I LINKSTA 0F31-S IS UNAVAILABLE (REPORTED BY CNM01)
FKV556I LOAD OF TA1N500 BY OPERATOR STARTED
FKV544I RELOAD WAS SUCCESSFUL FOR TA1N500 AND IS AVAILABLE
EZL504I LINKSTA 0F31-S IS AVAILABLE (REPORTED BY CNM01)
*EZL504I LINKSTA 0F31-S IS AVAILABLE (REPORTED BY CNM01)
*EZL504I NCP TA1N500 IS AVAILABLE (REPORTED BY CNM01)
```

Testing AON/SNA Advanced Peer-to-Peer Networking Monitoring

To test the AON/SNA Advanced Peer-to-Peer Networking, test checkpoint commands and the control point display.

Checkpoint Commands

To test the checkpoint commands:

1. From the command facility, enter AON.
2. Select **2** for SNA.
3. Select **6** for Advanced Peer-to-Peer Networking.
4. Select **1** for Issue checkpoint commands.
5. Select **3** for Checkpoint both databases.

   Sample result:

```
FKVK5100          Operator Command Interface: VTAM Commands         CNM01

Output of: F NET,CHKPT,TYPE=ALL

IST097I    MODIFY   ACCEPTED
IST1123I   MODIFY CHKPT TO DATASET TRSDB    WAS SUCCESSFUL
IST1123I   MODIFY CHKPT TO DATASET DSDB2    WAS SUCCESSFUL















Command ===>
F1=Help      F2=Main Menu   F3=Return                                 F6=Roll
F7=Backward  F8=Forward                                              F12=Cancel
```

Control Point Display Command

To test the control point display:

1. From the command facility, enter **AON**.
2. Select **2** for SNA.
3. Select **6** for Advanced Peer-to-Peer Networking.

4. Select **2** for Display control points.

   Sample result:

```
FKVKA200         SNA Automation: APPN CP Display                  CNM01

Type an action code.  Then press Enter.                           More:  +
 1=Details  2=Delete Topology  3=Delete Directory  4=Active Monitoring
 5=Timers   6=AutoView
    Control Point        Node Type
 _  ISTADJCP             ADJCP MAJOR NODE
 _  USIBMTA.TA1PT106     EN
 _  TA1CP213             *NA*
 _  TA1CP214             *NA*
 _  USIBMTA1.OPER1       EN
 _  USIBMTA.NTC0PUN6     *NA*
 _  USIBMTA.TA1CP210     EN
 _  APPN.TA1PT209        EN
 _  USIBMXXX.YYY00000    EN
 _  USIBMTA.TA1PT107     EN
 _  USIBMTA.TA1PT220     EN
 _  USIBMTA.TA1CP207     NN
 _  USIBMTA.TA1PT203     EN
 _  TA1CP208             *NA*

Command ===>
F1=Help      F2=Main Menu   F3=Return                  F5=Refresh    F6=Roll
F7=Backward  F8=Forward                                             F12=Cancel
```

Testing AON/SNA X.25 Monitoring

This section explains how to test X.25 automation for AON/SNA. X.25 automation includes the LUDRPOOL and X25MONIT functions.

Testing the LUDRPOOL Function

You can bypass this section if you do not use X.25 with dynamic reconfiguration. To perform this test, you must have dynamic reconfiguration LUs defined in your NCP.

To begin the test:

1. Enter LUDRPOOL.
2. AON/SNA displays the SNA Automation: X25 LUDRPOOL panel.

   Sample result:

```
FKVKX200       SNA Automation: X25 LUDRPOOL                       CNM01


    NCP name : ________

    Monitor  : 2           (1=Yes  2=No)

    Interval : 10
    Threshold: 000





Command ===>
F1=Help      F2=Main Menu   F3=Return                                F6=Roll
                                                                    F12=Cancel
```

3. Enter the name of your NCP in the NCP name field. AON/SNA updates the SNA Automation: X25 LUDRPOOL panel.

   Sample result:

```
FKVKX200       SNA Automation: X25 LUDRPOOL                        CNM01


    NCP name : TA1N500_

    Monitor  : 2            (1=Yes  2=No)

    Interval : 10
    Threshold: 000




FKV651I LUDRPOOL FOR NCP TA1N500 = 104
Command ===>
F1=Help      F2=Main Menu   F3=Return                                F6=Roll
                                                                    F12=Cancel
```

4. Change the value in the Monitor field from 2 to 1 to turn on Monitoring, and press Enter. AON/SNA updates the SNA Automation: X25 LUDRPOOL panel.

   Sample result:

```
FKVKX200       SNA Automation: X25 LUDRPOOL                        CNM01


    NCP name : TA1N500_

    Monitor  : 1            (1=Yes  2=No)

    Interval : 10
    Threshold: 000




EZL001I REQUEST LUDRSTAT WAS SUCCESSFUL FOR OPER1
Command ===>
F1=Help      F2=Main Menu   F3=Return                                F6=Roll
                                                                    F12=Cancel
```

5. Move the cursor to the command line and enter `TIMER`.
6. AON/SNA displays the active timers on the Timer Management panel.

   Sample result:

```
EZLK6000               TIMER MANAGEMENT    CNM01 NETOP1   08/19/19 08:41:38
                                                            1 TO    1 OF    1
Target:            Target Network ID:            Total Selected Timers:    1
                                                   Total Purged Timers:    0
Filter criteria:
Type one action code. then press enter.
  1|A=Add  2|C=Change  3|P=Purge  4=Add CHRON timer
    Timer ID  Scheduled           Type   Interval   Task      Save   Catchup
   _  TA1N500   08/19/19 11:02:36   AFTER            AUTX25MN
              FKVEOPFI TA1N500 10 000






Command ===>
F1=Help      F2=Main Menu   F3=Return                  F5=Refresh    F6=Roll
F7=Backward  F8=Forward                                             F12=Cancel
```

7. Look for a timer with the timer ID of the NCP name.
8. Press F3 to return to the SNA Automation: X25 LUDRPOOL panel.

To trigger threshold processing:

1. Enter 1 in the `Monitor` field on the X25 LUDRPOOL panel and change the value in the Threshold field to a value higher than the number available.
2. AON/SNA updates the SNA Automation: X25 LUDRPOOL panel.

Sample result:

```
FKVKX200       SNA Automation: X25 LUDRPOOL                        CNM01


    NCP name : TA1N500_

    Monitor  : 1            (1=Yes  2=No)

    Interval : 10
    Threshold: 200




FKV651I LUDRPOOL FOR NCP TA1N500 = 104
Command ===>
F1=Help      F2=Main Menu   F3=Return                               F6=Roll
                                                                   F12=Cancel
```

3. Move the cursor to the command line and enter DDF.
4. AON/SNA displays the CNM01 Network Status panel (the DDF menu). On the DDF menu, the X25 RESOURCES are now highlighted in pink.

   Sample panel:

```
FKVPNSNA
                    CNM01 NETWORK STATUS

 SUBAREA RESOURCES   APPN RESOURCES          X25 RESOURCES
   NCPS                CONTROL POINTS          X25 MACHINES
   CDRMS               END NODES               X25 PU SVC INOP
   CDRSCS
   LINES
   LINKS
   PUS
   APPLS

   MISCELLANEOUS RESOURCES
```

5. Move the cursor to X25 RESOURCES and press F8. AON/SNA displays the CNM01 Network Status - X25 Resources panel. This panel shows your NCP name in pink.

   Sample result:

```
FKVPNLX1                                       PAGE 1 OF 1
              CNM01 NETWORK STATUS - X25 RESOURCES

   TA1N500
```

6. Move the cursor to the NCP name and press F2. AON/SNA displays the Detail Status Display panel.

   Sample result:

```
                 ---- DETAIL STATUS DISPLAY ----
                                                     1 OF    1


      COMPONENT: TA1N500              SYSTEM   : CNM01

      COLOR    : PINK                 PRIORITY :      270

      DATE     : 08/19/19             TIME     : 09:01:51

      REPORTER : AUTX25MN             NODE     : CNM01

      DUPLICATE COUNT:

       1 'FKV653E LUDRPOOL FOR NCP TA1N500 = 104 : THRESHOLD = 200'
```

Testing the X25MONIT Function

To perform the test for the X25MONIT function, your system must have:

- Active X.25 switched virtual circuit (SVC) links
- At least one switched virtual circuit (SVC) link defined in the configuration file
- DDF customized for X.25
- Started the X25MONIT environment through the configuration file or the X25INIT command
- Access to an X.25 switched virtual circuit (SVC) device that can start a connection with a monitored switched virtual circuit (SVC) link

To run the X25MONIT test:

1. Enter `X25MONIT`. AON/SNA displays the Operator Command Interface: X.25 Monit panel.

   Sample result:

```
FKVKX100       Operator Command Interface: X.25 Monit              CNM01

Type an action code.  Then press Enter.
1=Add   2=Change  3=Delete
 Res Name              -------STATUS-------       ------SVCs------
   Mch Name Group   NCP Name MCH-Li MCH-PU MCH-LU  Type  Tot Act Busy Free Tmr.
 LINE1
 2  XL01001 X25S01B TA1N500   ACTIV  ACTIV  ACTIV   INOUT 7   0   0 7
 LINE2
 _  XL01002 X25S01A TA1N500   ACTIV  ACTIV  ACTIV   IN    1   0   0 7
 LINE3
 _  XL01003 X25S01C TA1N500   ACTIV  ACTIV  ACTIV   OUT   23  0   0 23
 LINE4
 _  XL01004 X25S01D TA1N500   ACTIV  ACTIV  ACTIV   INOUT 3   0   0 3
 LINE5
 _  XL01001 X25S01E TA1N500   ACTIV  ACTIV  ACTIV   OUT   3   0   0 3




Command ===>
F1=Help      F2=Main Menu   F3=Return   F5=Refresh    F6=Roll
F7=Backward  F8=Forward          F12=Cancel
```

2. Verify that the values for the Name, Group, NCP, Type, and Total columns are correct.
3. Check the values for the Active, Busy, and Free columns.
4. Start a connection from your X.25 device.
5. Press F5 to refresh the panel. AON/SNA decreases the value in the Free column by 1 and increases the value in the Busy column by 1.
6. Disconnect the X.25 device.
7. Press F5 to refresh the panel. AON/SNA decreases the values for the Busy column by 1 and increases the values for the Free column by 1.

Setting Up Focal-point Services

A *focal point* is a domain node that you define in the control file. The focal point domain is the central control point for information about other domains in your distributed network. AON routines forward messages to the focal point from the distributed domains. You can also issue commands and receive responses from the focal point NetView to distributed NetView programs using AON *gateways*.

By implementing AON focal point services, you can:

- Enable command facility (NCCF) operators to use the gateway pipelines to send commands to other NetView domains and receive responses, eliminating the need for personal NNT sessions for each NCCF operator.
- Manage the RACF passwords for gateway automation operators

- Set up and initiate automated user NNT sessions
- Set up automated user terminal access facility (TAF) full-screen sessions
- Display the status of gateway automation operators, user NNT sessions, and TAF full-screen sessions.

The AON router establishes and maintains connections between hosts and forwards messages and alerts from multiple hosts to a single host. Because of this, network operators can receive all network alert messages at a single console. The message destination is controlled within the control file. Routed messages identify the origination host. The router displays the status of host connections in the operator interface full-screen displays and also in DDF.

Automation Notification Forwarding Design

Automation notifications are messages that describe significant actions detected or taken by AON. These notifications are necessary to understand and operate the network. A forwarding facility sends notifications from one NetView program to another NetView program. Through a focal point, you can monitor several NetView programs and their networks from one NetView program. AON uses automation notifications to update DDF. By forwarding notifications, AON provides a consolidated, hierarchical view of an entire operating environment on the focal point DDF.

To forward notifications between different systems, designate a focal point where AON sends all notifications. Optionally, you can designate a backup for the primary focal point. Define the connectivity of the hosts in a tree-structured hierarchy, so that all notifications are forwarded to the focal point. Figure 8 on page 74 shows an example of a notification forwarding hierarchy.

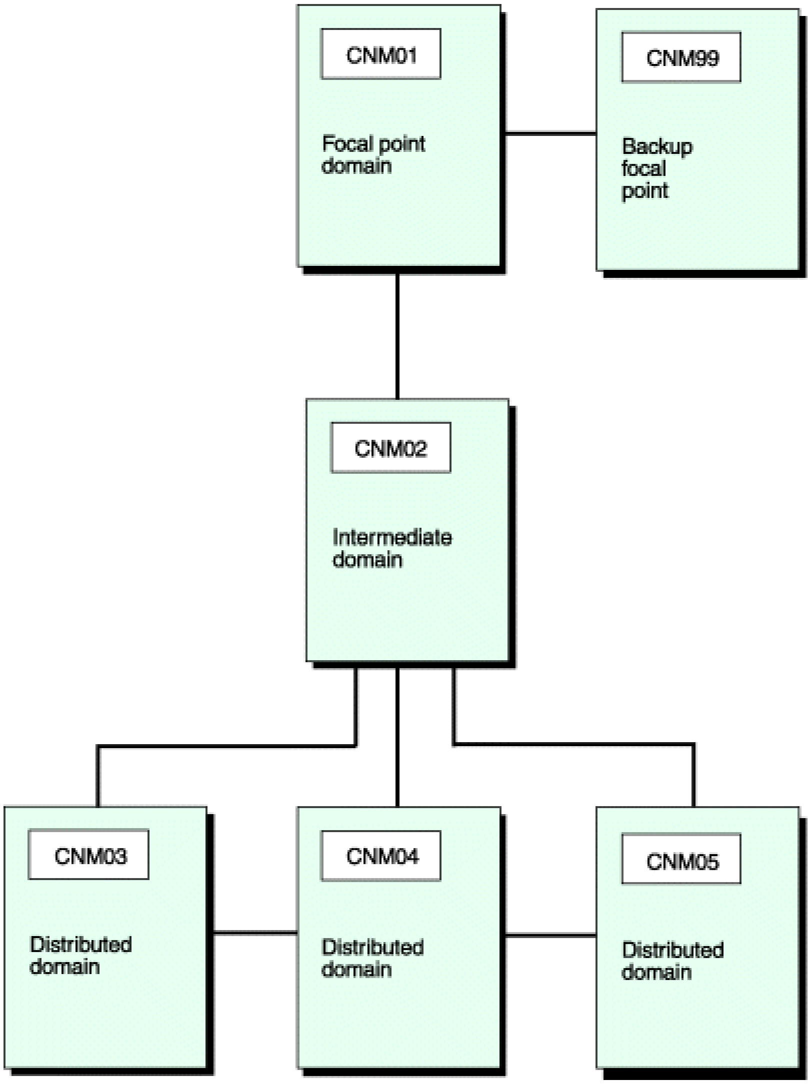


Figure 8. Notification Forwarding Hierarchy

In Figure 8 on page 74, notifications from domains CNM02, CNM03, CNM04, and CNM05 are forwarded to domain CNM01. Notifications sent from the distributed nodes pass through CNM02 and are routed to the focal point domain, CNM01. CNM02 also forwards its own notifications to the focal point. If domain CNM01 is not available, an alternate path is defined to the backup focal point, CNM99.

Define notification forwarding hierarchy in the FORWARD FOCALPT control file entry. Each domain has a single FORWARD FOCALPT entry that defines primary and backup focal points. The FORWARD FOCALPT entry for both the distributed and intermediate domains in Figure 8 on page 74 is:

```
FORWARD FOCALPT,PR=CNM01,BKUP=CNM99
```

No FORWARD FOCALPT entries are defined for domains CNM01 or CNM99. If no FORWARD FOCALPT entry is defined or if the domain specified in a FORWARD FOCALPT entry is the current domain, AON considers the current domain as the focal point and displays all notifications without forwarding them. For more information about the FORWARD FOCALPT control file entry, see the *IBM Z NetView Administration Reference*.

If the domain specified in the FORWARD FOCALPT entry is not available, AON logs notifications in the NetView log of the current domain. For example, in Figure 8 on page 74, if neither domain CNM01 nor its backup focal point, CNM99, is available, all notifications from domain CNM03 are logged in the NetView log on domain CNM03.

Gateways

To forward automation notifications and route commands and responses between NetView domains, AON uses automation operators referred to as *gateways*. Each domain has a single automation operator defined as an *outbound gateway operator*. The outbound gateway automation operator establishes and maintains all the connections to other domains. These connections are established when the outbound gateway operator logs on to the target domain. This logon process creates an NNT session on the target domain using the operator ID of the original outbound gateway operator.

On the target domain, the original outbound gateway automation operator is referred to as the *inbound gateway*. A domain can have one or more inbound gateways, depending on the number of domains connected to it.

Default gateway names are formed by combining the GAT prefix with the domain name. For example, the CNM01 domain outbound gateway automation operator is named GATCNM01. Similarly, any inbound gateway automation operator name is the GAT prefix concatenated with the inbound gateway domain name. For example, the gateway names for three domains, CNM01, CNM02 and CNM03, are shown in Figure 9 on page 75. Domain CNM01 is the focal point for notification forwarding for distributed hosts CNM02 and CNM03.

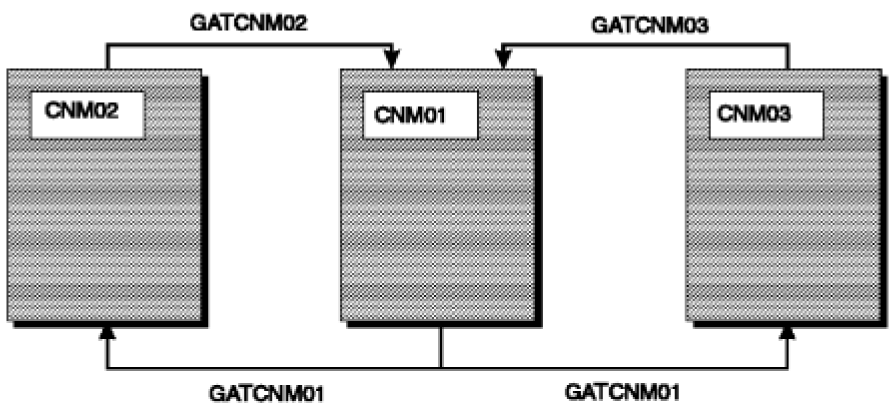


Figure 9. Gateway Names in a Distributed Network

In Figure 9 on page 75, using the default naming convention, the outbound gateway automation operator for domain CNM01 is called GATCNM01. GATCNM01 is also the inbound gateway automation operator for domains CNM02 and CNM03. Similarly, GATCNM03, an outbound gateway for domain CNM03, is an inbound gateway on the focal point domain, CNM01.

You can override default gateway names with the LOGONID parameter of the GATEWAY control file entry or with the AUTOOPS control file entry.

For each domain, only the outbound gateway task is defined in the control file using the AUTOOPS entry. In Figure 9 on page 75, the control file entry for domain CNM01 is:

```
AUTOOPS GATOPER,ID=GATCNM01
```

For more information about the AUTOOPS control file entry, see the *IBM Z NetView Administration Reference*.

In the GATEWAY control file entry, you define domains to which outbound gateway automation operators connect. In this same definition, you specify the password that the gateway automation operator uses to log on to other domains. This password must match the password in the target domain for this outbound gateway automation operator. In Figure 9 on page 75, the control file entries are for domain CNM01:

```
GATEWAY CNM02,PASSWORD=pwd_cnm01,DESC='AON NETVIEW'
GATEWAY CNM03,PASSWORD=pwd_cnm01,DESC='SYSTEM SY2 AON NETVIEW'
```

Define gateway automation operators to a NetView domain in the DSIOPF member of the DSIPARM data set. For example, in Figure 9 on page 75 the DSIOPF entries for domain CNM01 follow:

```
GATCNM01  OPERATOR PASSWORD=pwd_cnm01
          PROFILEN EZLPRFAO
GATCNM02  OPERATOR PASSWORD=pwd_cnm02
          PROFILEN EZLPRFAO
GATCNM03  OPERATOR PASSWORD=pwd_cnm03
          PROFILEN EZLPRFAO
```

For more information about the GATEWAY control file entry, see the *IBM Z NetView Administration Reference*.

Passwords

You do not need to define automation operators to RACF. However, you do need to define to RACF any ID that is used for an NNT session, even if the ID is used by an automation operator.

Because gateway automation operators can automatically log on to other NetView domains, AON must have a way of storing passwords and an automated process for maintaining them. AON stores passwords in an encrypted format in a VSAM data set. AON provides an interface for retrieving passwords when necessary. AON routines automatically change the passwords every 30 days. If you do not use RACF and the AON automated password management and retrieval, you must hard code passwords in the GATEWAY control file entry.

Note: The NetView program provides the option of specifying whether password checking is performed by the NetView program or by an SAF security product, such as RACF. The method of checking is specified by the SECOPTS.OPERSEC setting, described in the *IBM Z NetView Administration Reference*. If you specified that password checking is to be performed by the NetView program, be aware that any password defined to the NetView program is automatically converted to uppercase and stored in uppercase. If you specified that password checking is to be performed by using an SAF security product, you might be able to use a mixed case password.

When you implement AON automated RACF password management and retrieval, the following restrictions apply:

- Gateway automation operator IDs must be defined using the password NOINTERVAL RACF command. This command is documented in your RACF manuals.
- AON generates a random eight-character password for the gateway automation operator. Use the MASK parameter on the ENVIRON RACF control file entry to specify fewer characters for passwords.
- AON automatically changes passwords every 30 days.
- AON stores the passwords in encrypted format in a VSAM data set, which is used when AON retrieves the passwords. Restrict access to the command processor, GETPW, because it retrieves the password from the VSAM data set.

  Note: *Command authorization checking* is NetView security. It stops an operator from issuing a command. GETPW returns the password to the routine for logon. Without authorization, an operator can use GETPW to log on as the gateway operator.
- In a shared RACF data set environment (for example, if two NetView programs are running in the same host), the ENVIRON RACF control file entry must be coded using the OWNER or LIST parameters. The domain specified in either of these two parameters must be the owner of the RACF data sets.

For more information, see "Installing RACF Gateway Automation Operator Password Option" on page 85.

Note: All non-RACF users must define all of their automation operators to their SAF product. In addition, for automation operators that are also gateway operators, store the passwords in the AON password data set. All the considerations regarding RACF also apply to other SAF products.

Connection

When a domain receives the following message from the NetView program, its outbound gateway automation operator tries to connect to all the domains specified in the GATEWAY entries in its control file:

```
DSI112I NCCF READY FOR LOGON AND SYSTEM OPERATOR COMMANDS
```

When the outbound gateway automation operator logs on to the other domains as the inbound gateway, the inbound gateway sends a command to the outbound gateway automation operator on each domain to which it connects, requesting to log. For example, in Figure 9 on page 75, if the DSI112I message is received on domain CNM01, GATCNM01 attempts to log on to domains CNM02 and CNM03, which are defined as gateway entries in the control file on domain CNM01. After GATCNM01 logs on to domains CNM02 and CNM03 as the inbound gateway operator, it sends messages to GATCNM02 and GATCNM03 asking them to log on to CNM01. In this way all the domains establish outbound and inbound connections without user intervention.

In each domain, AON displays the status of gateways through the Dynamic Display Facility.

Command and Response Routing

After the gateways between NetView domains are established, automation or NCCF operators can use the outbound gateway to issue commands to another domain and receive a response from the other domain on the inbound gateway from that domain. The need for all operators to have personal NNT sessions for each domain to which they need access is eliminated. The AON SENDCMD command issues commands across the gateways.

To facilitate command and response routing between different NetView domains, you can also specify an adjacent NetView with the ADJNETV control file entry. This entry defines an adjacent NetView domain that acts as intermediary host for commands and responses between one NetView and another. You can also specify a backup adjacent domain to be used if the primary adjacent domain is unavailable. In Figure 10 on page 78, domains CNM02, CNM03 and CNM04 are distributed hosts with CNM01 defined as the focal point. To route commands and responses between domains CNM02 and CNM04, you specify domain CNM03 as the adjacent domain. Optionally, you can define domain CNM01 as the backup adjacent domain.

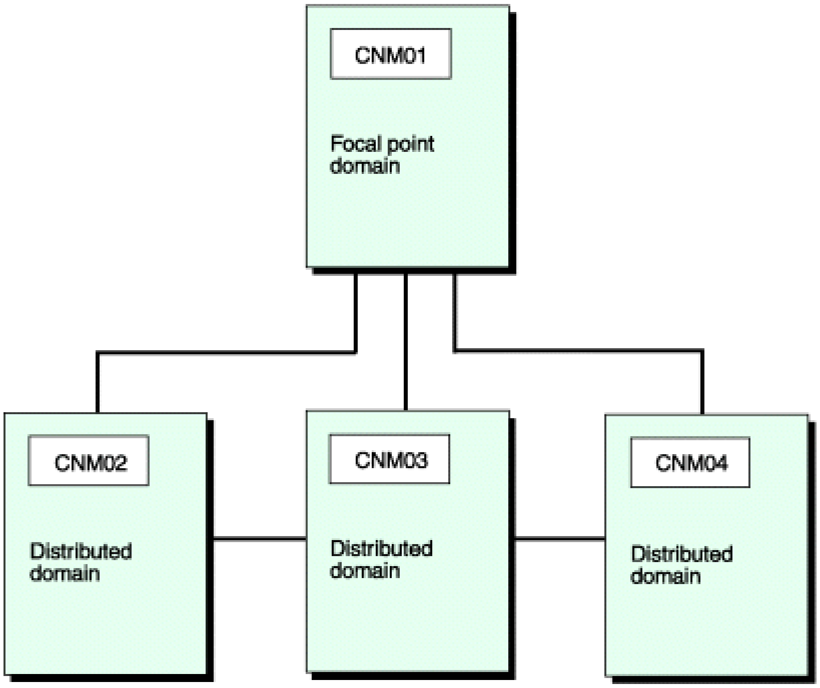


Figure 10. Adjacent NetView Programs

The control file entry for domain CNM02 (as specified in domain CNM04) is:

```
ADJNETV CNM02,DOMAIN=CNM03,ALTNETV=CNM01,DESC='PASSTHRU to CNM02'
```

and the control file entry for domain CNM04 in CNM02 is:

```
ADJNETV CNM04,DOMAIN=CNM03,ALTNETV=CNM01,DESC='PASSTHRU to CNM04'
```

NetView Definitions

To support notification forwarding, consider the following conditions:

- If an RRD statement does not already exist in the CNMSTUSR or C*xx*STGEN member, add one for each host with which the defined-host directly communicates. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
- Automation table statements are required for focal point services. Sample statements are supplied with AON and do not require modification.

Example of Focal Point Implementation

Figure 11 on page 79 shows a sample network of five NetView domains. The definitions required to implement the sample network follow.

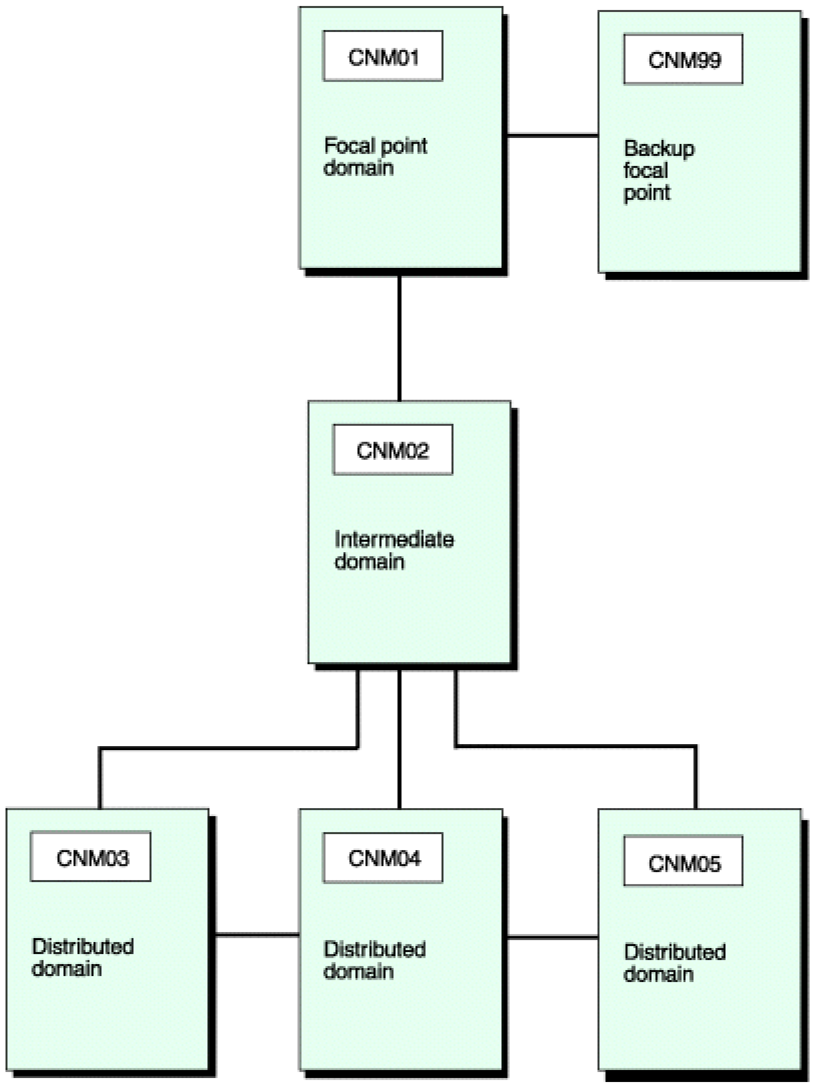


Figure 11. Notification Forwarding Hierarchy

In Figure 11 on page 79, CNM01 is the primary focal point domain and CNM99 is the backup focal point. Domains CNM03 and CNM05 include adjacent NetView definitions so that commands and responses can be routed between them.

Domains CNM01, CNM99, and CNM02 share the same RACF data set. CNM01 is the owning domain for the shared RACF support, this is specified on the ENVIRON RACF control file entry for the shared domains. Because CNM01 is the owning domain, its RACF password is specified on the RACF gateway definitions for CNM02 and CNM99.

Control File Entries

This section lists the AON control file entries needed to implement the sample network that is illustrated in Figure 11 on page 79.

Domain CNM01:

```
AUTOOPS GATOPER,ID=GATCNM01
ENVIRON RACF,LIST=(CNM01,CNM99,CNM02)
GATEWAY CNM02,PASSWORD=RACFNNT,DESC='CNM02 intermediate host'
GATEWAY CNM99,PASSWORD=RACFNNT,DESC='CNM99 Backup focal point'
ADJNETV CNM03,DOMAIN=CNM02,DESC='CNM03 distributed node'
ADJNETV CNM04,DOMAIN=CNM02,DESC='CNM04 distributed node'
ADJNETV CNM05,DOMAIN=CNM02,DESC='CNM05 distributed node'
```

Note: Because no FORWARD FOCALPT entry has been defined, AON treats domain CNM01 as the focal point and displays all automation notifications on this domain without attempting to forward them.

Domain CNM99:

```
AUTOOPS GATOPER,ID=GATCNM99
ENVIRON RACF,LIST=(CNM01,CNM99,CNM02)
FORWARD FOCALPT,PRI=CNM01
GATEWAY CNM01,PASSWORD=RACFNNT,DESC='CNM01 Focal point'
GATEWAY CNM02,PASSWORD=RACFNNT,DESC='CNM02 distributed node'
ADJNETV CNM03,DOMAIN=CNM02,DESC='CNM03 distributed node'
ADJNETV CNM04,DOMAIN=CNM02,DESC='CNM04 distributed node'
ADJNETV CNM05,DOMAIN=CNM02,DESC='CNM05 distributed node'
```

Domain CNM02:

```
AUTOOPS GATOPER,ID=GATCNM02
ENVIRON RACF,LIST=(CNM01,CNM99,CNM02)
FORWARD FOCALPT,PRI=CNM01,BKUP=CNM99
GATEWAY CNM01,PASSWORD=RACFNNT,DESC='CNM01 Focal point'
GATEWAY CNM99,PASSWORD=RACFNNT,DESC='CNM99 Backup Focal Point'
GATEWAY CNM03,PASSWORD=RACFNNT,DESC='CNM04 Distributed host'
GATEWAY CNM04,PASSWORD=RACFNNT,DESC='CNM05 Distributed host'
GATEWAY CNM05,PASSWORD=RACFNNT,DESC='CNM06 Distributed host'
```

Domain CNM03:

```
AUTOOPS GATOPER,ID=GATCNM03
FORWARD FOCALPT,PRI=CNM01,BKUP=CNM99
ADJNETV CNM01,DOMAIN=CNM02,DESC='Adjacent NetView CNM01'
ADJNETV CNM99,DOMAIN=CNM02,DESC='Adjacent NetView CNM99'
ADJNETV CNM05,DOMAIN=CNM04,ALTNETV=CNM02,DESC='Adjacent NetView CNM05'
GATEWAY CNM02,PASSWORD=RACFNNT,DESC='CNM02 Intermediate domain'
GATEWAY CNM04,PASSWORD=RACFNNT,DESC='CNM04 Adjacent Host'
```

Domain CNM04:

```
AUTOOPS GATOPER,ID=GATCNM04
FORWARD FOCALPT,PRI=CNM01,BKUP=CNM99
ADJNETV CNM01,DOMAIN=CNM02,DESC='Adjacent NetView CNM01'
ADJNETV CNM99,DOMAIN=CNM02,DESC='Adjacent NetView CNM99'
GATEWAY CNM02,PASSWORD=RACFNNT,DESC='CNM02 Intermediate domain'
GATEWAY CNM05,PASSWORD=RACFNNT,DESC='CNM05 Adjacent Host'
```

Domain CNM05:

```
AUTOOPS GATOPER,ID=GATCNM05
FORWARD FOCALPT,PRI=CNM01,BKUP=CNM99
ADJNETV CNM01,DOMAIN=CNM02,DESC='Adjacent NetView CNM01'
ADJNETV CNM99,DOMAIN=CNM02,DESC='Adjacent NetView CNM99'
ADJNETV CNM03,DOMAIN=CNM04,ALTNETV=CNM02,DESC='Adjacent NetView CNM03'
GATEWAY CNM02,PASSWORD=RACFNNT,DESC='CNM02 Intermediate domain'
GATEWAY CNM04,PASSWORD=RACFNNT,DESC='CNM04 Adjacent Host'
```

NetView DSIPARM DSIOPF Entries

This section lists the NetView DSIPARM DSIOPF member entries needed to implement the sample network illustrated in Figure 11 on page 79.

Domain CNM01:

```
GATCNM01 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM99 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

Domain CNM99:

```
GATCNM01 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

```
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM99 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

Domain CNM02:

```
GATCNM01 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM03 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM04 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM05 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM99 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

Domain CNM03:

```
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM03 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM04 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

Domain CNM04:

```
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM03 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM04 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM05 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

Domain CNM05:

```
GATCNM02 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM04 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
GATCNM05 OPERATOR PASSWORD=RACFNNT
         PROFILEN EZLPRFAO
```

NetView RRD Entries

This section lists the NetView RRD entries in the CNMSTYLE member that are required to implement the sample network illustrated in Figure 11 on page 79.

Domain CNM01:

```
RRD.CNM02 = *
RRD.CNM99 = *
```

Domain CNM99:

```
RRD.CNM01 = *
RRD.CNM02 = *
```

Domain CNM02:

```
RRD.CNM01 = *
RRD.CNM03 = *
RRD.CNM04 = *
RRD.CNM05 = *
RRD.CNM99 = *
```

Domain CNM03:

```
RRD.CNM02 = *
RRD.CNM04 = *
```

Domain CNM04:

```
RRD.CNM02 = *
RRD.CNM03 = *
RRD.CNM05 = *
```

Domain CNM05:

```
RRD.CNM02  = *
RRD.CNM04  = *
```

RACF Definitions

It is assumed that the VSAM RACF password database (NETVIEW.CNM01.PASSWORD) was successfully allocated when you ran the VSAM allocation job CNMSJ004 during NetView installation.

RACF Considerations

Because domains CNM01, CNM02, and CNM99 share the same RACF data set, you must define the following gateway operators in that data set:

- GATCNM01
- GATCNM02
- GATCNM99
- GATCNM03
- GATCNM04
- GATCNM05

You must also define the following gateway operators in the data sets indicated:

RACF data set
  Gateway operators

CNM03
  GATCNM02 and GATCNM04

CNM04
  GATCNM02, GATCNM03 and GATCNM05

CNM05
  GATCNM02 and GATCNM04

AON Considerations

You must issue the GETPW command on each domain with the following operands to initialize the VSAM RACF password data set with the passwords for gateway operators. The password field contains the actual password for the outbound gateway operator to log on to the specified domain.

For domain CNM01, issue:

```
* GETPW GATCNM01 CNM01,INIT=password
```

For domain CNM99, issue:

```
* GETPW GATCNM99 CNM01,INIT=password
```

For domain CNM02, issue:

```
* GETPW GATCNM02 CNM01,INIT=password
  GETPW GATCNM02 CNM03,INIT=password
```

```
GETPW GATCNM02 CNM04,INIT=password
GETPW GATCNM02 CNM05,INIT=password
```

For domain CNM03, issue:

```
GETPW GATCNM03 CNM02,INIT=password
GETPW GATCNM03 CNM04,INIT=password
```

For domain CNM04, issue:

```
GETPW GATCNM04 CNM02,INIT=password
GETPW GATCNM04 CNM03,INIT=password
GETPW GATCNM04 CNM05,INIT=password
```

For domain CNM05, issue:

```
GETPW GATCNM05 CNM02,INIT=password
GETPW GATCNM05 CNM04,INIT=password
```

Note: Entries marked with an asterisk (*) are necessary only if this is the owning RACF domain.

CNM01, CNM99, and CNM02 share the same RACF password data set. The shared relationship is defined on the ENVIRON RACF entry in one of two ways:

- `ENVIRON RACF OWNER=CNM01 SHARE=(CNM02,CNM99)`
- `ENVIRON RACF LIST=(CNM01,CNM02,CNM99)`

Specify one domain, CNM01 in this case, as the owning domain.

With shared RACF, you specify the outbound gateway operator ID and RACF owner domain in all sharing systems. Table 16 on page 83 illustrates how the GETPW command differs in a shared and nonshared RACF environment.

Table 16. Comparison of shared and nonshared RACF environment

Domain	Command	Nonshared RACF	Shared RACF
CNM01	GETPW	`GATCNM01 CNM99 pwd` `GATCNM01 CNM02 pwd`	`GATCNM01 CNM01 pwd`
CNM99	GETPW	`GATCNM99 CNM01 pwd` `GATCNM99 CNM02 pwd`	`GATCNM99 CNM01 pwd`
CNM02	GETPW	`GATCNM02 CNM01 pwd` `GATCNM02 CNM99 pwd`	`GATCNM02 CNM01 pwd`

Implementing Notification Forwarding

It is best to implement notification forwarding in a top-down approach, defining focal points first, then the distributed hosts. This approach works best because the focal point is ready to handle the forwarded messages when message forwarding is turned on in remote hosts.

With a top-down approach, if message forwarding is not yet implemented, AON displays messages at remote hosts to notification operators. After you have enabled message forwarding, AON routes messages to the focal point specified.

In summary, the tasks required to implement notification forwarding in a focal point environment include:

- Designing your notification forwarding hierarchy.
- Tailoring NetView definitions (in the CNMSTUSR or C*xx*STGEN member). For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
- Defining the focal point and backup focal point control file entries (FORWARD FOCALPT).

- Defining the outbound gateway operator entries (AUTOOPS)
- Defining the inbound gateway operator entries (GATEWAY and ADJNETV).
- Adding NetView outbound and inbound operator IDs (DSIOPF).
- Initializing the NetView outbound and inbound operator IDs and passwords.

Notification Forwarding Example

For this example, assume that the AON operator AONNET2 wants to forward a request (AON notify) to CNM99 from CNM01. The outbound gateway operator for CNM01 is GATCNM01. The outbound gateway operator for CNM02 is GATCNM02. The outbound gateway operator for CNM99 is GATCNM99. The inbound gateway operator for CNM02 that receives notifications from CNM01 is GATCNM01. The inbound gateway operator for CNM99 that receives notifications from CNM02 is GATCNM02.

The program sends the request to the CNM01 outbound gateway operator (GATCNM01). The outbound gateway operator determines if the request is for the CNM01 domain. In this case, it is not. The outbound gateway operator then routes the request to the inbound gateway operator in domain CNM02 (GATCNM01) from domain CNM01. When the inbound gateway operator receives the request, it sends the request to the outbound gateway operator for CNM02 (GATCNM02). The outbound gateway operator for CNM02 determines whether the forwarded request is for this domain. In this case, it is not. The outbound gateway operator then routes the request to the inbound gateway operator in domain CNM99 (GATCMM02) from domain CNM02. When the inbound gateway operator receives the request, it sends the request to the outbound gateway operator for CNM99 (GATCNM99). The outbound gateway operator for CNM99 determines whether the forwarded request is for this domain. In Figure 12 on page 84 the request is for this domain, so it issues the request.

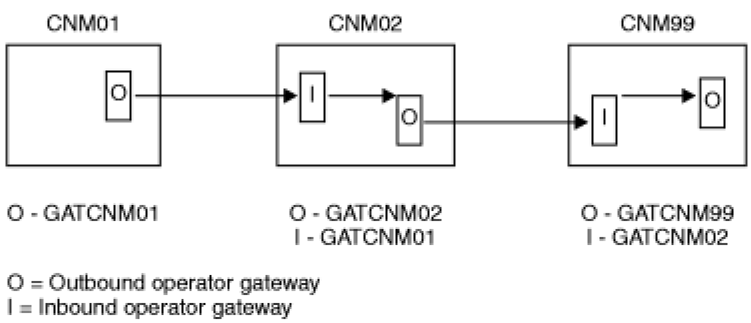


Figure 12. Notification Forwarding Example

Table 17 on page 84 lists the control file entries that are needed to accomplish the notification forwarding illustrated in Figure 12 on page 84.

Table 17. Example of the Required Control File Entries for Notification Forwarding

Domain	**Control file entries**

CNM01	`AUTOOPS  GATOPER,ID=GATCNM01`
	`GATEWAY  CNM02,DESC='NEXT DOMAIN',`
	`         PASSWORD=xxxx`
	`ADJNETV  CNM99,DOMAIN=CNM02`
	`FORWARD  FOCALPT,PRI=CNM99`

CNM02	`AUTOOPS  GATOPER,ID=GATCNM02`
	`GATEWAY  CNM99,DESC='NEXT DOMAIN',`
	`         PASSWORD=xxxx`
	`GATEWAY  CNM01, DESC='`
	`FORWARD  FOCALPT,PRI=CNM99`

Table 17. Example of the Required Control File Entries for Notification Forwarding (continued)	
Domain	**Control file entries**

CNM99	`AUTOOPS    GATOPER,ID=GATCNM99`
	`GATEWAY    CNM02,DESC='NEXT DOMAIN',`
	`           PASSWORD=xxxx`
	`ADJNETV    CNM01,DOMAIN=CNM02`
	`FORWARD    FOCALPT,PRI=CNM99`

Installing RACF Gateway Automation Operator Password Option

Note: Before you install the RACF gateway automation operator password option, ensure that the conditions as specified in "NetView RRD Entries" on page 81 are acceptable.

Use the following steps to install the RACF option:

1. Allocate a VSAM data set for the RACF gateway passwords (see "Allocate VSAM Data Set" on page 85).
2. Define gateway operator IDs to RACF (see "Define Gateway Operators to RACF" on page 85).
3. Use the GETPW command processor (see "Initialize Passwords in the Password Data Set" on page 85) to set the gateway operator ID passwords in the VSAM data set.
4. In a shared RACF data set environment, code the control file ENVIRON RACF entry.
5. Schedule a recycle of NetView.

Allocate VSAM Data Set

The CNMSJ004 job (contained in the NETVIEW.V6R3USER.INSTALL data set) defines the VSAM cluster for the RACF save password facility for gateway automation operators. You already ran this job during NetView installation. To rerun this job to allocate the VSAM RACF password database NETVIEW.CNM01.PASSWORD follow these steps:

1. Review AON IDCAMS member EZLSI101. This is where the VSAM cluster information is located for the RACF password data set.
2. If you are reallocating NETVIEW.CNM01.PASSWORD, edit CNMSID01 to delete the existing database so that you can allocate a new database.
3. Rerun CNMSJ004 to allocate the VSAM RACF password database.

   Before rerunning the job, update the DASD type, database name, and any other information that is unique to your environment. Ensure that you are allocating only the VSAM RACF password database.

Update the NetView Start Procedure

If you allocated the VSAM RACF password database NETVIEW.CNM01.PASSWORD, add the following DD statement to the NetView start procedure:

```
//EZLPSWD  DD   DSN=NETVIEW.CNM01.PASSWORD,
//         DISP=SHR,AMP='AMORG,BUFNI=10,BUFND=5'
```

Define Gateway Operators to RACF

The RACF administrator must define the gateway operator IDs using the PASSWORD NOINTERVAL RACF command. The password for the gateway operator is eight characters. You can specify fewer characters for passwords by using the MASK parameter of the ENVIRON RACF control file entry.

Initialize Passwords in the Password Data Set

In a RACF environment, the password you store in the AON data set depends on whether you are using shared RACF. In a nonshared environment, issue the GETPW command using the INIT option for each domain. This enables AON routines to retrieve and manage the RACF passwords for the gateway automation operators. In a shared environment, issue the GETPW command for only the owning domain. Table 16 on page 83 compares the GETPW command for shared and nonshared environments. See "GETPW - Gateway Password Maintenance" on page 86 for more information. "Passwords" on page 76 also contains additional information about password checking in a RACF environment.

GETPW - Gateway Password Maintenance

The GETPW command processor maintains a VSAM file containing passwords used by gateway automation operators when establishing NNT sessions. The records in the data set are keyed using a combination of the user ID and domain ID. Each record has the following fields:

- Current® password field
- New password field
- Date-password-last-changed field

The passwords are stored in encrypted format and changed every 30 days. For the details and syntax of the GETPW command, see the NetView online help.

Note: If a shared RACF is specified on the ENVIRON RACF entry, when a non-owning gateway operator wants to sign on to one of the shared domains, the operator issues the following command:

```
GETPW gatoperid owning_domain,READ
```

In the following example, GETPW stores the value specified with the INIT parameter in the VSAM data set, with a key of GATCNM01CNM02 (the user ID and the domain ID).

```
GETPW GATCNM01 CNM02,INIT=pswd001
```

In the following example, GETPW retrieves the password from the VSAM data set that allows GATCNM01 to log on to CNM02.

```
GETPW GATCNM01 CNM02,READ
```

In the following example, GETPW deletes the password from the VSAM data set that GATCNM01 uses to log on to CNM02.

```
GETPW GATCNM01 CNM02,DELETE
```

Defining RMTCMD Gateway Sessions

Each NNT gateway session can also have a RMTCMD session defined. Several functions (i.e.: Inform Policy and TCP/IP for MVS) use the RMTCMD session to route requests to other NetView domains.

If you have 3 domains as shown in Figure 13 on page 86, then in addition to the NNT gateway sessions you also have RMTCMD gateway sessions that use a task prefix of RMT.

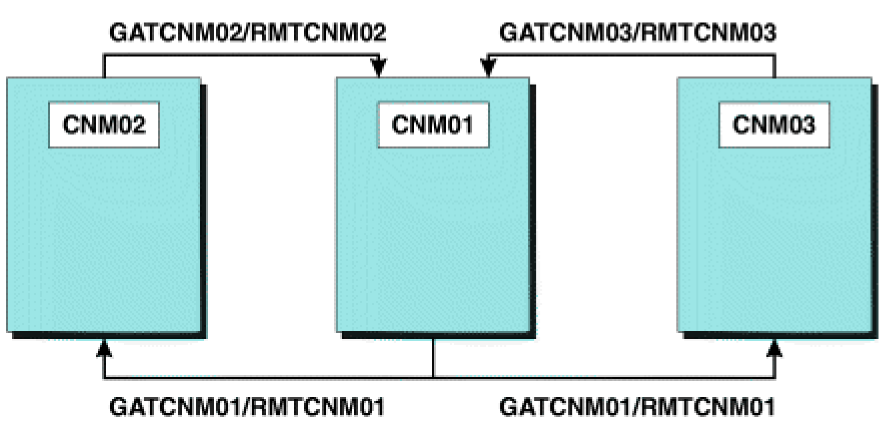


Figure 13. Gateway Names in a Distributed Network

The CDLOG statement is used to define the RMT operator and RMTCMD sessions. For details on CDLOG see the *IBM Z NetView Administration Reference*.

Examples:

To define a session from CNM01 to CNM03 in EZLCFG01 for domain CNM01:

```
CDLOG  GATCNM01.CNM03,
       SESSTYPE=RMT,
       TARGOP=RMTCNM01,
       INIT=YES,
       DESC='RMTCMD GATEWAY to CNM03'
```

This allows RMTCMD requests to flow from CNM01 to CNM03.

To define a session from CNM03 to CNM01 in EZLCFG01 for domain CNM03:

```
CDLOG  GATCNM03.CNM01,
       SESSTYPE=RMT,
       TARGOP=RMTCNM03,
       INIT=YES,
       DESC='RMTCMD GATEWAY to CNM01'
```

Add the appropriate operator ids in DSIOPF for each domain.

Chapter 3. Configuring the Operator Environment

The following topics describe aspects of the operator environment that you can customize:

- "Defining NetView Operators" on page 89
- "Specifying the Degree of Security Verification" on page 89
- "Defining Operator Data Sets" on page 89
- "Assigning Operators to Groups" on page 90
- "Suppressing Commands After Entry" on page 90
- "Defining PA and PF Keys" on page 90
- "Defining Hardcopy Printers" on page 91
- "Setting Initial Defaults" on page 91
- "Defining Domains Where This NetView Program Can Establish Cross-Domain Communication" on page 92

Defining NetView Operators

You can define your NetView operators either by using an SAF security product, through DSIPARM member DSIOPF, or both. For detailed information about defining NetView operators, refer to the *IBM Z NetView Security Reference*.

NetView operators, using the RMTCMD command, can issue commands from the NetView program running on your local system to a NetView program running on a remote system. When the operator issues a RMTCMD command and is not already logged on to the remote system, NetView logs the operator onto the remote system as a distributed autotask.

The operator can specify a logon ID on the RMTCMD command. However, if a logon ID is not specified, the NetView program uses the operator logon ID from the local system as the default logon ID for the distributed autotask session.

If you want operators to issue RMTCMD commands without specifying a logon ID for each command, ensure that each operator has a unique logon ID on all the systems to which RMTCMD commands are issued.

Specifying the Degree of Security Verification

You can define the degree of security verification to be performed when an operator logs on by using the SECOPTS statements in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Use the REFRESH command to refresh many types of security used while the NetView program is running. The REFRESH command can be used to change the security settings in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
SAF checking	*IBM Z NetView Security Reference*
REFRESH command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Defining Operator Data Sets

You can set up partitioned data sets (PDSs) which contain members that apply only to specific operators, for example, PF key definitions and command lists. To do this, follow these steps:

1. Decide on a naming convention for such data sets and allocate them. The default naming convention is `NETVIEW.OPDS.opid` where *opid* is the operator ID associated with each such data set.
2. In the CNMSTUSR or C*xx*STGEN member, set a common global variable called OpDsPrefix to your operator data set prefix. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. The default is `NETVIEW.OPDS`. You can use the **RESTYLE** command to enable the change without recycling the NetView program.
3. Set up the logon profile for each such operator to issue **OVERRIDE** commands that define data sets that are to be specific for that operator. LOGPROF1 (CNME1049) starts with the OpDsPrefix common global variable, or default naming convention, and appends the operator name to set up an operator data set for DSICLD and DSIOPEN. This enables CLISTs and PF-key definitions that are specific to this operator to be kept in this data set.
4. Ensure each such operator is authorized to read from the data sets intended for that operator. To save current PF-key settings, an operator must have write authority to the data set associated with the DSIOPEN DD. The **DISPFK** command displays and saves PF keys.
5. Add appropriate members to these data sets. See the *IBM Z NetView Command Reference Volume 1 (A-N)* for OVERRIDE and DISPFK for more information.

Assigning Operators to Groups

You can route messages to groups of operators. To define operator groups, use the ASSIGN statement:

```
ASSIGN.groupname.GROUP = list
```

Where:

groupname
Is the 1–7 character group name

list
Is the list of operator names separated by blanks or commas.

If you want information about...	Refer to...
Group names	*IBM Z NetView Automation Guide*

Suppressing Commands After Entry

If the operator types a suppression character before a command, the command is not shown on the terminal screen, hardcopy log, or NetView log. On the terminal screen, the operator sees the command as it is typed, but the NetView system does not echo the command to the screen after it is entered. The default suppression character in the CNMSTYLE member is the question mark (?). To change this suppression character, use the SUPPCHAR statement in the CNMSTUSR or C*xx*STGEN member, and update the statement for your environment. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. To prevent an operator from suppressing commands, comment out the SUPPCHAR statement in the CNMSTUSR or C*xx*STGEN member.

Note: If the text of one command is imbedded in another command, for example with EXCMD, enter the suppression character as the first character on the command line or the command buffer, as shown here:

```
?EXCMD OPER1,SDOM PASSWORD=XYZ
```

Defining PA and PF Keys

During logon to the NetView program, an operator runs the PFKDEF command list, CNME1010, which (as a default) references keys defined in the sample CNMKEYS. This command can also be included in the operator profile.

To change the NetView default PF key settings or the default line of text at the bottom of many NetView panels that describes PF key settings, modify CNMKEYS.

For specific information about modifying CNMKEYS, refer to the *IBM Z NetView Customization Guide*.

Defining Hardcopy Printers

If you print terminal activity as it occurs, define printers using a HARDCOPY statement in the CNMSTUSR or C*xx*STGEN member. For information about updating CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

A printer is not defined in the samples. The HARDCOPY statement has the following format:

```
HARDCOPY = luname1 luname2 …
```

Where:

luname
Is the LU name (1–8 characters) of the printer as it is defined to VTAM. Define as many printers as you need.

Several operators can share one printer, but each operator can print to only one at a time. If too many operators share the same printer, messages can be delayed by being queued at the printer. The NetView program cannot share a printer with another application or with another NetView program.

The hardcopy devices you define must be LU type 0 and LU type 1, or must use an LU type 0 or LU type 1 logmode entry. Printers attached to SNA controllers as LU type 1 logical units can use the M3287SCS logmode. LU type 2 and LU type 3 printers are not supported.

Note:

1. When you start a session, the NetView program checks the RU size you specified in the logmode. If you specify 0, the NetView program uses the default RU size of 4096 bytes. If you enter an RU size, it must be a minimum of 256 bytes.
2. The NORMQMAX value in the member specified by the SCRNFMT parameter of the DEFAULTS command or the default value supplied by the NetView product (3000) applies to hardcopy printers. Hardcopy printers can get backlogged because they are slow or because they run out of paper.

If you want information about...	Refer to...
NORMQMAX definition statement	*IBM Z NetView Administration Reference*

Setting Initial Defaults

The following DEFAULTS statements set initial NetView system defaults:

```
DEFAULTS.NetLog = Yes
DEFAULTS.SysLog = No
DEFAULTS.HcyLog = Yes
DEFAULTS.CMD = LOW
DEFAULTS.AUTOLOGN=yes
DEFAULTS.EVERYCON = yes
DEFAULTS.MAXABEND = 4
DEFAULTS.MAXLOGON = 5
DEFAULTS.AUTOSEC = BYPASS
DEFAULTS.MAXCPU  = 95
```

You can change values as needed for your system.

If you want information about...	Refer to...
DEFAULTS values	*IBM Z NetView Command Reference Volume 1 (A-N)* for the DEFAULTS command

Defining Domains Where This NetView Program Can Establish Cross-Domain Communication

The resource routing definition (RRD) statements in the CNMSTYLE member define the domains with which the NetView program can establish cross-domain sessions using NNT sessions. The CNMSTYLE member contains the following RRD statements:

```
*RRD.CNM01 = *
*RRD.CNM02 = *
```

Where:

CNM01
Is the network NETA NetView domain as it is coded on the DOMAIN keyword in the CNMSTYLE member.

CNM02, CNM99
Are the network NETA NetView domains of the cross-domain NetView systems.

Use the CNMSTUSR or C*xx*STGEN member to create an RRD statement for this NetView system and for each cross-domain NetView system so that this NetView system can establish cross-domain communication. When you include the RRD statement for this NetView system, you can use the same table of RRD statements for each NetView system. Specify each domain in a separate RRD statement. If you are using the RMTCMD command for cross-domain communication, RRD statements are not necessary. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you are using alert, message, and status forwarding, an RRD statement is required for each domain that is sending alerts, messages, or status to this domain and for each domain to which this domain is sending alerts, messages, and status.

Chapter 4. Extending the Command Environment

You can add or modify commands and command lists for your installation. The NetView program procedure language support includes command lists written in the NetView command list language and the REXX language. You can also write command processors and installation exits in a high-level language. The high-level language supported in the NetView environment is the Language Environment® for z/OS.

If you want information about...	Refer to...
Writing commands and command lists	*IBM Z NetView Customization Guide*

Using Language Processor (REXX) Environments in the NetView Environment

Before the TSO/E language processor can process an exec, a language processor environment must exist. A language processor environment is the environment in which a REXX exec runs. The following information describes how the NetView program uses these REXX environments and highlights issues to consider when estimating the number of language processor environments needed for your configuration.

The NetView program provides a number of parts that contain REXX source code. In addition, certain NetView components such as the MultiSystem Manager and AON consist of parts that contain compiled REXX code. All of the compiled REXX parts shipped with the NetView program have been compiled with the ALTERNATE option. If you access the REXX runtime library from the NetView environment, compiled REXX programs are run in compiled mode. Otherwise, the REXX alternate library is used and compiled REXX programs run in interpreted mode.

The NetView program also contains several parts that make use of the Data REXX function. Use the Data REXX function to include REXX instructions and functions in data files.

When a REXX command list is run in the NetView program, the REXX interpreter sets up a language processor environment for the NetView program. When the command list ends, this unique environment can be held for reuse by the same task. The NetView program retains these REXX environments to improve REXX environment initialization performance. As a result, it is very important to have a sufficient number of REXX environments available to the NetView program. If more blocks are required than are available, the NetView program issues the CNM416I REXX environment initialization error message.

Before running any REXX command lists in a z/OS environment, determine the number of concurrent REXX command lists that are usually active for a task. The NetView program retains up to three REXX environments and their associated storage until the operator logs off or the number of REXX environments retained is changed by the DEFAULTS or OVERRIDE command. Additionally, the NetView program always retains one REXX environment per task for Data REXX use. The MultiSystem Manager and AON utilize REXX command lists extensively.

The IRXANCHR table is a Time Sharing Option Extensions (TSO/E) table used to reserve storage for REXX environments. Both the NetView program and TSO/E refer to this table when allocating storage for each REXX environment that is activated.

When calculating the maximum number of language processor environments that the system can initialize in the NetView address space, consider the following items:

- Two entries in the REXX IRXANCHR table are required for each non-nested NetView or REXX command list to run. If a REXX command list is called from another REXX command list, a new environment is not required. The nested command list uses the environment of the primary command list.
- A recommended default number of REXX environment entries in IRXANCHR for the NetView program is twice the maximum number of command lists that can be scheduled to run concurrently under all

NetView tasks plus two additional entries for each concurrent active NetView task, including the main task.

The maximum number of environments the system can initialize in an address space depends on the maximum number of entries defined in the environment table, IRXANCHR, and on the kind of environments being initialized. To change the number of environment table entries, you can use the IRXTSMPE sample that TSO/E provides in SYS1.SAMPLIB or you can create your own IRXANCHR load module. The IRXTSMPE sample is a System Modification Program/Extended (SMP/E) user modification (USERMOD) to change the number of language processor environments in an address space. The prolog of IRXTSMPE has instructions for using the sample job. The SMP/E code that is included in the IRXTSMPE sample handles the installation of the load module.

Storage associated with each REXX environment can increase depending on the needs of the REXX command lists. Because each REXX command list can have different storage needs, REXX environments can grow to meet the needs of the most demanding REXX command list.

The following REXX environment values are set during NetView initialization:

- REXXENV. The number of inactive environments to be retained for each operator
- REXXSLMT. The amount of storage (in 1K increments) that a REXX environment can accumulate before being stopped after its current use is completed
- REXXSTOR. The amount of storage (in 1K increments) to be acquired by REXX environment initialization processing

Tuning the number of REXX environments and controlling how these environments are maintained within the NetView program improves performance, particularly if you are running MultiSystem Manager and AON. To limit REXX environment growth, use the DEFAULTS or OVERRIDE commands to modify the REXXENV, REXXSLMT and REXXSTOR values.

You can also override these default values by adding a DEFAULTS statement in the CNMSTUSR or C*xx*STGEN member. For example, the system default value for REXXSLMT is 250. To change this value to 300, add the following statement. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
DEFAULTS.REXXSLMT=300
```

The values of REXXENV, REXXSLMT and REXXSTOR do not apply to Data REXX environments. When Data REXX environments are built, the Data REXX environments are limited to one per task, and these environments last for the life of the task no matter how much storage they need.

If you want information about...	Refer to...
Estimating and tuning the number of REXX environments	"Estimating REXX Environments" on page 94
Enhancing REXX Performance	*IBM Z NetView Tuning Guide*
Introduction to the REXX Language	*IBM Z NetView Programming: REXX and the NetView Command List Language*
Language Processor Environments (IRXANCHR)	TSO/E Library
DEFAULTS and OVERRIDE commands and REXXENV, REXXSLMT and REXXSTOR	*IBM Z NetView Command Reference Volume 1 (A-N)* and the *IBM Z NetView Tuning Guide*

Estimating REXX Environments

The IRXANCHR table is a Time Sharing Option Extensions (TSO/E) table used to reserve storage for REXX environments. Both the NetView program and TSO/E refer to this table when allocating storage for each REXX environment that is activated. If IRXANCHR is set to accommodate 24 REXX environments, when the 25th TSO/E REXX EXEC or NetView REXX command list that needs a REXX environment tries to start, it does not run because it cannot access a REXX environment. When this happens and you are running a

NetView REXX command list, you receive message CNM416I with a return code of 20 and a reason code of 24. This error message means you have run out of REXX environments.

To ensure that your system has enough REXX environments for all processing, change the value for IRXANCHR. Because the NetView administrator is often not the same person as the TSO/E administrator, maintain a separate copy of the IRXANCHR table for NetView use.

Follow these steps to estimate the number of NetView REXX environments you require in IRXANCHR:

1. Obtain a copy of the IRXANCHR table from your TSO/E administrator. This table is stored in the TSOANCH member of the SYS1.SAMPLIB data set.
2. Determine the maximum number of REXX environments that a human NetView operator might need at one time. Not all operators need to use the same number of REXX environments.
3. Multiply the number in the previous step by the maximum number of human operators that are to use the NetView program at the same time.
4. Determine the total number of automated operator tasks you need to use, and add that total to the number in the previous step. (This assumes that each automated task uses only one REXX environment at a time.)
5. To the previous total, increase the number to include new operators you might add later and for work shifts that occasionally need extra operators.
6. Determine the maximum number of NetView tasks that might be active at one time and add this number to the previous total. This number represents the total number of data REXX environments.
7. Each REXX environment requires two entries in IRXANCHR. Multiply the total from the previous step by 2 and add one more to that doubled number. The result is the grand total.
8. Replace the TOTAL setting in the IRXANCHR table header in the TSOANCH member, which you obtained from the TSO/E administrator, with the grand total.
9. Change the Assembler DS statement that identifies the amount of storage needed for the total number of IRXANCHR table entries. Refer to the instructions in the TSOANCH sample for details.
10. Run the TSOANCH sample job to assemble your updated IRXANCHR load module.
11. If the load module is not link-edited as part of the TSOANCH job, link-edit your updated IRXANCHR table in a NetView private library and specify that library name in a STEPLIB DD statement in the NetView startup procedure.

In addition to adjusting the total number of REXX environments in IRXANCHR, you can tune the default number of REXX environments that the NetView program retains for each operator during an operator session. The NetView product ships with a default setting that retains three REXX environments for each operator (REXXENV=3). However, this number might have been adjusted on your system. To see the current setting, issue a LIST DEFAULTS command, and check the REXXENV setting.

Note: Data REXX environments are independent of the DEFAULTS and OVERRIDE command settings for REXX. Every active NetView task is assigned only one REXX environment that is dedicated to data REXX.

If an operator attempts to start four REXX command lists and each command list needs a REXX environment, the NetView program requests the environments, regardless of the REXXENV setting. However, when two or more of the REXX command lists for the operator complete, the NetView program frees only one of the four REXX environments now assigned to that operator. When REXXENV=3 and three or more REXX environments have been assigned to an operator, the NetView program keeps three of them active until the operator logs off or until a reset of the REXXENV value is processed. When the operator logs off, all REXX environments assigned to that operator are returned to the *free pool* where they can be assigned to another operator.

Waiting for a REXX environment to be assigned (rather than using one that is already retained) is a relatively small performance impact for human operators; however, the impact can be more significant on automated operator tasks. Automated tasks, which have the global command priority set to LOW and REXXENV is set to 1 or greater, do not have to wait for a REXX environment to be assigned.

Storage Considerations

The storage associated with each REXX environment can grow over time, depending on the needs of the REXX command lists that use the REXX environments. Because all REXX command lists do not have the same storage needs, the REXX environments eventually grow to meet the needs of the most demanding REXX command list.

- The storage associated with each REXX environment can grow over time, depending on the needs of the REXX command lists that use the REXX environments. Because all REXX command lists do not have the same storage needs, the REXX environments eventually grow to meet the needs of the most demanding REXX command list (REXXENV).
- To limit the number of REXX environments required by the NetView program, and thus reduce the amount of storage required for each task using REXX, use the NetView DEFAULTS and OVERRIDE commands (REXXSLMT).

Before running REXX command lists in an MVS environment, determine the amount of storage required to initiate a REXX environment. TSO/E REXX, by default, gets sufficient storage for an average REXX command list, with about six levels of nested calls. You can change the amount of storage acquired by using the DEFAULTS or OVERRIDE commands. REXX command lists that use many REXX variables or that nest more than six levels increase this storage as needed. Each REXX command list requires approximately 12K of storage to get started.

Using High-Level Languages with the NetView Program

To use high-level languages with the NetView program:

- Ensure that your Language Environment for z/OS runtime libraries are included in the link pack area (LPA), the LINKLSTxx, or in CNMPROC (CNMSJ009).

  Note: You can place some or all of the Language Environment for z/OS runtime library modules in LPALST*xx* and remove any Language Environment for z/OS runtime libraries in the LPALST*xx* from the STEPLIB of your NetView start procedure (CNMPROC) to improve performance of the NetView program. For more information, see the *z/OS Language Environment* library and the *IBM Z NetView Tuning Guide*.
- Ensure that all of the runtime libraries are APF-authorized.
- CNMPROC (CNMSJ009) includes an example of the runtime libraries in the STEPLIB of your start procedure. For example:

  ```
  //*        DD   DSN=CEE.SCEERUN,DISP=SHR
  ```

  If you are not running with the Language Environment for z/OS runtime libraries in PLPA or LINKLSTxx, uncomment the DD statement that applies to you and make any necessary changes. These changes take effect the next time the NetView program is started.

  You can also define I/O data set members for use with PL/I and C programs. The following examples are included in CNMPROC (CNMSJ009).

  ```
  //*PINFILE  DD   DSN=USER.HLL.INFILE,DISP=SHR
  //*POUTFILE DD   DSN=USER.HLL.OUTFILE,DISP=SHR
  //*CINFILE  DD   DSN=USER.HLL.INFILE,DISP=SHR
  //*COUTFILE DD   DSN=USER.HLL.OUTFILE,DISP=SHR
  ```

  Uncomment any that you want to use. Ensure that you have allocated the data sets before you start the NetView program.
- The CNMSTYLE member contains definitions that request the building of preinitialized language environments. Review the defaults and make any necessary changes in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. If you are not using either the PL/I or C program, set the REGENVS value to 0.

The following list shows the defaults in the CNMSTYLE member for PL/I:

```
HLLENV.IBMADPLI.REGENVS=2              // # of preinitialized environments
HLLENV.IBMADPLI.CRITENVS=0             // max # of env for enabled progs
HLLENV.IBMADPLI.DEFAULT=NOTPREINIT     // eligible programs PREINIT?
HLLENV.IBMADPLI.PSTACK=131072          // run time stack size
HLLENV.IBMADPLI.PHEAP=131072           // run time heap size
```

The following list shows the defaults in the CNMSTYLE member for C:

```
HLLENV.IBMADC.REGENVS=2                // # of preinitialized environments
HLLENV.IBMADC.CRITENVS=0               // max # of env for enabled progs
HLLENV.IBMADC.DEFAULT=NOTPREINIT       // eligible programs PREINIT?
HLLENV.IBMADC.PSTACK=131072            // run time stack size
HLLENV.IBMADC.PHEAP=131072             // run time heap size
```

If you want information about...	Refer to...
The PL/I sample command processors	*IBM Z NetView Programming: PL/I and C*
Using high-level languages with NetView	*IBM Z NetView Programming: PL/I and C*

Defining Commands and Command Lists

CNMCMD is a member of DSIPARM that is used to define commands. To modify existing command definitions:

1. Copy the existing command definition from CNMCMD or its included members to CNMCMDU.
2. Modify the copied CMDDEF statements in CNMCMDU.

Add any new command definitions to CNMCMDU.

Subsequent sections explain how to accomplish the following tasks:

- Add command processors.
- Specify a command type.
- Load a command module only when that command is run.
- Create synonyms for command keywords.
- Create a command or command list synonym.
- Issue a system or subsystem command from the NetView program.

Adding Command Processors

Add a CMDDEF statement to define the command verb for each command processor that you have written. Store your command processors in STEPLIB.

CMDDEF definition statements are located in CNMCMD. To avoid migration problems, place your command definition statements in CNMCMD %INCLUDE member CNMCMDU. In the NetView program, the **LIST** command is defined with the following statement:

```
CMDDEF.LIST.MOD=DSISHP
```

Where:

LIST
Is the name of the command.

DSISHP
Is the name of the module that contains the code to run the command.

Note:

1. When you are defining a user-written command processor, be sure to specify a unique module name on the MOD operand. Do not use a name that the system might recognize as a command, because the NetView program attempts to process that command instead of the user-written command processor.
2. Ensure that all CMDDEF statements begin in column 1.
3. Make all changes in uppercase.

You can use the ADDCMD command to dynamically add a command without restarting the NetView program. The command definition remains in effect until you restart the NetView program.

If you want information about...	Refer to...
The CMDDEF definition statement	*IBM Z NetView Administration Reference*
Command authorization	*IBM Z NetView Security Reference*

Specifying a Command Type

The TYPE operand has the following options:

R
A regular command

I
An immediate command

B
Both regular and immediate commands

D
A data services command

RD
A regular or data services command

P
A PIPE command stage

RP
A regular or PIPE command stage

BP
A regular, immediate, or PIPE command stage

H
A high priority command

Note:

1. A command list is always TYPE=R.
2. Do not change the command type for any CMDDEF statement that is supplied on the distribution tape.
3. When you add a CMDDEF statement for a user-written command processor, TYPE=R is assumed unless you specify otherwise.

In the samples, the RESET command is defined in the following way:

```
CMDDEF.RESET.MOD=DSIRSP
CMDDEF.RESET.TYPE=B
CMDDEF.RESET.CMDSYN=CANCEL
```

Note: If a module is intended to be used as a RESUME, LOGOFF, or ABEND routine, the first CMDDEF statement defining this module in CNMCMD must not be TYPE=I.

Loading a Command Module Only When That Command Is Run

Command modules that you supply do not have to be in active storage all the time the NetView program is running. To save storage, you might want to delay loading the command module for a rarely used command until that command is run. If you use a command often, however, you probably want to load the command module at initialization and keep it resident in active storage to save processing time required to load the module.

You designate whether a command module is kept resident in active storage by coding the RES operand of the CMDDEF statement. If you do not specify a RES operand, the command module is kept resident in active storage. If you want to load a command module when the command is run rather than at initialization, specify RES=N.

If you change command processors for testing purposes, you might want to specify RES=N on the CMDDEF statement. Specify RES=N to change a command processor without having to stop and restart the NetView program.

Before specifying RES=N for user command processors defined with CMDDEF definitions, verify that they do not depend on being loaded at the same location from one command call to the next.

Commands require RES=Y under the following conditions:

- If the code cannot be entered again or cannot be refreshed
- If an internal entry address is used as a system or VTAM exit address
- If the code is self-modifying
- If the control blocks are queued from modules
- If an address within a module is used as a parameter to another task

Note:

1. Do not change RES=Y to RES=N, or change from the default RES value to RES=N, on any CMDDEF statement that is supplied as a part of the NetView samples. If you change the residency (RES) of these modules, you can receive NetView abends. If a conflict in residencies exists between two CMDDEF statements, the default of RES=Y is chosen, and the module is resident.
2. If you specify RES=N for a command that is coded with TYPE=I or TYPE=B (the immediate commands), the command is still processed as if you coded RES=Y.

Creating a Synonym for a Command or Command List

To create a synonym for a command or a command list, use the CMDSYN definition statement. Operators can then enter either the original command name or the new command name.

To create a command synonym:

1. Add a CMDSYN definition on the CMDDEF statement. If you use multiple CMDSYN statements, specify *PREV* to continue the list of synonyms. Otherwise the synonyms replace synonyms defined in previous statements. For example:

   ```
   CMDDEF.command.CMDSYN=*PREV*,synonym1,synonym2
   ```

2. Inform the operators of the new command name.

Be careful not to use a name that is a VTAM command, another NetView command or command synonym, or a command in an application program that runs with the NetView program. Also, do not modify the command names on the CMDDEF statements that are supplied by the NetView product in CNMCMD. Some of these command processors depend on the name of the command to process correctly.

In the sample, you can create a synonym for the AUTOWRAP command in the following way:

```
CMDDEF.AUTOWRAP.CMDSYN=A
```

Now, the AUTOWRAP command is also named **A**. You can request AUTOWRAP by entering **A**.

Note: You can use the ADDCMD command to add or replace synonyms without having to recycle NetView.

Some CMDDEF statements in the samples already have CMDSYNs assigned to them, such as the following NetView command list:

```
CMDDEF.CNME0001.CMDSYN=ACQ
```

If you assign additional names, use *PREV* as one of the values on the CMDSYN statement. Otherwise, the new value replaces the previous value. When you assign a CMDSYN, ensure that the name is unique.

If you want information about...	Refer to...
Operands of the CMDDEF definition statement	*IBM Z NetView Administration Reference*
ADDCMD command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Suppressing Command Echoes

Echoes of commands that you type on the command line are sent to the screen when you press the ENTER key. Use command echo suppression to prevent echoes of certain commands from displaying on the screen. Suppression is useful when command echoes interfere with displays.

Note: Do not change the ECHO operand on any CMDDEF statements that are supplied by the NetView product. The NetView program uses this option to perform screen control when moving between components. If you change this operand, you can receive unexpected results at the terminal.

In the samples, the CLEAR command is defined with the following statements:

```
CMDDEF.CLEAR.MOD=DSICKP
CMDDEF.CLEAR.TYPE=H
CMDDEF.CLEAR.ECHO=N
CMDDEF.CLEAR.SEC=BY
```

Where:

ECHO=N
Specifies that the command is not echoed to the screen.

Note that the commands that are issued from command lists follow the &CONTROL statement rules.

If you want information about...	Refer to...
The &CONTROL statement rules	*IBM Z NetView Programming: REXX and the NetView Command List Language*

Creating Command Keyword Synonyms

Using synonyms for a command keyword can make the network operator's job easier. To create a command keyword synonym, code a parameter synonym (PARMSYN) on the CMDDEF statement for each keyword for which you are creating a synonym.

For example, you can add the following PARMSYN statements to alter keywords of the BGNSESS CMDDEF statement:

```
CMDDEF.BGNSESS.PARMSYN.OPCTL=OP
CMDDEF.BGNSESS.PARMSYN.APPLID=TO
CMDDEF.BGNSESS.PARMSYN.SRCLU=FROM
CMDDEF.BGNSESS.PARMSYN.SESSID=ID
CMDDEF.BGNSESS.PARMSYN.LOGMODE=LOG
```

Where:

OPCTL
Is the keyword for which you are creating a synonym.

OP
Is the new name for the keyword.

Now instead of entering the following command:

```
BGNSESS  OPCTL,APPLID=IMS1,SRCLU=TAF01000,SESSID=SESS1,LOGMODE=S3270
```

your operator can type this command to get the same results:

```
BGNSESS  OP,TO=IMS1,FROM=TAF01000,ID=SESS1,LOG=S3270
```

Issuing System and Subsystem Commands from the NetView Console

Place CMDDEF statements in CNMCMDU so that you can enter nonconflicting z/OS system and subsystem commands from a NetView console without prefixing the command with MVS. Each CMDDEF statement represents one z/OS subsystem command name that does not conflict with currently defined network command names.

For examples of CMDDEF statements that define MVS, JES2, and JES3 commands, refer to members CNMS6401, CNMS6402, and CNMS6403 in NETVIEW.V6R3M0.CNMSAMP. Comments are provided in these members to help you select any you might want to use.

The following format is the format for the CMDDEF statements:

```
CMDDEF.name.MOD=CNMCMJC
```

Where *name* is any z/OS command name.

Chapter 5. Configuring Optional NetView Services

You can include the following optional NetView services and functions:

- Command Statistics, described in "Enabling and Customizing Command Statistics" on page 109
- Management services (MS) transport function, described in "Defining Management Services Transport" on page 106
- High performance transport, described in "Defining High Performance Transport" on page 107
- Save/restore task for timers, global variables, and focal points, described in "Defining the Save/Restore Function" on page 107
- DB2® subsystem access, described in "Defining DB2 Subsystem Access" on page 108
- TSO command server, described in "Starting the TSO Command Server" on page 115
- UNIX command server, described in "Starting the UNIX Command Server" on page 117
- IBM Z NetView discovery library adapter (DLA), described in "Enabling the IBM Z NetView Discovery Library Adapter" on page 112
- NetView Web Services Gateway, described in "Configuring NetView Web Services Gateway" on page 103

Configuring NetView Web Services Gateway

Web Services Gateway provides you with an industry-standard open interface into the NetView program to issue commands and receive responses. It uses SOAP over HTTP or HTTPS as the transport mechanism to drive the request and to receive a reply.

These tasks are required to set up Web Services Gateway:

- "Configuring the NetView program" on page 103
- "Preparing the MVS System" on page 105
- "Preparing the Web Resources Files" on page 105
- "Verifying NetView Web Services" on page 105

If you want information about...	Refer to...
Using NetView Web Services Gateway	*IBM Z NetView Application Programmer's Guide*
CNMSTYLE statements	*IBM Z NetView Administration Reference*

Configuring the NetView program

Configuring the NetView program includes the following steps:

- "Configuring the CNMSTYLE Member" on page 103
- "Updating the NetView Startup Procedure" on page 104
- "Configuring Security" on page 104

Configuring the CNMSTYLE Member

To enable the Web Services Gateway function, copy the TOWER statement from the CNMSTYLE member to the CNMSTUSR or C*xx*STGEN member and remove the asterisk (*) before the NVSOA component.

The AUTONVSP autotask is used to attach the SOASERV command. To change the autotask name, modify the function.autotask statement in CNMSTYLE member CNMSTUSR or C*xx*STGEN:

```
(NVSOA)function.autotask.NVSOAPTSK = AUTONVSP
```


Review the CNMSTYLE statements in Table 18 on page 104 and make any necessary updates in the CNMSTUSR or C*xx*STGEN member. The *srvrname* is set to SOASRVR1 in the CNMSTYLE member. To change the server name, modify the NVSP.*srvrname* statements to specify the name of the Web Services server. You can start additional servers on different listening ports by varying the *srvrname* on the CNMSTYLE statements.

Table 18. CNMSTYLE statements

CNMSTYLE statement	Function
NVSP.*srvrname*.NUMTHRDS	Specifies the number of threads to be created to service incoming and outgoing IP connections. The minimum value is 1. If values greater than 2,147,483,647 are specified, these are set to the default value of 2,147,483,647.
NVSP.*srvrname*.PDS	Specifies the location where server web resources files are located.
NVSP.*srvrname*.PORT	Specifies the port on which the server is listening for connection requests. The minimum value is 0. Values greater than 2,147,483,647 default to 2,147,483,647.
NVSP.*srvrname*.SECURE	Specifies whether the transport is secured with SSL encryption.
NVSP.*srvrname*.TRC	Specifies the initial trace levels.
NVSP.*srvrname*.WAIT	Specifies the number of seconds to wait for a command response. The minimum value is 0. Values greater than 10,000,000 default to 10,000,000.

Updating the NetView Startup Procedure

Uncomment the following XML toolkit library in the CNMPROC (CNMSJ009) member:

```
//*****************************************************************
//*
//* IF YOU ARE STARTING THE NETVIEW WEB SERVICES SERVER THEN
//* YOU WILL NEED ACCESS TO THE IBM XML TOOLKIT RUNTIME LIBRARY.
//*
//* - IF YOU HAVE THIS LIBRARY ON YOUR SYSTEM BUT IT IS NOT
//*   ACCESSIBLE FROM THE PLPA OR LINKLST, THEN YOU MUST
//*   UNCOMMENT THE LINE BELOW.
//*
//* WHEN YOU UNCOMMENT THE LINE BELOW, MAKE SURE THAT THE DSN
//* ACTUALLY MATCHES THE NAME ON YOUR SYSTEM.  IN ADDITION,
//* MAKE SURE THAT THE DATA SET IS APF-AUTHORIZED.
//*
//* FOR THE LINE BELOW, THE FOLLOWING JCL SYMBOLIC IS ASSUMED:
//*      QIXM='IXM.V1R10M0', ** IBM XML TOOLKIT RUNTIME LIB.
//*
//*        DD
DSN=&QIXM..SIXMLOD1,DISP=SHR                      //*
                    //*****************************************************************
```

Alternatively you can add these libraries to the LPALSTxx member. For more information, see *IBM Z NetView Installation: Getting Started*.

Configuring Security

The Web Services server uses basic NetView operator ID and password or password phrase authentication for commands. Verify that the operator ID and password or password phrase that you plan to use are defined in the RACF product or in the DSIOPF member.

Commands are run under the authority of the operator ID specified on the SOAP request. SSL encryption is used to secure the transport to the server. For information about configuring security, see *IBM Z NetView Security Reference*.

Preparing the MVS System

The PROGxx member contains the names of program libraries that you want the system to concatenate to SYS1.LINKLIB and libraries that you want to define as authorized with the Authorized Program Facility (APF). Add the XML toolkit library to the PROGxx PARMLIB member:

```
APF ADD DSNAME(&QIXM..SIXMLOD1) VOLUME(xxxxxxxx)
```

The &QIXM variable must match the high-level qualifiers that are defined in the CNMPROC (CNMSJ009) sample.

Preparing the Web Resources Files

The web resources files that are used by the Web Services server are located in the following directory:

```
/usr/lpp/netview/v6r3/www/
```

See Table 19 on page 105 to update the files for your environment. The WSDL files automatically generate a proxy-client connection.

Table 19. Web Services Gateway files

File name	Purpose	Modifications
znvsoatx.htm	Text-based Web Services client	Update URLs for your environment. Locate the `<SELECT>` tag and modify the `<OPTION>`*your.web.services.server*`</OPTION>` tag.
znvsoa.htm	Graphic version of the Web Services client	Update URLs for your environment. Locate the `<SELECT>` tag and modify the `<OPTION>`*your.web.services.server*`</OPTION>` tag.
znvwsdl.wsdl	Provides web services definitions for different output formats	Update the soap:address location for your environment. You can do this by locating the `<soap:address location= >` tag.
znvwsdl1.wsdl	Provides web services definitions for different output formats	Update the soap:address location for your environment. You can do this by locating the `<soap:address location= >` tag.
znvwsdl2.wsdl	Provides web services definitions for different output formats	Update the soap:address location for your environment. You can do this by locating the `<soap:address location= >` tag.

Static and dynamic proxy SOAP clients, and also socket proxy client samples, are shipped in the following directory:

```
/usr/lpp/netview/v6r3/samples
```

Verifying NetView Web Services

Follow these steps to use the generic SOAP client to verify the output of the command that you sent:

1. Start your web browser.
2. Enable the **Access data sources across domains** option in the security settings for the domain in which your server is located. Enter one of the following addresses for the SOAP client:

   ```
   http://netviewhost:port/znvsoatx.htm
   https://netviewhost:port/znvsoatx.htm
   ```

The SOAP client HTML page is displayed.

3. In the Z NetView Generic SOAP Client panel, enter the Endpoint, for example:

```
http://netviewhost:port/znvsoa
```

4. Enter the SOAP method:

```
DoCmd
```

After you enter the method, the other fields are completed automatically.

5. Modify the tags or the text in the Edit Payload (XML) area as shown in the following example:

```
<Name>sysadmin</Name><Password>passwd</Password>
<NVCMD><cmd><![CDATA[nvcmd]]</cmd></NVCMD>
```

where *sysadmin* and *passwd* define the NetView operator ID and password or password phrase under which to run the command, and *nvcmd* is the NetView command to run.

6. Click **Make SOAP Request**. The output of the request is displayed in the **SOAP Response Payload** field.

If necessary, you can use the SOACTL command to enable tracing. The trace entries are written to the network log. For more information about the SOACTL command, see *IBM Z NetView Command Reference Volume 2 (O-Z)*.

Defining Management Services Transport

The management services (MS) transport function enables applications that are supplied by the NetView product or written by users to send data and to receive data from partner applications. Operations management and focal point applications are some examples of applications that use the MS transport.

The CNMSTYLE member contains the following task statement for the MS Transport function:

```
TASK.DSI6DST.INIT=Yes
```

DSI6INIT is the MS transport initialization sample and contains the following statement:

```
DSTINIT FUNCT=OTHER,XITDI=DSI6IDM
```

CNMCMSYS contains the following CMDDEF statements for the MS transport:

```
CMDDEF.REGISTER.MOD=DSI6REGP
CMDDEF.REGISTER.RES=N
CMDDEF.DSI6DSCP.MOD=DSI6DSCP
CMDDEF.DSI6DSCP.TYPE=D
CMDDEF.DSI6DSCP.PARSE=N
CMDDEF.DSI6LOGM.MOD=DSI6LOGM
CMDDEF.DSI6LOGM.TYPE=D
CMDDEF.DSIOSRCP.MOD=DSIOSRCP
CMDDEF.DSIOSRCP.TYPE=RD
CMDDEF.DSIOSRCP.PARSE=N
CMDDEF.DSIOARCP.MOD=DSIOARCP
CMDDEF.DSIOARCP.TYPE=RD
CMDDEF.DSIOARCP.PARSE=N
CMDDEF.DSIOARCP.SEC=BY
CMDDEF.DSIOURCP.MOD=DSIOURCP
CMDDEF.DSIOURCP.TYPE=D
CMDDEF.DSIOURCP.PARSE=N
CMDDEF.DSIOURCP.SEC=BY
CMDDEF.DSIOLGFP.MOD=DSIOLGFP
CMDDEF.DSIOLGFP.TYPE=RD
CMDDEF.DSIOLGFP.PARSE=N
CMDDEF.DSIOLGFP.SEC=BY
CMDDEF.DSI6SNDP.MOD=DSI6SNDP
CMDDEF.DSI6SNDP.TYPE=RD
CMDDEF.DSI6SNDP.PARSE=N
CMDDEF.DSI6SNDP.RES=N
CMDDEF.DSI6SNDP.SEC=BY
```

Defining High Performance Transport

Use the NetView high performance transport to send and receive large amounts of data using LU 6.2 communications.

The CNMSTYLE member contains the following task statement for the high performance transport function:

```
TASK.DSIHPDST.INIT=Y
```

DSIHINIT is the high performance transport initialization member. To establish nonpersistent conversations, uncomment the following statement in DSIHINIT:

```
*  PARTNER NETID=NETA,NAME=
CNM02,PERSIST=NO
```

Where:

NETID
Specifies the network ID of your system. If you specify the network ID as an asterisk (*), the network ID defaults to the one determined by VTAM based on the partner name of the remote node.

NAME
Specifies the name of the partner (logical unit or control point name) with which you are initiating a conversation.

PERSIST
Specifies whether all conversations between this NetView system and the remote node are persistent. If you do not specify the PERSIST keyword, the default is YES, meaning conversations are persistent.

Note: You do not have to code this statement at the remote node to use the high performance transport.

The DSIHPDST task uses the following CMDDEF statements in DSIPARM member CNMCMSYS:

```
CMDDEF.DSI6DSCP.MOD=DSI6DSCP
CMDDEF.DSI6DSCP.TYPE=D
CMDDEF.DSI6DSCP.PARSE=N
CMDDEF.DSI6LOGM.MOD=DSI6LOGM
CMDDEF.DSI6LOGM.TYPE=D
```

The following CMDDEF statements in DSIPARM member CNMCMSYS are used by DSIHSNDS and CNMHSMU:

```
CMDDEF.DSIOLGFP.MOD=DSIOLGFP
CMDDEF.DSIOLGFP.TYPE=RD
CMDDEF.DSIOLGFP.PARSE=N
CMDDEF.DSIOLGFP.SEC=BY
CMDDEF.DSI6SNDP.MOD=DSI6SNDP
CMDDEF.DSI6SNDP.TYPE=RD
CMDDEF.DSI6SNDP.PARSE=N
CMDDEF.DSI6SNDP.RES=N
CMDDEF.DSI6SNDP.SEC=BY
```

Defining the Save/Restore Function

Timers, global variables, and focal point information can be saved to VSAM and restored when the NetView program is restarted.

The CNMSTYLE member contains the following task statement:

```
TASK.DSISVRT.INIT=Y
```

The AAUDCPEX command definition statement in the CNMCMSYS member is used for the save/restore function:

```
CMDDEF.AAUDCPEX.MOD=AAUDCPEX
CMDDEF.AAUDCPEX.TYPE=D
```

```
CMDDEF.AAUDCPEX.PARSE=N
CMDDEF.AAUDCPEX.SEC=BY
```

The database for the save/restore function is defined using the CNMSJ004 job with input member CNMSI101.

To define security passwords for the save/restore function:

1. Stop the DSISVRT task.
2. Modify the definition statements in CNMSI101 that define the save/restore database, changing them to include the specification of VSAM cluster passwords. Rerun job CNMSJ004 using these modified statements to delete and redefine the save/restore database.
3. Update the CNMSTPWD member in DSIPARM to include the password that you specified when redefining the save/restore database. The following example shows the PWD statement that defines the password for the save/restore database:

   ```
   PWD.DSISVRT.P = password
   ```

   where:

   password
   Is the 1- to 8-character password for the database.
4. Restart the DSISVRT task.

Defining DB2 Subsystem Access

To use the DB2 program libraries, uncomment the following statement in sample CNMSJ009:

```
//*         DD   DSN=DSN510.SDSNLOAD,DISP=SHR
```

CNMSJSQL is a sample installation job that defines the plan for the NetView SQL stage to DB2. In the sample job, the name of the library on the SYSUT2 JCL statement must match the name specified in the BIND statement in the second step of the job. For example, the sample uses USER2.DBRMLIB. Modify this value to suit your system:

```
//SYSUT2 DD DSN=USER2.DBRMLIB(DSISQLDO),DISP=SHR
```

Change the IBMUSER value in the sample to identify the NetView program that is using SQL. *Do not* change the values for DSISQL*nn*, DSISQLDO, and DSISQLDP. Usually the NetView plan name is DSISQL*nn*, where *nn* is changed because of service or future releases. The CNMSJSQL sample is reshipped whenever a change to the DSISQLDO program causes the plan to be incompatible.

```
BIND PACKAGE(DSISQL04) MEM(DSISQLDO) ACT(REP) -
     ISOLATION(CS) LIB('USER2.DBRMLIB') OWNER(USER2)
BIND PACKAGE(DSISQL14) MEM(DSISQLDP) ACT(REP) -
     ISOLATION(CS) LIB('USER2.DBRMLIB') OWNER(USER2)
BIND PLAN(DSISQL04)  ACT(REP)   -
     PKLIST(DB2L01.DSISQL04.DSISQLDO,DB2L01.DSISQL14.DSISQLDP)  -
     ISOLATION(CS)  OWNER(USER2)
```

To access SQL databases using the SQL and SQLCODE pipe stages, the DSIDB2MT task is used to define the default DB2 subsystem. This task connects the NetView program to a specific DB2 subsystem so that any task in the NetView address space has access to that DB2. You can start the task using the NetView START command:

```
START TASK=DSIDB2MT,MOD=DSIDB2MT,MEM=DSIDB2DF,PRI=1
```

You can also start the SQL task automatically during NetView initialization. To do this, update the following task statement in the CNMSTUSR or C*xx*STGEN member (change INIT=N to INIT=Y). For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.DSIDB2MT.INIT=Y
```

Other DB2 subsystems can be specified by an operand on the SQL pipe stage. The SQL pipe stage can access DB2 subsystems regardless of whether DSIDB2MT is started, provided the subsystem name is specified on the SQL stage.

The DSIDB2DF member of DSIPARM defines the DB2 subsystem to which the NetView program connects. It uses one definition statement:

```
SUBSYSTEM=DB2
```

Enabling and Customizing Command Statistics

Statistics are provided for commands that run under the NetView program. You can view and save data for each invocation of a NetView command, NetView REXX, or CLIST program. The collected data includes the following metrics:

- The name of the command
- The task under which the command ran
- Total CPU time used
- Maximum storage used by the command at any one time
- Total I/O operations performed by the command

Command statistics can be collected for primary commands or primary and subordinate commands. A primary command is a command that is directly invoked by a NetView task, the automation table, or a timer. A subordinate command is a command invoked by a primary command.

By default, command statistics is disabled.

Using command statistics consists of the following items:

- Enabling Command Statistics
- Customizing Command Statistics

Enabling Command Statistics

To enable command statistics, complete the following actions:

- Determine whether you want statistics for all (both primary and subordinate) commands or just primary commands.
  - If you want command statistics monitoring to start immediately upon the completion of NetView initialization, ensure that the CMDMON.INIT.STATS statement is uncommented in your CNMSTUSR or CxxSTGEN member, and specify either PRIONLY or ALL for the value.
  - If you want to start command statistics dynamically, issue the CMDMON STATS=PRIONLY or CMDMON STATS=ALL command.
- Determine how many command statistics records you want to keep in NetView memory. The default value is `10000` records.

  If you want a different value, ensure that the CMDMON.DATA.MAXRECS statement is uncommented in your CNMSTUSR or CxxSTGEN member and that your value is specified.

  The in-memory storage is treated as 2 tables. When the first table is full, command statistics records are written to the second table, and the information in the first table is sent for processing. Sending a full table for processing is what is called capturing the data.
- Autotask AUTOCMON is the default autotask on which the command statistics data capture command runs. To change the autotask name, modify the function.autotask.CMDMON statement in the CNMSTUSR or CxxSTGEN member. Also review, and if necessary, update the autotask name in the DSIOPF member.

If you want information about...	Refer to...
Performance information for Command Statistics	*IBM Tivoli Z NetView Tuning Guide*
Command Statistics CNMSTYLE statements	*IBM Tivoli Z NetView Administration Reference*

Customizing Command Statistics

To customize the collection and processing of command statistics for your installation, complete the following steps:

- Review the Command Statistics Inclusion/Exclusion (CNMSCSIE) sample and determine if you want to add commands to include or exclude during command statistics data collection.

  To customize this sample, take the following actions:

  - Copy the CNMSCSIE sample into a DSIPARM dataset.
  - Add your commands to the appropriate inclusion or exclusion section in CNMSCSIE.
- Review the Command Statistics Data Processor (CNMSCSDP) sample. This sample is invoked when half of the specified number of records on the CMDMON.DATA.MAXRECS statement is reached.

  Determine if you want to use this sample with modifications or create your own command to run when the command statistics are captured. If you want to create your own command, ensure that the CMDMON.DATA.CAPTURE statement is uncommented in your CNMSTUSR or CxxSTGEN member and that your data processing command is specified.
- Review the Command Statistics Summary Data (CNMSCSSU) sample and determine if you want to customize it for your installation.
- Review the Command Statistics Data Formatter (CNMSCSFM) sample and determine if you want to change the format of the data.

Customizing the Command Statistics Inclusion/Exclusion Sample

Use the CNMSCSIE sample to specify the inclusion and exclusion lists for command statistics data collection.

The inclusion list begins on the line that follows the string `BEGIN.INCLUDE`, and ends on the line that precedes the string `BEGIN.EXCLUDE`. The exclusion list begins on the line that follows the string `BEGIN.EXCLUDE`, and ends at the end of the file. If no items are present in the inclusion list, all commands will be monitored, except those commands that appear in the exclusion list. However, if at least one item appears in the inclusion list, only those commands will be monitored, and will be overridden by the commands that appear in the exclusion list.

Comments can be specified in the following ways:

- A * character in column 1
- A // anywhere in the line that begins a comment and continues to the end of the line

Inclusion and exclusion list items cannot exceed 8 characters, and must appear on separate lines. If list items are longer than 8 characters or contain more than one space-delimited word, those list items will be ignored.

Any list item can contain any amount of leading or trailing white space.

The exclusion list overrides the inclusion list. In other words, if ABC* is specified on the inclusion list, and ABC12345 on the exclusion list, all commands that start with ABC except for ABC12345 will be monitored. Likewise, if ABC12345 is specified on the inclusion list, and ABC* on the exclusion list, no commands that start with ABC will be monitored, including ABC12345.

By extension, if a command appears on both the inclusion and exclusion lists, it will not be monitored. Similarly, if a command does not appear in the inclusion list, it will not be monitored.

You can specify a list item as a wildcard or an exact command. A wildcard is specified by adding a * to the end of the list item. All commands that start with the list item (up to the *) will be included or excluded based on the list in which they appear. List items without a * at the end are treated as exact command verbs.

Note: Wildcards in the middle of a list item are not supported, for example, A*Z.

Data REXX is supported. You can use Data REXX logic to include the contents of other members with a %INCLUDE statement.

Example include/exclude list

```
BEGIN.INCLUDE // the inclusion list starts here
  ABC* // include all ABC commands
  XYZ* // include all XYZ commands
  AUTOCNT // include NetView's AUTOCNT cmd
BEGIN.EXCLUDE // the exclusion list starts here
  ABC12345 // exclude ABC12345
  XYZ67890 // exclude XYZ67890

BEGIN.INCLUDE // the inclusion list starts here
// place items for the include list here, or leave this area
// as is to monitor all commands
BEGIN.EXCLUDE // the exclusion list starts here
// place items for the exclude list here, or leave this area
// as is to exclude no commands
```

If you want information about...	Refer to...
DATA REXX	*IBM Tivoli Z NetView Programming: REXX and the NetView Command List Language*

Customizing the Command Statistics Data Processor Sample

The Command Statistics Data Processor (CNMSCSDP) sample processes the collected command statistics data. By default, CNMSCSDP creates a new Comma Separated Value (CSV) file each day. Each time CNMSCSDP is invoked in the same day, the latest command statistics data is appended to the CSV file, which is called the DayFile in the sample.

The default file allocation size is 20 first extent cylinders on a 3390 DASD device, with 2 secondary extents. Approximately 1 cylinder is required for 3,000 command statistics records.

Review the comments at the beginning of CNMSCSDP to customize the sample for your installation.

Customizing the Command Statistics Summary Data Sample

The Command Statistics Summary Data (CNMSCSSU) sample issues the CMDMON DISPLAY command and provides CPU time, maximum storage, and I/O count totals on a command basis. It also provides a command count that indicates how many times the command was found in the CMDMON DISPLAY output. CNMSCSSU identifies each command based on a combination of the command name and the alternate name.

Note:

- If changes are made to the Command Statistics Data Formatter (CNMSCSFM) sample output format, the CNMSCSSU sample should also be changed to correctly pick up the new format.
- The totals that are provided by the CNMSCSSU sample are from a single point in time, but they can provide a quick view of which commands are utilizing the most resources.

Customizing the Command Statistics Data Formatter Sample

The Command Statistics Data Formatter (CNMSCSFM) sample is used to convert the raw statistical metrics and to format the output. It is used to format:

- Data records for the CMDMON DISPLAY command output
- Command Statistics Data Summary sample

The following table shows the format of the raw data when it comes into CNMSCSFM.

Table 20. Command Statistics raw data format

Column number	Data type	Description
1-8	Character	Command Name
9-16	Character	Alternate Name
17-25	Character	Parent Name
26-32	Character	NetView Task name
33-40	STCK value	Start Date and Time
41-48	STCK value	End Date and Time
49-56	Binary	CPU Time in Microseconds
57-60	Binary	Maximum Storage in Bytes
61-64	Binary	I/O Count
65-72	Character	Authorized User

Enabling the IBM Z NetView Discovery Library Adapter

An IBM Z NetView discovery library adapter (DLA) can collect mainframe and distributed data from the Z NetView RODM data cache. Using FTP, the DLA sends the data as a discovery library book to an IBM Tivoli Change and Configuration Management Database (IBM Tivoli CCMDB) file server or to another data store for which a discovery library reader exists. These discovery library books contain data about the resource instances and their relationships known to the system.

You can run this DLA on one centralized Z NetView installation, which can collect data in the RODM data cache from one or more NetView programs. If your enterprise distributes its TCP/IP topology information across multiple Z NetView installations, each of which maintains a RODM data cache, the DLA can be deployed at any or all of those Z NetView installations.

The following software must be running for this DLA to run successfully:

- RODM. RODM must be running and must be populated with discovery manager or MultiSystem Manager IBM Tivoli Network Manager data.
- TSO command server. This must be running if it is configured. See "Using a TSO Command Server" on page 114.

Also, at least one of the following components must be running:

- Discovery manager. The discovery manager configuration includes a hierarchy with the optional enterprise master NetView program at the top, and lower level NetView programs below it. The DLA discovers all resources from the selected discovery manager resource downward. For this reason, it is important to run the DLA at the highest level in the hierarchy under which you want all resources discovered.
- MultiSystem Manager IBM Tivoli Network Manager agent for distributed resources

Configuring the Discovery Library Adapter

The IBM Z NetView DLA dynamically creates and deletes two files and modifies one file. The DLA must be configured to enable this behavior. All configuration options are specified in the CNMSTYLE member and are described in the *IBM Z NetView Administration Reference*. Descriptions in these files indicate which configuration parameters are required and which parameters are optional. Each parameter is prefixed by the characters DLA. Make any changes in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

The DLA does not require any command line parameters.

These items outline the requirements:

- The file that is specified by the `DLA.xml_filename` parameter in the CNMSTYLE member (default value `NETVIEW.V6R3USER.DLA.CNMDLA`) is created. This parameter is required and must point to a sequential data set to which the DLA can write. This sequential data set, if it exists, is used to store the XML discovery library book. It remains available until the next time the DLA is run.

  Notes:

  1. If you change the `DLA.xml_filename` parameter between DLA runs, any data sets that you specified earlier and that were written to by the DLA are not deleted.
  2. The file that is specified by the DLA.XML_filename parameter in the CNMSTYLE member (NETVIEW.V6R3USER.DLA.CNMDLA by default) is created and sent by FTP to the host specified by the DLA.CMDB_FTP_SERVER statement. If the FTP transfer fails, you can manually send the file. For more information about manually sending files, see "Manually Accessing the IBM Tivoli CCMDB File Server" on page 113.

- The file that is specified by the `DLA.FTP_LOG_FILENAME` parameter in the CNMSTYLE member (default value `NETVIEW.V6R3USER.DLA.FTPSOUT`) is created. This parameter is required and must point to a sequential data set to which the DLA can write. This file buffers log information from the FTP file transmission. This sequential data set exists only while the DLA is running.
- The CNMSTATE file is used by the DLA to store information about the time stamp and status of the discovery library books that are transferred to the IBM Tivoli CCMDB file server. This required file is a sequential data set with a logical record length (LRECL) of 80 and a fixed-block record format. The `DLA.statefile` parameter in the CNMSTYLE member contains the location of this required file, and the default value is as follows:

  ```
  DLA.statefile=NETVIEW.V6R3USER.DLA.CNMSTATE
  ```

  Before you can run the DLA, the CNMSTATE file must be copied from the `NETVIEW.V6R3M0.DSIPARM(CNMSTATE)` data set to the location that is specified by the `DLA.statefile` parameter.

You can configure the DLA either while the IBM Z NetView program is stopped or before it is restarted. If required or optional configuration parameters are changed in the CNMSTYLE member after the IBM Z NetView program is started, use the following NetView command to process the changed parameters:

```
RESTYLE DLA
```

A NetView CHRON task for the DLA adapter is predefined to run on the AUTO2 task each Monday at 2:15 AM. Ensure that this is appropriate for your system. This parameter is initialized in the CNMSTUSR or C*xx*STGEN member using the auxInitCmd.DLAAUTO statement and can be changed there. By default, the CHRON task is not set; you must uncomment it to set it. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you uncomment the CHRON statement, or change the schedule after the NetView program is restarted, the RESTYLE DLA command does not pick up the CHRON change. To initiate or change a CHRON timer without restarting the NetView program, enter the TIMER command at the NetView command line and use the Timer Management panel. For more information about this panel, see the *IBM Z NetView User's Guide: NetView*.

Manually Accessing the IBM Tivoli CCMDB File Server

You can confirm that the target IBM Tivoli CCMDB file server is running by issuing a PING command to the fully qualified IP host name of the target IBM Tivoli CCMDB file server. IBM Tivoli CCMDB does not need to be running for the DLA to create and transfer a discovery library book.

Note: For testing purposes, the DLA transfers files to any FTP server for which you have access authority.

You can confirm access to the FTP service on the target file server by successfully sending a file to the server using FTP:

1. Access the TSO command line with this Interactive System Productivity Facility (ISPF) command:

```
=6
```

2. Open an FTP session to the target FTP server from the TSO command line on the source z/OS system with the following command:

```
FTP IP_hostname
```

3. Use the same target FTP server user ID and password that you specify in "Setting the IBM Tivoli CCMDB File Server Password" on page 114.
4. Change the source directory as needed, for example:

```
lcd ..
lcd user.init
```

5. Change the target directory as needed, for example:

```
cd ..
cd /tmp/dla/
```

6. Send an existing file with this command:

```
PUT localfile remotefile
```

Using a TSO Command Server

The DLA uses a TSO command server to launch the FTP client to transfer the discovery library book to the IBM Tivoli CCMDB file server. If a TSO command server is already running for functions other than this DLA, this DLA uses the TSO command server and leaves it running after the DLA completes. If a TSO command server is configured but not running, the DLA starts the TSO command server and uses it to transfer the discovery library book to the IBM Tivoli CCMDB file server; the DLA then stops the TSO command server.

To configure a TSO command server, if one is not already defined, see "Starting the TSO Command Server" on page 115 for information about CNMSJTSO.

Consider the following items that pertain to the TSO command server CNMSJTSO:

- Ensure that the following line in CNMSJTSO specifies the data set where member CNMFTP is located:

```
//SYSPROC  DD DSN=NETVIEW.V6R3M0.CNMCLST,DISP=SHR
```

- Ensure that the TSO command server has an adequate region size. Because the default region size is not adequate for this DLA, a region size of 0M (allocated dynamically) is recommended for most installations. Define it in the following way:

```
//&userid. EXEC PGM=IKJEFT01,REGION=0M,TIME=NOLIMIT,
                 PARM='CNMETSO,&ppiname.,&key.,NODEBUG'
```

- Check the default setting of the `strtsrv` environment variable by issuing this command:

```
LIST defaults
```

  Review the output. If you do not see the setting `STRTSERV: sbmtJob`, you can set the STRTSERV variable by issuing the following command from the NetView program:

```
DEFAULTS STRTSERV=sbmtJob
```

Setting the IBM Tivoli CCMDB File Server Password

The DLA uses the IBM Z NetView capability of securing passwords by storing them in the CNMSTPWD member. To ensure that CNMDLAR can successfully connect to the IBM Tivoli CCMDB file server using

FTP, uncomment `PWD.DLA.P` in the CNMSTPWD member, and assign to it the password for the user ID that is specified in the `DLA.ftp_uid` parameter in the CNMSTYLE member. These two parameters define the user ID and password that are used to open an FTP session to the IBM Tivoli CCMDB file server. They must be defined before CNMDLAR is run. The authority to view and change the CNMSTPWD member might be restricted to a limited class of operators.

Running the Discovery Library Adapter

The IBM Z NetView DLA can be run in either of two ways:

- Enter this command at the IBM Z NetView command line:

  ```
  CNMDLAR
  ```

- To run it automatically at a regularly scheduled time, uncomment the auxInitCmd.DLAAUTO statement in the CNMSTUSR or C*xx*STGEN member that processes the IBM Z NetView CHRON timer. For additional information about this statement, see "Configuring the Discovery Library Adapter" on page 112. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

When this DLA runs, results are recorded in the IBM Z NetView log (NETLOGA). More detailed results can be written to the NETLOG by setting the optional DLA.debug statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

After the DLA completes, it delivers a discovery library book to the target IBM Tivoli CCMDB file server. In accordance with IBM Tivoli CCMDB conventions, the book name includes a time stamp and identifies the IBM Z NetView program, and the host on which it resides, as the management software system, for example:

```
NETVZ61.hostname.2011-2-16T20-42-45Z.xml
```

This discovery library book is retained until the next time the DLA is run, after which it is replaced with the new book. If the automated FTP transfer of this book to the IBM Tivoli CCMDB file server fails (indicated by the DSI416I message), you can send it manually; for information, see "Manually Accessing the IBM Tivoli CCMDB File Server" on page 113. For proper collation, see the file name that is provided in the DSI366I message.

If this DLA runs more than once, the output file name contains the term `refresh`, as shown in the following example:

```
NETVZ61.hostname.2011-2-18T18-26-09Z.refresh.xml
```

To force the IBM Tivoli CCMDB Bulk Loader to do a create operation, erase the last time stamp value in the CNMSTATE file, as shown in the following example:

```
LAST_SUCCESSFUL_TRANSMISSION =
```

The DLA can take from 30 seconds to 15 minutes to run, depending on the number of TCP/IP-managed resources that are discovered.

Starting the TSO Command Server

You can start the TSO command server from the NetView program by issuing the START TSOSERV command. The TSO command server starts either as a submitted job or as an MVS started task, depending upon the setting of the STRTSERV parameter of the DEFAULTS command.

If multiple versions of the TSO command server JCL are required, the optional MEM parameter can be specified on the START TSOSERV command. The default member name is CNMSJTSO for submitted jobs and CNMSSTSO for MVS started tasks.

If the TSO command server is started as a submitted job, ensure that the sample job CNMSJTSO is contained in a DSIPARM data set. If the TSO server is started as a started task, ensure that the sample job

CNMSSTSO is copied into a data set defined in the IEFJOBS or IEFPDSI concatenation of master JCL, for example, SYS1.PROCLIB. This is required because CNMSSTSO contains a job statement. Also ensure that the sample MVS START command CNMSTSOS is contained in a DSIPARM data set. For more information about specifying whether the TSO server runs as a submitted job or as a started task, refer to the online help for the DEFAULTS STRTSERV command.

Before you start a TSO command server, you can customize the samples that are provided with the NetView program to fit your environment. If the SCNMLNKN data set is not defined as part of the LNKLST concatenation, you must include a STEPLIB DD statement for it in the TSO command server JCL. When a START TSOSERV command is issued, the NetView program issues an MVS START command (if DEFAULTS STRTSERV is set to STRTPROC) or submits a job (if DEFAULTS STRTSERV is set to SBMTJOB). For a started task, member CNMSTSOS contains the MVS START command that is used to start the procedure or job specified in the START TSOSERV command. For a submitted job, the member specified on the MEM keyword in the START TSOSERV command is submitted as an MVS job.

Member CNMSTSOS or the member containing the submitted JCL, for example, CNMSJTSO, can contain specific variables for which the NetView program performs substitutions before the task is started or the job is submitted. These variables have names that begin with an ampersand (&) followed by lowercase letters and end with a period (.). You can specify any of these substitution variables to customize how your TSO server is started. The following list describes these variables:

&sprcnm.
The member name containing the JCL that is to be started or submitted. This value is the value of the MEM keyword on the START TSOSERV command

- For a started task, if MEM is not specified, CNMSSTSO is used. This variable is specified in sample CNMSTSOS as the procedure or job to start.
- For a submitted job, if MEM is not specified, CNMSJTSO is used. This variable is not specified in sample CNMSJTSO because it is the same as the sample name.

&jobname.
The name to be used on the JOB statement on a submitted job. This value is dynamically built and is equivalent to the &ppiname. value, starting at the second character.

- For a started task, this variable is not specified in sample CNMSTSOS because it does not apply to the MVS START command.
- For a submitted job, this variable is specified in sample CNMSJTSO as the name on the JOB statement.

&userid.
The TSO user ID under whose authority the started task or submitted job runs. To easily identify which user ID the task or job is running for, the JOBNAME (for start) or the step name (for submit) specifies this name. The value is the value on the TSOSERV keyword on the START TSOSERV command.

- For a started task, this variable is specified in sample CNMSTSOS as the value for the JOBNAME keyword on the MVS START command.
- For a submitted job, this variable is specified in sample CNMSJTSO for the values for the USER keyword on the JOB statement and for the step name on the EXEC statement.

&ppiname.
The receiver name used for communication between the server and the NetView program across the PPI. This value is dynamically built and is used to communicate to the PPI receiver.

- For a started task, this variable is specified in sample CNMSTSOS on the PPINAME keyword on the MS START command. Sample CNMSSTSO uses this value as the first parameter passed to the CNMETSO program.
- For a submitted job, this variable is specified in sample CNMSJTSO as the first parameter passed to the CNMETSO program.

&key.
A random number used for correlation between the TSO server and the command requester. The value is used in the following ways:

- For a started task, this variable is specified in sample CNMSTSOS on the KEY keyword on the MVS START command. Sample CNMSSTSO uses this value as the second parameter passed to the CNMETSO program.
- For a submitted job, this variable is specified in sample CNMSJTSO as the second parameter passed to the CNMETSO program.

If you want information about...	Refer to...
START TSOSERV command	*IBM Z NetView Command Reference Volume 2 (O-Z)* for NCCF START
TSO server defaults	*IBM Z NetView Command Reference Volume 1 (A-N)* for DEFAULTS STRTSERV

Starting the UNIX Command Server

The UNIX command server enables UNIX commands to be entered from the NetView command line and returns the output of these commands to the NetView console. For more information, see "Defining the UNIX Command Server" on page 201.

Chapter 6. Configuring IP Management

Z NetView supports the management of heterogeneous networks that use both SNA and TCP/IP. To enable TCP/IP management in the NetView program, you must configure the host environment and the NetView program to support the required management functions.

Getting Started with IP Management

By following the procedures in this section, you can enable basic TCP/IP management in the NetView program.

About this task

After you complete these procedures, the NetView program is configured to provide the following capabilities:

IP management functions
Basic IP management includes the following functions:

- The ping command
- The tracerte command
- Simple Network Management Protocol (SNMP) management
- Critical port monitoring

IP packet trace collection and analysis
An examination of packet content can help you debug network problems. The NetView program provides real-time capture and formatting of both IP and Open Systems Adapter (OSA) packet trace data, including both headers and payload. Packet trace data uses the same format as the one used by the Interactive Problem Control System (IPCS). The formatter is directly integrated with the IP stack, preventing translation mismatches. You can use highly flexible tracing and formatting options to filter out unwanted data.

Both IPv4 and IPv6 packet tracing is supported. Packet trace data is also available in unformatted binary form for use by automation routines.

IP connection monitoring and management
The NetView program provides both real-time and historical connection data, including the following data:

- Stack name
- Local and remote addresses and ports
- Start and end times
- Termination code (for connections that are closed)
- Sent and received byte and segment counts
- Retransmit counts
- Connection state
- Network interface and host information
- TN3270 information
- Application Transparent Transport Layer Security (AT-TLS) information, if applicable
- Security-related information about encryption

Connection data is available in both a readable form and in a form for programmatic use. Host name translation, IPv4, and IPv6 addresses are all supported. Because of the cross-domain capabilities of the NetView program, you can also view connection data from remote z/OS hosts.

DVIPA monitoring and management

The NetView program collects and displays dynamic virtual IP address (DVIPA) information, including the relationships between a DVIPA and the TCP/IP connections that use it. The NetView program monitors the following kinds of DVIPA information:

- DVIPA definition and status
- Distributed DVIPA including server health
- DVIPA connections
- VIPA routes
- Distributed DVIPA connection routing

Additionally, changes within your sysplex that affect DVIPAs can generate events that enable the NetView program to provide up-to-date DVIPA information. Distributed DVIPA statistics are also provided to assist with workload balancing. These statistics can be used to monitor historical trends and for problem determination.

For more information about DVIPA data collection and DVIPA connections, see Managing Dynamic Virtual IP Addresses in *IBM Z NetView IP Management*.

Discovery and monitoring of z/OS systems and IP stacks in a sysplex

The NetView discovery manager provides a comprehensive set of sysplex monitoring tools and a view of the physical configuration of the sysplex. Information that is discovered by the discovery manager can be viewed in the local NetView program, the NetView sysplex master program, and the enterprise master NetView program.

The discovery manager can monitor the following resources:

- Central processor complex (CPC)
- Channel subsystem identifier
- Logical partition (LPAR)
- Sysplex
- Coupling facility
- z/OS image
- TCP/IP stack
- TCP/IP subplex
- IP interfaces
- NetView applications
- Telnet servers and ports

If you enable RODM and configure the appropriate TOWER statements, the discovery manager can also monitor the following resources:

- Open Systems Adapter (OSA) channels and ports
- HiperSockets adapters

For information about additional IP management capabilities, see "Advanced IP Management" on page 127.

Preparing the Host Environment

Before you configure the NetView program for IP management, you must configure your host environment to support the required functions.

Procedure

To prepare the host environment for IP management, complete the following steps:

1. Make sure that the following NetView system symbols are defined:

CNMNETID
The network ID. This ID is a string value up to 5 characters in length. This ID is used by CNMSTYLE processing to define the NETID variable and becomes the value of the CNMSTYLE.NETID common variable.

CNMRODM
The RODM name that is used by this NetView installation. This value must match the value of the NAME variable in your RODM start procedure (by default EKGXRODM). If the NAME variable is not specified, this value must match the default RODM name (EKGXRODM). The RODM name is used by CNMSTYLE processing to define the RODMname variable and becomes the value of the CNMSTYLE.RODMNAME common variable.

Tip: The CNMRODM system symbol is required only for monitoring OSA or HiperSockets adapters.

CNMTCPN
The TCP name for this NetView installation. This value must match the step name that is used by the TCP/IP task. This name is used by CNMSTYLE processing to define the TCPname variable and becomes the value of the CNMSTYLE.TCPNAME common variable.

DOMAIN
CNMDOMN
The unique domain name for NetView on this z/OS system.

Important: If multiple instances of NetView are running on the same z/OS system, additional configuration is required. For more information, see the *IBM Z NetView Installation: Getting Started*.

2. Make sure your security configuration grants the required access.

   The required configuration depends on the value of the SECOPTS.OPERSEC statement in CNMSTYLE.

   - If the SECOPTS.OPERSEC statement is set to SAFCHECK, SAFPW, or SAFDEF, you must configure access by using a System Authorization Facility product such as RACF:
     - Make sure that an OMVS segment is defined for each started SNMP and NetView task, and that the UID for these tasks is set to 0.
     - Define the following libraries to RACF program control:

```
RALT PROGRAM * ADDMEM('SYS1.LINKLIB'/******/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCEERUN'/******/NOPADCHK)
UACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCNMLNK1'/******/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SCNMLNKN'/******/NOPADCHK)
ACC(READ)
RALT PROGRAM * ADDMEM('hlq.SEAGALT'/******/NOPADCHK)
UACC(READ)
SETROPTS WHEN(PROGRAM) REFRESH
```

     For more information about RACF, see the z/OS Security Server RACF documentation at http://publib.boulder.ibm.com/infocenter/zos/v1r13/index.jsp.

   - If the SECOPTS.OPERSEC statement is set to NETVPW or MINIMAL, you must configure access by modifying the NetView command authorization table (typically the SECTABLE member in the DSIPARM data set). Make sure that the table authorizes the NetView user ID to run VARY TCPIP commands.

     For more information about the NetView command authorization table, see "Controlling Access to Commands" in the *IBM Z NetView Security Reference*.

3. To enable OSA tracing, edit the ATCSTR*xx* member to add and activate a transport resource list entry (TRLE) major node.

   Add an entry similar to the following example:

```
*************************************************
* OSAFBC0 QDIO OSA TRLE MAJOR NODE              *
* Change names and UCBs accordingly             *
*************************************************
OSAFBC0V VBUILD TYPE=TRL
```

```
OSAFBC0T TRLE   LNCTL=MPC,                                             X
               READ=FBC0,                                              X
               WRITE=FBC1,                                             X
               DATAPATH=(FBC2,FBC3),                                   X
               PORTNAME=OSAFBC0P,                                      X
               PORTNUM=0,                                              X
               MPCLEVEL=QDIO
```

Note that the DATAPATH statement must define a second unit control block UCB for trace input and output. If you have already defined a TRL major node with only one UCB, edit the DATAPATH statement to add a second UCB.

4. For packet trace instances, the NetView program must be authorized to access the z/OS Communications Server Real-time control NMI interface. To allow access to the interface, grant one of the following permissions to the SAF user ID under which the NetView program runs:

   a. UNIX superuser authority for the user ID assigned to the NetView job or started procedure. You can grant superuser authority by setting the z/OS UNIX user ID (UID) attribute to zero.
   b. Trusted or Privileged authority.
   c. READ access to the SAF resource names for NMI SERVAUTH resources, as described in *z/OS Communications Server: IP Programmer's Guide and Reference*, *Real-time control NMI: Configuration and enablement*.

Configuring the SNMP agent

NetView IP management requires the SNMP management functions that are provided by the Communications Server SNMP agent.

Procedure

To configure the SNMP agent and subagents, complete the following steps:

1. Update the TCP/IP profile configuration to enable the SNMP subagent.

   Use the following statements:

```
; Enable the SNMP TCP/IP subagent
SACONFIG ENABLED SETSENABLED OSAENABLED ; enables SNMP TCP/IP subagent
;
; Set the community name (must match community name used by SNMP agent)
SACONFIG ENABLED COMMUNITY communityname      ; replace with your value
;
; Enable packet trace service and OSA-Express Network Traffic Analyzer
; (OSAENTA) trace service
NETMONITOR PKTTRCSERVICE NTATRCSERVICE
;
; Enable real-time SMF NMI service
NETMONITOR SMFS TCPCONNS MINLIFET 0
;
; Set UDP ports
PORT
   161 UDP OMVS                ; OSNMPD Server
   162 UDP SNMPQE              ; SNMPQE Agent
```

2. Optional: If you want to start the SNMP TCP/IP subagent when TCP/IP starts, add the IOBSNMP procedure to the AUTOLOG statement in the TCP/IP profile.
3. Update the Telnet profile to enable the SNMP TN3270E Telnet subagent.

   Use the following statements:

```
; Enable the SNMP TN3270E Telnet subagent
TNSAConfig                  ; Start up TN SNMP subagent
   Enabled                  ; Subagent must be enabled
   Agent 161                ; Specify agent port to contact
   Cachetime 30             ; Rebuild MIB info if older than 30 seconds
   Community communityname  ; Community name (replace with your value)
```

4. To activate the configuration changes, stop and restart the TCP/IP and TN3270 address spaces.

For more information about restarting address spaces, see the documentation of the MVS STOP and START commands in the *MVS System Commands Reference* at http://publib.boulder.ibm.com/infocenter/zos/v1r13/index.jsp.

5. Update the `snmpd.conf` file in the `/etc` HFS directory.

   Make sure that the community name matches the name specified in the TCP/IP profile. Community names are case sensitive.

   Tip: If no `snmpd.conf` file exists in the `/etc` directory, you can copy the sample `snmpd.conf` file provided with z/OS Communications Server from the `/usr/lpp/tcpip/samples` directory. This sample defines the community name `publicv2c` for SNMP version 2c. This sample configuration file might need additional customization to provide adequate security for your environment.

   For more information about the `snmpd.conf` file, see the documentation of the SNMP agent (OSNMPD) in the *z/OS Communications Server: IP Configuration Reference* at http://publib.boulder.ibm.com/infocenter/zos/v1r13/index.jsp.

6. Copy the `osnmpd.data` sample file from the `/usr/lpp/tcpip/samples` HFS directory to the `/etc` HFS directory as `snmpd.data`.

   Modify the `sysDescr`, `sysContact`, `sysLocation`, and `sysName` statements as needed.

7. To obtain management information base (MIB) files for OSA adapters, go to https://www.ibm.com/servers/resourcelink and search for `OSA MIB`.

   Download the MIB files that are appropriate for your OSA adapters and copy the files to `/etc/netview/mibs` on the NetView system.

   Note: The IBM Resource Link website requires that you log in with an IBM ID. If you do not have an IBM ID, you can register to create one.

8. To start the SNMP agent:

   a) Copy the OSNMPDPR sample JCL procedure from the EZA.SEZAINST data set to a system procedure library as OSNMPD.

   b) Modify the sample, following the instructions in the header.

   c) Run the modified procedure.

9. To start the OSA subagent:

   a) Copy the IOBSNMP JCL sample procedure from the EZA.SEZAINST data set to a system procedure library.

   b) Modify the sample, following the instructions in the header.

   c) Run the modified procedure.

10. To verify that the SNMP agent and subagents are running correctly, follow these steps:

    a) Run the following command from the OMVS environment:

    ```
    osnmp -h hostname -c communityname walk sysordescr
    ```

    If the agent is running correctly, output similar to the following example is displayed:

    ```
    1.3.6.1.2.1.1.9.1.3.1 = z/OS SNMP Agent
    1.3.6.1.2.1.1.9.1.3.2 = z/OS TN3270 SNMP Subagent
    1.3.6.1.2.1.1.9.1.3.3 = z/OS TCP/IP SNMP Subagent
    1.3.6.1.2.1.1.9.1.3.4 = OSA subagent
    ```

    b) Run the following command from the OMVS environment:

    ```
    osnmp -c communityname -v walk ibmOSAExpChannelTable
    ```

    If the agent is running correctly, output similar to the following example is displayed:

    ```
    ibmOSAExpChannelNumber.4 = '0002'h
    ibmOSAExpChannelNumber.6 = '0001'h
    ibmOSAExpChannelNumber.10 = '000b'h
    ibmOSAExpChannelNumber.12 = '0004'h
    ibmOSAExpChannelNumber.14 = '0006'h
    ibmOSAExpChannelType.4 = 49
    ```

```
ibmOSAExpChannelType.6 = 49
ibmOSAExpChannelType.10 = 17
ibmOSAExpChannelType.12 = 48
```

Configuring NetView for IP Management

To enable NetView IP management functions, you must enable the IPMGT tower and complete the required configuration steps.

Before you begin

Before enabling IP management in the NetView program, make sure that your host environment is prepared as described in "Preparing the Host Environment" on page 120.

This procedure assumes that the NetView program is installed, and that the CNMSJ004 member is customized to define the VSAM TCPCONN data sets. If you plan to save packet trace data for later analysis, you must also update and run the CNMSJ033 sample to initialize the saved packet trace database.

Procedure

To enable IP management, complete the following steps:

1. To enable the IPMGT tower, copy the TOWER statement from CNMSTYLE to your CNMSTUSER or C*xx*STGEN member and remove the asterisk from the IPMGT keyword. The new TOWER statement should read as follows:

   ```
   TOWER = *SA  *AON  *MSM  *Graphics  NPDA NLDM TCPIPCOLLECT
     *AMI *DVIPA *TEMA IPMGT *NVSOA DISCOVERY *ACTIVEACTIVE
   ```

   Remember: Make any changes in a member included by CNMSTYLE, such as CNMSTUSR or C*xx*STGEN. For more information about modifying CNMSTYLE statements, see *Z NetView Installation: Getting Started*.

2. Add the following CNMSTYLE statements:

   ```
   * The RMTSYN statement specifies the protocol used for communications
   * by the PIPE ROUTE stage and by the RMTCMD and LABEL commands.
   * Uncomment as needed, and edit to specify your domain names
   *RMTSYN.USIBMNR.CNM01 = 9.9.9.100   // IP address &amp; default port
   *RMTSYN.USIBMNR.CNM02 = 9.9.9.200   // IP address &amp; default port
   *RMTSYN.USIBMNR.CNM03 = 9.9.9.300   // IP address &amp; default port

   * Enables TCP/IP communication for the 3270 NMC
   auxInitCmd.IPTSK = START TASK=DSITCPIP

   *  Enable connection and security management,  and packet trace management
   (TCPIPCOLLECT.TCPCONN)INIT.TCPCONN = Yes
   (TCPIPCOLLECT.CONNSEC)INIT.CONNSEC = Yes
   (TCPIPCOLLECT.PKTS)INIT.PKTS = Yes
   (TCPIPCOLLECT.PKTS)INIT.OPKT = Yes

   * Enable collection of trace and connection data
   function.autotask.PKTS.TCPIP = AUTTCPPT
   INIT.PKTS = Yes
   function.autotask.TCPCONN.TCPIP = AUTTCPCN
   INIT.TCPCONN = Yes
   TCPCONN.GTF.TCPIP = Yes
   TCPCONN.KEEP.TCPIP.A = */*,*/*
   CONNSEC.KEEP.TCPIP.A = */*,*/*
   PKTS.STORAGE.TCPIP = 50M
   OPKT.STORAGE.TCPIP = 50M

   * Specify sysplex member for resource discovery completion messages
   RTNDEF.BASE.AGENT.sysname = USIBNMR.CNM01

   * Start UNIX and TSO command servers
   auxInitCmd.SUNIX = EXCMD AUTO1 START UNIXSERV=*
   auxInitCmd.STSO = EXCMD AUTO1 START TSOSERV=tso_userid,MEM=CNMSSTSO,OP=NONE
   ```

where *tso_userid* is a valid TSO user ID or two single quotation marks (' '), indicating that the NetView operator ID is to be used as the TSO user ID. This TSO user ID provides the authority under which the TSO server job will run.

3. To enable the DVIPA tower, remove the asterisk from the DVIPA keyword in the CNMSTYLE TOWER statement.
4. To enable the DVIPA subtowers, add the following CNMSTYLE statement:

```
TOWER.DVIPA = DVTAD DVCONN DVROUT
```

   Note: DVIPA events and distributed DVIPA statistics require additional configuration. For more information, see *IBM Z NetView Installation: Configuring Additional Components*.
5. Move the following members from the supplied DSIPARM data set to the user DSIPARM data set:
   - CNMPOLCY
   - CNMSCM
6. Modify the CNMPOLCY member to specify the correct domain, LPAR, and community names.
7. Modify the CNMSCM member to specify the correct IP addresses, host names, and community names.
8. Copy the CNMSSUNX and CNMSSTSO sample jobs into a data set defined in the IEFJOBS or IEFPDSI concatenation of the master JCL (for example, SYS1.PROCLIB).
9. Modify the CNMSJTSO job to specify the correct data set names for your environment.

   (This job starts the TSO command server and is started during CNMSTYLE processing.)
10. Customize the CNMSJ032 job in the .CNMSAMP data set as needed.
11. Run the CNMSJ032 job to copy the required configuration files to the appropriate HFS directories.

Customizing RODM

To monitor OSA and HiperSockets adapters, you must customize RODM.

Before you begin

The DISCOVERY tower must be enabled. In addition, the INTERFACES subtower of the DISCOVERY tower must be enabled with the following CNMSTYLE statements:

```
# enable the INTERFACES subtower
TOWER.DISCOVERY = INTERFACES *TELNET
TOWER.DISCOVERY.INTERFACES =  OSA  HIPERSOCKETS
```

Procedure

1. Define the VSAM EKG data sets.

   See the CNMSJ004 job and the EKGSI101 member in the CNMSAMP data set and run the steps that are required for defining the RODM data sets.
2. Copy the EKGXRODM member from the CNMSAMP data set to the PROCLIB data set, and customize it for your environment.
3. Copy the EKGLOADP member from the CNMSAMP data set to the PROCLIB data set, and customize it for your environment.
4. Copy the CNMSJH12 member from the CNMSAMP data set to the JOBLIB data set, and customize it for your environment.

   You must uncomment the following data model statements required by Discovery Manager:

```
// DD DSN=NETVIEW.V6R3M0.CNMSAMP(FLCSDM1),DISP=SHR  <-CLASS/FIELD DEF
// DD DSN=NETVIEW.V6R3M0.CNMSAMP(FLCSDM2),DISP=SHR  <-CLASS/FIELD DEF
// DD DSN=NETVIEW.V6R3M0.CNMSAMP(FLCSDM3),DISP=SHR  <-CLASS/FIELD DEF
// DD DSN=NETVIEW.V6R3M0.CNMSAMP(FLCSDM4),DISP=SHR  <-CLASS/FIELD DEF
// DD DSN=NETVIEW.V6R3M0.CNMSAMP(FLCSDM5),DISP=SHR  <-CLASS/FIELD DEF
```

5. Start the EKGXRODM job.
6. Start the CNMSJH12 job to populate the NetView data model.

Verifying IP Management Capability

Follow these steps to verify correct operation of the NetView IP management functions.

Procedure

1. Stop and restart the NetView program.
2. To verify communication with TCP/IP hosts, issue the following commands from the Network Communications Control Facility (NCCF):

   `ping localhost`

   `ping` *`hostname`*, where *hostname* is a fully qualified TCP/IP host name that is reachable from the host where the NetView program is running

   Check the output of the ping command to verify that packets are successfully reaching the destination hosts.
3. Follow these steps to verify the operation of TCP/IP connection management functions:
   a) Start TCP/IP connection management by issuing the `TCPCONN START` command.
   b) From another computer, open a connection to the NetView host by issuing one of the following commands:
      - `telnet` *`hostname`*
      - `ftp` *`hostname`*

      where *hostname* is the TCP/IP host name of the host where the NetView program is running.
   c) On the NetView host, query TCP/IP connection data by issuing the `TCPCONN QUERY` command.

      Make sure that the command output includes information about the connection that you opened.
   d) Stop TCP/IP connection management by issuing the `TCPCONN STOP` command.
4. Verify the correct operation of the discovery manager:
   a) Issue the `CNMSSTAC` sample command.

      Verify that the output displays the discovered active TCP/IP stacks.
   b) If you have enabled HiperSockets discovery, issue the `CNMSHIPR` sample command.

      Verify that the output displays the discovered HiperSockets adapters.
5. To verify the correct operation of DVIPA monitoring, issue the `CNMSDVIP` command.

   Verify that the output displays the discovered DVIPA definition and status information.

Results

After you have verified the correct operation of the configured IP management functions, you can start using the NetView program to manage your IP networks. More information about configuring and using the NetView IP management functions is available in the NetView product documentation:

- For more information about configuring the NetView program for TCP/IP management, see *IBM Z NetView Installation: Configuring Additional Components*.
- For more information about configuring NetView user interface components, see *IBM Z NetView Installation: Configuring Graphical Components*.
- For more information about using NetView TCP/IP management functions, see *IBM Z NetView IP Management*.

Advanced IP Management

In addition to the basic IP management capabilities, NetView supports many additional IP management capabilities and configuration options.

Advanced IP management capabilities include the following:

Web services gateway
: The NetView web services gateway provides an industry-standard open interface to the NetView program for issuing commands and receiving responses. The gateway uses the Simple Object Access Protocol (SOAP) over HTTP or HTTPS as the transport mechanism. For more information about the Web services gateway, see *IBM Z NetView Installation: Configuring Additional Components*.

Web Application
: The NetView Web Application enables custom web applications and web pages to interact with the NetView program. For more information about the Web Application, see *IBM Z NetView Installation: Configuring Additional Components*.

Automated Operations Network (AON)
: The AON component was used for IP management in earlier versions of the NetView program. If you are upgrading from an earlier NetView version, you might need to enable the AON tower in order to complete migration. For more information about the AON, see *IBM Tivoli Z NetView Installation: Configuring Additional Components*.

MultiSystem Manager
: The MultiSystem Manager component of NetView enables you to monitor resources on your IP network using the IBM Tivoli Netcool/OMNIbus and Network Manager products. For more information about the MultiSystem Manager component, see *IBM Z NetView Installation: Configuring Graphical Components.*

Z NetView Enterprise Management Agent (NetView agent)
: The NetView agent provides information that you can use to manage your network from the Tivoli Enterprise Portal using sampled data. For more information about the NetView agent, see *IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent*.

Setting Required Statements for IP Management

To enable IP management in the NetView program, you must examine and either set or update the following statements. For statements in the CNMSTUSR or C*xx*STGEN member, see the information about using and changing CNMSTYLE statements in *IBM Z NetView Installation: Getting Started*.

Tip: If you have completed the procedure described in "Getting Started with IP Management" on page 119, some of these statements might already be set.

- Set the TCPname statement in the CNMSTUSR or C*xx*STGEN member to the TCP/IP job name. You can use a system symbolic (&CNMTCPN) in the SYS1.PARMLIB data set to set the value of the TCPname statement. MVS needs to be restarted after the system symbolic is defined.

  During initialization, the NetView program uses the value of the TCPname statement to set the DEFAULTS.TCPNAME value that is used by these NetView TCP/IP services. You can override the value set in the CNMSTYLE member by using the DEFAULTS command to change DEFAULTS.TCPNAME prior to starting (or restarting) the tasks, or you can override the value in the initialization members for the tasks. The DEFAULTS command can be issued by an operator or by a CLIST. This default applies to the NetView program, and cannot be overridden for a particular operator.
- By default, the NetView program supports both IPv4 and IPv6 addressing. To limit the NetView program to one addressing family, use the IPv6Env environment statement in the CNMSTUSR or C*xx*STGEN member. For information about the IPv6Env statement, see the *IBM Z NetView Administration Reference*.
- Examine the settings for the MAXPROCSYS and MAXPROCUSER statements in the BPXPRM*xx* member in the SYS1.PARMLIB data set. The MAXPROCSYS statement specifies the maximum number of processes that can be active at the same time. The MAXPROCUSER statement specifies the maximum number of processes with a single UID that are allowed to be active at the same time. The number of

TCP/IP-related processes that are spawned as a result of NetView commands can exceed the system-supplied defaults for these UNIX System Services settings. These limits might need to be increased. The settings can be temporarily increased by using the SETOMVS command and remain in effect until the next IPL.

Setting Policy Statements

The following table lists the policy statements for IP management, in alphabetical order by statement, and the member in which each is coded. For more information about these statements, see the *IBM Z NetView Administration Reference*.

Table 21. IP Management Policy Definitions

Description	**Statement**	**Member Where Coded**
Active monitoring	ACTMON	CNMIPMGT
Connection monitoring threshold definition	IPCONN	CNMIPMGT
Host definition	IPHOST	CNMIPMGT
Interface definition	IPINFC	CNMIPMGT
Nameserver definition	IPNAMESERV	CNMIPMGT
Socket definition	IPPORT	CNMIPMGT
Router definition	IPROUTER	CNMIPMGT
Telnet server definition	IPTELNET	CNMIPMGT
TN3270 server definition	IPTN3270	CNMIPMGT
Stack definition	TCP390	CNMPOLCY

Consider the following definitions for TCP390 statements:

- To enable commands that use SNMP, such as the IPTRACE and NVSNMP commands, define the community name.
- To receive DVIPA traps, define the same community name in the NetView program that you have configured in z/OS Communications Server for SNMP support. For more information about setting the community name for SNMP support, see the *z/OS Communications Server Configuration Reference*.
- Define stacks that the discovery manager does not find dynamically.

Note: To set community names for non-stack resources that use SNMP, use the CNMSCM member.

Enabling TCP/IP Services

The NetView program supplies several TCP/IP services that are provided as server and client functions. Server and client functions are available for the REXEC, RSH, and syslog services. The TN3270 service is only available as a client function. The REXEC and RSH services provide remote command processing support. The syslog service provides remote logging. A NetView operator can use the TN3270 service to establish a 3270 Telnet session with a Telnet server.

To enable the server function of these TCP/IP services, complete the following steps:

1. You can also specify the TCP/IP parameters for each task that is associated with these TCP/IP services in the CNMSTUSR or C*xx*STGEN member. When the task is restarted with the RESTYLE command, the values that are specified in the CNMSTYLE member are used by the TCP/IP service.

   Table 22 on page 129 shows the task and initialization statements for each TCP/IP service that is available as a server function.

Table 22. TCP/IP Services		
TCP/IP Service	**NetView Task**	**Task Initialization Statements in the CNMSTYLE Member**

REXEC	DSIRXEXC	`REXEC.TCPANAME`
		`REXEC.PORT`
		`REXEC.SOCKETS`
		`REXEC.PROTOCOL`

RSH	DSIRSH	`RSH.TCPANAME`
		`RSH.PORT`
		`RSH.SOCKETS`
		`RSH.PROTOCOL`

TCP/IP syslog	DSIIPLOG	`IPLOG.TCPANAME`
		`IPLOG.PORT`
		`IPLOG.SOCKETS`

2. If you are running the RSH server, place the DSIRHOST sample in DSIPARM and modify it to meet your security needs. In the following example of this file, all users on host1 except for user1 and all users on host2 can use RSH to send commands to the NetView program:

```
+host1
+host1 -user1
+host2
```

3. Ensure that the DSIRXEXC, DSIRSH, and DSIIPLOG tasks are started to complete the setup for the REXEC, RSH, and syslog servers. These tasks can be set to start automatically during initialization by changing INIT=N to INIT=Y in the following task statements in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.DSIIPLOG.INIT=Y
TASK.DSIRSH.INIT=Y
TASK.DSIRXEXC.INIT=Y
```

No special setup is needed to enable the client function of these TCP/IP services other than ensuring that the DEFAULTS.TCPNAME value is set correctly. The client commands (REXEC, RSH, IPLOG, and TN3270) can therefore be issued from the NetView program without the NetView server tasks being active.

Enabling TCP/IP Connection and Packet Trace Data Collection

TCP/IP connection data is collected in two ways in the NetView program:

- Using the TCPCONN START and TCPCONN QUERY commands. This method requires an autotask that uses a socket interface. Inactive connection data that is displayed in the NetView workspaces in the Tivoli Enterprise Portal is collected using this method.
- Using the TCPCONN QUERYACT command. This method uses the EZBNMIFR interface of z/OS Communications Server. Active connection data that is displayed in the NetView workspaces in the Tivoli Enterprise Portal and connection data that is displayed in IPSTAT panels are collected using this method.
- Review, and if necessary, update the IPMGT tower in the CNMSTUSR or C*xx*STGEN member for the IPTRACE function. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Packet trace data is collected using the PKTS START and PKTS QUERY commands. For each type of packet trace data that is collected, this method requires an autotask that uses a socket interface. Packet trace data that is displayed in IPTRACE panels is collected using this method. The IPTRACE packet trace data is formatted using the FMTPACKT command.

The rest of this topic describes how to set up connection and packet trace data collection that requires an autotask and uses the socket interface.

To use the socket interface to collect Open Systems Adapter (OSA) packet trace data, you must enable the NETMONITOR statement in your TCP/IP profile in z/OS Communications Server. To do that, add the following statement to your TCP/IP profile:

```
NETMONITOR TCPCONNSERVICE PKTTRCSERVICE NTATRCSERVICE
```

Where:

TCPCONNSERVICE
Enables the collection of TCP/IP connection data

PKTTRCSERVICE
Enables the collection of IP packet trace data

NTATRCSERVICE
Enables the collection of OSA packet trace data

If you do not want to collect data from one of these services, remove the appropriate parameter from the NETMONITOR statement.

For more information, see the *z/OS Communications Server IP Configuration Reference*. Especially note the SAF (RACF) requirement that the user ID (in this case associated with the NetView autotask described in the following text), or address space, must be permitted access to the relevant resources.

To configure the collection of TCP/IP connection and packet trace data in the NetView product, perform the following steps:

- Ensure that the TCP/IP connection management databases have been defined. The TCP/IP connection management databases are defined using the CNMSJ004 job with input member CNMSI101.
- Review and, if necessary, update the TCPIPCOLLECT tower in the CNMSTUSR or C*xx*STGEN member, along with the appropriate subtowers (TCPCONN for TCP/IP connection management, PKTS for packet trace management). See the *IBM Z NetView Administration Reference* for more information. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
- Review and, if necessary, update autotasks to handle the collection of connection and packet trace data. To do this automatically during NetView startup, use the CNMSTYLE statements that are shown in Table 23 on page 130 in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. In the `function.autotask` statements, *stackname* is the name of the TCP/IP stack.

Table 23. Stack Connection and Packet Trace Data Collector Autotasks

Autotask Name	Collector Autotask Function	CNMSTYLE Statement
AUTOOPKT	OSA packet trace data	`(TCPIPCOLLECT.OPKT)function.autotask.OPKT.`*stackname* `= AUTOOPKT`
AUTOPKTS	Packet trace data	`(TCPIPCOLLECT.PKTS)function.autotask.PKTS.`*stackname* `= AUTOPKTS`
AUTOTCPC	Connection data for a stack	`(TCPIPCOLLECT.TCPCONN)function.autotask.TCPCONN.`*stackname* `= AUTOTCPC`
AUTTRA*n*	Packet trace instances	`COMMON.PKTS.INSTANCE.POOL = AUTTRA,8`

  You can also define these autotasks interactively using the TCPCONN DEFINE and PKTS DEFINE commands. See the online help for more information.

- If you want to start data collection automatically during NetView startup, add the following statements to the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

- For connection data:

```
(TCPIPCOLLECT.TCPCONN)INIT.TCPCONN = Yes
```

- For packet trace data:

```
(TCPIPCOLLECT.PKTS)INIT.PKTS = Yes
```

- For OSA packet trace data:

```
(TCPIPCOLLECT.OPKT)INIT.OPKT = Yes
```

You can also start data collection interactively using the TCPCONN START and the PKTS START commands. For more information about these commands, see the online help.

- To stop data collection, use the TCPCONN STOP and PKTS STOP commands. For more information about these commands, see the online help.
- To define security passwords for the TCP/IP connection management databases:
  1. Stop the TCP/IP connection management by using the TCPCONN STOP command.
  2. Modify the definition statements in CNMSI101 that define the TCP/IP connection management databases, changing them to include the specification of VSAM cluster passwords. Rerun the CNMSJ004 job using these modified statements to delete and redefine the TCP/IP connection management databases.
  3. Update the CNMSTPWD member in DSIPARM to include the passwords that you specified when redefining the TCP/IP connection management databases. The following example shows the PWD statements that define the passwords for the TCP/IP connection management databases, where *p_password* is the 1- to 8-character password for the primary database and *s_password* is the 1- to 8-character password for the secondary database:

     ```
     PWD.DSITCONT.P = p_password
     PWD.DSITCONT.S = s_password
     ```

  4. Restart the TCP/IP connection management by using the TCPCONN START command.
- To enable automatic database maintenance for the TCP/IP connection data VSAM database, add a corresponding DBINIT command to the CNMSTUSR or C*xx*STGEN member. (See the comments in the TCPCONN section of the CNMSTYLE member for more information.) For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
- To filter the collection of connection data, add TCPCONN.KEEP and TCPCONN.DASD statements to the CNMSTUSR or C*xx*STGEN member. (See the comments in the TCPCONN section of the CNMSTYLE member for more information.) For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Enabling the Saving of Packet Trace Data

To enable the saving of packet trace data, follow these steps:

1. In the CNMSTYLE member, the `SAVED PACKET TRACE (FKXPKTS function)` section defines the following autotask with a default operator name of AUTOPSAV:

   ```
   function.autotask.FKXPKTS = AUTOPSAV
   ```

   If this operator name is not appropriate for your installation, change it in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
2. Customize the CNMSJM14 sample, which is a JCL job that provides a sample method for reorganizing the FKXPKTS database. Before you can use this JCL job, you must customize it for your installation; follow the instructions that are provided in the JCL comments. Before you run this JCL job, you must free the FKXPKTS file DD from the NetView program. After you run this job, reallocate the FKXPKTS file DD to the NetView program.

3. In the CNMKEYS member, review the key definitions for the PKTDTLS component, and make changes as needed.

Enabling the Saving of SNIFFER Trace Format Data Sets

To enable the saving of packet traces in Sniffer trace format, follow these steps:

1. In the CNMSTYLE member, review the `PKTS.SAVEHLQ` and `PKTS.SNFPROC` section, and adjust based on your installation requirements.
2. Copy the CNMSJSNF sample and rename as required to a SYSPROC concatenation data set so it will be available for the MVS START command.
3. Customize the CNMSJSNF procedure by following the instructions in the comments in the procedure.
4. If you are using cross-domain IP management to capture and save packet traces, the CNMSJSNF procedure must reside in a SYSPROC concatenated data set on the remote systems.

Enabling DVIPA Management

Management of dynamic virtual IP addresses (DVIPAs) includes the following items. By default, all DVIPA management is disabled.

- "DVIPA Data Collection" on page 132
- "DVIPA Events" on page 133
- "Distributed DVIPA Statistics" on page 135
- "User Interface Configuration for DVIPA Management" on page 136

DVIPA Data Collection

To enable DVIPA data collection for a domain, complete the following actions:

- Ensure that the DVIPA tower is uncommented on the TOWER statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. Note that enabling the DVIPA tower collects only DVIPA definition and status information.
- To enable additional DVIPA data collection, ensure that the TOWER.DVIPA subtowers (DVTAD, DVCONN, and DVROUT) are uncommented.
- The autotasks in Table 24 on page 132 are used by the DVIPA subtowers for data collection. To change the autotask name, modify the function.autotask statement in the CNMSTUSR or C*xx*STGEN member. Also review, and if necessary, update the autotask name in the DSIOPF member.

Table 24. DVIPA Data Collector Autotasks

Autotask Name	Collector Autotask Function	CNMSTYLE Statement
AUTOCT1	DVIPA definition and status	`(DVIPA)function.autotask.COLTSK1 = AUTOCT1`
AUTOCT2	Distributed DVIPA	`(DVIPA.DVTAD)function.autotask.COLTSK2 = AUTOCT2`
AUTOCT3	DVIPA connections	`(DVIPA.DVCONN)function.autotask.COLTSK3 = AUTOCT3`
AUTOCT4	VIPA routes and distributed DVIPA connection routing	`(DVIPA.DVROUT)function.autotask.COLTSK4 = AUTOCT4`

Sampling intervals for DVIPA data collection are 1 hour. If you want to modify these intervals, review the following CNMSTYLE statements and make any updates in the CNMSTUSR or C*xx*STGEN member:

- DVIPA.INTDVTAD

- DVIPA.INTDVCONN
- DVIPA.INTDVDEF
- DVIPA.INTDVROUT

For summary information about enabling data collection and data display for DVIPAs, see Appendix B, "Data Collection and Display for the NetView User Interfaces," on page 239.

DVIPA Events

To enable DVIPA events automation, complete the following steps:

1. Review the following DVIPA event automation samples to determine the z/OS Communications Server events that you want to monitor:
   - DVIPA TCPIP profile updates
     - CNMSDVCG monitors VIPADYNAMIC TCPIP profile changes.
   - Runtime DVIPA updates
     - CNMSDVTP monitors DVIPA traps.
     - CNMSDVEV monitors DVIPA SMF (non-SNMP) updates.
   - Sysplex Monitoring Messages
     - CNMSSMON monitors sysplex monitoring messages.
2. Remove or comment out the samples that you choose not to use.

   DVIPA SMF updates and DVIPA traps provide the same updates, but DVIPA traps requires SNMP.

   DVIPA runtime updates and VIPADYNAMIC TCPIP profile changes can be generated for the same actions.
3. The DVIPAUTO autotask is used for DVIPA event automation. To change the autotask name, modify the function.autotask statement in the CNMSTUSR or C*xx*STGEN member:

   ```
   (DVIPA)function.autotask.DVIPAUTM = DVIPAUTO
   ```

DVIPA SMF (Non-SNMP) Updates

To receive z/OS Communications Server DVIPA SMF runtime updates, add a NETMON SMFSERVICE DVIPA statement or an SMFCONFIG TYPE119 DVIPA statement to the z/OS Communications Server TCPIP profile. For more information about how to update the TCPIP profile, see the *z/OS Communications Server IP Configuration Reference*.

The NetView program can now receive the following z/OS Communications Server DVIPA SMF runtime updates:

- DVIPA Status Change
- DVIPA Removed
- DVIPA Target Added
- DVIPA Target Removed
- DVIPA Target Server Started
- DVIPA Target Server Ended

DVIPA Traps

To enable DVIPA traps, complete the following actions:

- Update the z/OS Communications Server snmpd.conf configuration file to send traps to the NetView program. For information about updating the snmpd.conf file, see the *z/OS Communications Server IP Configuration Reference*.

- Enable SNMP by starting the z/OS Communications Server SNMP agent (OSNMPD). For more information, see the *z/OS Communications Server IP Configuration Reference*.
- Configure the SNMP Trap Automation Task Configuration statements from the CNMSTYLE member in the CNMSTUSR or C*xx*STGEN member; for information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
  - Use the same port on which traps are sent (default 162).
  - Indicate in these statements the name of the DST task that is to catch the traps, or manually start this DST task.

    Note: If traps are being generated for IBM Tivoli Network Manager through MultiSystem Manager, use the same DST task name for both IBM Tivoli Network Manager and the NetView program.

The NetView program can now receive the following z/OS Communications Server DVIPA traps:

- ibmMvsDVIPAStatusChange
- ibmMvsDVIPARemoved

To receive additional z/OS Communications Server DVIPA traps, issue the following UNIX Systems Services command:

```
snmp -h host -r 0 -c communityname -v set ibmmvsdvipatrapcontrol.0 \'FC\'h
```

The following z/OS Communications Server DVIPA traps can now be received:

- ibmMvsDVIPATargetAdded
- ibmMvsDVIPATargetRemoved
- ibmMvsDVIPATargetServerStarted
- ibmMvsDVIPATargetServerEnded

DVIPA TCPIP Profile Updates

To receive z/OS Communications Server VIPADYNAMIC TCPIP profile updates, configure the following statements:

- Add a NETMON SMFSERVICE PROFILE statement to the z/OS Communications Server TCPIP profile. For more information about how to update the TCPIP profile, see the *z/OS Communications Server IP Configuration Reference*.
- Review and, if necessary, update an autotask in the CNMSTYLE member for each TCPIP stack that is to collect z/OS Communications Server VIPADYNAMIC TCPIP profile updates for the NetView program. Do this by adding the following statement to your CNMSTUSR or C*xx*STGEN member, where *stackname* is the name of the TCPIP stack, and *taskname* is the name of the autotask that collects TCPIP profile updates. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

  ```
  function.autotask.TCPSM.stackname=AUTOTCPS
  ```

  By default, the NetView program uses autotask AUTOTCPS to collect TCPIP profile updates from the TCPIP stack referred to by the &CNMTCPN system symbolic. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Notes:

1. The DISCOVERY tower does processing to receive TCPIP profile updates. If you disable the DISCOVERY tower but want to receive TCPIP profile updates, issue the following internal command:

   ```
   TCPSM START TCPNAME=tcpproc
   ```

2. For more information about the z/OS Communications Server TCP/IP stack profile network management interface, see the *z/OS Communications Server IP Programmer's Guide and Reference*.

Sysplex Monitoring Messages

To receive sysplex monitoring messages, modify the z/OS Communications Server TCPIP profile to include the SYSPLEXMONITOR parameter on the GLOBALCONFIG statement. When the SYSPLEXMONITOR parameter is specified and a problem occurs, z/OS Communications Server issues action messages. When the problem is resolved, z/OS Communications Server issues a Delete Operator Message (DOM). The NetView program can automate on both the messages and the DOMs to start rediscovery.

When SYSPLEXMONITOR RECOVERY is specified, the CNMSSMON sample processes only the EZZ9675E and EZZ9676E messages as events. When SYSPLEX MONITOR NORECOVERY is specified, the CNMSSMON sample processes all other messages as events. For more information, see the CNMSSMON sample.

DVIPA Event Delay Timers

DVIPA events are used to keep DVIPA information current in the NetView product. The following event delay timers are built into the DVIPA event processing to ensure that DVIPA information is not discovered for each DVIPA event, which can cause performance problems:

- DVIPA.Event.Delay
- DVIPA.Mast.Disc.Delay

DVIPA data collection intervals default to one hour. If you are using DVIPA events, you might want to increase the DVIPA data collection intervals by changing the following CNMSTYLE statements:

- DVIPA.INTDVDEF
- DVIPA.INTDVTAD
- DVIPA.INTDVCONN
- DVIPA.INTDVROUT

For more information about using DVIPA with a master NetView program, see Chapter 7, "Configuring NetView Sysplex Management," on page 151.

Distributed DVIPA Statistics

Distributed DVIPA statistics are kept each time data collection is run for the DVIPA.DVTAD subtower. To keep distributed DVIPA statistics and review the statistics, complete the following actions:

- Run the CNMSJ002 sample job to allocate the primary (CNMDVIPP) and secondary (CNMDVIPS) sequential data sets where the data is to be written.
- Ensure that the CNMDVIPP and CNMDVIPS DD statements are defined in your NetView startup procedure. For more information, see the CNMSJ009 sample job.
- Make sure that the DVTAD subtower is uncommented in the TOWER.DVIPA CNMSTYLE statement.
- Make sure that the Init.DVIPASTATS CNMSTYLE statement is set to Y to start the logging of distributed DVIPA statistics at NetView initialization. You can also use the DVIPALOG command to dynamically start and stop logging of distributed DVIPA statistics.
- The DVIPSTAT autotask is used to collect distributed DVIPA statistics and to write the records to a data set. To change the autotask name, modify the function.autotask statement in the CNMSTUSR or C*xx*STGEN member:

  ```
  (DVIPA)function.autotask.DVIPSTAT = DVIPSTAT
  ```

- Review, and if necessary, update the following CNMSTYLE statements to set filters for the distributed DVIPAs for which statistics are recorded:
  - DVIPA.STATS.DVIPA
  - DVIPA.STATS.PORT
  - DVIPA.STATS.TCPNAME.z

- Review and, if necessary, update the following CNMSTYLE statements that specify the number of records to write to the corresponding data sets:
  - DVIPA.STATS.Pri.MAXR (Primary data set)
  - DVIPA.STATS.Sec.MAXR (Secondary data set)

  One record is written for each distributed DVIPA target for each data collection, based on the specified filters.

  When the record count is reached, the data set starts logging records to the other data set. Messages are issued when the primary (CNM482I) and secondary (CNM483I) data sets become active. For the format of the record written to the data set, see *IBM Z NetView IP Management*. You can use the CNMSDVST sample to format the records.

User Interface Configuration for DVIPA Management

You can manage DVIPAs from the Z NetView Enterprise Management Agent and the 3270 interface.

Using the Z NetView Enterprise Management Agent

The Z NetView Enterprise Management Agent provides DVIPA management functions from the Tivoli Enterprise Portal. Before you can use the DVIPA management functions, the DVIPA data collection must be enabled and the DVIPA subtowers under the TEMA tower must also be enabled. For more information, see *IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent*.

Using the 3270 Interface

You can view DVIPA information by using the DVIPA 3270 commands. Before you can use these commands, the DVIPA tower and associated subtowers must be enabled. For information about using these commands, see *IBM Z NetView IP Management*.

Enabling the Discovery Manager

The discovery manager discovers sysplex and System z® resources. Discovery manager data can be viewed in 3270 sessions, the NetView management console, and the Tivoli Enterprise Portal by using the Z NetView Enterprise Management Agent. Not all data can be seen in all interfaces.

- "Discovery Manager Data Collection" on page 136
- "Multiple NetView Programs on a Single z/OS Image" on page 137
- "User Interface Configuration for Discovery Manager" on page 138

Discovery Manager Data Collection

Data collection by the discovery manager uses a combination of sampling and message automation.

By default, the discovery manager is enabled with the DISCOVERY tower in the CNMSTYLE member. This provides for data collection of the following kinds of resources:

- Sysplex
- Coupling facility
- z/OS image
- TCP/IP stack
- NetView application
- Central processor complex (CPC)
- Channel subsystem identifier
- Logical partition (LPAR)

To enable additional discovery manager data collection, complete the following actions:

- Ensure that the DISCOVERY tower is uncommented on the TOWER statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.
- To use SNMP to discover data, set DISCOVERY.SNMP to YES in the CNMSTUSR or C*xx*STGEN member.
- To enable additional discovery manager data collection, ensure that statements for the TOWER.DISCOVERY and TOWER.DISCOVERY.INTERFACES subtowers are uncommented. To see IP interface or Telnet server and port information, uncomment the statements for the appropriate TOWER.DISCOVERY subtowers. To see Open Systems Adapter (OSA) and HiperSockets adapter information, uncomment the appropriate statements for TOWER.DISCOVERY.INTERFACES subtowers.
- If you enable the DISCOVERY.INTERFACES subtower, ensure that the z/OS Communications Server SNMP agent (OSNMPD) is running.
- If you enable the DISCOVERY.INTERFACES.OSA subtower, ensure that the OSA SNMP subagent (IOBSNMP) is running.
- The autotasks that are shown in Table 25 on page 137 are used by the DISCOVERY tower and subtowers for data collection. To change an autotask name, modify the function.autotask statement in the CNMSTUSR or C*xx*STGEN member; for information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. Also review and, if necessary, update the autotask name in the DSIOPF member.

Table 25. Discovery Manager Autotasks

Autotask Name	**Collector Autotask Function**	**CNMSTYLE Statement**
AUTOAON	DISCOVERY tower resources	`function.autotask.Policy = AUTOAON`
AUTOCT5	IP interfaces, including OSA and HiperSockets adapters	`(DISCOVERY.INTERFACES)function.autotask.COLTSK5 = AUTOCT5`
AUTOCT6	Telnet servers and Telnet server ports	`(DISCOVERY.TELNET)function.autotask.COLTSK6 = AUTOCT6`
AUTOCT7	NetView applications	`(DISCOVERY)function.autotask.COLTSK7 = AUTOCT7`

Sampling intervals are used for IP interfaces (1 hour), Telnet servers and ports (1 hour), and NetView applications (5 minutes). If you want to modify these intervals, review the following CNMSTYLE statements and make any updates in the CNMSTUSR or C*xx*STGEN member:

- DISCOVERY.INTINTERFACE
- DISCOVERY.INTTELNET
- DISCOVERY.INTAPPL

Note: If you are using the Z NetView Enterprise Management Agent, and you have enabled the TEMA.HEALTH subtower, you might want to set the DISCOVERY.INTAPPL and NACMD.INTHEALTH intervals to the same value.

Message automation is used to update information when some discovery manager resources start and stop. Review the CNMSEPTL automation sample member for these events. The CNMSEPTL member is included when the DISCOVERY tower is enabled.

For summary information about enabling data collection and data display for discovery manager resources, see Appendix B, “Data Collection and Display for the NetView User Interfaces,” on page 239.

Multiple NetView Programs on a Single z/OS Image

If you are running multiple NetView programs on a single z/OS image, you should either turn off the DISCOVERY tower completely or specify DISCOVERY.NetViewOnly=YES in the CNMSTUSR or C*xx*STGEN member on the secondary NetView program. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If DISCOVERY.NetViewOnly=YES is specified, you can collect minimal information without extra CPU utilization and without the display of duplicate data in the 3270 interface and Tivoli Enterprise Portal at the master NetView program.

For example, if NetView programs CNM01 and CNM02 are running on the SYS1 z/OS image, the DISCOVERY tower is enabled, and DISCOVERY.NetViewOnly=NO is set for both NetView programs, 2 rows are displayed for a single TCP/IP stack on SYS1 in response to a CNMSSTAC DOMAIN=ALL command issued from the master NetView program.

User Interface Configuration for Discovery Manager

You can manage resources that are discovered using the discovery manager from the NetView management console, the Z NetView Enterprise Management Agent, and the 3270 interface.

Using the NetView Management Console

You can view sysplex and System z view topology from the NetView management console. For more information about viewing this topology, see *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: NetView Management Console*.

Using the Z NetView Enterprise Management Agent

The Z NetView Enterprise Management Agent provides TCP/IP stack, Telnet server and port, Open Systems Adapter (OSA), HiperSockets adapter, and NetView application information from the Tivoli Enterprise Portal. Before you can use the discovery manager functions, the data collection towers and subtowers and the TEMA subtowers must be enabled. For more information about enabling these functions, see *IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent*.

Using the 3270 Interface

You can view sysplex and System z information by using the discovery manager 3270 commands. Before you can use these commands, the DISCOVERY tower and associated subtowers must be enabled. For information about using these commands, see *IBM Z NetView IP Management*.

Defining Critical IP Port Monitoring

The NETSTAT command displays all resources that are currently active. However, the NETSTAT command can sometimes indicate that a resource is active when the port for that resource refuses a connection. Use the TESTPORT command to check a port that looks active but might be refusing connections because it is inactive. Such ports are known as hung listeners.

The TESTPORT command provides additional function for monitoring critical ports. To define ports that are to be monitored by the TESTPORT command and the intervals for monitoring them, use the COMMON.IPPORTMON.IPADD, COMMON.IPPORTMON.INTVL, and COMMON.IPPORTMON.PORTNUM statements in the CNMSTUSR or C*xx*STGEN member. For more information about these COMMON.IPPORTMON statements, see the *IBM Z NetView Administration Reference*. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

For additional information about the TESTPORT command, see the online help.

Enabling IP Functions with the IPMGT Tower

Many IP-related functions that were previously implemented as part of the NetView Automated Operations Network (AON) component are now implemented as base NetView services. These functions, which are referred to as the AON IP functions, no longer require AON enablement and configuration. Instead, they can be enabled by using the IPMGT tower. However, AON IP functions that are already enabled using the AON TCP subtower do not need to be reconfigured. The AON TCP subtower and the IPMGT towers are mutually exclusive.

To enable the IPMAN command for active monitoring of IP resources, use the IPMGT ACTMON subtower statement in the CNMSTUSR or C*xx*STGEN member. Also, add statements to the CNMIPMGT member for the resources that are to be monitored; for a list of these statements, see “Setting Policy Statements” on page 128. For information about using CNMSTYLE and tower statements, see *IBM Z NetView Installation: Getting Started*.

To enable automated responses to intrusions detected by the use of z/OS Communications Server Intrusion Detection Services (IDS), use the IPMGT IDS subtower statement in the CNMSTUSR or C*xx*STGEN member. For more information about IDS, see “Enabling Intrusion Detection Services” on page 146.

The following IP functions are also available with the IPMGT tower:

- PING panel
- TRACERTE panel
- NVSNMP panel (SNMP command panel)
- IPSTAT (IP connections) panel

Setting Up TCP/IP Support for the AON IP Functions

The NetView program provides the following AON IP functions. (TCP390 definitions are required only for remote TCP/IPs and to override discovered data.)

Note: The configuration that is described in this section is for IP-related functions that were previously implemented as part of the NetView Automated Operations Network (AON) component. These functions, which are referred to as the AON IP functions, are now implemented as base NetView services and no longer require AON enablement and configuration.

- Stack monitoring

  You can use the NetView program to automatically monitor stacks that have been discovered by the NetView program or defined in TCP390 policy definitions.
- IP connection management

  Using the 3270, Web browser, or NetView management console, you can manage IP connections into your stacks. All connections are supported, including TN3270, FTP, CICS, and SMTP. You can use operator filters to assist in reducing the amount of data. You can view connection byte counts and take action (for example, breaking the connection).
- SNMP functions

  Using 3270 panels, you can issue SNMP commands such as GET, SET, WALK, and Remote Ping. You can also use the MIB groups function to define several MIBs to be part of a group and use the NetView program to retrieve that group. Sample MIB groups are provided.
- IP resource manager

  Using the IP Resource Manager, you can display all of your critical IP resources and their current status from 3270 panels. You can manage those resources using pop-up windows.
- Proactive monitoring of IP resources

  You can proactively monitor critical IP resources, including:

  – IP stacks
  – Hosts
  – Interfaces
  – Routers
  – TN3270 servers

  For example, you can define a performance MIB to query within a router and define a threshold. When you exceed that threshold, the NetView program notifies you of the problem.
- IP connection monitoring and thresholding

You can monitor IP connections such as printer connections and apply policy definitions to determine if they are unavailable. You can choose to notify an operator or use automation to break the connection.

- Intrusion Detection Services

  You can use Intrusion Detection Services (IDS) support from the z/OS Communications Server to delete the following IDS events and take actions:

  – Scan detection
  – Attack detection
  – Traffic regulation for TCP connections
  – UDP receive queues

  Using the notification and inform policies, you can issue a message or generate an alert, an Event Integration Facility (EIF) event, email, or report based on a particular event type.

- IP trace

  Using 3270 panels or the Web browser, you can start and stop both component and packet traces.

You can use Table 26 on page 140 as you work through the installation tasks described in this section.

Table 26. AON IP Function Enablement

Task	**Job or Member Name**	**Reference**
Update the CNMSTUSR or C*xx*STGEN member to enable the IPMGT tower. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.	DSIPARM (CNMSTYLE)	*IBM Z NetView Installation: Getting Started*
Configure a UNIX command server	• DSIPARM (CNMSTYLE) • CNMSAMP (CNMSJUNX or CNMSSUNX) • DSIPARM (CNMPOLCY) • DSIPARM (CNMIPMGT)	"Defining UNIX Command Servers" on page 142
Configure SNMP support	DSIPARM (CNMSTYLE)	"Configuring SNMP Support for AON IP Functions" on page 142
Define at least one local MVS stack to the NetView program (See the note.)	DSIPARM (CNMPOLCY)	"Defining Local TCP/IP Stacks" on page 143
As needed, define stacks for remote NetView domains	• DSIPARM (CNMSTYLE) • DSIPARM (CNMPOLCY)	"Setting Up Cross-Domain Communication" on page 143
Set up monitoring of IP resources	DSIPARM (CNMIPMGT)	"Setting Up for Proactive Monitoring of IP Resources" on page 144
If needed, update community names	DSIPARM (CNMSCM)	"Resolving Community Names (Optional)" on page 145

Table 26. AON IP Function Enablement (continued)		
Task	**Job or Member Name**	**Reference**
Enable TCP/IP connection management	DSIPARM (CNMIPMGT)	"Managing TCP/IP Connections with AON IP Functions" on page 146
Define the TCP/IP connection monitoring policy	DSIPARM (CNMIPMGT)	"Defining TCP/IP Connection Monitoring Policy" on page 146
Enable Intrusion Detection Services	DSIPARM (CNMSTIDS)	"Enabling Intrusion Detection Services" on page 146
Enable TCP/IP component trace	DSIPARM (CNMSTYLE)	"Customizing TCP/IP Tracing" on page 147
Note: This can also be done by the NetView discovery manager.		

Defining AON IP Functions

Table 27 on page 141 summarizes the required and optional tasks necessary to implement a particular AON IP function.

Table 27. Customization by Function

Task	TCP/ IP Stack	UNIX Cmd Srvr	Rmt Stacks	Rmt Gtwys	Rmt Srvr Setup	Auto tasks	TN 3270 Srvr	SNMP Setup	SNMP Comm Name	Moni- toring Method
Stack monitoring	RD	O	O	O	O	R/O		R	O	
Ping command	RD		O	O	O					
Tracerte command	RD		O	O	O					
Manage IP connections	RD		O	O			O	R	O	
SNMP GET / SET/ and similar tasks	RD		O					R	O	
SNMP MIB groups	RD		O					R	O	
SNMP remote Ping	RD		O					R	O	
IP resource manager	RD	O	O					R	O	
IP trace support	RD		O							
Proactive monitoring	RD		O			R/O	O	R	O	O
IP connection monitoring and thresholding	RD					R/O		R	O	
Intrusion detection automation	RD	R				R/O				

Table 27. Customization by Function (continued)										
Task	**TCP/ IP Stack**	**UNIX Cmd Srvr**	**Rmt Stacks**	**Rmt Gtwys**	**Rmt Srvr Setup**	**Auto tasks**	**TN 3270 Srvr**	**SNMP Setup**	**SNMP Comm Name**	**Moni-toring Method**
Note: The abbreviations in the table have the following meanings: • R: Required task • RD: Required definition, coding by user is optional • O: Optional task • R/O: Required task, customization is optional										

Defining UNIX Command Servers

Determine the UNIX command servers that you need. UNIX command servers are optional for Intrusion Detection Services (IDS); see "Enabling Intrusion Detection Services" on page 146. For information about the requirements for the UNIX command server setup, see "Enabling the UNIX Command Server" on page 201.

To set up a UNIX command server:

1. Allocate an MVS initiator for the UNIX command server. If the command server is to be run as a started task, an MVS initiator is not required. Refer to the online command help for DEFAULTS and START for more information about running the UNIX command server as a started task.

   Note: The DEFAULTS.STRTSERV statement in the CNMSTYLE member specifies how the command server must be run.
2. Customize CNMSSUNX in CNMSAMP to enable the UNIX command server to run as a started task. This is the default.

   Note: If necessary, customize CNMSJUNX in CNMSAMP to enable the UNIX command server to run as a submitted job.
3. Create additional CNMSJ*xxx* or CNMSS*xxx* jobs for multiple TCP/IP stacks.
4. Authorize all IP management autotasks and any other operators to use the UNIX command servers. The default autotask names are AUTIPM1 and AUTIPM2 and are defined using the AUTOOPS policy definition in the CNMIPMGT member. See "Defining TCP/IP Autotasks" on page 144 for task names that need to be changed for your installation.

For the UNIX command server to start automatically for each stack during initialization, specify UNIXSERV=YES on the TCP390 policy definition in DSIPARM member CNMPOLCY. For more information, see "Defining Local TCP/IP Stacks" on page 143.

Configuring SNMP Support for AON IP Functions

Most AON IP functions are SNMP-based. See the comments in the CNMSTYLE member in DSIPARM to configure the NetView SNMP command.

To make updates, use the following statements in the CNMSTUSR or C*xx*STGEN member, and make any necessary modifications. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
COMMON.CNMSNMP.MIBS = all
COMMON.CNMSNMP.MIBPATH = /etc/netview/mibs:/usr/lpp/netview/v6r3/mibs
COMMON.CNMSNMP.timeout = 1
COMMON.CNMSNMP.retries = 5
```

For security considerations, see the *IBM Z NetView Security Reference*.

Defining Local TCP/IP Stacks

At a minimum, define at least one local MVS stack to the NetView program, if it is not defined automatically by the NetView discovery manager.

Use the TCP390 policy definition in DSIPARM member CNMPOLCY to define a local TCP/IP stack. The entire stack definition, or portions of it, are optional. The NetView program dynamically determines information about your stack, such as the IP address, host name, and TCP process name. If you want to use default values for all functions (such as Intrusion Detection Automation Services), you do not need to define a TCP390 statement for your stack. You need to define your stack if you want to override the default settings. The following example defines a local stack named NMPIPL10:

```
TCP390      NMPIPL10,
            IPADDR=9.67.50.52,
            COMMUNITYNAME=private,
            DOMAIN=LOCAL,
            UNIXSERV=YES,
            TCPNAME=TCPIP,
            HOSTNAME=NMPIPL10.raleigh.ibm.com,
             ...
```

Note:

1. The stack NMPIPL10.raleigh.ibm.com is known as NMPIPL10.
2. SNMP requests (for example stack monitoring processes) uses a community name of `private`.
3. The NetView domain managing this stack is the LOCAL domain. The local domain is the same domain where this policy definition resides.
4. This policy defines a UNIX command server. For more information, see "Defining UNIX Command Servers" on page 142.
5. You can specify either the IPADDR (recommended) or HOSTNAME parameter. If you do not specify the IPADDR parameter, the NetView program dynamically determines the IP address of the stack based on the HOSTNAME parameter. This takes additional processor cycles.
6. For security, if you specify COMMUNITYNAME, restrict access to the member containing your policy definitions.
7. TCPNAME defines the TCP/IP start procedure name. System symbolics can be used, for example, you can use TCPNAME=&CNMTCPN.

For more information about the TCP390 policy definition, see the *IBM Z NetView Administration Reference*.

Setting Up Cross-Domain Communication

Some IP functions (for example, connection management, SNMP functions, monitoring functions, IP tracing functions, and commands) support communication with remote NetView domains.

The discovery manager discovers all local stacks on a given system. Code TCP390 definitions for stacks on systems that are outside of your sysplex.

For example, to set up for cross-domain communication from NMPIPL10 (domain NTV70) to NMPIPL27 (domain NTV9D):

- In NTV70, define TCP/IP stacks on the remote domain. For example, to define a stack named NMPIPL27 in NetView domain NTV9D, use the following definition:

```
TCP390    NMPIPL27,
          IPADDR=9.67.50.41,
          DOMAIN=NTV9D,
          UNIXSERV=YES,
          TCPNAME=TCPIP,
          HOSTNAME=NMPIPL27.raleigh.ibm.com,
              ...
```

- In NTV9D, define the policy for the local stack:

```
TCP390    NMPIPL27,
          IPADDR=9.67.50.41,
          DOMAIN=LOCAL | lcldomid,
          UNIXSERV=YES,
          TCPNAME=TCPIP,
          HOSTNAME=NMPIPL27.raleigh.ibm.com,
           ...
```

For more information about TCP390 policy definitions, see the *IBM Z NetView Administration Reference*.

Defining TCP/IP Autotasks

Policy definitions are provided for two NetView autotasks that can be used for TCP/IP automation and management. Review the following statement in the CNMIPMGT member in DSIPARM:

```
AUTOOPS    TCPOPER,ID=(AUTIPM,2)
```

This defines AUTIPM1 and AUTIPM2 as the autotasks that are used by the AON IP functions. Modify this statement as necessary for your installation. If you change the autotask IDs, also make changes to DSIOPF to define your task IDs.

For more information about the AUTOOPS statement, see the *IBM Z NetView Administration Reference*.

Setting Up for Proactive Monitoring of IP Resources

You can use the NetView program to proactively monitor your critical IP resources based on policy definitions. You can use the following monitoring methods:

- "Pinging a Resource" on page 145
- "MIB Polling" on page 145
- "MIB Thresholding" on page 145

When proactive monitoring detects a failure, the following actions occur:

- The NOTIFY policy is used to determine the type of notification to be sent.
- A recovery monitoring timer is scheduled. This timer is scheduled as specified on the MONIT policy definitions. The timer checks the resource status and optionally sends a notification that the resource is still down.

Table 28 on page 144 shows the IP resource types you can monitor and the required policy definitions.

Table 28. Control File Entries

Resource	**Policy Definition**
Host	IPHOST
Stack	TCP390
Router	IPROUTER
Interface	IPINFC
Socket	IPPORT
Server	IPNAMESERV
Telnet server	IPTELNET
TN3270 server	IPTN3270
IP connection	IPCONN As shipped, IPCONN definitions are commented out.

For more information about policy definitions, see the *IBM Z NetView Administration Reference*.

Pinging a Resource

To perform a ping of a resource, code FORMAT=PING on its policy definition. If the ping response works, the resource status is NORMAL. If not, the resource status is DOWN.

MIB Polling

MIB polling uses SNMP to poll the interface table (ifTable) for the defined resource. The administration status is compared to the operational status. If one or more interfaces are expected to be up and are not, the resource is marked as degraded. Degraded does not mean that the resource is down.

To enable MIB polling, code FORMAT=SNMP, as shown in the following example:

```
IPHOST HOST01,SP=NMPIPL10,
OPTION=IP390
ACTMON=ALLHOSTS,
FORMAT=SNMP,
STATUS=(NORMAL,DEGR*),
HOSTNAME=host01.raleigh.ibm.com
```

The ACTMON=ALLHOSTS statement refers to the following statement that contains a common definition for monitoring all the IP hosts:

```
ACTMON ALLHOSTS,OPTION=IP390,INTVL=00:30,STATUS=NORMAL,
       FORMAT=PING
```

Using this definition, HOST01 is monitored every 30 minutes using a ping. The expected status is NORMAL (resource is active).

Additionally, you can add the following status parameter so that a status of degraded is not treated as a resource failure: STATUS=(NORMAL,DEGR*).

MIB Thresholding

MIB thresholding uses SNMP to query MIBs defined in the policy definition for the resource. You can define the MIB, its threshold value, and the threshold operator (greater than, less than, equal). When the proactive monitoring timer pops, the NetView program retrieves the MIB values and compares them to the policy definition for the resource. For example, to add a MIB threshold to the HOST01 resource and use the ipRoutingDiscards.0 MIB, code the following statements:

```
IPHOST HOST01,SP=NMPIPL10,
OPTION=IP390
ACTMON=ALLHOSTS,
FORMAT=SNMP,
STATUS=(NORMAL,DEGR*,THRESH*),
MIBVAR=(ipRoutingDiscards.0,GE,3),
HOSTNAME=host01.raleigh.ibm.com
```

In this example, if the MIB value is greater than or equal to 3, the resource status is set to THRESH. Notice also that THRESH is an acceptable status and therefore is not treated as a resource failure.

Resolving Community Names (Optional)

When monitoring resources using SNMP, the NetView program might need access to the community name for a resource. NetView SNMP reads MIB data from community-name protected resources using DSIPARM member CNMSCM.

To use CNMSCM for community name resolution, add an entry line for each host name to be resolved to a community name and then save the file. For more information, see the *IBM Z NetView Administration Reference*. To prevent unauthorized viewing or modification of CNMSCM, refer to the *IBM Z NetView Security Reference*.

For information about defining community names to TCP/IP, see the *z/OS Communications Server IP Configuration Reference*.

Managing TCP/IP Connections with AON IP Functions

TCP/IP connections can be established for any socket and can be established using a TN3270 server. With the AON IP functions, you can manage these connections from the 3270 interface, the NetView management console, or the Web browser.

To enable connection management, see Table 29 on page 146 and "Enabling TCP/IP Connection and Packet Trace Data Collection" on page 129.

Table 29. Connection Management

Connection	See
To a local TCP/IP stack	"Defining Local TCP/IP Stacks" on page 143 Set up your local stack definition with SNMP capability.
To a remote TCP/IP stack	"Setting Up Cross-Domain Communication" on page 143
From a TN3270 server	"Setting Up for Proactive Monitoring of IP Resources" on page 144 (TN3270 servers)

Because most stacks have numerous connections, consider limiting the amount of data that your operator must view. To do this, you can use one of the following methods:

- Code SESSTAT=NO on the IPPORT definition. All connection data for that port is ignored. Code this for ports that you do not want to manage.

  For example, if you do not want to manage connections with the NetView Web server interface task:

```
IPPORT DSIWBTSK,SP=NMPNET10,
                PORT=80,
                PROTOCOL=TCP,
                TCPNAME=NVPROCN,
                STATUS=NORMAL,
                FORMAT=PORT,
                SESSTAT=NO,
                DESC="NetView Web Browser Socket"
```

- For the Web browser, adjust the MAXCONN parameter. In the Web browser, you can limit the number of connections shown based on the value of the MAXCONN parameter on the TP390 DEFAULTS definition. Sample FKXCFG01 sets the limit to 1000. This limit applies to all connection types.

Defining TCP/IP Connection Monitoring Policy

You can monitor IP connections such as printer connections and apply policy definitions to determine if they are stopped. You can choose to notify an operator or use automation to break the connection.

To do this, customize the sample policy definitions in FKXCFG01:

```
*AUTOOPS  CONNOPER,ID=AUTCMON
*ACTMON IPCONN,INTVL=00:01:00
*NOTIFY IPCONN,ALERT=NO,MSG=YES,DDF=NO,INFORM=NO
*IPCONN TCPIP*,
```

Enabling Intrusion Detection Services

To enable Intrusion Detection Services (IDS), review the IDS statements in the CNMSTIDS member and update policy definitions in the CNMSTUSR or C*xx*STGEN member (see Table 30 on page 147). For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Table 30. IDS Support

Policy Definition	Update
TCP390	On the TCP390 stack policy definition, define your local TCP/IP stack and specify IDSAUTO=Y and optionally specify the IDSINTVL parameter.

NOTIFY	By default, the NOTIFY IDSAUTO policy is set up to issue alerts and messages for IDS events. You can optionally update this policy to enable IDS events to be forwarded to an Event Integration Facility (EIF) event receiver. For example:
	`NOTIFY   IDSAUTO,`
	`            ALERT=TEC,`
	`               MSG=YES,`
	`               DDF=NO`

NTFYOP	Use the NTFYOP policy definition to define which NetView operators receive IDS messages (class=64). For example:
	`NTFYOP   OPER1,`
	`         OPER='IDS-AUTO-SVCS',`
	`            CLASS=(64),HELDMSG=(I)`

INFORM	Update the Inform policy for email notifications (reports and responses to commands). For example:
	`GROUP    IDSOPERS,LIST=OPER1,OPER2,OPER3;`
	`INFORM     OPER1;`
	`  CONTACT CONNECTION=EMAIL,`
	`        INTERFACE=EZLESMTP,`
	`        ROUTE=IBPERSC@US.IBM.COM,`
	`        NAME=C. PERSON;`
	`INFORM    OPER2,...`
	`INFORM    OPER3,...`
	For more information, refer to sample EZLINSMP.

For more information about the policy definitions, refer to *IBM Z NetView Administration Reference*.

If Intrusion Detection Services is configured to use the Unix syslogd daemon, the syslog configuration file will need to include the following output definition statement:

```
*.xxxxxxxx.daemon.* /dev/console
```

In this statement, `xxxxxxxx` is replaced by the job name that is assigned to the syslogd daemon.

For details about syslogd configuration statements, see the *z/OS Communications Server: IP Configuration Reference*.

Customizing TCP/IP Tracing

Output writers can be used to store trace data for later analysis. You can customize the default output writer names for component or packet trace in the CNMSTYLE member. Review the following statements for your environment and make changes as appropriate in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

- Source JCL definitions for component and packet trace:

  ```
  COMMON.EZLTCPcTRACEwriter = CTTCP       // Component trace writer name
  COMMON.EZLTCPpTRACEwriter = PKTCP       // Packet trace writer name
  ```

  The Open Systems Adapter (OSA) trace does not have a default writer. You can use the packet trace writer for OSA packet trace data.
- Time interval to wait for a response:

```
COMMON.EZLIPTraceJCLWait   = 2     // ... for source JCL error response
```

For more information about IP tracing, see *IBM Z NetView IP Management*. For information about the IPTRACE command, which enables trace control through a panel interface, see the NetView online help.

Setting Up TCP/IP Connect Data Collection

TCP/IP connection management and connection security data is collected in several ways in the NetView program:

- Using the TCPCONN START and TCPCONN QUERY commands. This method requires an autotask that uses a socket interface. Inactive connection data that is displayed in the NetView workspaces in the Tivoli Enterprise Portal is collected using this method.
- Using the TCPCONN QUERYACT command. This method uses the EZBNMIFR interface of z/OS Communications Server. Active connection data that is displayed in the NetView workspaces in the Tivoli Enterprise Portal and connection data that is displayed in IPSTAT panels are collected using this method.
- Using the CONNSEC START and CONNSEC QUERY commands. This method collects security-related information for active connections. It can be used in conjunction with TCPCONN QUERYACT data in the Tivoli Enterprise Portal to view encryption and digital certificate information for active connections. The QUERY option makes this data available as NetView messages.
- Review, and if necessary, update the IPMGT tower in the CNMSTUSR or CxxSTGEN member for the IPTRACE function. For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*.

Packet trace data is collected using the PKTS START and PKTS QUERY commands. For each type of packet trace data that is collected, this method requires an autotask that uses a socket interface. Packet trace data that is displayed in IPTRACE panels is collected using this method. The IPTRACE packet trace data is formatted using the FMTPACKT command.

The following topics describe how to set up connection, security and packet trace data collection that requires an autotask and uses the socket interface.

- "Using the Socket Interface to Collect Trace and Connection Data" on page 148
- "Configuring TCP/IP Connection Data Collection" on page 149

Using the Socket Interface to Collect Trace and Connection Data

To use the socket interface to collect Open Systems Adapter (OSA) packet trace data and TCP/IP connection data, you must enable the NETMONITOR statement in your TCP/IP profile in z/OS Communications Server. To do that, add the following statement to your TCP/IP profile:

```
NETMONITOR TCPCONNSERVICE PKTTRCSERVICE NTATRCSERVICE ZERTSERVICE
```

where:

TCPCONNSERVICE
Enables the collection of TCP/IP connection data.

PKTTRCSERVICE
Enables the collection of IP packet trace data.

NTATRCSERVICE
Enables the collection of OSA packet trace data.

ZERTSERVICE
Enables the collection of security information for active connections.

If you do not want to collect data from one of these services, remove the appropriate parameter from the NETMONITOR statement.

In addition, if you want to collect zERT connection security data, add the ZERT parameter to the GLOBALCONFIG statement in the TCP/IP profile:

```
GLOBALCONFIG ZERT
```

For more information, see the *z/OS Communications Server IP Configuration Reference*. Especially note the SAF (RACF) requirement that the user ID (in this case associated with the NetView autotask described in the following text), or address space, must be permitted access to the relevant resources.

Configuring TCP/IP Connection Data Collection

About this task

To configure the collection of TCP/IP connection, security and packet trace data in the NetView product, perform the following steps:

Procedure

1. Ensure that the TCP/IP connection management databases have been defined.

   The TCP/IP connection management databases are defined using the CNMSJ004 job with input member CNMSI101.
2. Review and, if necessary, update the TCPIPCOLLECT tower in the CNMSTUSR or CxxSTGEN member, along with the appropriate subtowers (TCPCONN for TCP/IP connection management, PKTS for packet trace management, CONNSEC for security information management).

   For more information, see the *IBM Z NetView Administration Reference*. For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*.
3. Review and, if necessary, update autotasks to handle the collection of connection and packet trace data.

   To do this automatically during NetView startup, use the CNMSTYLE statements that are shown in Table 1 in the CNMSTUSR or CxxSTGEN member. For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*. In the function.autotask statements, stackname is the name of the TCP/IP stack.

   Table 31. Stack Connection and Packet Trace Data Collector Autotasks

Autotask Name	**Collector Autotask Function**	**CNMSTYLE Statement**
AUTOOPKT	OSA packet trace data	(TCPIPCOLLECT.OPKT)function.autotask.OPKT.stackname = AUTOOPKT
AUTOPKTS	Packet trace data	(TCPIPCOLLECT.PKTS)function.autotask.PKTS.stackname = AUTOPKTS
AUTOTCPC	Connection data for a stack	(TCPIPCOLLECT.TCPCONN)function.autotask.TCPCONN.stackname = AUTOTCPC
AUTTRA*n*	Packet trace instances	COMMON.PKTS.INSTANCE.POOL = AUTTRA,8
AUTOCSEC	Security data for a stack	(TCPIPCOLLECT.CONNSEC)function.autotask.CONNSEC.stackname = AUTOCSEC

4. You can also define these autotasks interactively using the TCPCONN DEFINE, PKTS DEFINE, and CONNSEC DEFINE commands.

For more information, see the online help of those commands or the *IBM Z NetView Command Reference Volume 1 (A-N)* and *IBM Z NetView Command Reference Volume 2 (O-Z)*.

5. If you want to start data collection automatically during NetView startup, add the following statements to the CNMSTUSR or CxxSTGEN member.

   For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*.

   - For connection data:

     ```
     (TCPIPCOLLECT.TCPCONN)INIT.TCPCONN = Yes
     ```

   - For packet trace data:

     ```
     (TCPIPCOLLECT.PKTS)INIT.PKTS = Yes
     ```

   - For OSA packet trace data:

     ```
     (TCPIPCOLLECT.OPKT)INIT.OPKT = Yes
     ```

   - For security data:

     ```
     (TCPIPCOLLECT.CONNSEC)INIT.CONNSEC = Yes
     ```

6. You can also start data collection interactively using the TCPCONN START, the PKTS START, and the CONNSEC START commands.

   For more information, see the online help of those commands or the *IBM Z NetView Command Reference Volume 1 (A-N)* and *IBM Z NetView Command Reference Volume 2 (O-Z)*.

7. To stop data collection, use the TCPCONN STOP, PKTS STOP and CONNSEC STOP commands.

   For more information, see the online help of those commands or the *IBM Z NetView Command Reference Volume 1 (A-N)* and *IBM Z NetView Command Reference Volume 2 (O-Z)*.

8. To define security passwords for the TCP/IP connection management databases, complete the following steps:

   a. Stop the TCP/IP connection management by using the TCPCONN STOP command.

   b. Modify the definition statements in CNMSI101 that define the TCP/IP connection management databases, changing them to include the specification of VSAM cluster passwords. Rerun the CNMSJ004 job using these modified statements to delete and redefine the TCP/IP connection management databases.

   c. Update the CNMSTPWD member in DSIPARM to include the passwords that you specified when redefining the TCP/IP connection management databases. The following example shows the PWD statements that define the passwords for the TCP/IP connection management databases, where p_password is the 1- to 8-character password for the primary database and s_password is the 1- to 8-character password for the secondary database:

      - PWD.DSITCONT.P = p_password
      - PWD.DSITCONT.S = s_password

   d. Restart the TCP/IP connection management by using the TCPCONN START command.

9. To enable automatic database maintenance for the TCP/IP connection data VSAM database, add a corresponding DBINIT command to the CNMSTUSR or CxxSTGEN member.

   See the comments in the TCPCONN section of the CNMSTYLE member for more information. For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*.

10. To filter the collection of connection and security data, add TCPCONN.KEEP, TCPCONN.DASD and CONNSEC.KEEP statements to the CNMSTUSR or CxxSTGEN member.

    See the comments in the TCPCONN and CONNSEC section of the CNMSTYLE member for more information. For information about changing CNMSTYLE statements, see the *IBM Z NetView Installation: Getting Started*.

Chapter 7. Configuring NetView Sysplex Management

To manage your sysplex from a single point of control with the Z NetView program, the NetView programs in your sysplex must participate in a z/OS XCF group and must be defined as having one of the following roles: master NetView program, master-capable NetView program, or basic NetView program. Additionally, discovery manager and DVIPA management data needs to flow to the master NetView program for display.

Using the NetView program for sysplex management requires the following configuration:

- "Configuring XCF Services" on page 151
- "Configuring DVIPA Management for Display at a Master NetView Program" on page 155
- "Configuring Discovery Manager for Display at a Master NetView Program" on page 156

For information about managing sysplexes with the NetView program, see *IBM Z NetView IP Management*.

Configuring XCF Services

The NetView program uses the z/OS cross-system coupling facility (XCF) to maintain an XCF group of NetView programs within a sysplex. The sysplex contains a master NetView program to which other group members forward resource data. XCF services are used in exchanging control information among the NetView programs and also in helping the NetView programs detect both other NetView programs that are members of the group and the identity of the group master.

Defining the NetView Role in a Sysplex

The role that a NetView program has is determined by a rank value that is configured by using the XCF.RANK statement in the CNMSTUSR or C*xx*STGEN member. A NetView program can have one of the following roles:

- A master NetView program is one to which other NetView programs in the sysplex forward discovered data. The user interfaces at the master NetView program can display data at a sysplex level. A master NetView program is also capable of acting as an enterprise master that has some types of data forwarded to it from systems outside its sysplex. For configuration information, see "Defining an Enterprise Master NetView Program" on page 153.
- A master-capable NetView program can become a master NetView program. These can serve as a backup to the master NetView program.
- A basic NetView program forwards data to the master NetView program but cannot assume the master role.

For more details about these roles, see the information about managing the NetView program as a sysplex application in *IBM Z NetView IP Management*.

The following values are valid:

- 250 indicates a master NetView program.
- 1 - 249 indicates a master-capable NetView program.
- 0 indicates a basic NetView program.
- -1 indicates a NetView program that is not a member of an XCF group. The data for this NetView program can only be viewed locally.

For information about using CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Many XCF CNMSTYLE statements are provided for configuring NetView XCF services, and default values are supplied for all statements. Minimal configuration is needed for a NetView program to use XCF. The only configuration that is required is to specify the NetView program that is to be the master by setting the rank for that NetView program to 250.

The default value of the XCF.RANK statement is 1, which means that a NetView program is master-capable by default. For any NetView program that either should be a basic NetView program or should not be in the XCF group at all, you need to change the XCF.RANK value.

To control the order in which master-capable NetView programs take over the master role when a master NetView program leaves the group, you must configure the XCF.RANK value for the master-capable NetView programs. The master-capable NetView program with the highest XCF.RANK value has the highest priority in taking over as the master NetView program.

A number of optional functions can also be configured for a master NetView program, including the following functions:

- Switching NETCONV connections or sessions from a former master NetView program to a new master NetView program
- Initiating NETCONV connections or sessions to specified NetView management console servers
- Initiating system jobs that should run on the system of the master NetView program, such as Resource Object Data Manager (RODM) jobs or jobs that are related to the Z NetView Enterprise Management Agent
- Running a user-defined CLIST when a NetView program assumes the master role
- Dynamically activating a dynamic virtual IP address (DVIPA) on the TCP stack of the master NetView program

For a complete list and descriptions of XCF-related definition statements, see the *IBM Z NetView Administration Reference*.

Consider the following items:

- NETCONV connections. XCF.TAKEOVER.CONVSNA*xx* and XCF.TAKEOVER.CONVIP*xx* statements define NetView management console servers to which a NetView program is to establish NETCONV connections if it assumes the master role. The NetView program always issues NETCONV commands when assuming the master role. The XCF.TAKEOVER.NETCONVS statement controls whether the NetView program also attempts to take over existing NETCONV connections or sessions from the previous master. These connections or sessions do not have to be defined with CONVSNA*xx* or CONVIP*xx* statements, nor are there any issues with taking over a connection or session that is defined on a CONVSNA*xx* or CONVIP*xx* statement.
- Monoplex configurations. If a master-capable NetView program is running on a monoplex or XCF-local system, there is no master NetView program until the INITWAIT period has expired. This might cause problems if a master NetView program needs to be available sooner. In these configurations, set the XCF.RANK value to 250 to make the master NetView program available at initialization.
- Operator control of rank. The XCF.RANK and XCF.TAKEOVER.DURATION definitions can be overridden by the PLEXCTL operator command. Operators can dynamically change the rank. If an operator sets the rank by using the PLEXCTL command, the operator request overrides the DURATION value that is in effect at the current master.
- RMTCMD requirements. When a NetView program assumes the master role, it sends RMTCMD commands to other NetView programs in the sysplex as part of setting up data forwarding. RMTCMD support needs to be enabled on the NetView programs in the sysplex group. Both SNA and IP are supported.

  If a NetView program has a TCP/IP stack with more than one IP address, add the following RMTSYN statement to the CNMSTYLE member or an included member for this NetView program to identify which stack IP address is to be used to communicate with this NetView program:

  ```
  RMTSYN.domain$$P.domain = IP_address optional_parameters
  ```

  Where:

  domain
  : The domain name of the local NetView program.

IP_address
: The IPv4 address to be used for communication with the local NetView program. If an alias is defined for the local NetView program on the remote master NetView program or on the enterprise master NetView program, that RMTSYN value or the resolved IPv4 address must match the IP address that is specified here. Aliases are defined using the RMTALIAS and RMTSYN statements.

optional_parameters
: Optional values for the local NetView program. For more information, see *IBM Z NetView Administration Reference*.

The RMTALIAS statement is not supported for this use of the RMTSYN statement.

For more information about the use of XCF by a NetView program and about the roles that a NetView program can assume in the sysplex group to which it belongs, see *IBM Z NetView IP Management*.

Defining an Enterprise Master NetView Program

A master NetView program can establish connections to NetView programs that are outside the sysplex in which it resides and have data forwarded to it. This master NetView program is known as an enterprise master NetView program. Before NetView V6R1, this type of configuration was used to support a single RODM data cache for an enterprise. As of NetView V6R1, this function supports an Active/Active enterprise master NetView program for the GDPS Continuous Availability solution. You can have a single enterprise master NetView program for the discovery manager and Active/Active functions, or you can have separate enterprise master NetView programs for these two functions. Because XCF services are not available outside the scope of a sysplex, you must configure the master NetView program to identify the NetView programs to be contacted that are outside the sysplex. Additional configuration is necessary for an Active/Active enterprise master NetView program.

If a contacted NetView program is a member of an XCF group in another sysplex, information about all the members in the group is passed back to the enterprise master NetView program that contacted it. Only one contact NetView program needs to be configured for each XCF group in a sysplex, although you can configure additional backup NetView programs. Note that these definitions also need to be available to any master-capable NetView programs in the XCF group of the enterprise master NetView program.

Use the following CNMSTYLE statements in the CNMSTUSR or C*xx*STGEN member to define the systems that are to be contacted when the enterprise master NetView program assumes the master role in the sysplex or stand-alone system in which it resides:

Statement
: **Description**

ENT.SYSTEMS.*name*
: List the RMTCMD aliases that are used by the enterprise master NetView program to contact other NetView programs within the enterprise. The other NetView programs can be stand-alone systems or members of an XCF group in another sysplex.

ENT.INT.*name*
: Set the pacing interval for data discovery commands that are sent to contacted NetView programs when a NetView program assumes the enterprise master role for discovery manager data.

For more information about the statements and their default values, see the *IBM Z NetView Administration Reference* or the comments in the CNMSTYLE sample. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

For more information about the GDPS Continuous Availability solution, see *IBM Z NetView for Continuous Availability Configuring and Using the GDPS Continuous Availability Solution*.

Coordinating with TCP/IP Configuration for DVIPA Support

TCP/IP communication with the master NetView program (for example, using the RMTCMD command) can be defined for the master NetView program. DVIPA support associates the RMTCMD address with the master NetView program so that RMTCMD requests always go to the master program.

When the master program moves in the sysplex, the address is dropped from the stack of the old master program and is dynamically defined on the stack of the new master program. When moving from a master NetView program to a master-capable NetView program, the same IP address can be used to reach the master NetView program.

To use DVIPA to represent the NetView application, define a block of IP addresses (VIPARANGE) in the TCP/IP configuration.

For information about enabling DVIPA support, see "Enabling DVIPA Management" on page 132.

Considerations for Master Switches

The master role in the sysplex can move from one NetView program to another for several reasons:

- Stopping of the master NetView program
- The master NetView program leaving the group because a STOP XCFGROUP command was issued
- Manual intervention by using the PLEXCTL operator command
- XCF.RANK CNMSTYLE statement changes in the CNMSTUSR or C*xx*STGEN member that are implemented by the RESTYLE XCF command

If more than one active NetView program is master-capable, the ranks of the NetView programs that are specified by the XCF.RANK statements in the CNMSTYLE member are used to determine the program that becomes the master program, with the higher rank having precedence. If more than one NetView program has the highest defined rank, a character comparison of the NetView domain name is used.

The XCF.TAKEOVER.INITWAIT and XCF.TAKEOVER.DURATION statements in the CNMSTYLE member can affect the process. These statements are intended to prevent unnecessary, rapid master changes as systems become active and join the sysplex group. If the sysplex group does not have a master (this can occur when a master-capable NetView program starts and joins the group before the intended master joins), the master-capable program waits for the interval specified by the XCF.TAKEOVER.INITWAIT statement before assuming the master role. This gives the intended master program time to start and join the group. When a master-capable NetView program assumes the master role, it retains that role for the time specified by the XCF.TAKEOVER.DURATION statement. Note that the duration value might prevent a NetView program that starts with a rank of 250 from assuming the master role.

When the master role changes, the following actions that are controlled by CNMSTYLE definition statements occur:

- The new master dynamically defines the DVIPA using the value that is specified (if one is defined) in XCF.MASTDVIPA statement in the CNMSTYLE member.
- If the interval that is specified on the XCF.TAKEOVER.DELAY statement is non-zero, a timer is set using the specified interval. The NetView program waits until the timer expires to send CNMEERSC discovery commands to other NetView programs in the sysplex group.
- XCF.proc.PROCSTR statements are processed. The NetView program checks the job specified on the statement (using an `MVS D A` command). If the job is not active, the NetView program generates an `MVS START` command for the job using the specified procedure string.
- If the XCF.TAKEOVER.NETCONVS statement is set to YES, the NetView program starts any NETCONV sessions that might be active on the former master NetView program. Note that if the former master NetView program is active, the new master NetView program sends `NETCONV ACTION=STOP` commands to the former NetView program to end the connections prior to restarting them.
- Any new NETCONV connections specified with XCF.TAKEOVER.CONVIP*nn* or XCF.TAKEOVER.CONVSNA*nn* statements are started.
- If a command list is specified on the XCF.TAKEOVER.CLIST statement, it is run.

Multiple NetView Groups

You can use the XCF.GROUPNUM statement to control the name of the XCF group that the NetView program joins. The GROUPNUM value is appended to DSIPLX to form the group name. This makes it possible to have multiple NetView XCF groups within a sysplex, allocating the NetView program among different groups.

Each NetView program can join only one DSIPLX*nn* group (the one defined by the XCF.GROUPNUM statement), although it can join additional user-defined groups. Each DSIPLX*nn* group has a master and is independent of any other DSIPLX*nn* group. Group members cannot use XCF services to communicate among groups.

Suppose that an installation runs separate automation and network NetView programs on each LPAR in their sysplex, and that the NetView programs are to be placed in separate XCF groups. The network NetView programs, which are named NVNE1, NVNE2 and NVNE3, are to be placed in an XCF group with the default group name of DSIPLX01. The automation NetView programs, which are named NVAU1, NVAU2, and NVAU3, are to be placed in an XCF group with a group name of DSIPLX02. NVNE1 and NVAU1 are the masters of their respective groups. To accomplish this configuration, the following XCF configuration statements are required:

- In the CNMSTUSR or C*xx*STGEN member for NVNE1:
  - `XCF.RANK=250`
- In the CNMSTUSR or C*xx*STGEN member for NVAU1:
  - `XCF.RANK=250`
  - `XCF.GROUPNUM=02`
- In the CNMSTUSR or C*xx*STGEN member for NVAU2:
  - `XCF.GROUPNUM=02`
- In the CNMSTUSR or C*xx*STGEN member for NVAU3:
  - `XCF.GROUPNUM=02`

Note that no configuration is required for NVNE2 or NVNV3.

This configuration results in the following XCF groups:

- The DSIPLX01 group has members NVNE1, NVNE2, and NVNE3.
- The DSIPLX02 group has members NVAU1, NVAU2, and NVAU3.

Configuring DVIPA Management for Display at a Master NetView Program

For DVIPA information to be displayed at the master NetView program, data must be forwarded from other NetView programs in the sysplex to the master NetView program. The master NetView program must be able to receive this forwarded information and process it for display at the appropriate user interface.

Additional configuration for displaying DVIPA information at the master NetView program can include the following items:

- Time to wait before the master NetView program updates DVIPA information in the Z NetView Enterprise Management Agent data space
- Processing of DVIPA events
- Capability to forward distributed DVIPA statistics to the master NetView program

To ensure that DVIPA information is forwarded to the master NetView program for display, ensure that the following configuration is complete on the non-master NetView programs:

- The DVIPA tower and any subtowers are enabled.
- A RMTCMD session or connection with the master NetView program is active and has an origin operator that matches the autotask that is defined in the CNMSTYLE function.autotask.XCF statement.

  Note: You can use the RMTCMD QUERY LCLAUTOS command to display the RMTCMD session or connection.

- For logging of distributed DVIPA statistics at the master NetView program, the CNMSTYLE DVIPA.STAT.Logto statement has a value of MasterOnly or ALL.

To ensure that DVIPA information is received and processed by the master NetView program, ensure that the following configuration is complete at the master NetView program:

- The DVIPA tower is enabled. The CNMSDVDS automation sample is included by the DSITBL01 sample when the DVIPA tower is enabled.
- DSIIF004I message automation, which is shipped in the CNMSDVDS automation sample, is active. The DSIIF004I message sends DVIPA data to the master NetView program
- BNH867I message automation, which is shipped in the CNMSDVDS automation sample, is active. The BNH867I message is used for distributed DVIPA statistics.
- DSIIF003I message automation, which is shipped in the CNMSDVDS automation sample, is active. The DSIIF003I message is used to notify the master NetView program that a DVIPA event was received for the specified system.
- If needed, the display of the Z NetView Enterprise Management Agent workspaces is enabled with the following actions:
  - Enable the necessary TEMA subtowers.
  - Review the DVIPA.Mast.EMARf.Delay statement in the CNMSTYLE member to specify how long NetView waits to write data to the Z NetView Enterprise Management Agent data space. This delay allows multiple updates to the data space to occur at one time.

Note: DVIPA connection and distributed DVIPA connection routing data is not forwarded to the master NetView program due to potentially large volumes of data. You can issue the 3270 commands and samples for these functions from the master NetView program to retrieve real-time data from other NetView domains.

Configuring Discovery Manager for Display at a Master NetView Program

For discovery manager information to be displayed at the master NetView program, data must be forwarded from other NetView programs in the sysplex to the master NetView program. The master NetView program must be able to receive this forwarded information and process it for display at the appropriate user interface.

To ensure that discovery manager information is forwarded to the master NetView program for display, ensure that the following configuration is complete on the non-master NetView programs:

- The DISCOVERY tower and any subtowers are enabled.
- A RMTCMD session or connection with the master NetView program is active and has an origin operator that matches the autotask that is defined in the CNMSTYLE function.autotask.XCF statement.

  Note: You can use the RMTCMD QUERY LCLAUTOS command to display the RMTCMD session or connection.

To ensure that discovery manager information is received and processed by the master NetView program, ensure that the following configuration is complete on the master NetView program:

- The DISCOVERY tower is enabled. The CNMSEPTL automation sample is included by the DSITBL01 sample when the DISCOVERY tower is enabled
- DSIIF002I message automation, which is shipped in the CNMSEPTL automation sample, is active. The DSIIF002I message sends discovery manager data to the master NetView program
- If necessary, TEMA subtowers are enabled to display the Z NetView Enterprise Management workspaces

Note: DVIPA connection and distributed DVIPA connection routing data is not forwarded to the master NetView program due to potentially large volumes of data. You can issue the 3270 commands and samples for these functions from the master NetView program to retrieve real-time data from other NetView domains.

Chapter 8. Defining and Maintaining Data Logs and Databases

This chapter includes the following steps that you can use to define the data logs:

- Maintaining the session monitor database
- Defining the JES job log
- Printing the consolidated audit, NetView, and z/OS log (Canzlog)
- Defining the Canzlog archive data sets
- Defining the network log
- Defining sequential access method logging support
- Printing the network log
- Installing the interactive problem control system

Maintaining the Session Monitor Database

The NetView session monitor component collects and stores data in a VSAM database defined during the installation of the NetView product. This database continues to grow in size unless you take action to periodically purge older data. You can do this by using the NetView NLDM PURGE or PURGEDB command with NetView timers.

A suggested strategy is to purge all records older than a particular age (for example 30 days), and to purge a subset of records older than a lower value (for example, 7 days). The criteria for identifying records in this second set is defined using PURGE exception statements in the CNMSTYLE member.

To set up an exception list in the CNMSTYLE member, use PEXLST*xx* statements in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
*NLDM.PEXLST01.A = NCP* *
*NLDM.PEXLST01.B = SSCP-SSCP
*NLDM.PEXLST02.A = HOST1 NCP*
*NLDM.PEXLST02.B = CP-CP
```

Change these statements to define the session monitor records that you do not want purged if they are less than 30 days old. For example, to exclude SSCP-SSCP and CP-CP records, specify the following statements:

```
NLDM.PEXLST01.A = SSCP-SSCP
NLDM.PEXLST01.B = CP-CP
```

To place these statements into effect, use the RESTYLE NLDM command.

The first step to automating the purge process is to setup a NetView timer to purge all records older than 30 days:

```
EVERY 001,PPT,ID=NLDMP1,NLDM PURGEDB RT/SESS BEFORE -30
```

The second step to automating the purge process is to setup a NetView timer to purge more recent records (older than 7 days):

```
EVERY 001,PPT,ID=NLDMP2,NLDM PURGEDB RT/SESS PEXLST01 BEFORE -7
```

If you want information about...	Refer to...
EVERY, PURGEDB, RESTYLE commands	*IBM Z NetView Command Reference Volume 1 (A-N)*

If you want information about...	Refer to...
PEXLST*xx* statement	*IBM Z NetView Administration Reference*

Defining the JES Job Log

If you are starting the NetView program specifying SUB=MSTR, the JES joblog is allocated by default when the NetView task DSIRQJOB requests a job ID for the NetView job. If you do not want the JES joblog, use the JesJobLog statement in the CNMSTUSR or C*xx*STGEN member, and modify it in the following way. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
JesJobLog=No
```

The default is Yes.

If you want information about...	Refer to...
JesJobLog statement	*IBM Z NetView Administration Reference*

Printing the Canzlog Log (Using the Canzlog API)

You can use the Canzlog API to create a REXX procedure to copy consolidated audit, NetView, and z/OS log (Canzlog) data to another file for subsequent printing. Figure 14 on page 160 contains an example procedure that extracts JES2 job start and job stop messages.

```
/* REXX Writing a subset of Canzlog data to another dataset.          */
/* Example shows extraction of start and stop messages on a JES2
   system.  Program ends if excessive time is used; see help for
   DEFAULTS CZBRWAIT and for PIPE CZR.                                */
msg2 = '$HASP395'          /* JES 2 end of job                        */
msg1 = '$HASP373'          /* JES 2 start of job                      */
ourBFS = 'MVSMSGS MSGID=('msg1 msg2')'
outputFile = 'USER.INIT(SEMSGS)'     /* adjust to suit                */
msgCount = 5000                      /* how many messages to fetch    */

'CNMECZFS CZSET' ourBFS              /* establish filter              */
IF RC = 0 THEN
 do
   "PIPE (NAME PRTTWO END %)",
   "| RR: CZR TIME FFFFFF" msgCount "| > '"outputFile"'",
   "% RR: | VAR report"
   parse var report actualCount . actualTime .
   say 'CZR got' actualCount 'messages, in ',
   actualtime * 0.004096 'seconds'
 end
Exit
```

Figure 14. Example Procedure to Copy Canzlog Data to Another File

Usage Notes:

1. You can use the DEFAULTS CZBRWAIT command to specify the number of seconds to wait for Canzlog browse responses.
2. Messages are copied to the SEMSGS member in the USER.INIT data set.
3. The CNMECZFS command sets up the filter to use for Canzlog data.
4. The PIPE CZR stage retrieves messages from the Canzlog log. The messages retrieved are identical to the original message at the time it was logged.

Printing the Canzlog Log (Using the PRINT command)

You can print messages from the Canzlog log to a data set by issuing the PRINT command from the NCCF operator's console, BROWSE window, or CANZLOG panel. If the PRINT command is issued in the BROWSE window or the CANZLOG panel, it inherits the filter specification from the BROWSE window and the CANZLOG panel. Users can specify the prefixes and details for the messages being printed, in addition to the messages themselves. Figure 1 contains the output for MVS messages that include message details and prefixes:

```
CNM102I CANZLOG PRINT FROM DOMAIN NTVAA AT '08/30/19 03:55:08'
FILTER:  MVS (z/OS) messages FILTER=MVSMSGS
PREFIXES: DATE TIME MTYPE SOURCE ORIGIN
DETAIL FOR MVS MESSAGE: OPDT MTYP JOBN OPID MCS DESC SYSN DOMN CONS
DETAIL FOR NETVIEW MESSAGE: OPDT MTYP JOBN OPID ATHR SYSN DOMN
------------------------------------------------------------------------------
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=0000
SYSN= NMP177   CONS=
08/28/19 04:09:01 E          NMP177   CNM617I NetView subsystem T630
initialized successfully
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=0000
SYSN= NMP177   CONS=
08/28/19 04:09:01 E          NMP177   CNM617I NetView subsystem CNMR
initialized successfully
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=1000
SYSN= NMP177   CONS=INTERNAL
08/28/19 04:09:01 E          NMP177   IEE252I MEMBER IEFSSN00 FOUND IN
SYS1.PARMLIB
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=1000
SYSN= NMP177   CONS=INTERNAL
08/28/19 04:09:01 E          NMP177   IEFJ003I DUPLICATE SUBSYSTEM SMS NOT
INITIALIZED
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=1000
SYSN= NMP177   CONS=INTERNAL
08/28/19 04:09:01 E          NMP177   IEFJ003I DUPLICATE SUBSYSTEM JES2 NOT
INITIALIZED
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=1000
SYSN= NMP177   CONS=INTERNAL
08/28/19 04:09:01 E          NMP177   IEFJ003I DUPLICATE SUBSYSTEM CNMR NOT
INITIALIZED
MVSMSG OPDT=08/28/19 04:09:01 MTYP=E JOBN=         MCS=0100 DESC=1000
SYSN= NMP177   CONS=
08/28/19 04:09:01 E          NMP177   DSN3100I -DB61 DSN3UR00 - SUBSYSTEM DB61
READY FOR START COMMAND
```

Figure 15. Message details and prefixes of an MVS message output

Figure 2 contains the output for NetView messages with message details and prefixes:

```
CNM102I CANZLOG PRINT FROM DOMAIN NTV9B AT '08/30/19 03:08:46'
FILTER:  Local NetView messages FILTER=NVMSGS
PREFIXES: DATE TIME MTYPE
DETAIL FOR MVS MESSAGE: OPDT MTYP JOBN OPID MCS DESC SYSN DOMN CONS
DETAIL FOR NETVIEW MESSAGE: OPDT MTYP JOBN OPID ATHR SYSN DOMN
--------------------------------------------------------------------------------
NVMSG  OPDT=01/12/19 14:00:05 MTYP=- OPID=DSILOGMT ATHR=Yes DOMN=NTV9B
01/12/19 14:00:05 - DWO854I Canzlog is active.
NVMSG  OPDT=01/12/19 14:00:05 MTYP=- OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:05 - -*--*--*--*--*--*--*--*--*--*--*--*--*--*- IBM
                    Z NetView V6R3 starting 12 Jan 2016 at
                    14:00:05 using CNMSTYLE style. NetView initialization
                    in progress for 4.31 seconds.
NVMSG  OPDT=01/12/19 14:00:05 MTYP=- OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:05 - TRACE
                    ON,TASK=ALL,OPTION=(DISP,PSS,QUE,STOR,UEXIT,SAF),SAFF=
                    (ALL),MONOPER=NONE
NVMSG  OPDT=01/12/19 14:00:05 MTYP=- OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:05 - DSI244I NETVIEW TRACE ACTIVE FOR TASK = ALL : MODE =
                    INT, SIZE = 4000 WITH OPTIONS = QUE PSS DISP STOR
                    UEXIT SAF, SAF TRACE = FAILURES FOR REQUEST TYPES =
                    AUTH EXTRACT FASTAUTH LIST STAT TOKENMAP TOKENXTR
                    VERIFY
NVMSG  OPDT=01/12/19 14:00:06 MTYP=- OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:06 - NetView converted 76 CMDMDL type statements from
                    DSICMDU. These statements will be overridden by any
                    similar statements in CNMCMD. Converted statements
                    are kept in a PIPE KEEP DSICMDU at PPT for two hours
                    after NetView startup.
NVMSG  OPDT=01/12/19 14:00:06 MTYP=C OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:06 C DSI234I DUPLICATE COMMAND 'ALCACHE' DETECTED
NVMSG  OPDT=01/12/19 14:00:06 MTYP=C OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:06 C CMDDEF.ALCACHE.MOD=BNJCACHE
NVMSG  OPDT=01/12/19 14:00:06 MTYP=C OPID=NTV9BPPT ATHR=No  DOMN=NTV9B
01/12/19 14:00:06 C DSI234I DUPLICATE COMMAND 'ARCLOG' DETECTED
```

Figure 16. Message details and prefixes of a NetView message output

Defining Canzlog Archive Data Sets

The consolidated audit, NetView, and z/OS log (Canzlog) contains NetView and system messages. You can set up the NetView program to archive Canzlog data. Archived data can be browsed with the active Canzlog log (active and archived data are one logical file).

The archiving function uses the following sequential data sets, which are defined by using CNMSTYLE statements:

- Message
- Index
- Primary index

Archive Data Sets

Only one NetView program at a time can archive Canzlog data on an instance of the z/OS system. All archive data sets for a particular collection have the same high level qualifier that is defined in the CNMSTYLE member.

Message data sets

Contains message data copied from the Canzlog log.

Each message data set is a sequential data set with fixed-length records. Each data set contains 8 megabytes (8M) of Canzlog data in 8192 1024-byte records.

A message data set is named after the first (oldest) record written to the file and has the following format:

```
hlq.CZYYMMDD.Thhmmssc
```

- The name of a message data set begins with the high level qualifier (*hlq*) that you specify on the `ARCHIVE.HLQ` statement in the CNMSTYLE member.
- The time stamp (Universal Time) includes the following information:
  - *YY*: year
  - *MM*: month
  - *DD*: day
  - *hh*: hour
  - *mm*: minute
  - *ss*: second
- The suffix (*c*) is a single character suffix (A-Z, 0-9, $, @, #) that is included only when the second and subsequent message data sets contain an oldest message that occurred in the same second as the oldest message in a previous message data set. This is used to make the message data set name unique.

Usage note: You can delete message data sets. If you delete these data sets and then later restore them, NetView operators can browse the data by specifying date and time ranges (using the TO and FROM operands on the BROWSE command). You can also create named filters to assist operators with browsing the message data sets. For information about creating named filters, see the CANZLOG command in *IBM Z NetView Command Reference Volume 1 (A-N)*.

Index data sets
Contain records that describe message data sets in the archive.

An index data set is a sequential data set with fixed-length records. This data set contains no more than 4096 80-byte records.

Each record describes one message data set and includes the following information:

- Version of the NetView program that wrote the data
- STCK value that indicates the date and time at which the first (oldest) message in the file was produced
- Low-level qualifier of the message data set

An index data set name has the following format:

```
hlq.DXyymmdd.HRhhMNmm
```

- The name of an index data set begins with the high level qualifier that you specify on the `ARCHIVE.HLQ` statement.
- The timestamp (Coordinated Universal Time, UTC) of the first (oldest) message that was written to the first message data set contained within the index data set:
  - *yy*: year
  - *mm*: month
  - *dd*: day
  - *hh*: hour
  - *mm*: minute

Usage notes:

1. Do not edit or delete the index data set.
2. Any index data sets that were created with the format `hlq.DXyymmdd.HRhh` are still valid and should not be deleted or edited, as with any other index data set.

Primary index data set
Describes the index data sets in the archive.

The primary index is a sequential data set with fixed-length records. It contains one 80-byte record for each index data set in the archive.

The primary index data set name has the following format:

```
hlq.NV.CANZLOG.INDEX
```

- The name of the primary index data set begins with the high level qualifier that you specify on the `ARCHIVE.HLQ` statement.

CNMSTYLE Statements

Use the statements in Table 32 on page 164 to define the Canzlog archive data sets. You can update archive-related information while the NetView program is active by using the RESTYLE ARCHIVE command. You can update the archive data set information even if the particular instance of the NetView program is not currently archiving Canzlog data.

Table 32. CNMSTYLE Statements That Define the Canzlog Archive Data Sets

Statement	Purpose
ARCHIVE.BROWSE.DATASPACES	Specifies the maximum number of data spaces that the NetView program can allocate for browsing the archived Canzlog data. This statement is exclusive with the ARCHIVE.BROWSE.MAXDSPSIZE CNMSTYLE statement.
ARCHIVE.BROWSE.MAXDSPSIZE	Specifies the maximum number of megabytes that the NetView program can allocate for browsing the archived Canzlog data. Specifying the ARCHIVE.BROWSE.MAXDSPSIZE statement is more granular than specifying the ARCHIVE.BROWSE.DATASPACES statement. This statement is exclusive with the ARCHIVE.BROWSE.DATASPACES CNMSTYLE statement.
ARCHIVE.HLQ	Specifies the high-level qualifier for data sets that contain Canzlog archive data. Also indicates that this instance of the NetView program can browse archived data, if there is data stored in an archive data set using the specified high-level qualifier.
ARCHIVE.INDEX.BLOCKSIZE	Specifies the block size for the primary index and index data sets
ARCHIVE.INDEX.DATACLAS	Specifies the data class for newly allocated archive index data sets
ARCHIVE.INDEX.MGMTCLAS	Specifies the management class for newly allocated archive index data sets
ARCHIVE.INDEX.SPACE	Specifies space allocation parameters for the primary index and index data sets
ARCHIVE.INDEX.STORCLAS	Specifies the storage class for newly allocated archive index data sets
ARCHIVE.INDEX.UNIT	Specifies the unit name (device type) for the volume serial number specified on the ARCHIVE.INDEX.VOLUME statement
ARCHIVE.INDEX.VOLUME	Specifies the volume serial number for the index data sets
ARCHIVE.MESSAGE.BLOCKSIZE	Specifies the block size for the message data sets

Table 32. CNMSTYLE Statements That Define the Canzlog Archive Data Sets (continued)	
Statement	**Purpose**
`ARCHIVE.MESSAGE.DATACLAS`	Specifies the data class for newly allocated archive message data sets
`ARCHIVE.MESSAGE.MGMTCLAS`	Specifies the management class for newly allocated archive message data sets
`ARCHIVE.MESSAGE.SPACE`	Specifies space allocation parameters for archive message data sets
`ARCHIVE.MESSAGE.STORCLAS`	Specifies the storage class for newly allocated archive message data sets
`ARCHIVE.MESSAGE.UNIT`	Specifies the unit name (device type) for the volume serial number or numbers specified on the `ARCHIVE.MESSAGES.VOLUMES` statement
`ARCHIVE.MESSAGE.VOLUMES`	Specifies 1 - 10 volume serial numbers for message data sets
`ARCHIVE.WRITE`	Specifies whether this instance of the NetView program is enabled for archiving. Note that only one NetView system in an LPAR archives data, even if more than one is enabled.

If you want information about...	Refer to...
ARCHIVE statements	*IBM Z NetView Administration Reference*

Deleting Archived Canzlog Data

When older message data sets are no longer needed for browsing, you can delete them or move them to a backing store. Do not delete any index data sets. You can determine the starting time and date for each message data set from the data set name. For information about index data sets and message data sets, see "Archive Data Sets" on page 162.

After deleting older message data sets, consider using the DEFAULTS CZTOPDAT command to change the default date for the FROM keyword for browsing Canzlog data. Doing this can prevent I/O errors if operators try to access deleted data.

Note: Certain actions, such as the TOP subcommand of the BROWSE command, cause the NetView program to automatically update the FROM keyword default date when I/O errors occur.

To restore deleted data sets, move them back into the space that is defined by the high-level qualifier. You can browse the restored data in the following ways:

- By specifying an appropriate date and time range with the FROM and TO keywords of the BROWSE command
- By using the CANZLOG panel to define and save a browse filter that includes an appropriate date and time range along with any other filter specifications that are needed.

To browse Canzlog data on a different system than the one where it was created, you must also copy the index data sets; see the information about capturing message log data in the *IBM Z NetView Troubleshooting Guide*.

Defining the Network Log

The network log is defined using the CNMSJ004 job with the CNMSI101 input member , and is used by the DSILOG task. The CNMSTYLE member determines whether the NetView program starts the network log facility task during initialization using the following statement:

```
TASK.DSILOG.INIT=Yes
```

If you change the TASK.DSILOG.INIT value to No (in CNMSTUSR or C*xx*STGEN), an operator must start DSILOG before any operator can use log browse. Otherwise, *domain_name*BRW is not able to complete initialization.

Defining Passwords for the Network Log

To define security passwords for the network log:

1. Stop the DSILOG task.
2. Modify the definition statements in CNMSI101 that define the network log, changing them to include the specification of VSAM cluster passwords. Rerun the CNMSJ004 job using these modified statements to delete and redefine the network logs.
3. Update the CNMSTPWD member in DSIPARM to include the passwords that you specified when redefining the network logs. The following example shows the PWD statements that define the passwords for the network logs:

   ```
        PWD.DSILOG.P = p_password
        PWD.DSILOG.S = s_password
   ```

   Where:

   p_password
      Is the 1- to 8-character password for the primary log.

   s_password
      Is the 1- to 8-character password for the secondary log.
4. Restart the DSILOG task.

Switching Recording Between Primary and Secondary Logs

Recording starts with the primary log and automatically switches to the secondary log when the primary log fills. With the LOGINIT statement, you can specify whether recording automatically switches back to the primary log when the secondary log fills. You can also specify that recording is to resume where it left off or restart at the beginning of the primary log.

In DSILOGBK, this LOGINIT statement is:

```
LOGINIT AUTOFLIP=YES,RESUME=YES
```

In the sample, when one log becomes full, recording automatically switches to the other log. The full log can then be printed or dumped while recording continues. CNMPRT (CNMSJM04) prints the log. If you do not want recording to switch automatically to the primary log, specify AUTOFLIP=NO. If you have only one log, recording always stops when the log is full.

In the sample, when the NetView program is started, recording resumes where it left off If you want recording to start at the beginning of the primary log, specify RESUME=NO.

If you want information about...	Refer to...
Printing logs	“Printing the Network Log ” on page 167

Printing the Network Log

You can use the CNMPRT (CNMSJM04) job to print the network log. The CNMSJM04 job was copied to your PROCLIB data set as CNMPRT during installation. The NetView startup procedure, the CNMPROC (CNMSJ009) procedure, also has commented-out JCL for printing the log.

Note: You can also use sample CNMS6214 to print the log.

To change the defaults used to print the network log, pass control statements to PGM=DSIPRT using the DSIINP DD statement. You can do this using one of two methods:

1. Create the following statements for a job stream or an instream procedure:

```
//DSIINP    DD *
            PASSWD=password
            OPER1,OPER2,NETOP1
            RANGE_DELIM=delimiter
            DATE_FORMAT=date_format
            TIME_FORMAT=time_format
            CONT_RANGE=[start_date] [start_time] delimiter [end_date] [end_time]
            TIME_RANGE=[start_time] delimiter [end_time]
            DATE_RANGE=[start_date] delimiter [end_date]
            TRANSTBL MOD=DSIEBCDC
```

2. Create a statement similar to the following to define a data set member to contain the print control statements, and put the preceding print control statements in this member. Ensure that the print control statements do not contain sequence numbers.

```
//DSIINP   DD   DSN=SYS1.PARMLIB(MEMBER),DISP=SHR
```

Only the second method applies for system-started JCL procedures.

Usage for Print Control Statements:

1. If you defined passwords for the network log, add a PASSWD statement:

   PASSWD= *password*

2. You can limit the output that is produced by specifying one or more operator IDs or tasks, separated by commas or blanks. For example, to limit the output to records related to operators OPER1, OPER2, and NETOP1, specify the following statement:

```
OPER1,OPER2,NETOP1
```

3. You can specify the range delimiter that you want to use when specifying a date and time range:

   RANGE_DELIM= – | *delimiter*

   where: delimiter is a 1-character symbol that is used to separate the start date and time from the end date and time. The *delimiter* must be different from any delimiter used in either the DATE_FORMAT or TIME_FORMAT statements. The default is a dash (-).

   Note: The RANGE_DELIM statement must precede the DATE_FORMAT or TIME_FORMAT statements.

4. You can specify the date format that you want to use when specifying a date value in subsequent range parameters:

   DATE_FORMAT= MM/DD/YY | *date_format*

   where:

   date_format
   Specifies the order of the month (MM), day (DD), and year (YY), and also 1-character non-alphanumeric delimiters between the values. This delimiter must be different from the delimiter in effect for TIME_FORMAT and RANGE_DELIM.

Note:

a. You can specify the month, day, and year in any order.

b. If you specify the month, day, and year, this indicates the month, day within the month, and year. If you specify only the year and day, this indicates the year and the day of the year (Julian date).

c. You can specify one or multiple M, D, and Y characters for the date. For example, the following format specifies the day, followed by the month and year using a dash for the delimiter:

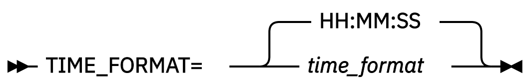
```
DATE_FORMAT=DD-MM-YYYY
```

The following format specifies a Julian date using a period for a delimiter:

```
DATE_FORMAT=YYYY.DDD
```

d. If you omit the DATE_FORMAT parameter, the default is MM/DD/YY.

5. You can specify the time format that you want to use when specifying a time value in subsequent range parameters:



where:

time_format
Specifies the order of the hour (HH), minutes (MM), and seconds (SS), and also 1-character non-alphanumeric delimiters between the values. This delimiter must be different from the delimiter in effect for DATE_FORMAT and RANGE_DELIM.

Note:

a. You can specify the hour, minutes, and seconds in any order.

b. You can specify one or multiple H, M, and S characters for the time. For example, the following format specifies the hour followed by the minutes, using a colon for the delimiter:

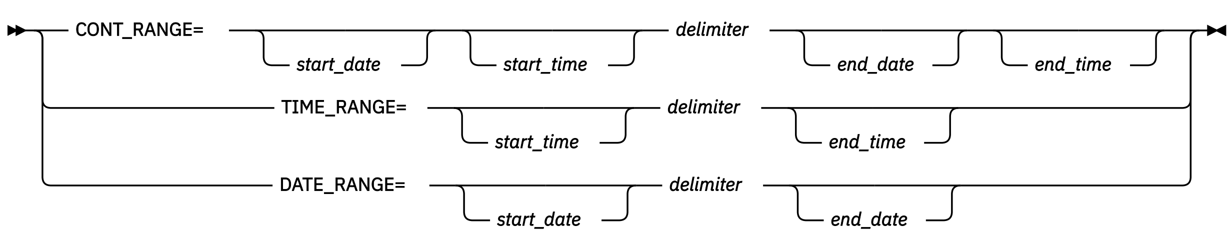
```
TIME_FORMAT=HH:MM:SS
```

c. If you omit the TIME_FORMAT parameter, the default is HH:MM:SS.

6. You can limit the output by specifying a starting and ending date and time range.

Use the CONT_RANGE parameter to specify a continuous range of time from one point in time to another.

Use the TIME_RANGE parameter to limit entries to a specific range of time for each day. Use the DATE_RANGE parameter to limit entries to a specific range of dates. You can specify both a TIME_RANGE and DATE_RANGE parameter.



where:

start_date | end_date
Specifies the date in the format defined by the DATE_FORMAT control parameter.

The default *start_date* is the earliest date in the log. The default *end_date* is the last date in the log.

start_time | end_time
Specifies the time in the format defined by the TIME_FORMAT parameter.

The default *start_time* is 00:00:00 (midnight).

The default *end_time* is 23:59:59 (one second before midnight).

delimiter
Specifies the delimiter as defined on the RANGE_DELIM parameter. If you omitted the RANGE_DELIM parameter, use a dash (-) for the delimiter.

Note:

a. Specify DATE_FORMAT, TIME_FORMAT, and RANGE_DELIM before CONT_RANGE, DATE_RANGE, and TIME_RANGE.
b. Do not specify CONT_RANGE if you specify either TIME_RANGE, DATE_RANGE, or both.

Examples:

a. To limit entries to those dated from August 1, 2019, until August 10, 2019, use:

```
DATE_RANGE=8/1/19-8/10/19
```

b. To limit entries to those that occur from 7:00 A.M. to 5:00 P.M. (within DATE_RANGE, if specified), use:

```
TIME_RANGE=7:00:00-17:00:00
```

c. To limit entries to those that occur from August 7, 2019, starting at 10:00 A.M., until August 10, 2019, ending at noon, use:

```
CONT_RANGE=8/7/19 10:00:00 - 8/10/19 12:00:00
```

7. To support a non-EBCDIC character set, use a TRANSTBL statement in the CNMSTUSR or C*xx*STGEN member to specify the same module that is specified in the TRANSTBL statement in the CNMSTYLE member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

   ▶▶─ TRANSTBL ── MOD ── = ── *module* ─▶◀

   For example, if your system supports kanji, use the following statement:

```
TRANSTBL MOD=DSIKANJI
```

8. Any statement with an asterisk (*) in column 1 is considered a comment, and is ignored by DSIPRT.

Defining Sequential Access Method Logging Support

NetView sequential access method log support makes it possible to:

- Define a primary and secondary output data set
- Define one or more sequential log tasks
- Interface to the sequential log subtask

Basic Sequential Access Method (BSAM) is the sequential logging access method used.

The information discussed here only shows how to define sequential log tasks and data sets to your system.

If you want information about...	Refer to...
Deciding whether you want to use sequential logging support and how to use it	*IBM Z NetView Customization Guide*

Allocating and Defining a Sequential Log Data Set

For each sequential data set that NetView processes, there must be a corresponding DCB and DD statement in the NetView start procedure. The characteristics of the data set and device-dependent information can be supplied by either source. The DD statement must also supply data set identification, device characteristics, and, if necessary, space allocation requests.

The NetView program defines the data control block (DCB) information with a subset of its parameters to ensure that it can use BSAM to write variable blocked records to a physical sequential data set. You can tailor other parameters, such as BLKSIZE, to meet your needs. The following parameters are coded on the NetView DCB statement and cannot be coded on the DD statement:

DSORG=PS
RECFM=VB
MACRF=(R,W)
KEYLEN=0

Another way to allocate a sequential log data set is by using the ALLOCATE command, which can dynamically allocate a sequential log. The log is accessible by all NetView tasks just as if you had coded a DCB and DD statement in the NetView start procedure.

If you want information about...	Refer to...
The ALLOCATE command	*IBM Z NetView Command Reference Volume 1 (A-N)*

Block Size (BLKSIZE)

BLKSIZE is the maximum size of a block of records that can be written. A minimum of 150 bytes is required. If you do not specify the BLKSIZE, or if its value is less than 150 bytes, the NetView system sets the BLKSIZE to 4096 bytes, without notification. You can use the NetView program to tailor the BLKSIZE of the data set according to the needs of the data. If the NetView program is given an acceptable BLKSIZE, but the size is not valid for a particular data set, unpredictable results can occur.

The BLKSIZE for the primary and secondary data sets must be the same. The BLKSIZE of the primary data set is used to set the BLKSIZE of the secondary data set. The NetView program sets the LRECL 4 bytes less than the BLKSIZE. If the NetView program attempts to log a record that is too large for the BLKSIZE you have defined, message CNM484I is issued, the record is truncated, and processing continues.

BLKSIZE affects the performance of the sequential log function. The size of the output buffer and the frequency of sequential log requests determine the number of I/O requests.

Note:

1. A date and time header record is written to your sequential log at the beginning of each block of records. You can alter the format of this record by coding the XITBO exit routine.
2. The first 2 bytes of this record contain a flag that is used when the log is resumed. Do not change these 2 bytes if you ever resume this log.

If you want information about...	Refer to...
XITBO (BSAM output exit routine)	*IBM Z NetView Programming: Assembler*

Data Set Disposition (DISP)

You can define the data disposition (DISP). DISP controls the status of the data set and shows what is to be done with it at the end of the job. Allowing the data set to be shared permits read access to the sequential log data set by other jobs.

Defining the Sequential Logging Function

To use the sequential logging function, use the following task statements in the CNMSTUSR or C*xx*STGEN member, and modify them as needed. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
*TASK.SQLOGTSK.MOD=DSIZDST
TASK.SQLOGTSK.MEM=SQLOGMEM
```

```
TASK.SQLOGTSK.PRI=2
TASK.SQLOGTSK.INIT=N
```

SQLOGMEM is the name of the member in DSIPARM that specifies the initialization parameters for the sequential logging task SQLOGTSK. The following definitions are the initialization definitions:

DSTINIT FUNCT=OTHER
Include this statement, and code FUNCT=OTHER.

DSTINIT DSRBO=1
The system default is 3, but, for this task, use only 1.

DSTINIT PBSDN=SQLOGP
This is the primary log DDNAME and must be the same name specified on the DD statement in the CNMPROC (CNMSJ009) member or defined by the ALLOCATE command.

DSTINIT SBSDN=SQLOGS
This is the secondary DDNAME and must be the same name specified on the DD statement in CNMPROC (CNMSJ009) or defined by the ALLOCATE command.

DSTINIT XITBN=*xxxxx*
This is the data set initialization routine.

DSTINIT XITBO=*xxxxx*
This is the sequential log output exit routine.

LOGINIT AUTOFLIP=YES
This permits the NetView system to switch from a secondary data set that is out of space to the primary data set. The NetView system always switches from the primary to the secondary if the out-of-space condition occurs on the primary data set.

LOGINIT RESUME=NO
This tells the NetView system not to resume processing of the sequential log data sets at task startup. If you code RESUME=YES, the NetView program determines which of the two data sets (PBSDN or SBSDN) was last used for sequential logging. Later logging of data is appended to that data set. After the initial RESUME, any switching of data sets, for a manual switch or automatic switch (AUTOFLIP), begins writing records at the top of the output data set. The previous data is erased.

Note: Code RESUME=NO for the first use of the log data set. This causes the NetView program to initiate the data set.

DSIPARM member CNMCMENT contains the following CMDDEF statements for the sequential logging function:

```
CMDDEF.DSIBSWCP.MOD=DSIBSWCP
CMDDEF.DSIBSWCP.TYPE=D
CMDDEF.DSIBSWCP.SEC=BY
CMDDEF.DSIZBSQW.MOD=DSIZBSQW
CMDDEF.DSIZBSQW.TYPE=RD
CMDDEF.DSIZBSQW.PARSE=N
CMDDEF.DSIZBSQW.SEC=BY
```

If you want information about...	Refer to...
The DSTINIT statement	*IBM Z NetView Administration Reference*
The NetView installation exits	*IBM Z NetView Programming: Assembler*

CNMPROC (CNMSJ009)

Figure 17 on page 172 is an example of a sequential log task, USRSQLOG, using a tape (TAPEOUT) as the primary output data set, and a DASD data set as the secondary data set. The DD statement gives the NetView program access to the sequential log data sets. This example also illustrates how to use BLKSIZE and DISP with DD statements.

Note: Because of device dependencies, certain combinations of primary and secondary database definitions might not be allowed in your system environment.

```
//*CNMSJ009 JOB 'ACCOUNTING INFORMATION','NETVIEW STARTUP PROC',
//*  CLASS=A,MSGCLASS=A,MSGLEVEL=(1,1)
⋮
//NETVIEW  EXEC PGM=&PROG,TIME=1440,
//         REGION=&REG.K,PARM=(&BFSZ.K,&SLSZ),
//         DPRTY=(13,13)
              DCB=(BLKSIZE=200)
⋮
//BNJ36SE  DD   DSN=&VQ1..SA&SA..BNJ36SE,
//         DISP=SHR,AMP=AMORG
⋮
//TAPEOUT  DD  DSN=data_set_name,DISP=(,KEEP),
//         VOLUME=(PRIVATE,RETAIN,,,SER=tape#),
//         UNIT=unit addr,
//         LABEL=(,NL),
//         DCB=(BLKSIZE=200)
//DASDOUT  DD  DSN=data_set_name,DISP=SHR,
//         VOLUME=SER=serial_number,
```

Figure 17. Example of a Sequential Log Task

Installing the Interactive Problem Control System Exit Routine

The interactive problem control system (IPCS) is a component of MVS that you can use for diagnosing software failures. IPCS makes it possible to:

- Format and display dump data
- Locate modules and control blocks
- Validate control blocks
- Check certain system components

IPCS also provides a verb exit interface whereby a verb exit routine can be written to generate a unique diagnostic report that is not currently available in IPCS. For more information about IPCS, refer to the *Interactive Problem Control* library.

The NetView program supplies an IPCS verb exit routine, CNMIPCS, that you can use to analyze dumps of the NetView program from an MVS system.

The NetView IPCS code must be installed in the data set defined by a CNMLINK DD statement. If you place CNMLINK in the LNKLST, the IPCS automatically has access to the code. If you have not included CNMLINK in LNKLST, remember to STEPLIB to this code in the TSO LOGON procedure that you use with IPCS.

The NetView IPCS code supports an ISPF panel interface. These panels need to be installed in the data set defined by an SCNMPLIB DD statement in NetView program. If you want TSO users to use the panel interface, concatenate this data set to the ISPPLIB DD statement in the appropriate TSO LOGON procedure.

The following example shows this process:

```
//IPCSPROC EXEC PGM=IKJEFT01,DYNAMNBR=70,REGION=3072K
//STEPLIB  DD   DSN=NETVIEW.V6R3M0.CNMLINK,DISP=SHR
//         DD   DSN=SYS1.MIGLIB,DISP=SHR
//ISPPLIB  DD   DSN=ISP.SISPPENU,DISP=SHR
//         DD   DSN=ISF.SISFPENU,DISP=SHR
//         DD   DSN=NETVIEW.V6R3M0.SCNMPLIB,DISP=SHR
//....
```

Note: If you have included the CNMLINK data set in your STEPLIB concatenation, and subsequently include it in the LNKLST concatenation, remove the STEPLIB statement from your TSO LOGON procedure.

If you want information about...	Refer to...
IPCS	*IBM Z NetView Troubleshooting Guide*

Chapter 9. Centralizing Operations

This section includes steps that you can use to centralize your operations.

Forwarding Data to Architectural Focal Points

NetView architectural focal point support is based on the focal point architecture described in the SNA library. With this architecture, the sender of the data is an *entry point application* and the receiver is a *focal point application*. The data is broken down into *categories*, for example ALERT and OPS-MGMT are categories of data. The entry point and focal point applications can be provided with the NetView program or defined by users. Data is sent (forwarded) from an entry point to its focal point over the MS transport. The entry points and focal points need not be NetView programs, for example an entry point NetView program can send alerts to a non-NetView product such as AS/400. Products that conform to the architecture can serve as a focal point or entry point for the NetView program.

When defined, an entry point NetView program can send data to its focal point over the MS transport, and a focal point NetView program can receive data from its entry points over the MS transport. The following sections explain the definitions necessary to define the NetView program as an architectural entry point application and an architectural focal point application for the OPS-MGMT, ALERT, and user-defined categories.

Because data is sent to architectural focal points over the MS transport, when switched lines are used, the NetView program does not perform the dial to establish the connection. Dialing is done by the VTAM program. Also, the NetView program does not control whether the MS transport uses persistent or nonpersistent sessions. Use the VTAM program to make this decision.

If you want information about...	Refer to...
Architectural focal point concepts and applications	*IBM Z NetView Automation Guide*

Forwarding Operations Management Data through LU 6.2

The operations management support function enables applications that are supplied by IBM or written by users to send architectural operations management commands and requests to remote systems for processing, and to receive operations management reports from those remote systems.

In cooperation with the focal point support function, operations management support also enables a served application in an entry point node to be informed of the identity of the focal point for unsolicited operations management data. The served application sends operations management data to a focal point using the management services (MS) transport.

The CNMSTYLE member contains the following MS transport task statement:

```
TASK.DSI6DST.INIT=Yes
```

If you want information about...	Refer to...
The MS transport	"Defining Management Services Transport" on page 106
The operations management support function	*IBM Z NetView Application Programmer's Guide*

To define a focal point for operations management data, use the DEFFOCPT and DEFENTPT statements. Use the DEFFOCPT or DEFENTPT statement at the entry point, but you do not need to use either statement at the focal point.

DEFFOCPT Statement

The DEFFOCPT statement defines primary and backup focal points for operations management data. To define a focal point for operations management data, add or uncomment the DEFFOCPT statements in DSI6INIT.

Note: DSI6INIT is the initialization member for the DSI6DST task.

The following statements are the relevant DEFFOCPT statements in DSI6INIT:

```
* DEFFOCPT  TYPE=OPS_MGMT,PRIMARY=NETA.CNM02,BACKUP=NETB.CNM99
* DEFFOCPT  TYPE=OPS_MGMT,BACKUP=CNM03
```

Where:

PRIMARY
Specifies the name of the domain that is used as the primary focal point.

TYPE=OPS_MGMT
Specifies that operations management data is sent to the focal point.

BACKUP
Specifies the name of the domain that is used as the backup focal point.

OVERRIDE
Specifies that all DEFFOCPT statements are used at initialization regardless of whether any focal point details for this category are found in the VSAM save/restore database.

Uncomment these statements and change the primary and backup focal point names to match your network names.

DEFENTPT Statement

Use the DEFENTPT EPONLY statement in DSI6INIT to set up the operations management function as an entry point or a focal point. The DEFENTPT statement only applies to the operations management category. The DEFENTPT statement is:

```
* DEFENTPT   EPONLY=NO
```

Where:

EPONLY=NO
Specifies that this host is a focal point for operations management data, and an entry point. NO is the default.

If you define a focal point using the DEFFOCPT statement at this host, the DEFENTPT statement is automatically set to EPONLY=YES.

If you use the DEFENTPT statement to define your host as an entry point, you can use the CHANGE keyword on the FOCALPT command to define a focal point (without using a DEFFOCPT statement). Here, issue the FOCALPT CHANGE command from a focal point or a FOCALPT ACQUIRE command from the entry point to establish a focal point relationship for operations management data.

Forwarding Alerts through LU 6.2

The alert function requires that the DSI6DST task be active. The hardware monitor BNJDSERV task must also be active.

You can use the hardware monitor recording filters to choose which alerts the NetView program needs to forward. The ROUTE filter selects alerts for forwarding. However, an alert must pass the ESREC and AREC filters before it goes to the ROUTE filter.

You can use the SRFILTER command to specify filter settings from the hardware monitor, or you can use the SRF action to specify them from the automation table. For more information about the SRFILTER command, refer to the online help.

A forwarded alert is filtered a second time on the focal-point system. The alert is always logged as an alert in the hardware monitor database of the focal point system (it cannot be blocked with the SRFILTER command or the automation table SRF action). The ROUTE filter cannot forward the alert a second time.

Setting Up an Alert Focal Point

The architectural alert support permits the hardware monitor to act as an ALERT-NETOP application. This enables the hardware monitor to receive alerts over LU 6.2 from entry point applications. You do not need to perform any setup to start this function, other than to ensure that the DSI6DST and BNJDSERV tasks are active.

Setting Up an Alert Entry Point

The architectural alert support permits the NetView hardware monitor ALERT-NETOP application to act as an EP-ALERT (entry point for category ALERT) application. This enables ALERT-NETOP to forward alerts over LU 6.2 to the current alert focal point.

By default, ALERT-NETOP sends alerts over LUC as described in "Forwarding Alerts through LUC" on page 179.

To send alerts over LU 6.2 (the recommended alert forwarding method), ensure that the following statement in the CNMSTYLE member is not commented out:

```
NPDA.ALERTFWD = SNA-MDS-LOGONLY
```

For information about the LOGONLY, AUTHRCV, and SUPPRESS options, refer to the NPDA.ALERTFWD statement in the *IBM Z NetView Administration Reference*.

Uncommenting the NPDA.ALERTFWD statement enables ALERT-NETOP to act as an entry point. This enables ALERT-NETOP send alerts to its focal point. To define the focal point that receives these forwarded alerts, uncomment the following DEFFOCPT statement in DSI6INIT, replacing the primary focal point name of NETA.CNM02 with your specified focal point name.

```
*  DEFFOCPT TYPE=ALERT,PRIMARY=NETA.CNM02
```

You can specify one to eight backup focal points.

If your specified alert focal point is typically going to be a non-NetView product, such as an AS/400, the non-NetView product might not receive all alerts that the NetView program sends, because the NetView program might send alerts that do not conform to the SNA library architecture (and the receiving product does not know how to process them) or the non-NetView product does not have various subsets of the architecture. Refer to the *IBM Z NetView Automation Guide* for more information.

After the ALERTFWD and DEFFOCPT statements are specified, when you next restart the NetView program the hardware monitor ALERT-NETOP application forwards all alerts it receives to the alert focal point defined in the DEFFOCPT statement, if the focal point is available.

Setting up an Intermediate Node Alert Focal Point

When the ALERT-NETOP application receives alerts that were sent from entry points over LU 6.2, ALERT-NETOP can forward these alerts again to its alert focal point over either LU 6.2 or LUC. Only alerts received over LU 6.2 can be sent again; alerts received over LUC are never sent again. Because ALERT-NETOP receives alerts from entry points and forwards alerts to its focal point, the NetView program is an *intermediate node* alert focal point.

Setting up an Intermediate Node to Forward Alerts through LU 6.2

To set up an intermediate node to forward alerts over LU 6.2, see "Setting Up an Alert Entry Point" on page 175. Notice that the setup for an intermediate node to forward alerts over LU 6.2 is exactly the same as the setup for an entry point to forward alerts over LU 6.2. This is because the intermediate node is itself an entry point.

CNMSTYLE

When a NetView intermediate node receives an alert over LU 6.2, alert data is recorded to the hardware monitor database. You might want the alert to simply pass through the intermediate node without alert data being recorded on the database. To specify this, the following ALRTINFP statement is defined in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
NPDA.ALRTINFP.RECORD = Yes
```

See the *IBM Z NetView Administration Reference* for more information about the ALRTINFP statement. ALRTINFP applies only when alerts are received over LU 6.2 and then sent again over LU 6.2. Use the default of ALRTINFP RECORD, which records alert data to the hardware monitor database.

Setting up an Intermediate Node to Forward Alerts through LUC

To set up an intermediate node to forward alerts over LUC, see "Forwarding Alerts through LUC" on page 179. While it is possible to have an entry point NetView program forwarding alerts over LU 6.2 and an intermediate node NetView program forwarding alerts over LUC, it is recommended that all NetView nodes use LU 6.2 to forward alerts.

Additional Considerations for Forwarding Alerts through LU 6.2

Additional considerations for forwarding alerts over LU 6.2 include:

- TAF

  Operators at the centralized host can perform problem determination by accessing the remote host using terminal access facility (TAF) or cross-domain function.

  Additional TAF source LUs might be required, depending on the number of operators that access remote hosts using the terminal access facility. For more information, see "Defining the Terminal Access Facility" on page 183.
- Hardware monitor

  The hardware monitor tasks must be active to forward alerts. Enter the STARTCNM NPDA command to start the hardware monitor tasks if they are not active.

  The hardware monitor must be active on the focal point host for GMFHS to provide the correct status for native resources. The hardware monitor must also be active on every distributed system that supports service points used to collect status for the native resources.

Forwarding Alerts Using TCP/IP

When you want to receive alerts over a TCP/IP connection, initialize the DSIRTTR task. The DSIRTTR task works with DSICRTR. The following statements are the related statements in the CNMSTYLE member:

RTT.PORT
: Specifies the port that is used by the status focal point host for TCP/IP communication. The default is 4021.

RTT.SOCKETS
: Specifies the maximum number of sockets this status focal point host can use for connecting to programmable workstations. The default is 50.

RTT.TCPANAME
: Specifies the TCP/IP application procedure name that the status focal point host uses. This is a required keyword for the TCP/IP function.

You can start the DSIRTTR task automatically during NetView initialization by using the following task statement in the CNMSTUSR or C*xx*STGEN member and changing INIT=N to INIT=Y. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
TASK.DSIRTTR.INIT=Y
```

Forwarding User-Defined Data through LU 6.2

Like all architectural focal point functions, user-defined entry point and focal point applications require that the DSI6DST task, described in "Forwarding Operations Management Data through LU 6.2" on page 173, be active.

Setting Up a User-Defined Focal Point

Your user-defined focal point application must register with the MS transport. When registered, entry point applications can forward data to it.

Setting Up a User-Defined Entry Point

Your user-defined entry point application must register with the MS transport with interest in your user-defined category of data (the focal point application name). When registered, the MS-CAPS application notifies your entry point application of your focal point netid.nau name, which MS-CAPS obtains from the DEFFOCPT statement. Your entry point application can then begin forwarding data to your focal point application. If your focal point application becomes unavailable, for example because of a line break, MS-CAPS notifies your entry point application that you have no focal point and MS-CAPS tries to acquire a backup focal point.

To define the focal point for your user-defined category, uncomment the following DEFFOCPT statement in DSI6INIT, replacing the primary focal point name of NETA.CNM02 with your preferred focal point netid.nau name and replacing USERCAT with your user-defined category name (which is identical to your user-defined focal point application name).

```
*  DEFFOCPT TYPE=USERCAT,PRIMARY=NETA.CNM02,OVERRIDE
```

You can specify one to eight backup focal points if you wish.

Defining the Entry Points in the Sphere of Control of a Focal Point

The *sphere of control* of a focal point is all of the entry points that have an established relationship with a registered focal point.

Using the sphere of control function, operators at a focal point can manage all focal point-entry point relationships, which includes the ability to perform the following tasks:

- Display all entry points in the sphere of control of a focal point.
- Delete entry points from the sphere of control of a focal point.
- Dynamically refresh the sphere of control environment.

The focal point sphere of control environment is defined in the sphere of control configuration file DSI6SCF. This file defines:

- Entry point names
- Primary focal point categories
- Primary focal point names
- Backup focal point names (optional)

The sphere of control manager (SOC-MGR) at the focal point reads the configuration file under the following circumstances:

- During NetView initialization to set up the focal point-entry point sphere of control environment
- When an operator issues the FOCALPT REFRESH command to update the sphere of control environment

Note: The SOC-MGR does *not* read the configuration file at initialization when both of the following conditions exist:

- Save/restore data exists
- DSISVRT is active

Define which entry points are to be explicitly obtained into the sphere of control of a focal point in the sphere of control configuration file DSI6SCF. Add a one-line statement in DSI6SCF for each entry point node. The format for each statement in the configuration file is:

```
EPNAME        FPCAT      PRIMARY FP      BACKUP FP
```

Where:

EPNAME
Is the name of the network and LU or VTAM CP name (netid.nau) where the entry point resides. For the NetView program, the LU name is the NetView domain name. *netid* is optional. If you specify an asterisk (*) for *netid*, VTAM determines the *netid* of the LU.

Note: If two nodes in two different networks have the same LU name, the one that VTAM finds can vary depending on the configuration of nodes that are active at a time.

FPCAT
Defines the focal point category. This definition makes it possible for you to specify the initial primary backup focal point settings for the specified category. The following categories are valid:

OPS MGMT
Specifies that the category is operations management.

ALERT
Specifies that the category is alert.

SPCS
Specifies that the category is SPCS.

user_defined
Specifies that the category is a user-defined category.

PRIMARY FP
Is the name of the network and LU or VTAM CP name (netid.nau) where the focal point resides.

BACKUP FP
Is the name of the network and LU or VTAM CP name (netid.nau) where the backup focal point resides. The backup focal point is optional.

The following example shows the entries in a sphere of control configuration file:

```
*  EPNAME           FPCAT        PRIMARY FP       BACKUP FP
*--------------    --------     -------------    ---------------
  NETA.CNM69        OPS_MGMT     NETA.CNM99       NETB.CNM18
  NETC.CNM01        OPS_MGMT     NETA.CNM99       NETB.CNM18
  NETC.CNM02        ALERT        NETA.CNM99       NETB.CNM18
  NETB.CNM20        OPS_MGMT     NETA.CNM99       NETB.CNM18
  NETB.CNM18        OPS_MGMT     NETA.CNM99       NETC.CNM02
  NETB.CNM16        ALERT        NETA.CNM01       NETB.CNM18
*
```

During initialization, the SOC-MGR reads the entries in the configuration file. If the focal point specified under PRIMARY FP is the same as the node on which you are running, the SOC-MGR attempts to explicitly obtain the entry point into its sphere of control.

For example, if the configuration file in the preceding example resides on NETA.CNM99, the SOC-MGR on NETA.CNM99 attempts to obtain all of the entry points listed under EPNAME, except NETB.CNM16, into its sphere of control.

Because the SOC-MGR ignores any statements where the primary focal point specified is a node other than the node on which you are running, you can define focal point-entry point relationships for your network in one configuration file, and use the same file on all systems to start the sphere of control environment.

If you want information about...	Refer to...
How sphere of control works with architectural focal points	*IBM Z NetView Automation Guide*

Forwarding Data to NetView Focal Points

The NetView program provides focal point support for the alert category that uses a private NetView-to-NetView protocol. These focal point methods are unique to the NetView program. With this NetView focal point support, the entry points and focal points must all be NetView programs. The NetView focal point support provides less function than the architectural focal point support because the NetView focal point support cannot use the services that are provided with the architectural focal point support. For example, NetView focal points cannot use the services provided by the MS-CAPS application (including the SOC-MGR support).

For information about the NetView forwarding function, refer to *IBM Z NetView Automation Guide*. After it is defined, an entry point NetView program can forward data to its focal point over the LUC transport and a focal point NetView program can receive data from its entry points over the LUC transport.

Forwarding Alerts through LUC

Note: Consider using the LU6.2 method to forward alerts. For more information, see "Forwarding Operations Management Data through LU 6.2" on page 173.

The LUC alert forwarding method is unique to the NetView program, and the entry point and focal point must be NetView programs.

The alert forwarding function of the NetView program enables centralized network management of distributed hosts. See the following information about setting up alert focal points and distributed hosts for alert forwarding.

If you want information about...	Refer to...
Using the alert forwarding function	*IBM Z NetView Automation Guide*

Setting Up an LUC Alert Focal Point

To forward alerts from a distributed host, see "Setting Up a Distributed Host" on page 180.

If you are using nonpersistent sessions, see "Establishing Nonpersistent Sessions" on page 181.

DSICRTTD

Define enough DSRBOs to handle alert forwarding, cross-domain communications, and distributed database retrieval. In DSICRTTD, DSRBO is a DSTINIT parameter that specifies the projected number of concurrent user requests for services from this data services task.

The value defined in the samples is 5, which allows one DSRBO for alert forwarding and four DSRBOs for any cross-domain communication involving this host or distributed database retrieval that is done from this host.

Note: The term *any cross-domain communication involving this host* means any cross-domain sessions initiated by this host, or any cross-domain sessions established with this host from another host over an LUC session.

To determine the number of DSRBOs that this alert focal point host needs, consider the number of cross-domain conversations where this host can be involved at a time, and the number of operators performing distributed database retrieval from this host.

Change the value of the DSRBO to the number required for this host.

CNMSTYLE

When LUC.CTL=GLOBAL is specified in the CNMSTYLE member, the NetView program ignores the specific LU names in the LUC.CNMTARG statements. If you have coded LUC.CTL=SPECIFIC, use the CNMSTUSR or C*xx*STGEN member to add a LUC.CNMTARG statement for each domain with which this host communicates using an LUC session. The following LUC.CNMTARG statements are in the CNMSTYLE member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
* LUC.CNMTARG.A=CNM01LUC
* LUC.CNMTARG.B=CNM02LUC YES
* LUC.CNMTARG.C=B01NVLUC NO
```

The second parameter on the LUC.CNMTARG statements overrides the default value for persistent sessions specified in the LUC.PERSIST statement.

LUC conversations are used for alert forwarding and distributed database retrieval, and hardware monitor and session monitor cross-domain conversations.

Setting Up a Distributed Host

If you are using nonpersistent sessions, see "Establishing Nonpersistent Sessions" on page 181.

CNMSTYLE

Ensure that the NPDA.ALERTFWD statement is commented out in the CNMSTYLE member or that it specifies NV-UNIQ. If you need to make changes, use the CNMSTUSR or C*xx*STGEN member; for information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
NPDA.ALERTFWD = NV-UNIQ
```

The NV-UNIQ option specifies that the NetView program forwards alerts over LUC. This is the default when the NPDA.ALERTFWD statement is commented out.

Alerts sent over LUC are forwarded only once, from the entry point (distributed host) to the focal point. The focal point cannot forward these alerts again, neither with LUC nor LU 6.2 alert forwarding. If you want the receiving focal point to forward the alerts it receives from entry points, use LU 6.2 alert forwarding.

DSICRTTD

Define enough DSRBOs to handle alert forwarding, cross-domain communications, and distributed database retrieval. The value defined in the samples is 5, which allows one DSRBO for alert forwarding and four DSRBOs for any cross-domain communication involving this host or distributed database retrieval that is done from this host.

To determine the number of DSRBOs this distributed host needs, consider the number of concurrent cross-domain conversations for this host.

Change the value of the DSRBO value to the number required for this host.

To specify the names of the primary and optional backup alert focal points, uncomment and change the focal point names to match your configuration in the following DEFFOCPT statement:

```
*  DEFFOCPT PRIMARY=CNM02LUC,TYPE=ALERT,BACKUP=CNM99LUC
```

Do not code the BACKUP operand in the DEFFOCPT statement if you are not using a backup host.

Additional Considerations for Forwarding Alerts through LUC

Additional considerations for alert forwarding include:

- Hardware monitor

  The hardware monitor tasks must be active to forward alerts. Enter the **STARTCNM NPDA** command to begin the hardware monitor tasks if they are not active.

The hardware monitor must be active on the focal point host for GMFHS to provide the correct status for native resources. The hardware monitor must also be active on every distributed system that supports service points used to collect status for the native resources

- NV-UNIQ/LUC alert focal point

  When an alert is forwarded with the NV-UNIQ/LUC method, the NetView program first forwards it to the primary focal point. If unsuccessful, the NetView program forwards it to the backup focal point. Note that the NetView program first tries to establish a session with the primary focal point, regardless of whether a persistent session with a backup focal point exists. If you do not define a backup, or if the NetView program cannot forward the alert to either the primary or backup focal point, only the entry point NetView logs the alert.

  NV-UNIQ/LUC alert focal points do not support focal point nesting. When an NV-UNIQ/LUC alert focal point receives alerts that were forwarded from a NetView entry point using LUC, the NV-UNIQ/LUC alert focal point does not forward such alerts again. Alerts that have been forwarded once with LUC cannot be forwarded a second time. If you need intermediate node alert focal points, consider using the SNA-MDS/LU 6.2 alert forwarding mechanism.

- Terminal access facility

  Operators at the centralized host can perform problem determination by accessing the remote host using terminal access facility (TAF) or cross-domain function.

  Additional TAF source LUs might be required, depending on the number of operators that access remote hosts using the terminal access facility. For more information, see "Defining the Terminal Access Facility" on page 183.

- Activating the links

  If you use leased lines, activate the links between alert focal points and distributed hosts. See the z/OS Communications Server library for additional information.

- CNM router

  The CNM router must be active at distributed and alert focal point hosts for LUC alert forwarding to work.

Establishing Nonpersistent Sessions

NetView LUC alert forwarding uses LUC sessions to forward alerts from distributed hosts to the focal point and to perform distributed database retrieval. Also, the hardware monitor and session monitor use LUC sessions to retrieve cross-domain data. Nonpersistent session support gives you the option of deactivating low-usage LUC sessions.

To define NetView-to-NetView LUC sessions as nonpersistent:

- Change the value of the session inactivity interval, or timeout interval, in the NetView constants module, DSICTMOD, using CNMS0055. The NetView program brings the session down when the interval of inactivity between sessions exceeds this value.

  Note: Reassemble DSICTMOD using CNMS0055 after making changes.

- In the CNMSTUSR or C*xx*STGEN member, define the domains to which your sessions are nonpersistent by coding the LUC.PERSIST statement, or specify NO on the LUC.CNMTARG statement for these domains. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

  You can define all LUC sessions as nonpersistent by using a global definition on the DSTINIT statement.

  You can also override this global name by individual domain on the CNMTARG statement. When you define PERSIST=NO on an individual LU statement, the LUC session from the host domain to the domain specified on the LU keyword is nonpersistent, and is brought down if it is inactive for the number of seconds specified in the timeout interval in DSICTMOD.

Examples

1. You are in domain CNM01, and want to establish a NetView-to-NetView LUC session with domain CNM02 that is stopped after 10 seconds of inactivity. Perform the following tasks:
   - In CNMSTUSR or C*xx*STGEN, specify:

     ```
     LUC.CNMTARG.B=CNM02LUC NO
     ```

   - In DSICTMOD, change the nonpersistent timeout interval from 0 to 10.
   - Reassemble DSICTMOD using CNMS0055.
2. You want all sessions originating from this domain to be stopped after 30 seconds of inactivity. Perform the following tasks:
   - In CNMSTUSR or C*xx*STGEN, specify:

     ```
     LUC.PERSIST=NO
     ```

   - In DSICTMOD, change the nonpersistent timeout interval from 0 to 30.
   - Reassemble DSICTMOD using CNMS0055.

   Note: You can also specify NO on an individual LUC.CNMTARG statement in CNMSTUSR or CxxSTGEN, overriding the default LUC.PERSIST statement in CNMSTYLE.

Defining Advanced Peer-to-Peer Networking Session Configurations

The NetView session monitor provides information about Advanced Peer-to-Peer Networking session configurations and session flow control data.

The location of the NetView program is very important in ensuring that all Advanced Peer-to-Peer Networking data is collected and available for viewing. Accomplish this by setting up LUC sessions.

LUC sessions must exist between endpoint nodes and interchange nodes within the same network. Without these LUC sessions, the session monitor at the endpoint node is missing some or all session configuration information. For example, at a subarea end node without an LUC session to the interchange node, the session monitor only has virtual route data, and does not have any RSCV data. With an LUC session to the interchange node, the session monitor at this subarea end node has RSCV data and virtual route data.

LUC sessions must also exist between interchange nodes in adjacent networks. Without these LUC sessions, the session monitor at the interchange node cannot get Advanced Peer-to-Peer Networking session configuration data from the adjacent network.

If the session monitor is placed at an interchange node, LUC sessions to the end nodes are not necessary because the interchange node has session configuration data. With this placement of the NetView program, LUC sessions are needed only to other interchange nodes in adjacent networks.

LUC sessions are necessary for obtaining session configuration and route data not available in the local NetView program. Use the following general rules for setting up LUC sessions:

- Set up an interchange node-to-interchange node (where interchange nodes are in different networks) LUC session if your session monitor is at one of these interchange nodes, and you need to see Advanced Peer-to-Peer Networking session configuration data from the adjacent network for sessions passing through the interchange node.
- Set up an interchange node-to-session end point node LUC session if your session monitor is at the session end node, and you need to see Advanced Peer-to-Peer Networking session configuration data from an end point in an adjacent network. This situation also requires LUC sessions between the interchange nodes.
- Set up an intermediate node-to-interchange node LUC session if your session monitor is at an intermediate node that is not an interchange node or a session end node. Since this intermediate node is not performing any boundary functions, it receives no session awareness (SAW) data. The LUC session is required for the SDOMAIN command.

If you want information about...	Refer to...
Data availability scenarios for Advanced Peer-to-Peer Networking sessions	The *IBM Z NetView Automation Guide*

Defining the Terminal Access Facility

The terminal access facility (TAF) lets an operator control any combination of subsystems from one terminal. The operator does not have to log off or use a separate terminal for each subsystem. The subsystem can be in the same domain or in another domain.

You can have two types of TAF sessions: operator-control sessions and full-screen sessions. Table 33 on page 183 illustrates the subsystems you can control through the NetView program using TAF, and the applicable session types.

Table 33. Subsystems Controlled Through TAF

Subsystem	Operator-control	Full-screen
CICS	X	X
IMS	X	X
HCF DPPX	X	X
HCF DPCX		X
TSO		X
DSX		X
NPM		X
SSP (THRU TSO)		X
TPF V2R4	X	X

In operator-control sessions, TAF acts like an SNA 3767 (LU type-1) terminal in session with CICS/VS, IMS/VS, or HCF, except TAF ends the session when a permanent error sense code (for example, 081C) is received. In this type of session, any transaction you can enter from a 3767 terminal attached directly to one of these subsystems can also be entered from the command facility panel. Operator-control sessions are also called 3767-type sessions or LU1 sessions.

Note: Data entered during an operator-control session is not translated from lowercase to uppercase.

In full-screen sessions, TAF acts like an SNA 3270 (LU type-2) terminal in session with CICS, IMS, HCF Version 2 Release 1, TSO, or a cross-domain NetView system. TAF lets full-screen applications operating on these subsystems use a NetView panel. The NetView operator can also enter commands and data as though the terminal were directly connected to the subsystem. Full-screen sessions are also called 3270-type sessions or LU2 sessions.

Defining Additional Source LUs

A01APPLS (CNMS0013) defines five operator-control sessions and ten full-screen sessions. The following definitions are the first three definitions for operator-control sessions:

```
TAF01000  APPL  MODETAB=AMODETAB,EAS=9,                X
                DLOGMOD=M3767
*               STATOPT='TAFAPPL 000'
TAF01001  APPL  MODETAB=AMODETAB,EAS=9,                X
                DLOGMOD=M3767
*               STATOPT='TAFAPPL 001'
TAF01002  APPL  MODETAB=AMODETAB,EAS=9,                X
                DLOGMOD=M3767
*               STATOPT='TAFAPPL 002'
```

The following definitions are first three definitions for full-screen sessions:

```
TF01#000 APPL  MODETAB=AMODETAB,EAS=9,                    X
               DLOGMOD=M2SDLCNQ
*              STATOPT='DYNAMIC TAF 000'
TF01#001 APPL  MODETAB=AMODETAB,EAS=9,                    X
               DLOGMOD=M2SDLCNQ
*              STATOPT='DYNAMIC TAF 001'
TF01#002 APPL  MODETAB=AMODETAB,EAS=9,                    X
               DLOGMOD=M2SDLCNQ
*              STATOPT='DYNAMIC TAF 002'
```

The names (such as TAF01F00) are the SRCLU (secondary LU) names used to start TAF sessions. The default SRCLU names used by the BFSESS and BOSESS command lists are derived from the operator's application (APPL) name. If you want all of your operators to use these command lists without specifying an SRCLU value, separate full-screen and operator-control SRCLU statements are required for each operator. The derived name is TAF, followed by the fourth and fifth characters of the APPL name, followed by O (operator-control) or F (full-screen), followed by the seventh and eighth characters of the APPL name.

When an operator issues a BGNSESS command, an SRCLU is dynamically allocated to that operator by a command list. Each operator requires a separate SRCLU. If you need more than five concurrent operator control session users or more than 10 concurrent full-screen session users, define additional SRCLUs. If you code a password on an SRCLU APPL statement (PRTCT=*nnnnn*), the password must be the same as the NetView password for that domain.

The MODETAB parameter points to AMODETAB (CNMS0001), the logmode table for both operator-control and full-screen sessions. The DLOGMOD operand points to an entry in AMODETAB (CNMS0001). Each entry is preceded by a description of the device it supports. Make sure the DLOGMOD operands for your SRCLUs point to the proper entries. To take advantage of graphics or color, use a logmode that includes query. To take advantage of larger screens, the screen size values in the TAF logmode must match the values specified in the logmode for the NetView terminal. For IBM 3290 terminals, use logmode MSDLCQ. TAF sessions always use SDLC logmode types, even on BSC terminals. For a complete list of logmode entries, review AMODETAB (CNMS0001).

Before establishing a full-screen session, TAF checks the bind parameters that the application sends. If the bind indicates that the application can write to an alternative screen, the alternative screen size in the TAF bind must match the alternative screen size in the NetView bind with the terminal.

For an operator-control session, the maximum RU size that can be received by TAF from the subsystem is 16 Kilobytes.

When defining a TAF terminal to an application (for example, IMS/VS), take one of the following actions:

- Use a bind that does not allow writing to an alternative screen.
- Use an alternative screen size to match the screen size of the NetView terminal used to start the TAF session.

Accessing the Customer Information Control System Using TAF

If you are accessing the Customer Information Control System (CICS) using TAF, define the SRCLUs to CICS.

An example of the parameters you can use to define an operator-control session to CICS is:

```
DFHTCT TYPE=INITIAL,APPLID=CICS1,.....
DFHTCT TYPE=TERMINAL,                                       X
      TRMIDNT=LU1,                                          X
      TRMTYPE=3767,                                         X
      RUSIZE=256,                                           X
      BUFFER=256,                                           X
      TIOAL=256,                                            X
        .
        .
        .
      NETNAME=TAF01O00,              (SRCLU)                X
        .
```

```
        .
        .
```

Note: Each RUSIZE, BUFFER, and TIOAL cannot exceed 256 bytes for each operator-control session. Refer to the CICS documentation for more information.

An example you can use to define a full-screen session to CICS is:

```
DFHTCT TYPE=TERMINAL,                                     X
      TRMIDNT=LU2,                                        X
      TRMTYPE=LUTYPE2,                                    X
        .
        .
        .
      NETNAME=TAF01F00,                  (SRCLU)          X
      LOGMODE=M2SDLCNQ,                                   X
        .
        .
        .
```

The NETNAME parameter refers to an SRCLU.

Accessing the Information Management System Using TAF

If you are accessing Information Management System (IMS) using TAF, define the SRCLUs to IMS.

An example of the parameters you can use to define an operator-control session to IMS is:

```
COMM  APPLID=IMS1,......          (APPLID definition)
TYPE UNITYPE=SLUTYPE1             (SRCLU OPCTL definition)
TERMINAL NAME=TAF01O00            (VTAM LU/NODE name)
NAME TAF01O00                     (IMS/VS LTERM name)
```

An example you can use to define a full-screen session to IMS is :

```
COMM      APPLID=IMS1,......        (APPLID definition)
TYPE      UNITYPE=SLUTYPE2         (SRCLU FLSCN definition)
TERMINAL  NAME=TAF01F00,                                     X
          MODEL=2,                                           X
          FEAT=(NOCD),                                       X
          OPTIONS=TRANRESP
NAME      TAF01F00
```

Note: If you specify a SEGSIZE or OUTBUF operand on the TYPE statement to IMS, it must match the RU size in the logmode table defined to VTAM.

Accessing TSO Using TAF

If you are accessing TSO using TAF, you must know the LU name for TSO, which is usually different from the ACB name. The LU name is the label on the first (principal) APPL statement defining TSO to VTAM in your VTAMLST. Refer to A01MVS (CNMS0047) for this label.

Note: Ensure the minor node names that define the TSO applications TSO001 - TSO999 are a derivative of the major node name that you used to define the TSO application statement.

Accessing CLSDST(PASS) Applications Using TAF

When using the BGNSESS command, operators need to use the application name (LU name) when this name is different from the ACB name. Aliases *cannot* be used.

When an operator logs on to an application that uses CLSDST(PASS), the application name is used by TAF to anticipate the LU name that is used for the operator session. It is required that the application name be an initial substring of the eventual operator session LU name. If, for example, CNMAA is an initial substring of CNMAA001, an operator session with LU name CNMAA001 is accepted by TAF for the application CNMAA. This pattern matches that used by TSO, the NetView program, and certain other applications to derive LU names for operator sessions. Use of long application names (especially eight character names) limits your ability to use TAF.

Using TAF with Default LU Names

If TAF is to be used on LUs with default names, add APPL statements to define the LUs available for use. These names must be defined if you want BGNSESS to choose SRCLU values. The LU naming convention TF*aa*#*nnn* is shown here:

aa
: are the last two characters of the domain ID

nnn
: is a decimal number in the range of 000–999

Because BGNSESS selects LUs sequentially beginning with the lowest available number *nnn*, only define the maximum number of LUs you expect to run concurrently on your system for domain *aa*. For example, if your system has a maximum of 50 LUs running with default names for domain NC, include APPL statements defining TFNC#000 through TFNC#049.

The following example is an APPL statement for a VOST LU:

```
TF01#001 APPL  MODETAB=AMODETAB,EAS=9,                               X
               DLOGMOD=M2SDLCNQ
*              STATOPT='DYNAMIC TAF 001'
```

For additional examples, refer to CNMS0013 (A01APPLS).

Chapter 10. Defining Automation

This chapter includes the following topics that describe setting up NetView automation facilities:

- "Updating the Automation Table" on page 187
- "Revising MVS Messages" on page 188
- "Enabling MVS Command Revision" on page 189
- "Enabling Workload Management to Manage the NetView Program" on page 190
- "Enabling SMF Record Type 30 Automation" on page 191

The Automated Operations Network (AON) facility is also used for automation. For information, see "Defining AON" on page 37.

Updating the Automation Table

The automation table is installed and operational as part of the base NetView installation. The following sections describe additional customization procedures that you might consider for your environment.

If you want information about...	Refer to...
Automation table	*IBM Z NetView Automation Guide*

Defining Frame Relay and LMI Support

Frame relay defines the physical interface between customer equipment and network connection point. NCP Version 6 accommodates the frame relay high speed switching protocol. The NetView program can receive and act on the information generated from NCP.

You can enable frame relay switching equipment (FRSE) and local management interface (LMI) support by uncommenting the statements in the NetView automation table, DSITBL01. The following statements allow alerts and frame relay information to flow through the automation table.

```
*IF MSUSEG(0000) ¬= '' THEN
*   BEGIN;
*    IF MSUSEG (0000.52.07 7) = HEX('01') &
*       (MSUSEG (0000.52.0E) ¬= ''  |
*       MSUSEG (0000.52.0F) ¬= '') THEN
***********************************************************************
*  ADD OR CHANGE STATEMENTS BELOW TO WRITE YOUR OWN COMMAND PROCESSOR *
***********************************************************************
*       BEGIN;
*       END;
*   END;
*IF MSUSEG(1332) ¬= '' THEN
*   BEGIN;
*    IF MSUSEG (1332.52.07 7) = HEX('01') &
*       (MSUSEG (1332.52.0E) ¬= ''  |
*       MSUSEG (1332.52.0F) ¬= '') THEN
***********************************************************************
*  ADD OR CHANGE STATEMENTS BELOW TO WRITE YOUR OWN COMMAND PROCESSOR *
***********************************************************************
*       BEGIN;
*       END;
*   END;
```

To write your own network management application, write the logic in a command processor. You can include logic in this command processor to create objects in RODM for display by the NetView management console. This command processor is not provided with the NetView program.

Note: Be sure to add a CMDDEF statement in the CNMCMD %INCLUDE member CNMCMDU of DSIPARM for your command processor.

Handling Undeleted MVS Messages

You can handle MVS action messages. These messages are either WTOR messages or messages that have a descriptor code that matches the setting of the MVSPARM.ActionDescCodes statement in the CNMSTYLE member.

Defining VSAM Database Automation

The hardware monitor, session monitor, TCP/IP connection, and save/restore databases can be automatically purged or reorganized. To do this, use the following CNMSTYLE statements in the CNMSTUSR or C*xx*STGEN member, and enable VSAM database maintenance automation by removing the asterisk at the beginning of the auxInitCmd statements. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
*auxInitCmd.DB1=DBINIT NLDM NONE CYL 50 50 Y PURGE 2 Y PURGE 2 02:00:00 1
*auxInitCmd.DB2=DBINIT NPDA NONE CYL 50 50 Y PURGE 5 Y PURGE 5 02:30:00 1
*auxInitCmd.DB4=DBINIT SAVE NONE CYL 50 50 Y REORG 0 Y REORG 0 03:30:00 1
*auxInitCmd.DB5=DBINIT TCPCONN NONE CYL 50 50 Y PURGE 2 Y PURGE 2
* 04:00:00 1
```

To change the default values for these statements, follow the format specified in the help for the DBINIT command list, CNME2009.

You can change the DSITBL01 processing. Search for DBFULL in the DSITBL01 member. The defaults shipped in the DSITBL01 member show that if the database fills up twice in a 15-minute period, VSAM database automation is stopped. If the database fills up twice in a 15-minute period, allocate more space for the database. One suggestion is to make the time period greater than the time it takes to reproduce the database using the DBFULL command, but less than the time it takes to fill a newly-reproduced database.

Forwarding Alerts and Messages to an Event Integration Facility Event Receiver

The CNMSIHSA sample contains automation table statements that can be used to forward alerts and messages to the NetView Event/Automation Service address space. From there, the alerts and messages can be sent to an Event Integration Facility (EIF) event receiver, such as the Tivoli Netcool/OMNIbus program.

To enable alerts and message routing:

- Customize the CNMSIHSA sample.
- Uncomment the following statement in the DSITBL01 member:

  ```
  *%INCLUDE CNMSIHSA
  ```

If you want information about...	Refer to...
Enabling the Event/Automation Service	“Enabling the Event/Automation Service” on page 203

Revising MVS Messages

You can use the message revision table to intercept MVS messages and make changes to certain aspects, including the following aspects:

- Message text
- Color
- Route codes
- Descriptor codes

- Display and system log attributes

The message revision table is active even when the base NetView program is not active, but the SSI address space is required. However, loading or querying the message revision table or gathering statistics depends on the base NetView address space being active.

Note that you can start message revision, if it is not already running, when the base NetView program starts. To do that, use the SSI.ReviseTable statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
Message revision table	*IBM Z NetView Automation Guide*

Enabling MVS Command Revision

Use the MVS command revision function to intercept MVS commands before they are processed. Command sources include the MVS console and the NetView MVS command. You can make changes to the command text, write a message to the command issuer, and then run the command or suppress the command. You can also transfer the command to the NetView program for more involved actions.

Configuring the NetView Program

A NetView operator ID is required for the autotask that manages MVS commands that are directed to the NetView program. By default, the NetView program uses the MVSCMDS operator ID. To use a different operator ID, update the following statement in the CNMSTUSR or C*xx*STGEN member:

```
function.autotask.MVSCmdRevision=operid
```

For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Preparing the MVS System

The command revision exit uses the NetView subsystem interface (SSI). Ensure that the NetView subsystem address space is started and that the command revision exit is enabled.

To enable the command revision exit for processing on MVS:

1. Ensure the load module DSIRVCEX is in a load library in the MVS LINKLST concatenation. If required, issue the following command to enable it:

   ```
   F LLA,REFRESH
   ```

2. Update an MPFLST*xx* member in PARMLIB by adding the following statement:

   ```
        .CMD USEREXIT(DSIRVCEX)
   ```

   To activate the change, issue the following command, where *xx* is the suffix of the MPFLST member:

   ```
   SET MPF=xx
   ```

Activating Command Revision

The command revision function requires the NetView SSI address space to be active.

To activate the command revision function:

1. Create a command revision table or include UPON statements for command revision in your existing message revision table.
2. Issue the REVISE command.

Note that you can start command revision, if it is not already running, when the base NetView program starts. To do that, use the SSI.ReviseTable statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
Creating the command revision table	*IBM Z NetView Automation Guide*
Using the REVISE CMD command	*IBM Z NetView Command Reference Volume 2 (O-Z)*

Enabling Workload Management to Manage the NetView Program

You can optionally use the z/OS Workload Manager (WLM) to manage NetView task performance in relation to other tasks and applications running on the system or sysplex. The NetView program uses WLM to balance the workload between NetView tasks. When WLM is enabled, NetView calls WLM during task initialization and passes it the task information to allow WLM to assign it to the appropriate service class. Each service class can be given different performance goals and importance.

Preparing WLM for the NetView Environment

Before NetView support for WLM can be enabled, prepare WLM for the NetView environment. Ensure that the following definitions are in place:

- Log on to TSO using your USERID that is authorized to update WLM policies and open the Workload Manager dialog.
- Create a new definition that contains as a minimum:

  Service Policy
  Select option **1** on the Service Definition menu, specifying a **service policy name**, and press Exit to save your changes.

  Workload
  Select option **2** on the Service Definition menu, specifying a **workload name**, and press Exit to save your changes.

  Automation (Z System Automation) Service Class
  Select option **1** on the Service Class Selection List menu. Specify a **service class name** (for example NETVAUTO). This service class is used for NetView automation tasks. Insert a new period. The following values are examples:

  - Processing velocity of 50%
  - Importance of 1

  Press Exit to save your changes.

  Note: If you already have an STCHI service class defined, consider using this class.

  Default Service Class
  Select option **4** on the Service Definition menu, specifying a **service class name** (for example NETVDFLT). This service class is used for NetView tasks that are not assigned to another service class. Insert a new period. The following values are examples:

  - Processing velocity of less than 50%
  - Importance of 2

  Press Exit to save your changes.

  Note: If you already have an STCME service class defined, consider using this class.

  Classification Rule
  Select option **6** on the Service Definition menu. This displays the Subsystem Type Selection List for Rules menu. Specify a subsystem type of NETV. From this menu, select action **1** to insert a rule and select action **2** to insert subrules. Figure 18 on page 191 shows an example of specifying rules and subrules. The following values are examples:

- Specify a default service class name as previously defined.
- Classify tasks by AOST type (rule) as the transaction class (TC), then by Z System Automation task names (subrule) as the user ID (UI) value. For each task, specify the service name as previously defined for IBM Z System Automation autotasks.

Press Exit to save your changes.

```
  Subsystem-Type  Xref  Notes  Options  Help
 --------------------------------------------------------------------------
                Modify Rules for the Subsystem Type      Row 1 to 18 of 18
 Command ===> ____________________________________________ SCROLL ===> PAGE

 Subsystem Type . : NETV        Fold qualifier names?   Y  (Y or N)
 Description  . . . NetView z/OS

 Action codes:  A=After    C=Copy         M=Move     I=Insert rule
                B=Before   D=Delete row   R=Repeat   IS=Insert Sub-rule
                                                              More ===>
          -------Qualifier-------------           -------Class--------
 Action    Type       Name     Start                Service     Report
                                          DEFAULTS: NETVDFLT    ________
  ____  1  TC         AOST      ___                 ________    ________
  ____  2    UI         SA390*___                   NETVAUTO    ________
  ⋮
```

Figure 18. Inserting WLM Rules

Save and Activate the Definitions

Select **Utilities** on the Service Definition menu. Select option **1** to install the definition, then select option **3** to activate the service policy.

Enabling WLM Support

After completing the MVS workload management definitions, use the WLM statement in the CNMSTUSR or C*xx*STGEN member. Remove the asterisk (*) from the beginning of the statement, and change the SubSystemName value if necessary to correspond to the system instance name specified in the WLM service classification rules. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
*WLM.SubSystemName=&DOMAIN
```

Verifying WLM Support

To verify that the NetView program is defined to MVS workload management, use the LIST command or the LISTWLM command.

To display the WLM service class name of the WLM service class assigned to each NetView subtask, enter:

```
LIST STATUS=TASKS WLM=YES
```

To display a windowed list of active NetView subtasks with their assigned WLM service class name, enter:

```
LISTWLM
```

This list is sorted in ascending order by WLM service class name, task type, and task ID.

To list the WLM service class for a single task, use the LIST command.

If WLM is not in use by the NetView program, the WLM service class is shown as `Not Available` by the LIST and LISTWLM commands.

Enabling SMF Record Type 30 Automation

The MVS IEFACTRT SMF installation exit receives control from the system when a job or job step ends, either normally or abnormally. The NetView program provides an IEFACTRT sample exit (CNMSMF3E) that passes data across the PPI to a receiver which issues a message that can be automated using the NetView automation facilities.

To set up the CNMSMF3E sample as an IEFACTRT exit routine, requires the following tasks:

- “Configuring the NetView program” on page 192
- “Preparing the MVS System” on page 193

If you want information about...	Refer to...
SMF record type 30 details	*z/OS MVS System Management Facilities (SMF)*
MVS dynamic exits	*z/OS MVS Initialization and Tuning Reference*
IEFACTFT exit details	*z/OS MVS Installation Exits*

Configuring the NetView program

Configuring the NetView program includes the following steps:

- “Configuring SMF30 statements in CNMSTYLE” on page 192
- “Defining the AUTOSMF3 autotask” on page 192
- “Defining the DSISMF3F command” on page 192
- “Defining automated processing for message BNH874I” on page 192
- “Updating the automation table” on page 193

Configuring SMF30 statements in CNMSTYLE

Review the following CNMSTYLE statements and make any necessary updates to the CNMSTUSR or C*xx*STGEN member:

```
************************************************************************
*       SMF 30 record receiver
*
* Uncomment the following (and, optionally, supply a preferred
* operator ID) to start the SMF 30 record receiver function.
*
************************************************************************
* function.autotask.SMF30 = AUTOSMF3
*
AUTOTASK.?SMF30.Console = *NONE*
AUTOTASK.?SMF30.InitCmd = CNMSMF3R
```

Defining the AUTOSMF3 autotask

The supplied DSIOPF member in the DSIPARM data set includes a definition for the AUTOSMF3 autotask. Ensure that this definition is included in your DSIOPF member.

Defining the DSISMF3F command

The CNMCMENT member in the DSIPARM data set includes a definition for the DSISMF3F command. This command is driven by the CNMSMF3R PPI receiver to format the BNH874I message. Ensure that this definition is included in your CNMCMENT member.

You can either use the shipped DSISMF3F command to format the BNH874I message or you can customize the CNMSMF3F sample and then link it as DSISMF3F.

Defining automated processing for message BNH874I

The CNMSMF3A sample command list is issued by the automation table when the BNH874I message is issued. Update this command list to take appropriate action.

Updating the automation table

The DSITBL01 member in the DSIPARM data set includes sample automation to issue the CNMSMF3A command when a BNH874I message is received. Ensure that similar automation statements are included in your automation table.

Preparing the MVS System

Preparing the MVS System includes the following tasks:

- "Assemble and Link-Edit the CNMSMF3E Exit" on page 193
- "Update SYS1.PARMLIB Library" on page 193

Assemble and Link-Edit the CNMSMF3E Exit

The CNMSMF3E sample is an IEFACTRT SMF exit that processes type 30 SMF records and sends them across the PPI to the NetView program for automation. After making any necessary changes, assemble and link-edit the CNMSMF3E exit into a library in the LNKLST concatenation. Refresh the LNKLST to dynamically activate the new exit.

Update SYS1.PARMLIB Library

Make the following updates to the SYS1.PARMLIB library:

1. Update the SMFPRMxx member:

   - Ensure that the TYPE operand of the SYS specification includes record type 30, for example:

     ```
     SYS(TYPE(30,37,38,39))
     ```

   - Ensure that the EXITS operand of the SYS (system-wide) specification includes IEFACTRT, for example:

     ```
     SYS(EXITS(IEFACTRT,IEFUJI,IEFU83,IEFU84,IEFU85))
     ```

   More than one exit routine can be defined for the IEFACTRT exit, so there can be more than one EXIT statement for the SYS.IEFACTRT exit.

2. Update the PROGxx member to include the CNMSMF3E exit, for example:

   ```
   EXIT ADD EXITNAME(SYS.IEFACTRT) MODNAME(CNMSMF3E)
   ```

If you want information about...	Refer to...
Updating SYS1.PARMLIB library	*IBM Z NetView Installation: Getting Started*

Verifying the SMF 30 processing

To verify that the SMF 30 processing is functioning correctly:

1. Issue the DISPPI command to verify that the CNMSMFR receiver is active, for example:

   ```
   DISPPI RCVRID=CNMSMFR
   ```

   During CNMSTYLE processing, the CNMSMF3R command is run by the AUTOSMF3 autotask. This starts the CNMSMFR receiver.

2. End a job or job steps such that SMF type 30 records (subtype 4 or 5) are generated.
3. Examine the NetView log for the following messages:

   - CNM493I messages that includes the CNMSMF3A command, indicating that the automation table entry was processed

- BNH874I messages after submitting a job or starting a started task that ended with completion codes

Chapter 11. Setting Up UNIX System Services for the NetView Program

The NetView program uses z/OS UNIX System Services for the following functions:

- UNIX command server
- Some AON/TCP functions
- Event/Automation Service
- Event correlation

If you are not planning to use NetView z/OS UNIX functions, continue with Chapter 12, "Enabling the NetView Program with Other Products," on page 211.

Table 34 on page 195 lists the tasks necessary to prepare UNIX for z/OS to enable NetView functions.

Table 34. Tasks to Prepare UNIX System Services

Task	UNIX Command Server	Event / Automation Service	AON/TCP Functions	Event Correlation
Modify UNIX system services parameters	X		X	X
Update security	X	X	X	X
Add or change environment variables	X		X	X

TCP/IP Considerations

Each of the applications that uses z/OS UNIX System Services is a z/OS UNIX sockets application. Any z/OS sockets application needs to reference TCP/IP configuration data. The method of accessing this data is defined by the z/OS version of TCP/IP that is running. See the *z/OS Communications Server IP Configuration Guide* for general information about how z/OS UNIX sockets applications interact with TCP/IP. This book also discusses how a z/OS UNIX sockets application:

- Gains affinity to a TCP/IP stack
- Resolves names to IP addresses
- Finds required TCP/IP configuration data sets

The following sections show examples of using the UNIX System Services RESOLVER_CONFIG environment variable for resolving the TCP/IP configuration data.

Note:

1. Using RESOLVER_CONFIG is recommended for the following reasons:
   - In a multiple-stack environment, you can specify which TCP/IP stack the NetView program is to use.
   - RESOLVER_CONFIG is higher in the search order and easier to change than, for example, the SYSTCPD DD statement.
2. RESOLVER_CONFIG use applies only to the UNIX command server and the Event/Automation Service, particularly when the Event/Automation Service is started in the UNIX System Services shell and not in the NetView base.

A commented SYSTCPD DD statement is provided in the Event/Automation Service startup procedure to identify the location of TCP/IP configuration data. A SYSTCPD statement is not required for the Event/

Automation Services if you use the RESOLVER_CONFIG UNIX System Services environment variable. See the *z/OS Communications Server IP Configuration Guide* for information about how a z/OS UNIX sockets application uses the SYSTCPD DD statement, and determine whether you need to use a SYSTCPD statement in the Event/Automation Service startup procedure.

Note:

1. For all of the z/OS UNIX sockets applications provided with the NetView program, the stack affinity is determined by UNIX System Services configuration definitions. See the *z/OS Communications Server IP Configuration Guide* for considerations for multiple instances of TCP/IP.
2. The UNIX command server is an indirect z/OS UNIX sockets application. The application does not use z/OS UNIX sockets. Some of the UNIX System Services commands that are run using the command server are z/OS UNIX sockets commands. Because of this, these commands require access to TCP/IP configuration data.
3. The AON/TCP application uses the UNIX command server to process commands. As a result, this application is also an indirect z/OS UNIX sockets application. However, part of the AON/TCP application runs in the NetView address space. This part of the AON/TCP application is an z/OS sockets application. The NetView part of the AON/TCP application gets its stack affinity from configuration statements in the NetView configuration member CNMPOLCY in DSIPARM.
4. In a z/OS UNIX INET (single stack) environment, the socket application program is always associated with the single TCP/IP stack. In the z/OS UNIX Common INET (CINET) environment, your application is associated with multiple TCP/IP stacks unless the application specifically associates with a particular stack.

   Some NetView socket applications do not bind themselves to a particular stack in a multi-stack environment. If you require that NetView socket applications (such as the native SNMP command) or user-written socket applications running in the NetView address space be bound to a particular stack in a CINET environment, consider establishing affinity using the following method. The BPXTCAFF program can be used to bind affinity to a stack of your choice. To do this, the BPTXTCAFF program must run prior to the program that initializes the NetView address space. One way of doing this is to add this statement in the NetView startup JCL just prior to the statement used to start NetView:

   ```
   //STEP0   EXEC PGM=BPXTCAFF,PARM=MYSTACK
   ```

   where the PARM= parameter is a pointer to the stack to which you want to establish affinity. This example shows what the startup JCL might look like after the step is added. This example is for reference only.

   ```
                            ⋮
   //*****************************************************************
   //STEP0    EXEC PGM=BPXTCAFF,PARM=MYSTACK
   //*****************************************************************
   //*
   //* PROCEDURE TO START-UP THE NETVIEW APPLICATIONS.
   //* THE REGION SIZE IS 4096K IN THIS SAMPLE AS SPECIFIED IN THE
   //* REG PARAMETER ABOVE.  TO CALCULATE THE CORRECT REGION SIZE
   //* FOR YOUR NETWORK, PLEASE REFERENCE THE NETVIEW STORAGE ESTIM
   //* MANUAL.
   //*
   //*
   //NETVIEW  EXEC PGM=&PROG,TIME=1440,
   //         REGION=&REG.K,
   //         PARM=(&BFSZ.K,&SLSZ,'&DOMAIN','&DOMAINPW','&ARM',
   //         '&SUBSYM','&NV2I'),
   //         DPRTY=(13,13)
                            ⋮
   ```

   For additional information about BPXTCAFF and CINET environments, see *UNIX System Services Planning*.
5. For the Event/Automation Service (E/AS), to request transport affinity when multiple transports are used, set the _BPXK_SETIBMOPT_TRANSPORT environment variable to the name of the preferred

transport before starting E/AS. This variable can also be set in the PARM= parameter of the E/AS started procedure. For example, the following PARM sets up transport affinity to TCPIP03:

```
//STEP1  EXEC PGM=&PROG,TIME=1440,REGION=&REG,
//            PARM=('ENVAR("_BPXK_SETIBMOPT_TRANSPORT=TCPIP03")',
//            '/INITFILE=&INITFILE')
```

Modifying UNIX System Services System Parameters

Member BPXPRMxx in SYS1.PARMLIB contains system values and the file information required for the startup of z/OS UNIX System Services. This member contains MOUNT statements that cause the specified HFS-type data set to be mounted during z/OS UNIX System Services initialization.

If necessary, add a MOUNT statement in member BPXPRM*xx* for the target HFS data set:

```
MOUNT FILESYSTEM('<HFS Pathname>')
      TYPE(HFS)
      MODE(READ)
      MOUNTPOINT('<PathPrefix>/usr/lpp/netview')
```

Note: You might have already added this statement during NetView SMP/E installation.

<HFS Pathname> is the name of the target HFS data set that was allocated during NetView SMP/E installation and was used to install the NetView z/OS UNIX System Services code into HFS directories. If you did not allocate this target HFS data set, you do not need to add this MOUNT statement to your BPXPRMxx member. If you specified a <PathPrefix> during the installation of NetView (for example, /service/), specify the full path name to your mount point directory as your MOUNTPOINT value (for example: ‘`/service/usr/lpp/netview`’).

Note: The steps in the NetView program directory direct you to mount your target HFS data set in read/write (RDWR) mode. After completing these configuration steps, remount your target HFS data set in read (**READ**) mode to protect the data installed in your NetView HFS directories.

To ensure that sufficient resources are available for all UNIX applications, including Java™ applications, consider the following settings in BPXPRM*xx*:

```
MAXTHREADS(10000)
MAXTHREADTASKS(5000)
MAXASSIZE(2147483647)
```

Examine the settings for the MAXPROCSYS and MAXPROCUSER statements in the BPXPRM*xx* member in the SYS1.PARMLIB data set. The MAXPROCSYS statement specifies the maximum number of processes that can be active at the same time. The MAXPROCUSER statement specifies the maximum number of processes with a single UID that are allowed to be active at the same time. The number of TCP/IP-related processes that are spawned as a result of NetView commands can exceed the system-supplied defaults for these UNIX System Services settings. These limits might need to be increased. The settings can be temporarily increased by using the SETOMVS command and remain in effect until the next IPL.

If you want information about...	Refer to...
BPXPRM*xx*	z/OS library

Creating Directories and Copying MIB Source Files

The CNMSJ032 member (CNMSAMP) performs the following actions:

- Create directories in your z/OS UNIX System Services environment
- Copy MIB source files to a working directory
- Copy properties files and sample rule files for the event correlation service

Review the comments in the job profile and make any changes before running this job. This job must be run by a userid that has superuser authority (for example, ROOT).

Run the CNMSJ032 job. CNMSJ032 creates the following directories:

- /etc/netview/v6r3
- /etc/netview/mibs
- /tmp/netview/v6r3
- /etc/netview/v6r3/properties
- /etc/netview/v6r3/usercode
- /tmp/netview/v6r3/logs
- /var/netview/v6r3/rulefiles

MIB source files are then copied from the /usr/lpp/tcpip/samples directory to the /etc/netview/mibs directory. You can also place other MIB source files in /etc/netview/mibs.

CNMSJ032 also copies the event correlation properties file to /etc/netview/v6r3/properties and the sample rules file to /var/netview/v6r3/rulefiles. You can edit the properties file to customize the correlation engine. For information about the properties file statements, see the *IBM Z NetView Administration Reference*.

Verify the return codes:

- A return code of 0 indicates that the MIB source files were copied successfully.
- A return code of 4 indicates that the MIB source file or files already exist in the /etc/netview/mibs directory. Because of that, the MIB source file or files were not copied. Review the held output report (SYSTSPRT) to see which MIB source files were not copied and manually migrate the files to the current release of the NetView program.
- If you receive a return code greater than 4, check the z/OS library for information to correct the problem and resubmit the job.

Updating UNIX System Services Environment Variables

Table 35 on page 198 shows the z/OS UNIX System Services environment variables that need to be added or modified for the UNIX command server and AON/TCP functions.

Table 35. z/OS UNIX System Services Environment Variables by NetView Function

Environment Variable	Value[1]	UNIX Command Server	AON/TCP Functions
PATH	`/usr/lpp/netview/v6r3/bin`	X	X
RESOLVER_CONFIG	`//'TCPIP.INIT(TCPDATA)'`		X

Note:

1. Can vary by installation.

Note:

1. *PathPrefix*`/usr/lpp/netview/v6r3` is the default directory into which the NetView program is installed.

   If you specify a different value during installation for *PathPrefix* (for example, /service), make the appropriate substitutions to the example path names.

2. For performance considerations, avoid using the STEPLIB environment variable.

   For more information, see the z/OS library.

The correlation engine can function without requiring any environment variables to be set. However, if needed for your installation, you can define environment variables that override default values of the locations of the correlation engine code, properties files, rules files, and log files. Environment variables can be set in UNIX System Services initialization files if the correlation engine is started as a daemon or started from a shell. Table 36 on page 199 shows the environment variables that you can set.

Table 36. Environment Variables That Can be Set in UNIX System Services Initialization Files

Environment Variable	Value
LOGPATH	The path to the directory that contains the log file. The default value is `/tmp/netview/v6r3/logs/`.
RULEPATH	The path to the directory that contains the XML rule file. The default value is `/var/netview/v6r3/rulefiles/`.
PROPPATH	The path to the directory that contains the property files for the correlation engine and logging service. The default value is `/etc/netview/v6r3/properties/`.
PROPFILE	The name of the correlation engine properties files. The default value is `correlator.properties`.
RULEFILE	The name of the rule file to load. The default value is `znvrules.xml`.
LOGFILE	The name of the log file. The default value is `nvcorrelation.log`.
INSTPATH	The path to the directory that contains the correlation engine code. The default value is `/usr/lpp/netview/v6r3/`.
USERPATH	The path to the directory that contains user-compiled Java code for customized actions. The default value is `/etc/netview/v6r3/usercode/`.

The variables can be set in the CNMJZCE startup JCL in a manner similar to the variables for the UNIX command server, described in “Specifying NetView z/OS UNIX Environment Variables” on page 200. The STDENV DD statement can be used to define the variables in-stream (as is done in the shipped sample), or else the DD statement can refer to a z/OS data set name or to a partitioned data set. When starting the correlation engine from a shell or as a daemon, the environment variables can be set in a manner described in “Managing NetView z/OS UNIX Functions from z/OS UNIX” on page 200.

Some of the environment variables correspond to properties that are contained in the properties files that are used by the correlation engine. These are the location and name of the rule file and the log file. The following order of precedence is used:

- Properties file
- Environment variables
- Defaults

The properties file that is used by the correlation engine can be passed as a parameter on the startup script. If the parameter is supplied, it overrides the value of the PROPPATH and PROPFILE environment variables.

The logging service that is used by the correlation engine has its own properties file, specified by the JLOG.CONFIGURATION property in the properties file. This file contains entries for the default logging level and the location and name of the log file. The LevelLogger section has the following entry:

```
logger.level.level=
```

The level value can be specified to control the number of log entries created. Generally, more entries are generated to aid in debugging problems. Levels are arranged in a hierarchy. The level specified is the lowest level logged. All higher levels are also logged. The following values are valid for level, in order of severity:

- FATAL
- ERROR
- WARN

- INFO
- DEBUG_MIN
- DEBUG_MID
- DEBUG_MAX

The default value is INFO. Information entries indicate the status of the correlation engine and the connections being made. DEBUG_MIN logs entries that show the flow of events into and out of the correlation engine. DEBUG_MID traces module entry/exit and includes information about the events being sent and received. DEBUG_MAX traces additional internal processing in the engine, such as parsing of events.

The logging level in use can be changed by issuing the CORRSERV LOGLEVEL command from the NetView program. Edit the JLOG.CONFIGURATION file only to change the default level.

You can also use the JLOG.CONFIGURATION file to specify the name and location of the log file. The following entries are relevant:

```
handler.file.filedir=
handler.file.filename=
```

These are shipped as $(LOGPATH) and $(LOGFILE). The values are substituted for the values of the LOGPATH and LOGFILE environment variables, if set, or to the defaults of `/tmp/netview/v6r3/logs/` and `nvcorrelation.log` if the environment variables are not set. If you want to use a non-default value and do not want to set environment variables, these values can be changed in the configuration file.

Specifying NetView z/OS UNIX Environment Variables

To manage NetView z/OS UNIX functions directly from the NetView program, add and modify the environment specification in the STDENV DD statement in the UNIX command server JCL sample CNMSJUNX or CNMSSUNX.

You can define the STDENV DD in the UNIX command server JCL in several ways:

- z/OS UNIX path name, for example:

```
//STDENV    DD PATH='/etc/netview/v6r3/stdenv',PATHOPTS=ORDONLY,
//             PATHMODE=SIRWXU
```

- Instream data within the JCL, for example:

```
//STDENV    DD DATA
PATH=/bin:/usr/lpp/netview/v6r3/bin:/usr/lpp/tcpip/bin
RESOLVER_CONFIG=//'TCPIP.INIT(TCPDATA)'
⋮
/*
```

  Note: Environment variable definitions are limited to 72 bytes in length in JCL.

- z/OS data set name or partitioned data set (PDS) member name. The data set can be a fixed or variable block data set with a record length large enough to accommodate the largest environment variable definitions. For example:

```
//STDENV    DD DSNAME=NETVIEW.DSIPARM(STDENV),DISP=SHR
```

Managing NetView z/OS UNIX Functions from z/OS UNIX

To manage NetView z/OS UNIX functions from z/OS UNIX, add or modify the environment variables in z/OS UNIX. The environment variables are defined in one of the following profiles:

- The UNIX user profiles, for the user who is starting and stopping functions
- The default UNIX profile (for example `/etc/profile`)

Variables defined in this way can be exported in the following way:

```
export name=value
```

or

```
name=value
export name
```

Enabling the UNIX Command Server

The UNIX command server enables UNIX commands to be entered from the NetView command line and returns the output of these commands to the NetView console. The UNIX commands entered from the NetView program are run in the UNIX System Services environment of the z/OS operating system.

Defining the UNIX Command Server

To enable the running of UNIX for z/OS commands from the NetView program, a dedicated PPI receiver (CNMEUNIX) is needed to receive commands and data from the NetView program. A server process running in a UNIX for z/OS address space waits on this PPI receiver for incoming commands and data.

CNMEUNIX runs as a UNIX for z/OS kernel process. The UNIX command server consists of three parts that must be installed in the UNIX hierarchical file system (HFS). The default directory into which the installation installs the parts is `<PathPrefix>/usr/lpp/netview/v6r3/bin`.

Setting a region size for the UNIX command server of 0MB allows the UNIX command server to access all available memory. System-defined installation exits might limit this amount. If you receive an out-of-memory condition for the UNIX command server address space, adjust the installation exit values.

If the UNIX command server is started as a submitted job, ensure that the CNMSJUNX sample job is contained in a DSIPARM data set. If the UNIX command server is started as a started task, ensure that the CNMSSUNX sample job is copied into a data set defined in the IEFJOBS or IEFPDSI concatenation of master JCL, for example, SYS1.PROCLIB. This is required because CNMSSUNX contains a job statement. Also ensure that the sample z/OS START command CNMSUNXS is contained in a DSIPARM data set. For more information about specifying whether the UNIX command server runs as a submitted job or as a started task, see the online help for the DEFAULTS STRTSERV command.

Before you start a UNIX command server, you can customize the samples that are provided with the NetView program to fit your environment. If the SCNMLNKN data set is not defined as part of the LNKLST concatenation, you must include a STEPLIB DD statement for it in the UNIX command server JCL. When a START UNIXSERV command is issued, the NetView program issues an MVS START command (if DEFAULTS STRTSERV is set to STRTPROC) or submits a job (if DEFAULTS STRTSERV is set to SBMTJOB). For a started task, member CNMSUNXS contains the MVS START command that is used to start the procedure or job specified in the START UNIXSERV command. For a submitted job, the member specified on the MEM keyword in the START UNIXSERV command is submitted as an MVS job.

Member CNMSUNXS or the member containing the submitted JCL, for example, CNMSJUNX, can contain specific variables for which the NetView program performs substitutions before the task is started or the job is submitted. These variables have names that begin with an ampersand (&) followed by lowercase letters and end with a period (.). You can specify any of these substitution variables to customize how your UNIX command server is started. The following list describes these variables:

&sprcnm.

The member name containing the JCL that is to be started or submitted. This value is the value of the MEM keyword on the START UNIXSERV command

- For a started task, if MEM is not specified, CNMSSUNX is used. This variable is specified in sample CNMSUNXS as the procedure or job to start.
- For a submitted job, if MEM is not specified, CNMSJUNX is used. This variable is not specified in sample CNMSJUNX because it is the same as the sample name.

&jobname.

The name to be used on the JOB statement on a submitted job. This value is always set to CNMEUNIX.

- For a started task, this variable is not specified in sample CNMSUNXS because it does not apply to the MVS START command.
- For a submitted job, this variable is not specified in sample CNMSJUNX because the name is always set to CNMEUNIX.

&userid.

The TSO user ID under whose authority the started task or submitted job runs. To easily identify which user ID the task or job is running for, the JOBNAME (for start) or the step name (for submit) specifies this name. The value is always set to CNMEUNIX.

- For a started task, this variable is specified in sample CNMSUNXS as the value for the JOBNAME keyword on the MVS START command.
- For a submitted job, this variable is not specified in sample CNMSJUNX.

&ppiname.

The receiver name used for communication between the server and the NetView program across the PPI. The value is always set to CNMEUNIX.

- For a started task, this variable is not specified in sample CNMSUNXS because the PPI name is always CNMEUNIX.
- For a submitted job, this variable is not specified in sample CNMSJUNX because the PPI name is always CNMEUNIX.

If your installation is a RACF controlled environment, there are additional RACF requirements. For more information, see the *IBM Z NetView Security Reference*.

Starting the UNIX Command Server

To start the UNIX command server from the NetView program, enter the following command from the command facility:

```
START UNIXSERV=*
```

If multiple versions of the UNIX command server JCL are required, the optional MEM parameter can be specified on the START UNIXSERV command. You can use the MEM parameter to specify members other than the CNMSJUNX for submitted jobs or CNMSSUNX for started tasks.

After starting UNIXSERV, you receive the following message:

```
DSI633I START COMMAND SUCCESSFULLY COMPLETED
```

If you want information about...	Refer to...
START command	NetView online help for NCCF START
Issuing UNIX for z/OS commands from the NetView program	NetView online help for PIPE UNIX
Security considerations for the UNIX command server	*IBM Z NetView Security Reference*

Starting from UNIX for z/OS

Because the UNIX command server runs as a UNIX daemon, you can start the UNIX command server from UNIX for z/OS. For example, you can use the following command:

```
_BPX_JOBNAME='CNMEUNIX' /usr/lpp/netview/v6r3/bin/cnmeunix > /tmp/nvunix.out 2>&1&
```

Assigning a value for _BPX_JOBNAME enables the named address space to be displayed on an SDSF active tasks display or when a **D A** command is issued from an z/OS console.

Note: You can add this command to a UNIX for z/OS initialization script file such as `/etc/rc`.

The directory containing the UNIX command server code needs to be included in the PATH environment variable (set up in `/etc/profile`).

UNIX command server diagnostic information is written primarily to `stdout` but, under certain circumstances, messages can be written to `stderr`. The diagnostic information written to `stdout` contains the error data that is returned in the secondary output stream of the PIPE UNIX stage and might include the name of the UNIX service that failed.

Verifying that the UNIX Command Server is Active

Regardless of how the UNIX command server is started, verify that the UNIX command server is running by entering the DISPPI command from the NetView command facility. The CNMEUNIX PPI receiver needs to exist and be active as shown in the following example:

```
DWO948I RECEIVER  RECEIVER  BUFFER      QUEUED      TOTAL       STORAGE
DWO949I IDENTITY  STATUS    LIMIT       BUFFERS     BUFFERS     ALLOCATED
DWO950I --------  --------  ----------  ----------  ----------  ----------
DWO952I NETVALRT  INACTIVE        1000           0           0           0
DWO951I NETVRCV   ACTIVE           500           0          24           0
DWO951I    :        :
DWO951I    :        :
DWO951I CNMEUNIX  ACTIVE          1000           0           3           0
DWO951I    :        :
DWO951I    :        :
DWO968I END OF DISPLAY
```

Enabling the Event/Automation Service

The Event/Automation Service serves as a gateway for event data between the IBM Z NetView management environment, managers and agents that deal with Event Integration Facility (EIF) events, and SNMP trap managers and agents. With this gateway function, you can manage all network events from the management platform of your choice.

Event/Automation Service runs as a separate z/OS address space. The default startup procedure is IHSAEVNT.

Event/Automation Service consists of the following services that convert event data into different formats and forward the data to event management tools:

- The alert adapter and message adapter services convert Z NetView alerts and messages into EIF events before forwarding the event data to a manager that handles EIF events, such as Tivoli Netcool/OMNIbus. As a result, all network events can be managed from a distributed event manager.
- The confirmed alert and message adapter services convert alerts and messages into EIF events and forward them to an event server. The event server then replies with a confirmation that indicates acceptance of the EIF event.
- The alert-to-trap service converts Z NetView alerts into SNMP traps before forwarding the trap data to an SNMP manager. Event/Automation Service performs the function of an SNMP subagent, and sends the converted alert data to an SNMP agent for eventual forwarding to an SNMP manager.
- The event receiver service converts events from an agent that uses EIF events into alerts before forwarding the alert to the Z NetView program through the Alert Receiver PPI mailbox. As a result, all network events can be managed from the hardware monitor.
- The trap-to-alert service converts SNMP traps that arrive from SNMP managers into alerts before forwarding the alert to the Z NetView program through the Alert Receiver PPI mailbox.

Preparing the z/OS Host Components

The Event/Automation Service can be configured to start from either the z/OS system console using job IHSAEVNT, or from a UNIX System Service command shell.

Preparing to Start as a Job

To start the Event/Automation Service from a job, follow these steps:

1. Copy the IHSAEVNT sample from NETVIEW.V6R3M0.CNMSAMP to a system procedure library.
2. Allocate a data set to contain your configuration parameters. The DCB attributes must match those of the NETVIEW.V6R3M0.SCNMUXCL data set.
3. For any services that you plan to use (see "Getting Ready to Start the Event/Automation Service" on page 205), copy the members that you need to change from the SCNMUXCL data set to the data set allocated in the previous step.
4. If you use an SAF product, such as RACF, define the procedure IHSAEVNT to have superuser authority in the OMVS segment of the security product.

Preparing to Start in a UNIX System Service Command Shell

To configure the Event/Automation Service to be started from a UNIX system service command shell, follow these steps:

1. Add NETVIEW.V6R3M0.CNMLINK to the STEPLIB environment variable for your shell session.
2. Create a file in the Hierarchical File System (HFS) named IHSAC000 that has run permission and has the sticky bit on.
3. For any services that you plan to use (see "Getting Ready to Start the Event/Automation Service" on page 205), copy the members that you need to change from the SCNMUXCL data set to the HFS directory `/etc/netview/v6r3`. Rename the PDS members to their corresponding names as shown in "Event/Automation Service Configuration Files" on page 204. These names are case sensitive.
4. Copy the IHSAMSG1 file from the NETVIEW.V6R3M0.SDUIMSG1 data set to the `/usr/lpp/netview/msg/C` HFS directory. Rename this member to its corresponding name as shown in "Event/Automation Service Configuration Files" on page 204. This name is case sensitive.

Event/Automation Service Configuration Files

This table describes the locations of the configuration parameters for the components of the Event/Automation Service.

Table 37. Event/Automation Service configuration files

SCNMUXCL Member Name	HFS Configuration File Name	Used By
IHSAINIT	global_init.conf	All services for global initialization
IHSAACDS	alert_adpt.cds	Alert adapter service
IHSABCDS	confirm_alert_adpt.cds	Confirmed alert adapter service
IHSAACFG	alert_adpt.conf	Alert adapter service
IHSABCFG	confirm_alert_adpt.conf	Confirmed alert adapter service
IHSAATCF	alert_trap.conf	Alert-to-trap service
IHSALCDS	alert_trap.cds	Alert-to-trap service
IHSAECDS	event_rcv.cds	Event receiver service
IHSAECFG	event_rcv.conf	Event receiver service

Table 37. Event/Automation Service configuration files (continued)		
SCNMUXCL Member Name	**HFS Configuration File Name**	**Used By**
IHSAMCFG	message_adpt.conf	Message adapter service
IHSANCFG	confirm_message_adpt.conf	Confirmed message adapter service
IHSAMFMT	message_adpt.fmt	Message adapter service
IHSANFMT	confirm_message_adpt.fmt	Confirmed message adapter service
IHSATALL	trap_alert_all.cds	Trap-to-alert service
IHSATCDS	trap_alert.cds	Trap-to-alert service
IHSATCFG	trap_alert.conf	Trap-to-alert service
IHSATMSM	trap_alert_msm.cds	Trap-to-alert service
IHSATUSR	trap_alert_user.cds	Trap-to-alert service
IHSAMSG1	ihsamsg1	All services

Getting Ready to Start the Event/Automation Service

The Event/Automation Service is composed of the following services:

- Alert adapter service
- Message adapter service
- Confirmed alert adapter service
- Confirmed message adapter service
- Event receiver service
- Trap-to-alert service
- Alert-to-trap service

By default, the alert adapter, message adapter, and event receiver services are started when you start the Event/Automation Service. You can prevent these services from starting automatically, or you can allow the confirmed alert adapter, confirmed message adapter, trap-to-alert and alert-to-trap service to start automatically. See the NOSTART statement in the *IBM Z NetView Administration Reference* for information about how to prevent a service of the Event/Automation Service from starting.

To make changes, modify the Event/Automation Service procedure IHSAEVNT. Add the user data set that you created for modifying configuration files from the SCNMUXCL data set to the IHSSMP3 DD statement in the procedure. If you do not have the LE/370 libraries defined in your system concatenation, uncomment the LE/370 statement in the procedure and set it to the correct data set.

If you need to configure Language Environment for z/OS runtime options for all Event/Automation Service tasks, you can define a CEEOPTS file containing these runtime options in the Event/Automation Service start procedure.

For example, to set the TZ and _TZ environment variables to the value EST5EDT for all Event/Automation Service tasks, you would code the following line in a sequential data set named USER.LEPARM:

```
ENVAR('TZ=EST5EDT','_TZ=EST5EDT')
```

In the IHSAEVNT sample start procedure for the Event/Automation service, you would uncomment the CEEOPTS data definition (DD) example, and change the value of the DSN keyword as follows:

```
//CEEOPTS   DD   DISP=SHR,DSN=USER.LEPARM
```

For more information about the CEEOPTS DD statement and Language Environment for z/OS runtime options, see the *z/OS Language Environment Programming Reference*.

Getting Ready to Start the Alert Adapter Service

You must provide an event server location for the Alert Adapter service. This is done using the ServerLocation and optionally the ServerPort statements in the alert adapter configuration file (SCNMUXCL IHSAACFG member or alert_adpt.conf HFS file). See the ServerLocation and ServerPort statements in the *IBM Z NetView Administration Reference* for information about how to provide the event server information.

Note: If your ServerPort statement is set such that PortMapper is required, make sure that the Portmapper service is running on the event server at the IP location specified on the ServerLocation statement. By default, this statement is set to require the Portmapper service.

The routing of alerts from the NetView address space to the Event/Automation Service alert adapter service is disabled by default. To route NetView alerts to the Event/Automation Service alert adapter service, the hardware monitor TECROUTE and AREC filters must be set to PASS. This allows all alerts to be routed to the Event/Automation Service alert adapter service. For information about setting hardware monitor filters, see the SRFILTER command in the NetView online help.

The alert adapter service has a number of other settings that can be customized. For information about customizing the alert adapter service, see the *IBM Z NetView Customization Guide*.

Getting Ready to Start the Message Adapter Service

You must provide an event server location for the Message Adapter service. This is done using the ServerLocation and optionally the ServerPort statements in the message adapter configuration file (SCNMUXCL IHSAMCFG member or message_adpt.conf HFS file). See the ServerLocation and ServerPort statements in the *IBM Z NetView Administration Reference* for information about how to provide the event server information.

Note: If your ServerPort statement is set such that PortMapper is required, make sure that the Portmapper service is running on the event server at the IP location specified on the ServerLocation statement. By default, this statement is set to require the Portmapper service.

The routing of messages from the NetView address space to the Event/Automation Service is disabled by default. To route NetView messages to the Event/Automation Service, add statements to your automation table to select specific messages and route them using a PIPE stage. The CNMSIHSA sample member contains automation table statements to route messages.

To enable message routing from the NetView program, customize the CNMSIHSA sample to route any messages that you want. Then, uncomment the statement that includes CNMSIHSA in the DSITBL01 automation table. For more information about customizing CNMSIHSA, see the *IBM Z NetView Automation Guide*.

The message adapter service has a number of other settings that can be customized. For information about how to further customize the message adapter service, see the *IBM Z NetView Customization Guide*.

Getting Ready to Start the Confirmed Alert Adapter Service

You must provide an event server location for the confirmed alert adapter service. This is done using the ServerLocation and optionally the ServerPort statements in the confirmed alert adapter configuration file (SCNMUXCL member IHSABCFG or HFS file confirm_alert_adpt.conf). See the ServerLocation and ServerPort statements in the *IBM Z NetView Administration Reference* for information about how to provide an event server location.

The routing of alerts from the NetView address space to the Event/Automation Service confirmed alert adapter service is disabled by default. To route NetView alerts to the Event/Automation Service confirmed alert adapter service, follow these steps:

- Set the hardware monitor AREC, ESREC, and TECROUTE filters to PASS. This allows all alerts to be routed to the Event/Automation Service confirmed alert adapter service. For information about setting hardware monitor filters, see the SRFILTER command in the NetView online help.
- Use NetView automation to drive a command that uses the PIPE PPI stage with the TECRTCFM keyword to forward alert data to the Event/Automation Service. The Event/Automation Service can then send the data to the confirmed alert adapter. For example automation statements, see the CNMSIHSA sample. For examples on how to add user-specified data to the alert data to be passed to the confirmed alert adapter, see the CNMEALUS sample.

The confirmed alert adapter service has a number of other settings that can be customized. For information about customizing the confirmed alert adapter service, see the *IBM Z NetView Customization Guide*.

Getting Ready to Start the Confirmed Message Adapter Service

You must provide an event server location for the confirmed message adapter service. This is done using the ServerLocation and optionally the ServerPort statements in the confirmed message adapter configuration file (SCNMUXCL member IHSANCFG or HFS file confirm_message_adpt.conf). See the ServerLocation and ServerPort statements in the *IBM Z NetView Administration Reference* for information about how to provide an event server location.

The routing of messages from the NetView address space to the Event/Automation Service is disabled by default. To route NetView messages to the Event/Automation Service, add statements to your automation table to select specific messages and route them using the PIPE PPI stage with the TECRTCFM keyword. Sample member CNMSIHSA contains automation table statements to route messages.

To enable message routing from the NetView program, customize samples CNMSIHSA and CNMEMSUS to route any messages that you want. Then, uncomment the statement that includes CNMSIHSA in the DSITBL01 automation table. For more information about customizing CNMSIHSA and CNMEMSUS, see the *IBM Z NetView Automation Guide*.

The confirmed message adapter service has a number of other settings that can be customized. For information about how to further customize the confirmed message adapter service, see the *IBM Z NetView Customization Guide*.

Getting Ready to Start the Event Receiver Service

The event receiver service functions properly without customization. However, this service has a number of settings that can be customized using the event receiver configuration file (SCNMUXCL member IHSAECFG or HFS file event_rcv.conf). For information about how to customize the event receiver service, see the *IBM Z NetView Customization Guide*.

Note: If your UsePortmapper statement is set such that PortMapper is required, ensure that the Portmapper service is running on the z/OS host where the Event/Automation Service is running. By default, this statement is set to require the Portmapper service.

Getting Ready to Start the Trap-to-Alert Service

The trap-to-alert service functions properly without further customization. However, this service has a number of other settings that can be customized using the trap-to-alert configuration file (SCNMUXCL member IHSATCFG or HFS file trap_alert.conf). For information about customizing the trap-to-alert service, see the *IBM Z NetView Customization Guide*.

Note: If you have an SNMP manager that uses port 162 on the same system as the Event/Automation Services, customize the trap-to-alert service to use another port or use the sample trap forwarding daemon provided with the Event/Automation Service to forward traps. For information about how to use the sample trap forwarding daemon, see the *IBM Z NetView Customization Guide*.

Getting Ready to Start the Alert-to-Trap Service

The alert-to-trap service functions properly without further customization. However, this service has a number of other settings that can be customized using the alert-to-trap configuration file (SCNMUXCL member IHSAATCF or HFS file alert_trap.conf). For information about customizing the trap-to-alert service, see the *IBM Z NetView Customization Guide*.

Because the alert-to-trap service functions as an SNMP subagent, the SNMP agent provided with TCP/IP must be started and properly configured so that the alert-to-trap service can pass traps to the agent. See your TCP/IP documentation for information about how to enable the SNMP agent daemon.

The routing of alerts from the NetView address space to the Event/Automation Service alert-to-trap service is disabled by default. To route NetView alerts to the alert-to-trap service, set the hardware monitor TRAPROUTE and AREC filters to PASS. This allows all alerts to be routed to the alert-to-trap service. For information about setting hardware monitor filters, see the SRFILTER command in the NetView online help.

Starting the Event/Automation Service

You can start the Event/Automation Service from an z/OS system console using job IHSAEVNT , or from a UNIX System Service command shell.

Starting the Event/Automation Service Using Job IHSAEVNT

To start the Event/Automation Service, enter the following command at the system console:

```
S IHSAEVNT
```

Messages similar to those in Figure 19 on page 208 are displayed.

```
IHS0075I Event Automation Services started. Subtask initialization is in progress for IHSATEC
IHS0124I Event Receiver task initialization complete.
IHS0124I Alert Adapter task initialization complete.
IHS0124I Message Adapter task initialization complete.
```

Figure 19. Messages for Starting the Event/Automation Service

The other Event/Automation Service services are not automatically started when the Event/Automation Service is started. For information about how to start and stop individual services when the Event/Automation Services is started, see the NOSTART statement in the *IBM Z NetView Administration Reference*.

Starting the Event/Automation Service Using the UNIX System Services Command Shell

After performing the required steps listed previously for starting the Event/Automation Service from a UNIX System Services command shell, start the Event/Automation Service by entering **IHSAC000** from the command shell:

You see messages similar to those in Figure 20 on page 208.

```
IHS0075I Event Automation Services started. Subtask initialization is in progress for IHSATEC
IHS0124I Event Receiver task initialization complete.
IHS0124I Alert Adapter task initialization complete.
IHS0124I Message Adapter task initialization complete.
```

Figure 20. Messages for Starting the Event/Automation Service from a UNIX System Services Command Shell

The other Event/Automation Service services are not automatically started when the Event/Automation Service is started. For information about how to start and stop individual services when the Event/

Automation Services is started, see the NOSTART statement in the *IBM Z NetView Administration Reference*.

Enabling Event Correlation

Event correlation is a NetView function that runs under UNIX Systems Services outside the NetView address space. It works with the NetView automation table to correlate NetView messages and alerts that are mapped to events. You can use correlation to relate and process multiple messages or MSUs for automation. For more information, see the *IBM Z NetView Automation Guide*.

The connection between the correlation engine and the NetView program is made through TCP/IP sockets. If a large volume of events are to be sent for correlation, consider optimizing local socket performance in Communications Server and your TCP/IP setup.

Installing the Event Correlation Engine

Use the CNMSJ032 sample job to configure the directories used by the correlation engine. For more information, see "Creating Directories and Copying MIB Source Files" on page 197. When the job runs successfully, you can customize the properties and rule files. For information about the properties file statements, see the *IBM Z NetView Administration Reference*.

Creating the Event Correlation Engine Rules

By default, a sample rule file is placed in the following location:

```
/var/netview/v6r3/rulefiles
```

You can customize this file as needed. You can also specify another rule file or directory location by changing the RULEFILE property in the properties file or by setting the RULEPATH and RULEFILE environment variables. For the rule statement syntax, see the *IBM Z NetView Automation Guide*.

Updating the XML Correlation Rules

You can use the CORRSERV command to refresh the rule definitions. For more information, see the online help for the CORRSERV command.

Starting the Correlation Engine

Before you start the correlation engine, verify that the Java program is installed on the UNIX System Services, the system path is updated so that the system can find the Java command, and the rule definitions are in place.

You can start the correlation engine in the following ways:

- At system initialization. To do this, modify the /etc/rc file to start the corrstart.sh shell file under /usr/lpp/netview/v6r3/bin. The following statement is an example:

  ```
  _BPX_JOBNAME='NVCORRD' /usr/lpp/netview/v6r3/bin/corrstart.sh &
  ```

- From an OMVS ID by running the UNIX System Service command shell corrstart.sh

After you start the correlation engine, start the DSICORSV task.

When the correlation engine has started, you can use the NetView CORRSERV command to control the correlation engine, including stopping it. For more information about the CORRSERV command, see *IBM Z NetView Command Reference Volume 1 (A-N)*.

If you stop the correlation engine, you can start it again from the command shell corrstart.sh.

Chapter 12. Enabling the NetView Program with Other Products

The following products complement the NetView program to provide a comprehensive set of enterprise management functions:

- "Tivoli Netcool/OMNIbus" on page 211
- "IBM Z System Automation" on page 211
- "Tivoli Business Service Manager" on page 212
- "OMEGAMON Products" on page 213

If the installation of any complementary products includes command definitions, these command definitions must now be placed in CNMCMDO. Their format matches the command definitions in CNMCMD.

Tivoli Netcool/OMNIbus

IBM Tivoli Netcool/OMNIbus is a service level management system that collects enterprise-wide event information from a wide variety of IT and network environments in real time and presents a consolidated view of this information to operators and administrators. Tivoli Netcool/OMNIbus tracks alert information in a high-performance, in-memory database, and presents information of interest to specific users through filters and views that can be configured individually.

If you want information about...	Refer to...
Setting up the interface between Netcool/OMNIbus and the NetView program	"Enabling the Event/Automation Service" on page 203

IBM Z System Automation

The IBM Z System Automation application, which is based on the NetView program, provides a single point of control for a full range of systems management functions. System Automation functions include monitoring, controlling, and automating a large range of system elements that spans both the hardware and software resources of your enterprise. System Automation plays a key role in supplying high-end automation solutions and can be used to automate I/O, processor, and system operations. It enables high availability for critical business applications through policy-based, self-healing capabilities. Users with single z/OS systems and Parallel Sysplex® clusters can use System Automation to ease management, minimize costs, and maximize application availability.

As shipped by the NetView program, IBM Z System Automation is disabled. To enable System Automation, take these actions:

- Follow the instructions in the IBM Z System Automation information.
- If you are running AON and IBM Z System Automation in the same NetView address space, see "Enabling Workload Management to Manage the NetView Program" on page 190.
- If you configure IBM Z System Automation to start system services (for example, TCP/IP and UNIX System Services), consider that some of these services might be required by AON initialization. You can use the COMMON.EZLINITDELAY statement in the CNMSTUSR or C*xx*STGEN member to specify the amount of time to wait before initializing AON. This enables IBM Z System Automation to start the system services that might also be required by AON. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

If you want information about...	Refer to...
IBM Z System Automation	IBM Z System Automation library or to http://www.ibm.com/systems/z/os/zos/features/system_automation.

System Operations

The system operations component monitors and controls system operations applications and subsystems such as the Z NetView program, System Display and Search Facility (SDSF), job entry subsystem (JES), Resource Measurement Facility (RMF), Time Sharing Option (TSO), RODM, VTAM, DB2, CICS, IMS, OMEGAMON, Tivoli Business Service Manager , and Tivoli Workload Scheduler. With system operations, you can automate Parallel Sysplex applications. System Automation can automate applications that are distributed over a sysplex by virtually removing system boundaries for automation through its automation manager/automation agent design. System Automation reduces the complexity of managing a Parallel Sysplex cluster through goal-driven automation and concepts such as grouping and powerful dependency support, which you can use to model your configuration. Single systems are also fully supported; the automation scope is then just one system. System Automation uses enterprise monitoring to interoperate with products such as IBM Tivoli Netcool/OMNIbus or the IBM Tivoli Service Request Manager® product in case of an incident and to update the health status information that is displayed on the Tivoli Enterprise Portal through the IBM Tivoli Monitoring infrastructure.

Processor Operations

The processor operations component monitors and controls processor hardware operations. It provides a connection from a focal point processor to a target processor. With the NetView program on the focal point system, processor operations automates operator and system consoles for monitoring and recovering target processors. You can use processor operations to power on, power off, and reset multiple target processors; to load the initial programs; to set the time-of-day clocks; to respond to messages; to monitor status; and to detect and resolve wait states.

I/O Operations

The I/O operations component provides a single point of control for managing connectivity in your active I/O configurations. This component can detect unusual I/O conditions and can be used to view and change paths between a processor and an input/output device, which can involve using dynamic switching, either the enterprise systems connection (ESCON) or fiber channel connection (FICON®) switch. You can change paths by controlling channels, ports, switches, control units, and input/output devices through an operator console or API.

Tivoli Business Service Manager

IBM Tivoli Business Service Manager is an enterprise management product that monitors the data processing resources that are critical to a business application. Mission-critical business systems typically span host and distributed environments; include many interconnected application components, both commercial and custom; and rely on diverse middleware, databases, and supporting platforms.

Tivoli Business Service Manager provides end-to-end business systems management to organize related components and give business context to management decisions. A unique, configurable business system view enables management and control of the multiple integrated software components required to deliver a specific business service. The product also shows and allows the manipulation of the relationships between applications, so that you can more easily detect inefficiencies or diagnose problems in critical business systems.

If you want information about...	Refer to...
Configuring the NetView management console for the Tivoli Business Service Manager Console	*IBM Z NetView Installation: Configuring Graphical Components*

OMEGAMON Products

The NetView program interoperates with IBM OMEGAMON products through the NetView Web application and the Z NetView Enterprise Management Agent. For installation and configuration information, see *IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent*.

From the NetView Web application, you can retrieve performance data for a TCP/IP connection from OMEGAMON for Networks using the Tivoli Enterprise Monitoring Server Web Services interface.

The Z NetView Enterprise Management Agent provides linking to OMEGAMON product workspaces. If you have the appropriate OMEGAMON products installed and configured, the links between the Z NetView Enterprise Management Agent workspaces and the OMEGAMON workspaces are operable. For more information, see *IBM Z NetView Installation: Configuring the NetView Enterprise Management Agent*.

IBM Tivoli Change and Configuration Management Database

A Z NetView Discovery Library Adapter (DLA) for TCP/IP extracts data about TCP/IP and sysplex resources from the RODM data cache and sends the managed resource information to IBM Tivoli Change and Configuration Management Database (IBM Tivoli CCMDB) to be stored in the configuration management database (CMDB). Source data for distributed and zSeries TCP/IP resources and connections is collected by the MultiSystem Manager IBM Tivoli Network Manager agent. Data for sysplex resources is collected by the NetView discovery manager.

NetView TCP/IP data in the configuration management database enables applications such as IBM Tivoli CCMDB to locate resources discovered by other providers, such as an FTP server in a TCP/IP network. Operators and network analysts can use this resource correlation to solve outages and improve configuration and change management. Of particular value is the correlation between TCP/IP z/OS resources that are discovered by the Z NetView program and z/OS resources that are discovered by the z/OS and z/VM® DLAs.

For more information about the NetView DLA for TCP/IP, see "Enabling the IBM Z NetView Discovery Library Adapter" on page 112.

IBM System z Advanced Workload Analysis Reporter

The Z NetView product can be integrated with IBM System z Advanced Workload Analysis Reporter (IBM zAware).

IBM zAware provides analytical data for z/OS systems that can be used to identify unusual or unexpected activity or problems. IBM zAware analyzes message traffic and assigns a number value to messages, known as an *anomaly score*, that indicate to what extent (if any) the messages are outside the normal patterns of operation. Messages that fall outside normal patterns have higher anomaly scores. IBM zAware organizes anomaly scores by 10-minute intervals. For detailed information about IBM zAware, see the *IBM System z Advanced Workload Analysis Reporter (IBM zAware) Guide*, SC27-2623.

The NetView program provides sample command lists (CLISTs) that you can use as a model interface to the IBM zAware API. With these CLISTs, you can take advantage of NetView functions, such as automation, to process information from the server. You can then perform tasks such as generating events when unusual activity is detected. The CLISTs query the IBM zAware server and generate multiline messages, which are logged and sent to NetView automation.

These CLISTs are distributed in the CNMSAMP sample. The CLISTs, including the names of the corresponding CNMSAMP members, are as follows:

ZAIGET (CNMS8050)
: Communicates with the IBM zAware server, over the server API, by sending queries and getting responses back. Successful query results are stored in a NetView *safe*, which is a named area for messages that is associated with a group of CLISTs. This CLIST is intended to be called from a higher-level CLIST, such as the ZAIPROC CLIST.

ZAIPROC (CNMS8051)
Calls the ZAIGET CLIST, retrieves data from the NetView safe, and formats the retrieved data into multiline messages that are presented to the NetView product. This CLIST builds the query for the IBM zAware server that is passed to the ZAIGET CLIST. The caller can specify the sysplex and LPAR and the 10-minute interval for which data is to be retrieved. This CLIST can be called directly or can be driven by the ZAITIMER CLIST.

ZAITIMER (CNMS8052)
Calls the ZAIPROC CLIST, specifying the most recent complete 10-minute interval for which data is to be retrieved. This CLIST checks for unusual conditions in the specified interval. This CLIST is intended to be driven every 10 minutes by using the NetView CHRON or EVERY command.

IBM zAware configuration

To enable integration of the NetView program with IBM zAware, you must configure the Secure Socket Layer and the NetView program.

The IBM zAware server API is a TCP/IP interface that requires Secure Socket Layer (SSL) protocol. To implement SSL, configure the z/OS AT-TLS support. For information about configuring AT-TLS support, see the *z/OS Communications Server: IP Configuration Guide*.

Configure the NetView program as follows:

- Place the CLISTs in a data set in the NetView DSICLD DD concatenation. The CLISTs can then be called from the NetView command environment.
- The ZAIGET CLIST must know the name of the TCP stack on which it is running and whether to use IPv4 or IPv6 sockets. The default TCP name is TCPIP, and the default stack type is IPv4. You can change the defaults in one of the following two ways:
  - In the ZAIGET CLIST, find the following section of code:

    ```
    /* Initialization of variables                                */
    /**************************************************************/
    rtrncod = 0
    traceme = ''
    tcpName = 'TCPIP'
    AF = 'INET'
    closestream = 'NO'
    data.0 = 0
    xdata.0 = 0
    ```

    Change the value of `tcpName` to the name of your stack. To use IPv6, change the value of `AF` to `INET6` (or, for IPv4, leave the value as `INET`).
  - In the ZAIPROC CLIST, find the ZAIGET call statement:

    ```
    call zaiget 'userid='apiUser 'password='apiPass
    ```

    Add the `TCPNAME=`*`stackname`* and `AF=`*`stacktype`* parameters to this statement to specify the stack name and type.

IBM zAware message output

The CLISTs issue two key messages.

For information about these and other messages that are issued by these CLISTs, see the ZAIPROC sample.

- A ZAI0001I message formats output from an INTERVAL request to the IBM zAware server for a specified interval and LPAR.
- A ZAI0005I message contains output from an LPAR request that summarizes anomaly scores for all the 10-minute intervals for a specified date and LPAR.

Messages that are generated by these samples can be viewed in the NetView consolidated log using the BROWSE CANZLOG command, for example, as shown in Figure 21 on page 215. You can use these messages to perform problem determination by looking at other system messages that were issued before the interval for which the ZAI00001I message was generated.

```
Canzlog  TAG=(NVMSG,MVSMSG,DOM)                  09/12/19 16:07:53 -- 16:10:08
16:07:53 ZAI0001I Interval Results.
          System  :                  UTCPLXSB-SQ0
          Interval:       2019-09-12T09:00:00.000Z
          Anomaly :                          99.6
         -----------------------------------------------------------------------
         Anomaly   Message   Count       Cluster         Contribution  Rarity
         0.999000    ING815I   3      UNCLUSTERED         11.511       87.0
         0.999000  IOSHC112E   2      UNCLUSTERED          11.61       75.0
         0.994000    ING819I   4      UNCLUSTERED          5.368       17.0
         0.992000   HZS0002E   2      UNCLUSTERED          4.938       21.0
         0.979000    IRR010I   3      UNCLUSTERED          3.906       27.0
         0.978000  ICH70001I   2      UNCLUSTERED          3.84        23.0
         0.977000    IEF285I   1      UNCLUSTERED          3.828        3.0
         0.975000    AOF570I   1      UNCLUSTERED          3.717        5.0
         0.970000    IEA989I   1      UNCLUSTERED          3.527        4.0
                 -------------------- END OF DATA --------------------
16:08:07 CANZLOG
16:08:52 DOM TCB
16:08:52 DOM ASID
16:08:56 DOM SMID
16:10:08 CANZLOG
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>
```

Figure 21. Example ZAI0001I message

You can also automate these messages by using NetView automation functions. You can use the NetView automation table to generate events for messages or intervals with high anomaly scores.

The following table provides the mapping between the ZAI message attributes and the IBM zAware XML elements:

Table 38. ZAI message attributes

ZAI message attribute name	XML element name	Description
System	sys_id	Indicates the name of the system that was specified on the INTERVAL request, and the name of the sysplex to which the system belongs.
Interval	start_time	Indicates the beginning of the first interval for which data is available for the specified system on the date in the LPAR request. The start time is provided in the XML dateTime data type format in Coordinated Universal Time (UTC), as follows: *YYYY-MM-DD*T*hh:mm:ss.ttt*Z

Table 38. ZAI message attributes (continued)		
ZAI message attribute name	**XML element name**	**Description**
Anomaly (summary in the message header)	Anomaly_score	Indicates the anomaly score for this interval. The interval anomaly score is the percentile of the sum of the anomaly scores for the individual messages that were issued within an interval. The possible interval anomaly scores are as follows: **0 - 99.4** The interval contains messages and message clusters that match or exhibit relatively insignificant differences in expected behavior, as defined in the model. A score of 0 indicates intervals that exhibit no difference from the system model. Intervals with scores that are greater than 0 but less than 99.5 contain some messages that are unexpected or issued out of context. Scores in this range indicate intervals that do not vary significantly from the system model. **99.5** The interval contains some rarely seen, unexpected, or out-of-context messages. This score indicates intervals that have some differences from the system model but do not contain messages of much diagnostic value. **99.6 - 100** The interval contains rarely seen messages (these messages appear in the model only once or twice), or many messages that are unexpected or issued out of context. This score indicates intervals with more differences from the system model; these intervals can contain messages that might help in diagnosing anomalous system behavior. **101** The interval contains messages that merit investigation, such as the following: • Unique message IDs that were not detected previously in the model • Unusual or unexpected messages • Messages that are defined as critical by the IBM rules • A much higher volume of messages than expected
Anomaly (column)	anomaly	Indicates the rarity of this message ID within the selected interval. The anomaly score is a combination of the interval contribution score for this message and the rule, if any, that is in effect for this message. Higher scores indicate greater anomaly, so messages with high anomaly scores are more likely to indicate a problem.
Message	message_id	Indicates the message number, such as ING8151 and IOSHC112E.
Count	num_instances	Indicates the number of times that this message was issued within this 10-minute interval.

Table 38. ZAI message attributes (continued)

ZAI message attribute name	XML element name	Description
Rarity	bernoulli	Indicates how frequently the message ID is issued within a sampled set of 10-minute intervals in the system model. The value range is 1 - 101, where lower numbers indicate a higher frequency and higher numbers indicate a lower frequency, as follows: • A value of 1 indicates that the message is issued in almost all intervals in the model. • A value of 100 indicates that the message is issued in almost none of the intervals in the model. • A value of 101 indicates that this message ID was not issued in any interval in the model.
Contribution	Poisson	Indicates how closely the message ID distribution in current data matches the Poisson distribution of that message ID in data during the training period for the system model. This value is provided only for message IDs that are not part of a cluster. The higher the Poisson value, the greater the difference from expected behavior.
Cluster	cluster_status	Indicates whether this message is part of an expected pattern of messages that are associated with a routine system event (for example, starting a subsystem or workload). These patterns or groups are called *clusters*. A message that is issued out of context (without the other messages in the same cluster) might indicate a problem. The values are as follows: **New** This message has not previously been detected in the model. **In context** This message was issued as expected, within a cluster to which this message belongs. **Out of context** This message is expected to be issued as part of a specific cluster but was issued in a different context. **UNCLUSTERED** This message is not part of a defined cluster.

Calling the ZAIGET sample

You can write a CLIST to call the ZAIGET sample to retrieve data from the IBM zAware server.

You can write a CLIST to query the IBM zAware server and parse the returned XML data. To do that, call the ZAIGET sample from your CLIST to retrieve data from the IBM zAware server. The XML response to the query is placed in a safe that is named ZAIGET; this safe is available while the REXX environment that calls the ZAIGET CLIST is available. The data can be retrieved from the safe by using the NetView PIPE SAFE stage.

The ZAIPROC CLIST sample shows how to call the ZAIGET sample and parse the returned XML data. The format of the call is as follows:

```
call zaiget 'userid='user 'password='pw 'tcpname='stackname 'af='stacktype 'url='url
```

The ZAIGET keywords are not case sensitive, but the values that are specified on the keywords are case sensitive.

AF

Optional: Indicates whether IPv4 or IPv6 sockets are to be used. A value of `INET` indicates IPv4, which is the default value. A value of `INET6` indicates IPv6.

PASSWORD

Specifies the password for the user ID that is authorized for the IBM zAware server. This parameter is required.

TCPNAME

Indicates the name of the TCP stack on which the ZAIGET sample is running. This parameter is required.

URL

Specifies the HTTP GET request that includes the query parameters for the IBM zAware query. If the URL parameter is not specified, the ZAIGET CLIST looks for a URI in the ZAWARE.URL task global variable. If the task global variable is not set, this parameter is required. The format of the URI is as follows:

```
GET https://server_ip_address/zAware/authuser/Analysis?reqtype=request_type
  &intervalid=interval_id&time=time&plexname=sysplex_name&LPARname=system_name
```

For detailed information about the GET request, see the *IBM System z Advanced Workload Analysis Reporter (IBM zAware) Guide*.

USERID

Specifies the user ID that is authorized for the IBM zAware server. This parameter is required.

Chapter 13. Installing the Globalization Feature

If you ordered this feature, follow the steps in this chapter to perform these tasks:

- Install a globalization feature
- Create translated messages

Note: If you use REXX in the NetView environment, the language specified for TSO/E REXX must be compatible with the language specified for the NetView program.

Installing a Globalization Feature

To install a globalization feature:

1. Load the globalization feature from the distribution tape following the instructions in the NetView program directory.
2. If you customized or added messages, add a %INCLUDE statement for your customized member at the beginning of the CNMTRMSG member, or move your translations into the CNMTRUSR member.
3. If you are migrating from a previous release of the NetView program and you customized messages, or you want to modify the vV6.3.0 messages to reflect your customization, see "Creating Translated Messages" on page 220 for more information about how to modify NetView messages.
4. Use the following statement in the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

   ```
   transTbl = DSIKANJI
   ```

   The NetView program supports EBCDIC or Kanji character sets. All the NetView workstations in the domain must support the character set you decide to use. Multilingual support is not available. Coexistence of Japanese and English domains is allowed. The NetView program sends messages in English between these domains.

   The system console supports only the EBCDIC character set. So, do not generate Japanese messages or commands that are sent to the system console in user-written command lists, command processors, installation exit routines, or subtasks. This restriction also applies to messages sent to the NetView program through the subsystem interface. The NetView program does not support Japanese characters in its command strings, regardless of origin.

   If you are using REXX in the NetView environment, the language specified for TSO/E REXX must be compatible with the language specified for the NetView program.

   To enable the printing of a non-EBCDIC character set, CNMPRT (CNMSJM04) must have a TRANSTBL statement that specifies the same module that is specified in the TRANSTBL statement in the CNMSTYLE member.
5. To begin the Japanese-support automatically when the NetView program is started, use the following statement in the CNMSTUSR or C*xx*STGEN member, and remove the asterisk (*) from the beginning of the statement. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

   ```
    transMember = CNMTRMSG
   ```

   In the CNMTRMSG member, uncomment the CNMMSJPN member in the following way:


```
"*  %INCLUDE CNMMSJPN"
to
"%INCLUDE CNMMSJPN"
```

Note: If the TRANSMSG statement is not included in the CNMSTYLE member, a NetView operator can issue the command to initiate message translation.

6. Add the 939 code page to the GMFHS data model DUIFSTRC.

```
Global_NLS_Parameters_Class       MANAGED OBJECT CLASS;
  PARENT IS Presentation_Services_Global_Parameters_Class;
  ATTRLIST
   CodePage                             INTEGER INIT(939);
END;
OP Global_NLS_Parameters_Class.CodePage
                                        HAS_SUBFIELD NOTIFY;
View_Parent_Class                  MANAGED OBJECT CLASS;
```

Note: You can only place characters from code pages other than 037 in the DisplayResourceName field for any data model you create within RODM. For example, you can change the following line in sample DUIFSNET:

```
"DisplayResourceName ::=[CHARVAR] 'V01LG01';"
```

You can change it in the following way:

```
"DisplayResourceName ::=[CHARVAR] 'some_other_characters';"
```

Remember to add the shift-out and shift-in characters around any DBCS characters you add.

7. To enable GMFHS to send Japanese text to a NetView management console topology console, add the following parameter to member DUIGINIT:

```
JAPANESE=ON
```

Creating Translated Messages

To create your own messages for translation:

1. Create your translation entries in the CNMTRUSR sample.

   Note: When a message is processed that has a message ID that starts with an asterisk (*), the asterisk is ignored during table comparisons and is always copied to the first character in the translated message. For example, an entry for EZL501I is matched by message IDs EZL501I and *EZL501I, and the resulting output is the same except for the leading asterisk in the second case.

2. Uncomment the %INCLUDE statement for the CNMTRUSR sample in the CNMTRMSG sample.

   Note: To modify any messages that are supplied by IBM, copy them to the beginning of the CNMTRUSR sample, and then modify the copies. If two or more messages have the same identifier, the NetView program uses the message that occurs first in the member. The rules for writing your own message translations are listed in "Formatting of Globalization Feature Message Skeletons" on page 221.

3. Issue the following command from the command facility to do syntax checking and load the message translations:

```
TRANSMSG MEMBER=CNMTRMSG
```

4. Use the following statement in the CNMSTUSR or C*xx*STGEN member. Remove the asterisk at the beginning of the statement, and replace CNMTRMSG with the name of the DSIMSG member containing the translated messages. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

```
transMember = CNMTRMSG
```

   This automatically loads the message translations the next time you start the NetView program.

Messages issued during normal operation are translated as specified in the loaded translation member.

Formatting of Globalization Feature Message Skeletons

You can write your own message translations for any message output by DSIPSS, including all messages that can be displayed on the command facility screen. Some messages that are displayed on full-screen panels cannot be translated. The following rules are used to define single-byte character set (SBCS) and double-byte character set (DBCS) message translations in a member of DSIMSG:

- Each message translation is defined only once. If you define it more than once, the first copy is used for translation.
- The message translations are saved in 72-byte records.
- To code a comment, code an asterisk (*) at column 1 of a record.
- Each message translation contains a message identifier and a message text.
- The message identifier (*msgid*) is the first token delimited by a blank in the translation. The *msgid* is divided into *msgid1* and optionally *msgid2*. For single line messages, or for lines of a multiline message which can be uniquely identified by their first token, only *msgid1* (for example, DSI633I) is needed. Other multiline message lines can be specified as *msgid1*.*msgid2* or *msgid1*.* where *msgid1* refers to the first token of the first line of the message, *msgid2* refers to the first token of the target line, and the asterisk (*) refers to all lines of the message identified by *msgid1*. Examples can be found in sample CNMMSENU. Each identifier (*msgid1* or *msgid2*) can contain a maximum of 12 characters. The *msgid* must start in column 1 of a record.
- The message text follows the message identifier. The format of the text is shown here:

  ```
  W1 &6 W2 W3 &4 and so on
  ```

  Where:

  W*x*
  Is globalization feature text.

  &*n*
  Is the message insert substituted from the corresponding token of the English message, which can be qualified, as described below.

  &*n* represents the *n*th token of the English message as described in "Counting English Message Inserts for Globalization Feature Message Skeletons" on page 222. If the specified insert number does not have a corresponding token in the English message, the value is null. Valid insert numbers are 1–128.
- You can specify a single token or a token range. Specify a token range by putting a dash (-) after the first token number followed by the second token number. This indicates that the specified range is to be placed into the translated text, including all blanks which precede each token in the range. Specifying a range with only one token (for example &5-5) places that token into the translated text with all its leading blanks. Omitting the range specification, (for example &5) causes leading blanks to be dropped. The special token E means "to the end". For example, &6-E specifies the sixth token (with leading blanks) to the end of the message.
- You can use an optional length field with a message insert number through the notation * in the following way:

  ```
  &n*m
  ```

  Where:

  n
  Is the insert number (or range).

  m
  Is the length of the insert value to be displayed. The valid length ranges from 1–99. If the actual token is longer than *m*, the token is truncated. If it is shorter, the token is padded with blanks.

You can use this notation for column alignment.

- Dates and times in a message token or token range can be translated into the format which is customized by the DEFAULTS or OVERRIDE command. To do this, specify the input date or time format enclosed in apostrophes after the length above, or if the length is not included, after the token number and range. For information about these formats, refer to the online help for the DEFAULTS and OVERRIDE commands. The NetView program scans the token or tokens in the message for a set of characters which matches this format. If a match is found, the matching text is replaced by the customized format, using the same number of characters that were in the original message. If more characters are required or if no match is found, the token or tokens are inserted unchanged. Examples can be found in sample CNMMSENU.
- The delimiter following the message insert must be either a blank or a period (.). A period concatenates the value of the insert and the following text. For example, if you specify &3 and the third token of the message is ABC, then `&3.DEF` defined in a translation is represented as ABCDEF. If you specify `&3 DEF` it is represented as ABC DEF.
- If a message translation is longer than one line, the continuation lines must start in column 2. The data in these lines, excluding the blanks following the last nonblank character in the preceding line, are concatenated. If the text is a DBCS and concatenation results in a shift-in character followed by a shift-out character, the redundant shift-in/shift-out is removed.
- Translated regular/HELD/REPLY messages cannot exceed 256 bytes, even with the English tokens inserted.
- Translated immediate messages cannot exceed 10 characters less than the screen width. For example, if you have a 24 x 80 character screen, immediate messages cannot be longer than 70 characters. Be sure to code your immediate messages based on the smallest terminal screen in your network.
- No individual line of a translated MLWTO message can exceed the width of a screen. Be sure to code your MLWTO messages based on the smallest terminal screen in your network. The characters exceeding the limits are truncated, and, for a DBCS string, a shift-in character is added where appropriate.

Counting English Message Inserts for Globalization Feature Message Skeletons

Command facility English messages are constructed from predefined message words and dynamically assigned (by the message building modules) message inserts. Only the predefined message text can be translated. If a globalization feature message translation contains English message inserts, positions of these inserts in the globalization feature message translation are shown by placing the token numbers of these inserts from the English message at the places where they are displayed. The token numbers of the message inserts of the English message can be determined according to the following rules:

- The blank, comma, single quotation mark, and quoted string (a string enclosed by quotation marks, as defined in the following items) delimiters are used to separate the message into tokens. The message identifier is the first token. Each word of the message text as delimited by blanks, commas, quotation marks, and quoted strings is a separate token.
- A blank followed by a comma is interpreted as a token and uses a token number.
- A character that is not a delimiter, followed by a comma and then a blank, is interpreted as two spaces and does not use a token number.
- A quoted string begins with a single quotation mark that follows a delimiter (blank, comma, or quotation mark). If a single quotation mark is found outside a quoted string, it is considered to be a delimiter.
- A quoted string ends with a single quotation mark followed by one of these characters:
  - Blank ( )
  - Colon (:)
  - Comma (,)
  - Exclamation point (!)
  - Period (.)

- Question mark (?)
- Semicolon (;)
- Character X

Note: If a single quotation mark is found within a quoted string (and is not followed by one of these characters), the quotation mark is interpreted as part of the quoted string.

- All words in a quoted string are interpreted as a single token and use one token number.

Use care in counting tokens to distinguish between quotation marks as ordinary delimiters and as quoted string delimiters. For example, *X'03'*, contains 3 tokens - *x, 03,* and a null token; whereas, *'03'* contains only 1 token, *03*, because it is a quoted string.

The following examples show the token numbers of message inserts in command facility English messages and how to put the token numbers into the translated globalization feature message translations.

1. `DSI422I SENSE CODE = X'code' REASON = error_message_text`

   Where the *error_message_text* can contain a maximum of four tokens.

   &1 : DSI422I
   &2 : SENSE
   &3 : CODE
   &4 : =
   &5 : X
   &6 : code
   &7 : REASON
   &8 : =
   &9 : 1st token of the error message
   &10: 2nd token of the error message
   &11: 3rd token of the error message
   &12: 4th token of the error message

   The message variable *code* has token number &6 and the string insert *error_message_text* has the token numbers &9 and beyond. The message translation can be:

   ```
   DSI422I <ABC  DEF> = &9 &10 &11 &12 <GHIJ> = X'&6.'
   ```

   where:

   < Shows shift-out character
   > Shows shift-in character

   Suppose the English message issued is:

   ```
   DSI422I SENSE CODE = X'00000014' REASON = INVALID STATION
   ```

   The following translated message might be displayed on the operator screen:

   ```
   DSI422I <ABC  DEF> = INVALID STATION   <GHIJ> = X'00000014'
   ```

2. `DSI198I 'command' COMMAND NOT ALLOWED TO RUN UNDER` *`tasktype`* `TASK`

   This list shows the tokens:

   &1 : DSI198I
   &2 : command
   &3 : COMMAND
   &4 : NOT
   &5 : ALLOWED
   &6 : TO

&7 : RUN
&8 : UNDER
&9 : tasktype
&10: TASK

The message translation is:

```
DSI198I <ABC> &9 <DEF> '&2.' <GHI>
```

Suppose the English message that is issued is:

```
DSI198I 'HOLD SCREEN' COMMAND NOT ALLOWED TO RUN UNDER NNT TASK
```

The following translated message is displayed on the operator screen:

```
DSI198I <ABC> NNT <DEF> 'HOLD SCREEN' <GHI>
```

Chapter 14. Preparing to Use the NetView REST Server

The IBM Z NetView Representational State Transfer (REST) Server (hereafter referred to as the server) enables users to leverage NetView capabilities through RESTful APIs. These APIs are used in IBM-provided web applications, and can be used in user-written web applications to issue NetView commands and retrieve NetView data.

The server can be run stand-alone, and if PTF UJ04218 for APAR OA59600 is applied, as an application under IBM WebSphere® Liberty (Liberty) or Apache Tomcat.

Before the IBM Z NetView RESTful APIs can be used in any of those environments, there are a few installation/configuration steps common to each environment which must be completed.

1. The IBM Z NetView REST Server requires an MVS user ID, which must also be defined as a NetView program operator ID. This ID is required to facilitate communication between the server and the NetView program. See the *IBM Z NetView Security Reference* for information about how to define this ID to the SAF product and associate it with the address space under which the server is run.
2. Unix directories must be created to contain customized files.
3. The server `application.yml` file must be customized for your environment.
4. The NetView program configuration must also be customized to facilitate communication between the server and the NetView program.

Preparing the Server Environment

Before the IBM Z NetView RESTful APIs can be used in any of the environments, prepare the server environment.

Creating Installation Specific Unix Directories

About this task

The following Unix directories are required to contain various files. These directories must have permissions that allow files within them to be modified by an administrator and the server.

Procedure

1. Create the server home directory.

   For example, `mkdir -p /u/netview/restsrvr`.
2. Create the server log directory.

   For example, `mkdir /u/netview/restsrvr/logs`.

Customizing the Server application.yml File

About this task

Complete the following steps to customize the server `application.yml` file:

Note: The `application.yml` file content is encoded in ASCII. It can be edited using the ISPF "z/OS UNIX Directory List Utility".

Procedure

1. Copy the sample server configuration file to the server home directory created in the previous topic.

   For example,

```
cp /usr/lpp/netview/v6r3/restsrvr/samples/application.yml \
 /u/netview/restsrvr/application.yml
```

2. Edit the copied `application.yml` file.

   The following list identifies configuration options that must be customized and how they affect the server. If a property is not mentioned, do not change it.

   - Zowe™ Configuration Options

     API Catalog
     A URI that points to the Zowe API Catalog. When specified, this URI enables the NetView APIs to be integrated with the Zowe API Catalog.

     Login
     A URI that points to the Zowe Login API. Similar to API Catalog, this URI is used to verify that the Zowe token provided is valid.

     Query
     A URI that points to the Zowe Query API. This URI is used in the case that you have logged into Zowe, and are providing the NetView REST Server with a valid Zowe token. This API is used to verify that the token is valid.

   - Server Options

     Port
     This option determines which port the server will run under on the system that is hosting the server. The default port number is 3275.

   - Local Configuration Options

     Logging
     Under `logging: level: root:`, you can change the logging level to ERROR, WARN, INFO, DEBUG, or TRACE. DEBUG can be used for problem determination.

Customizing the NetView Program for Communication with the Server

About this task

Complete the following steps to customize the NetView program for communication with the server:

Procedure

1. Define the autotask which processes requests from the server.

   Simply uncomment these lines in CNMSTYLE, and do not change them in any other way.

```
*function.autotask.RESTOP = AUTOREST
*AUTOTASK.?RESTOP.Console = *NONE*
*AUTOTASK.?RESTOP.InitCmd = CMDSERV NAME=EJNREST AUTHSNDR=N
```

2. Optional: Customize the MVS START command template.

   This step only applies when running the server as a stand-alone application.

   DSIPARM sample member EJNSRSTS enables you to specify a job name to be associated with the server started task.

Running the Server as a Stand-alone Application

Additional Configuration Steps

About this task

Complete the following additional configuration steps to run the server as a stand-alone application:

Procedure

1. Create the keystore directory.

   For example, `mkdir /u/netview/restsrvr/keystore`.
2. Load the keystore.

   See "Security for the NetView REST Server" in the *IBM Z NetView Security Reference* for information on how to generate the keystore.

   Note: The server `application.yml` file contains entries related to the keystore. Ensure those values are consistent with the values used when creating the keystore.
3. Customize EJNSSRST in CNMSAMP for your environment.

   Copy the sample JCL procedure EJNSSRST from the NetView sample library to a cataloged JCL procedure library, such as SYS1.PROCLIB. Customize it for your environment. The sample contains comments describing items which might need to be modified.

Starting and Stopping the NetView REST Server

Starting the Server

The NetView REST Server can be started from the NetView program by issuing a START RESTSERV=* command. When the START command completes, message DSI360I is issued, indicating that an MVS START command has been issued to start the REST Server started task. Message DSI633I is issued upon successful initialization of the started task.

Stopping the Server

The REST Server can be stopped by issuing the NetView program STOP RESTSERV=* command. This causes an MVS STOP command to be issued.

Diagnosing Startup Issues

If the REST Server fails to start after the NetView program START RESTSERV=* command indicates successful completion, helpful messages can be found in the following sources:

- The job log for the started task
- The server log file
- The MVS system log

Running the Server under Liberty

If PTF UJ04218 for APAR OA59600 is applied, the server is also distributed as a Web application ARchive (WAR) file, which can be deployed in the Liberty server.

Additional Configuration Steps

About this task

Complete the following additional configuration steps to run the server under Liberty:

Procedure

1. Copy the war file to the Liberty Apps Directory.

   ```
   cp /usr/lpp/netview/v6r3/restsrvr/lib/NetViewServer-1.0.0-exec.war \
   <Liberty_Server_Directory>/apps
   ```

   where

   Liberty_Server_Directory
   The name of the path where Liberty is installed.

2. Create a symbolic link for ejnnvcmd in the Liberty Apps Directory.

   ```
   ln -s /usr/lpp/netview/v6r3/restsrvr/bin/ejnnvcmd \
    <Liberty_Server_Directory>/ejnnvcmd
   ```

   where

   Liberty_Server_Directory
   The name of the path where Liberty is installed.

3. Add an SSL Certificate and keystore.

   Run the **securityUtility** command from the Liberty bin directory, for example:

   ```
   securityUtility createSSLCertificate \
    --server=<server_name> \
    --password=<password>
   ```

   Replace *server_name* and *password* with appropriate values.

   Be sure to keep the output as it contains entries to be added to the Liberty `server.xml` file.

4. Modify the Liberty `server.xml` file as follows:

   - In the featureManager section, ensure that a feature property specifies the value servlet-3.0 instead of jsp-2.3.
   - Add the entries specified in the output from the **securityUtility** command.
   - Ensure that the httpEndpoint property contains the host="*" attribute.
   - Add a webApplication entry for the NetView REST Server similar to

     ```
     <webApplication contextRoot="server_name"
      location="NetViewServer-1.0.0-exec.war" />
     ```

     Replace *"server_name"* with an appropriate name.

5. Add the NVRESTCONFIG environment variable to the Liberty `server.env` file.

   Add the following line:

   ```
   NVRESTCONFIG=dir/application.yml
   ```

   where

   dir/application.yml
   The directory and name of the customized NetView REST Server `application.yml` file.

6. Before the NetView REST Server is started, the Angel and Liberty Server processes need to be restarted.

Using the IBM Z NetView REST Server with Apache Tomcat

If PTF UJ04218 for APAR OA59600 is applied, the server is also distributed as a Web application ARchive (WAR) file, which can be deployed in Apache Tomcat.

Additional Configuration Steps

About this task

Complete the following steps to use the IBM Z NetView REST Server with Apache Tomcat:

Warning:

Apache Tomcat uses a logging.properties file to control the log levels. This will override the value specified in the REST Server `application.yml` file.

Caution must be exercised when specifying logging.properties with a threshold value of FINE, FINER, FINEST or ALL. Doing so has the potential of causing user credentials to be exposed.

Procedure

1. Copy the IBM Z NetView Web application ARchive (WAR) file to the Apache Tomcat webapps folder:

   ```
   cp /usr/lpp/netview/v6r3/restsrvr/lib/NetViewServer-1.0.0-exec.war \
   <apache-tomcat-directory>/webapps/
   ```

2. Create a symbolic link for ejnnvcmd to Apache's bin folder:

   ```
   ln -s /usr/lpp/netview/v6r3/restsrvr/bin/ejnnvcmd \ <apache-tomcat-
   directory>/bin/ejnnvcmd
   ```

3. Set up environment variables.

   Ensure these environment variables have been set:

   ```
   _BPXK_AUTOCVT=ON
    NVRESTCONFIG=<Rest_Server_Configuration_Directory>
   ```

   where

   Rest_Server_Configuration_Directory
   The directory that contains the REST Server `application.yml` file with write permission.

4. Edit the `server.xml` file in the conf directory as follows:

   ```
   <Connector port="<port>" keystoreType="PKCS12"
   keystorePass="<password>"
   keystoreFile="<location_of_keystore>/localhost.keystore.p12"
   sslProtocol="TLS" clientAuth="false"
   secure="true" scheme="https"
   SSLEnabled="true"
   protocol="HTTP/1.1"
   />
   ```

   where

   port
   The default port number is 8080. If 8080 is already in use, you will want to change it.

   password
   This needs to match the value specified in the Rest Server `application.yml` file.

   location_of_keystore
   This needs to match the value specified in the Rest Server `application.yml` file.

   SSLEnabled
   This ensures secure encryption and transport protocols are used for communicating on the selected port.

Starting the Server

From within the Apache bin folder, issue the command: **catalina.sh start**.

Accessing the REST Server from a Browser

Users on the same network as the server can access the NetView REST Server through a web browser. The URI format is as follows: `https://[server domain]:[REST Server Port]`.

The default REST Server port number is 3275. This URI is then followed by the page you are trying to reach, for example, `/swagger-ui.html` which will take you to a page where you can view the server's Swagger documentation. Swagger documentation provides information about how to use the APIs as well as acting as a prototyping platform where you can test to run the APIs with custom parameters.

Installing the Zowe Sample Application

After installing Zowe, copy `app.component.ts` and `app.component.html` from the samples directory to the Zowe Sample Angular app.

```
cp /usr/lpp/netview/v6r3/restsrvr/samples/zowe/NetViewSample/app.componet.* \
 <zowe-directory>/sample-angular-app/webClient/src/app/
cp /usr/lpp/netview/v6r3/restsrvr/samples/NetViewRestServer.yml \
 <zowe-directory>/components/apimediation/api-defs/
```

Before being able to see the changes in the sample application, you will need to follow Zowe's instructions for modifying applications.

For further instructions about Zowe, see the *Program Directory* for IBM Z NetView.

Note: Line 24 in `app.component.ts`:

```

BASE_URL = 'https://localhost:3275/ibm/netview/v1';
```

If the port matches and the REST Server is on the same system as Zowe, no changes are needed. Otherwise, change *localhost:3275* to the domain and port that matches the REST Server.

Adding REST Server to the Zowe API-Mediation Layer

Copy the sample `/usr/lpp/netview/v6r3/restsrvr/samples/NetViewRestServer.yml` to `/api-mediation/api-defs/NetViewRestServer.yml`.

Configuration for Automation Table Testing

The NetView REST Server supports an API call to test an automation table. This call allocates a data set with a name containing a time stamp to hold the results of the test. The data set allocation does not specify the volume on which the data set is to reside or the device type on which the data set is to be allocated. To use the API to test an automation table, ensure the system is configured to select a volume on which the data set can reside.

Chapter 15. Configuring and Enabling the NetView Program for Service Management Unite Dashboards

IBM Service Management Unite Automation V1.1.7 provides IBM Z NetView dashboards that include:

- NetView overview dashboard that displays all NetView domains accessible to an Enterprise Master NetView program
- NetView health dashboard that displays key metrics about a NetView domain and its tasks
- Automation dialogs that provide the capability to create, validate, and test automation table statements, as well as manage automation table members
- Sysplex connection distribution dashboard that displays connection distribution metrics and graphical views for DDVIPAs and ports

To view and make use of these dashboards, configuration and enablement of the following NetView functions is required:

- NetView REST Server
- Sysplex Master or Enterprise Master NetView program
- AUTOTEST function
- Distributed DVIPA Statistics

NetView REST Server

Service Management Unite Automation leverages the RESTful APIs that are provided by the NetView REST Server.

For more information about the NetView REST Server, see the following chapters:

- Chapter 14, “Preparing to Use the NetView REST Server,” on page 225
- *Using the NetView REST Server* in the *IBM Z NetView Application Programmer's Guide*

Sysplex Master or Enterprise Master NetView Program

If you want to interact with multiple NetView domains from a single Service Management Unite, you need to configure the Service Management Unite program to point to a sysplex master or enterprise master NetView program.

Sysplex Master NetView Program

A sysplex master NetView program can move between NetView programs in the sysplex when the sysplex master NetView is stopped. Consider the following conditions before choosing this configuration option:

- You will need to designate one NetView program to always be the sysplex master program.
  - This requires the XCF.RANK CNMSTYLE statement to be set to 250.

For more information about a sysplex master NetView program, see Chapter 7, “Configuring NetView Sysplex Management,” on page 151.

Enterprise Master NetView Program

An enterprise master NetView program can be a stand-alone system or be a system within a sysplex.

To configure the enterprise master NetView program, you must take the following actions:

- Enable the DISCOVERY tower
  - To minimize discovery of TCPIP resources, configure the following statement in your CNMSTYLE user member:

    DISCOVERY.NetViewOnly=Yes
    This automatically sets XCF.RANK = 0 so the NetView domain cannot ever be any type of master NetView program.
- Set XCF.RANK = 250 in the CNMSTYLE user member.
- Follow further enterprise configuration and enablement directions in Chapter 7, “Configuring NetView Sysplex Management,” on page 151.

AUTOTEST function

The AUTOTEST function requires that the DSIATOPT task be started. The NetView program does not start this task by default. To start DSIATOPT during NetView initialization, add the following statement to your CNMSTYLE user member:

```
TASK.DSIATOPT.INIT=Y
```

Distributed DVIPA Statistics

To see information in the Distributed DVIPA Statistics dashboard, the following configuration is required:

- z/OS Communications Server VIPADISTRIBUTE TCPIP profile definitions. For more information, see *z/OS Communications Server: IP Configuration Reference*.
- Enabling DVIPA Management. For more information, see “Enabling DVIPA Management” on page 132.

Appendix A. Running Multiple NetView Programs in the Same LPAR

You can run multiple NetView programs in the same logical partition (LPAR), with some or all of them controlling NetView management consoles. In this environment, all instances of the NetView program must be at the same version and release level.

One reason you might want to run two NetView programs on the same system is to divide the work in your network. One NetView program can be used to perform systems automation functions, using a combination of NetView and IBM Z Systems Automation programs. A second NetView program can be used to perform network management and network automation functions, using a combination of NetView functions, MultiSystem Manager, the SNA Topology manager, AON, and IP management.

Configuring the Two NetView Programs

To configure multiple NetView programs to run on the same LPAR, you first need to decide which is the primary NetView program. In this case, the *primary NetView* program is the one that owns the CNMI interface and other tasks that cannot be duplicated. The *secondary NetView* programs are the ones that must be configured to coexist with the primary NetView program.

Follow these steps to configure a NetView V5 or later program as a secondary NetView program:

Table 39. Steps to Configure a Secondary NetView Program

Step	Description
Define a separate subsystem name in the IEFSSNxx member of SYS1.PARMLIB for the secondary NetView program and NetView subsystem.	This subsystem name corresponds to the first four characters of the secondary NetView program and NetView subsystem procedure names. You must also add the INITRTN(DSI4LSIT) parameter to each subsystem defined for a V6 or later NetView program.
Create the SSI procedure for the secondary NetView program and modify the SSI definitions if you plan to use the SSI interface instead of extended multiple console support consoles to exchange commands or messages with MVS.	You can run with one SSI procedure. Do this step only if you want multiple SSI procedures (one for each NetView program). The first four characters of this procedure name must correspond to the subsystem name for the secondary NetView program. • Create a separate SSI address space for the secondary NetView program. • Match the version and release of each SSI procedure (for the primary and the secondary NetView program) to the NetView version and release. • Specify a unique command designator for each SSI procedure using the MVSPARM.Cmd.Designator statement in the CNMSTYLE member. The designator must be unique within a sysplex. • Specify NOPPI in the SSI startup procedure of the secondary NetView program. MVS allows only one SSI to provide the PPI function.
Allocate new VSAM databases for the secondary NetView program and RODM.	Run NetView sample CNMSJ004, changing the data set names to conform with your naming convention.

Table 39. Steps to Configure a Secondary NetView Program (continued)	
Step	**Description**
Allocate a new DSIPARM data set for the secondary NetView program.	Copy the contents of the DSIPARM data set from the primary NetView program to the DSIPARM data set for the secondary NetView program.
Create and modify the secondary NetView startup procedure (CNMSJ009).	• Change PROG=BNJLINTX to PROG=DSIMNT on the EXEC statement. • Change the VSAM and DSIPARM data set names to specify the data sets allocated for the secondary NetView program. • Assign a domain name for the secondary NetView program. The first four characters of this procedure name must correspond to the subsystem name chosen for the secondary NetView program.
Create and modify the secondary GMFHS startup procedure (CNMSJH10).	• Change the CNMPARM DD statement to specify the DSIPARM data set you created for the secondary NetView program. This data set contains the DUIGINIT initialization member for GMFHS created when the secondary NetView program is started. • Assign a domain name for GMFHS.
Create and modify the secondary RODM startup procedure (EKGXRODM).	• Change the NAME parameter to identify the secondary RODM. • Change the EKGLOGP, EKGLOGS, EKGMAST, EKGTRAN, and EKGD*nnn* DD statements to specify the new VSAM data sets allocated for the secondary NetView program. • Change the EKGCUST DD statement to specify the data set that contains the customization and initialization member for the secondary RODM. **Note:** If you are using an SAF product, such as RACF, define and authorize the user IDs used to connect to the secondary RODM. For example, you can authorize the secondary NetView program, and the DSIQTSK running in the secondary NetView program, to be able to connect to RODM. For more information, refer to *IBM Z NetView Security Reference*.
Create a secondary RODM load job (CNMSJH12).	• Specify the secondary RODMNAME. • Comment out any SNA Topology Manager data model samples (FLBTRDM*x*) so that they are not loaded into the RODM data cache.
Update DSIPARM member DUIGINIT (GMFHS initialization) to configure the secondary GMFHS.	If you are using system variables, some or all of these modifications might not be necessary. • Change RODMNAME to match the secondary RODM. • Change DOMAIN to specify the secondary NetView domain, or enter the secondary domain name as a GMFHS startup parameter. • If you are using an SAF product such as RACF, add the access *userid* (RODMID statement) used to connect to the secondary RODM.

If necessary, update your automation table for the secondary GMFHS.	To forward non-SNA alerts to a GMFHS other than the one associated with the secondary NetView program, modify the GMFHSDOM value in the following statement:
	`IF (MSUSEG(0000)¬= '' | MSUSEG(0002) ¬= '') & HIER ¬= ''`
	`   THEN EXEC (CMD('DUIFECMV GMFHSDOM=xxxxx')`
	`        ROUTE(ONE DUIFEAUT)) CONTINUE(Y);`
	where *xxxxx* is the primary NetView domain name.

Table 39. Steps to Configure a Secondary NetView Program (continued)	
Step	**Description**

Configure the secondary NetView program using the CNMSTUSR or C*xx*STGEN member. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.	• Disable the AAUTCNMI, DSIROVS, and DSIKREM tasks by changing the CNMI statement in the following way:
	`CNMI = NO`
	• Change the PPI receiver name (DEFAULTS.PPIPREFX=&NV2I.) for the CNMCALRT task to any value other than the one that is used for the primary NetView program.
	• Set the alias for the secondary GMFHS job name and procedure name that are used with CNME2101 by modifying the following statements:
	`COMMON.DUIFHNAM = GMFHS`
	`COMMON.DUIFHPRC = CNMGMFHS`
	• Set the alias for the secondary RODM job name and procedure name that are used with CNME1098 by modifying the following statements:
	`COMMON.EKGHNAM = RODM`
	`COMMON.EKGHPRC = EKGXRODM`
	• Change the domain name (DOMAIN = C&NV2I.01) to a unique value if it was not changed in the startup procedure.
	• If you are starting the NetView program by specifying SUB=MSTR, the JES joblog is allocated by default when the NetView task DSIRQJOB requests a job ID for the NetView job. If the JES joblog is not wanted, change the joblog constant.
	• Disable the SNA Topology Manager by changing the Graphics subtower statement in the following way:
	`TOWER.Graphics = *SNATM`
	• For the Z NetView Enterprise Management Agent, enable the TEMA tower on only one NetView program for each LPAR.
	Note: If you decide to divide the TEMA functions between NetView programs, the TEMA.SESSACT subtower is supported on only one NetView program for each LPAR (because of the DSIAMLUT session monitor task).
	• To limit the discovery manager function because it is being done by another NetView program on the LPAR, either turn off the DISCOVERY tower completely or specify DISCOVERY.NetViewOnly=YES in the CNMSTUSR or C*xx*STGEN member on the secondary NetView program.
	• If the RMTCMD command or the CNMTAMEL task on two NetView programs are to use TCP/IP and both are using the same TCP/IP stack, specify the ports that are to be used for these functions to avoid conflicts. The definition statements are RMTINIT.PORT and TAMEL.PORT.

Update DSIPARM member DSICNM to configure the secondary NetView program.	Insert an asterisk prior to the `0 MONIT` statement and remove the asterisk prior to the `0 SECSTAT` statement. **Note:** Only one status monitor can receive status updates from VTAM. Other status monitors are secondary status monitors and show all resources in NEVACT state.

Table 39. Steps to Configure a Secondary NetView Program (continued)	
Step	**Description**
Update DSIPARM member DSIQTSKI to configure the secondary NetView program.	• Change the CMDRCVR ID to something other than DSIQTSK if you want the secondary NetView program to have its own PPI command receiver task. • For RODM access and control, specify a valid REP statement with the RODM name and user ID for connection to RODM.
Create a new VTAM APPL definition for the secondary NetView program.	If necessary, change the PPT APPL statement from PPO to SPO. Only one NetView program can have the primary program operator (PPO) interface. Unsolicited VTAM messages are only sent to this NetView program.
If users want to issue SNMP (CNMESNMP) commands, NetView instances need to provide necessary directory and file accesses for users.	Consider whether users in multiple instances of NetView are to issue SNMP (CNMESNMP) commands and whether the SNMP commands issued in each instance of NetView are to use the same or different sets of MIB files and configuration files as those are used by SNMP commands in another NetView instance. • If NetView instances are to share a directory, such as `/var/snmp`, and configuration files, all applicable user profiles should be connected to the same existing security group with permission to create directories, search directories, and write to configuration or index files. If the security group does not exist, users need to create one. • If a directory, such as `/var/snmp`, is not to be shared with another NetView instance, another NetView instance should be configured to use a different path for persistent files. A CEEOPTS file definition (DD statement) can be used to define a separate path for persistent SNMP files, such as configuration and index files. For more information, see *IBM Z NetView Security Reference*.

NetView Task Restrictions

The VTAM and MVS products restrict how multiple NetView programs can run on the same system. Some NetView tasks are assigned unique names that cannot be changed because VTAM can only recognize one instance of that task, with the specific assigned name. The following tasks cannot be duplicated when running multiple NetView programs:

Table 40. Tasks that Cannot be Duplicated When Running Two NetView Programs	
Task	**Description**
AAUTCNMI	Only one NetView program can own the communication network management interface (CNMI). The CNMI is a VTAM interface that the NetView program and other network management products use to receive and send alerts and other information. Because you cannot rename the AAUTCNMI task, only one NetView program can activate the task. Do not have other NetView programs activate AAUTCNMI. Other NetView programs can access the data of the CNMI owner through a cross-domain session.

Task	Description
DSIAMLUT	The DSIAMLUT task is used by the NetView session monitor to receive session information from VTAM. VTAM can only recognize one DSIAMLUT task, and the task cannot be renamed. Thus, only one NetView program can activate DSIAMLUT. You can still start the session monitor on other NetView programs, but VTAM session information must be forwarded from the NetView program on which DSIAMLUT is active.
DSICRTR	VTAM can recognize only one DSICRTR task with an active APPL definition. However, you can define multiple DSICRTR tasks on the same VTAM. For the first NetView program, in the DSICRTR initialization member, code FUNCT=CNMI. For additional NetView programs, code FUNCT=OTHER in the DSICRTR initialization member. These NetView programs do not receive any information over the CNMI interface.
DSIMCAT	Use the DSIMCAT task to automate MVS and subsystem commands that are entered from any MVS console or console interface. Only one NetView program on the same system can have the DSIMCAT task active. Additional NetView programs cannot start this task.
DSIKREM	The DSIKREM task communicates with remote 3172 and 3174 consoles. Because this task uses the CNMI, it is bound by the limit of one per VTAM program. The second NetView program cannot start this task.

Table 40. Tasks that Cannot be Duplicated When Running Two NetView Programs (continued)

Any application or NetView task name that has a domain-qualified name works when running multiple NetView programs. Because each NetView program is assigned to a different domain, the fully qualified network name of each application or task (which includes the domain ID) is unique.

Using the Subsystem Router in a Sysplex Environment

If you are running multiple NetView systems or if you are defining a sysplex environment, use the ConsMask statement in the CNMSTUSR or C*xx*STGEN member to ensure that you have a unique subsystem router task name. For more information about the ConsMask statement, see the *IBM Z NetView Administration Reference*. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*.

Starting the NetView Program Before Starting JES

If you plan to start the NetView program and the SSI under the master subsystem before you start JES, the following rules apply:

- Start the PROC with the START command using the parameter SUB=MSTR.
- When you start the NetView program with the SUB=MSTR parameter, use the START TASK=DSIRQJOB command. This is needed for the SUBMIT or ALLOCATE (for INTRDR or SYSOUT files only) commands to complete successfully.
- Store the procedure in the data set SYS1.PROCLIB, not in a user PROCLIB supported by JES.
- The procedures must contain only a single job step.
- You cannot reference SYSIN, SYSOUT, or VIO data sets. If you are using the sample start procedures, comment out all references to the symbolic SOUTA=A in CNMPROC (CNMSJ009).
- JES needs to remain coded as the primary subsystem. But in the IEFSSN member for JES, code the NOSTART parameter so that MVS does not automatically start JES at initialization.
- You cannot specify AMP=AMORG on a log data set.
- After DSIRQJOB receives a job ID from JES, if JES ends abnormally or ends without notifying DSIRQJOB to release the job ID, DSIRQJOB and NetView program cannot be stopped before JES becomes active

again. If JES is ended abnormally or ended by a user from the command line, the user can use the NetView MVS Command Management to circumvent this.

These are the steps to set up MVS Command Management to stop DSIRQJOB when a command is enter to abnormally end JES (for example $PJES2,ABEND or $PJES2,TERM).

1. Activate NetView MVS Command Management. Refer to the *IBM Z NetView Automation Guide* for information about how to activate NetView MVS Command Management.
2. If a Command Inclusion List is used, ensure that either the $PJES2,ABEND command or the or $PJES2,TERM command is in the list. If a Command Exclusion List is used, make certain that the command is not excluded. If a Console Inclusion/Exclusion List is used, make certain that the console that issues the command is included (or not excluded),
3. Give authority to DSIMCAOP to issue the NetView STOP Command.
4. Modify CNMENCXY so it issues STOP TASK=DSIRQJOB when the incoming MVS command is either $ $PJES2,ABEND or $PJES2,TERM.

Appendix B. Data Collection and Display for the NetView User Interfaces

Table 41 on page 239 shows the requirements for data collection and display for the NetView user interfaces.

Table 41. NetView Data Collection and Display

Data Discovered	Collection Tower or Subtower	Event, Sampled (Sampling Interval in Seconds), or Real-time	User Interface	Display Tower or Subtower
DVIPA definition and status	DVIPA	Sampled (3600) and event	3270 session	DVIPA
			Tivoli Enterprise Portal	TEMA.DVDEF
Distributed DVIPA	DVIPA.DVTAD	Sampled (3600) and event	3270 session	DVIPA.DVTAD
			Tivoli Enterprise Portal	TEMA.DVTAD
DVIPA connection	DVIPA.DVCONN	Sampled (3600)	3270 session	DVIPA.DVCONN
			Tivoli Enterprise Portal	TEMA.DVCONN
VIPA routes	DVIPA.DVROUT	Sampled (3600) and event	3270 session	DVIPA.DVROUT
			Tivoli Enterprise Portal	TEMA.DVROUT
Distributed DVIPA connection routing	DVIPA.DVROUT	Sampled (3600)	3270 session	DVIPA.DVROUT
			Tivoli Enterprise Portal	TEMA.DVROUT
Sysplex	DISCOVERY	Event	NetView manageme nt console	GRAPHICS
Coupling facility	DISCOVERY	Event	NetView manageme nt console	GRAPHICS

Table 41. NetView Data Collection and Display (continued)				
Data Discovered	**Collection Tower or Subtower**	**Event, Sampled (Sampling Interval in Seconds), or Real-time**	**User Interface**	**Display Tower or Subtower**
z/OS image	DISCOVERY	Event	NetView manageme nt console	GRAPHICS
NetView application	DISCOVERY	Event and sampled (300)	3270 session	DISCOVERY
			NetView manageme nt console	GRAPHICS
			Tivoli Enterprise Portal	TEMA
TCP/IP stack	DISCOVERY	Event	3270 session	DISCOVERY
			NetView manageme nt console	GRAPHICS
			Tivoli Enterprise Portal	TEMA.SYSPLEX
TCP/IP subplex	DISCOVERY	Event	3270 session	DISCOVERY
			NetView manageme nt console	GRAPHICS
IP interface	DISCOVERY.INTERFACES	Sampled (3600)	3270 session	DISCOVERY.INTERFACES
			NetView manageme nt console	GRAPHICS
Telnet server and port	DISCOVERY.TELNET	Event and sampled (3600)	3270 session	DISCOVERY.TELNET
			NetView manageme nt console	GRAPHICS
			Tivoli Enterprise Portal	TEMA.TELNET

Table 41. NetView Data Collection and Display (continued)				
Data Discovered	**Collection Tower or Subtower**	**Event, Sampled (Sampling Interval in Seconds), or Real-time**	**User Interface**	**Display Tower or Subtower**
Open Systems Adapter (OSA) interface (including VLAN)[“1” on page 242]	DISCOVERY.INTERFACES.OSA	Sampled (3600)	3270 session	DISCOVERY.INTERFACES.OSA
			NetView manageme nt console	GRAPHICS
OSA[“1” on page 242]	DISCOVERY.INTERFACES.OSA	Sampled (3600)	NetView manageme nt console	GRAPHICS
OSA channel and port[“1” on page 242]	DISCOVERY.INTERFACES.OSA	Sampled (3600)	3270 session	DISCOVERY.INTERFACES.OSA
			Tivoli Enterprise Portal	TEMA.OSA
HiperSockets interface (including VLAN)[“1” on page 242]	DISCOVERY.INTERFACES. HIPERSOCKETS	Sampled (3600)	3270 session	DISCOVERY.INTERFACES. HIPERSOCKETS
			NetView manageme nt console	GRAPHICS
HiperSockets adapter[“1” on page 242]	DISCOVERY.INTERFACES. HIPERSOCKETS	Sampled (3600)	NetView manageme nt console	GRAPHICS
Interface and TCP/IP stack HiperSockets configuration [“1” on page 242]	DISCOVERY.INTERFACES. HIPERSOCKETS	Sampled (3600)	3270 session	DISCOVERY.INTERFACES. HIPERSOCKETS
			Tivoli Enterprise Portal	TEMA.HIPERSOCKETS
NetView task	TEMA.HEALTH	Sampled (30)	3270 session	
			Tivoli Enterprise Portal	TEMA.HEALTH

Table 41. NetView Data Collection and Display (continued)				
Data Discovered	**Collection Tower or Subtower**	**Event, Sampled (Sampling Interval in Seconds), or Real-time**	**User Interface**	**Display Tower or Subtower**
TCP/IP inactive connection	TCPIPCOLLECT. TCPCONNTEMA. CONINACT	Sampled (3600)	3270 session	TCPIPCOLLECT.TCPCONN
			Tivoli Enterprise Portal	TCPIPCOLLECT. TCPCONNTEMA. CONINACT
TCP/IP active connection		Real-time	3270 session	
	TEMA.CONNACT	Sampled (900)	Tivoli Enterprise Portal	TEMA.CONNACT
Session data	NLDM	Sampled (900)	3270 session	NLDM
			Tivoli Enterprise Portal	TEMA.SESSACT

Note:

1. RODM must be running for the data to be displayed.

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